

PLANT EXTRACTS MARKET

GLOBAL FORECAST TO 2027

BY TYPE (OLEORESINS, ESSENTIAL OILS, FLAVONOIDS, ALKALOIDS, CAROTENOIDS),
SOURCE (HERBS & SPICES, FRUITS & VEGETABLES, FLOWERS), APPLICATION,
FORM, AND REGION

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
APEDA	Agricultural and Processed Food Products Export Development Authority
ASE	Accelerated Solvent Extraction
CAGR	Compound Annual Growth Rate
CFIA	Canadian Food Inspection Agency
CPI	Consumer Price Index
CRN	Council of Responsible Nutrition
CU	Customs Union
DGFT	Directorate General of Foreign Trade
DNA	Deoxyribonucleic Acid
DSHEA	Dietary Supplement Health and Education Act
DTC	Direct to Consumer
EFEQ	European Federation of Essential Oils
EFSA	European Food Safety Authority
EMA	European Medicines Agency's
EU	European Union
EUFIC	The European Food Information Council
FAO	Food and Agriculture Organization
FDA	Food and Drug Administration
FDI	Foreign Direct Investments
FDR	Food and Drug Regulations Act
FSA	Food Standards Agency
FSANZ	Food Standards Australia and New Zealand
FSSA	Food Safety and Standards Act
GDP	Gross Domestic Product
HHP	High hydrostatic pressure
HMPC	The Committee on Herbal Medicinal Products
ICMR	Indian Council of Medical Research
IFIC	International Food Information Council
IIC	Integrity Ingredients Corporation

IFRA	The International Fragrance Association
ISO	International Organization for Standardization
MAE	Microwave Assisted Extraction
NHP	Natural Health Product
OECD	Economic Co-operation and Development
OTC	Over-the-Counter
PPI	Purchaser Price Index
PSE	Pressurized Solvent Extraction
R&D	Research and Development
RoW	Rest of the World
SENASA	National Food Safety and Quality Service
SFDA	State Food and Drug Administration
SFE	Supercritical Fluid Extraction
SME	Small and Medium-sized Enterprise
UAE	Ultrasound-assisted Extraction
UK	United Kingdom
UN	United Nations
US	United States
USD	United States Dollar
USDA	United States Department of Agriculture
UV	Ultraviolet
WHO	World Health Organization

1 INTRODUCTION

1.1 STUDY OBJECTIVES

MARKET INTELLIGENCE

- To determine and project the size of the plant extracts market, with respect to type, application, source, form, and regional markets, ranging from 2022 to 2027
- To identify the attractive opportunities in the market by determining the largest and fastest-growing segments across regions
- To provide detailed information about the key factors influencing the growth of the market (drivers, restraints, opportunities, and industry-specific challenges)
- To provide the regulatory framework for major countries related to the plant extracts market
- To analyze the micro-markets with respect to individual growth trends, prospects, and their contribution to the total market

COMPETITIVE INTELLIGENCE

- To identify and profile the key players in the plant extracts market
- To provide a comparative analysis of market leaders based on the following:
 - Product offerings
 - Business strategies
 - Strengths and weaknesses
 - Key financials
- To understand the competitive landscape and identify the major growth strategies adopted by players across the key regions
- To analyze the value chain and products across the key regions and their impact on the prominent market players
- To provide insights on key product innovations and investments in the global plant extracts market

1.2 MARKET DEFINITION

A plant extract is a substance or an active ingredient with desirable properties; this extract is removed from the tissue of a plant, usually by treating it with a solvent to be used for a particular purpose. Extracts may be used in various sectors of activities, including pharmaceuticals for therapeutic properties, food & beverage products for functional properties, and cosmetics with functional properties for beauty and well-being.

As defined by Alfa Chemistry, a plant extract is a substance or active substance of a desired property extracted from plant tissue by treatment for a specific purpose. Plant extracts can be used in a variety of areas, including health foods, therapeutic drugs, cosmetics, processing aids, additives - chemical alternatives, and so on.

1.3 STUDY SCOPE

FIGURE 1 MARKET SEGMENTATION



Other types include anthocyanins, saponins, tannins, and lignin.

Other applications include feed, home care, personal care, and toiletries.

Rest of the World includes South America, Africa, and the Middle East.

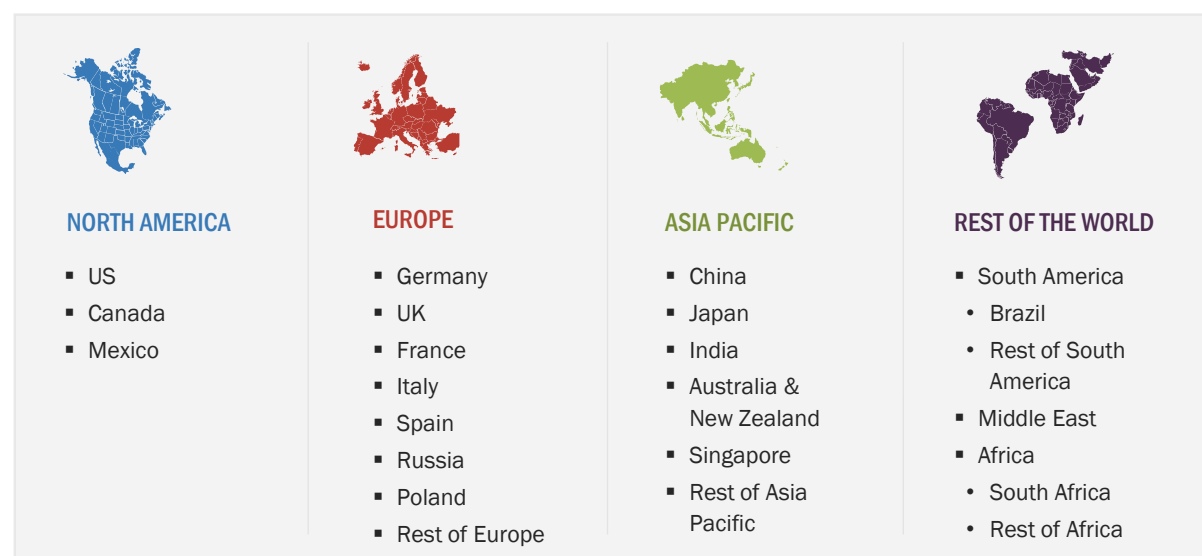
Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

1.3.1 INCLUSIONS AND EXCLUSIONS

SCOPE	INCLUSION	EXCLUSION
Type	- Oleoresins - Essential oils - Flavonoids - Alkaloids - Carotenoids - Other types	No exclusion
Source	- Fruits & vegetables - Herbs & spices - Flowers	No exclusion
Form	- Dry - Liquid	No exclusion
Application	- Food & beverages - Cosmetics - Pharmaceuticals - Dietary supplements - Other applications	Applications in textile, timber, and leather

Region	- North America - Europe - Asia Pacific - Rest of the world	No exclusion
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1.4 REGIONS COVERED



Rest of Europe includes Ireland, the Netherlands, Sweden, Turkey, Belgium, and other EU & non-EU countries.

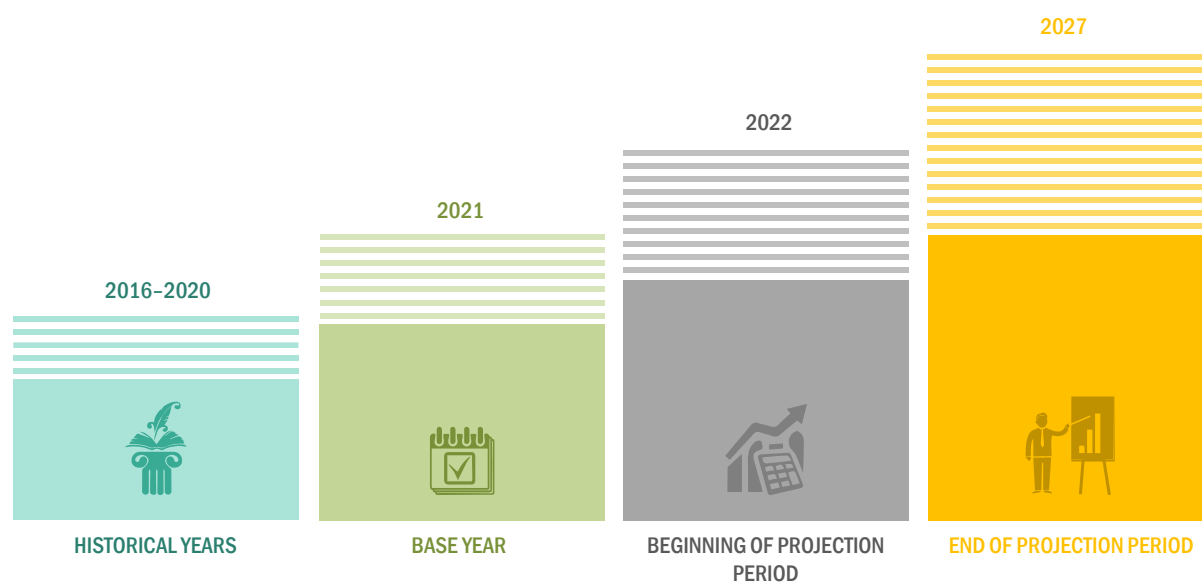
Rest of Asia Pacific includes Vietnam, Malaysia, Thailand, Indonesia, the Philippines, and South Korea.

Rest of South America includes Argentina, Chile, and Colombia.

Rest of Africa includes Egypt, Kenya, Israel, and Nigeria.

Source: Secondary Research, Primary Interviews, Industry Journals, and Related Research Publications

1.5 YEARS CONSIDERED



- The fiscal year considered for most company profiles is 2022. In case the data was unavailable for the base year for specific companies, the data from the previous year was considered.
- The period considered for studying the competitive landscape ranges from January 2018 to January 2022.
- Wherever required, up to two or three digits have been considered post the decimal point to differentiate between the market sizes of the segments.

1.6 CURRENCY CONSIDERED

The currency used to indicate the market size (in terms of value), and the company revenues in this report are US dollars.

- ISO 4217 codes are used to depict currency names; the market size is represented using USD in terms of value.
- The market has been studied in terms of USD million; the denomination is based on the size of the values, enabling uniform accommodation of these values in tables, figures, or any form of study of a particular segment.
- The revenues were taken from their annual reports/company websites for companies reporting their revenues in USD.
- For companies reporting their revenues in other currencies, the average annual currency conversion rate was used for the particular year to convert the values into USD.

TABLE 1 USD EXCHANGE RATES CONSIDERED, 2017–2021

Country/Region (Currency)	2017	2018	2019	2020	2021
Europe (EUR)	0.923	0.848	0.893	0.877	0.846
UK (GBP)	0.808	0.750	0.784	0.779	0.727
Denmark (DKK)	6.864	6.319	6.670	6.538	6.290
India (INR)	67.809	68.422	70.394	74.102	73.936
Japan (JPY)	116.667	110.424	109.008	106.725	109.817
Switzerland (CHF)	1.024	0.979	0.994	0.939	0.914

Source: IRS

1.7 STAKEHOLDERS

- Supply side: Plant extract producers, suppliers, distributors, importers, and exporters
- Demand side: Food & beverage manufacturers, cosmetics manufacturers, nutraceutical manufacturers, pharmaceutical industries, and dietary supplement manufacturers
- Regulatory bodies: Government agencies and Non-governmental Organizations (NGO)
- Commercial R&D institutions and financial institutions
- Associations, regulatory bodies, and other industry-related bodies:
 - Food and Drug Administration (FDA)
 - United States Department of Agriculture (USDA)
 - Herb Society of America
 - American Herbal Products Association
 - The International Herb Association
 - American Spice Trade Association
 - European spices Association
 - Spices Board
 - Essential Oil Association of India
 - The International Federation of Essential Oils and Aroma Trades
 - European Flavors & Fragrances Association

1.8 SUMMARY OF CHANGES

- Addition/Refinement in the market overview: Updated market-related drivers, restraints, and opportunities have been added to provide an in-depth overview of the market. Additional validated data gives better insights into the current and future trends in the market.
- Addition/Refinement in the industry trends chapter: Updated Average Selling Price (ASP) trends, key stakeholders and buying criteria, regulatory bodies and associations, trends, and disruptions impacting customers, and case study analysis have been included in the study.
- Assumptions based on the COVID-19 impact have been included while estimating the market size.
- Market segmentation based on form has been included in the report.
- Market segmentations based on source and type have been updated.
- Company financials and product portfolios have been updated in the company profiles section for 2021/2022. The micro-quadrants have been updated as well. Companies' recent developments have been updated in the report.
- The company evaluation quadrant for both key players and start-ups/SMEs has been provided (with respect to 25 companies) according to the latest company developments. Key player strategies and revenue analysis of the top five market players have been included.
- Adjacent markets for the plant extracts market have also been provided.

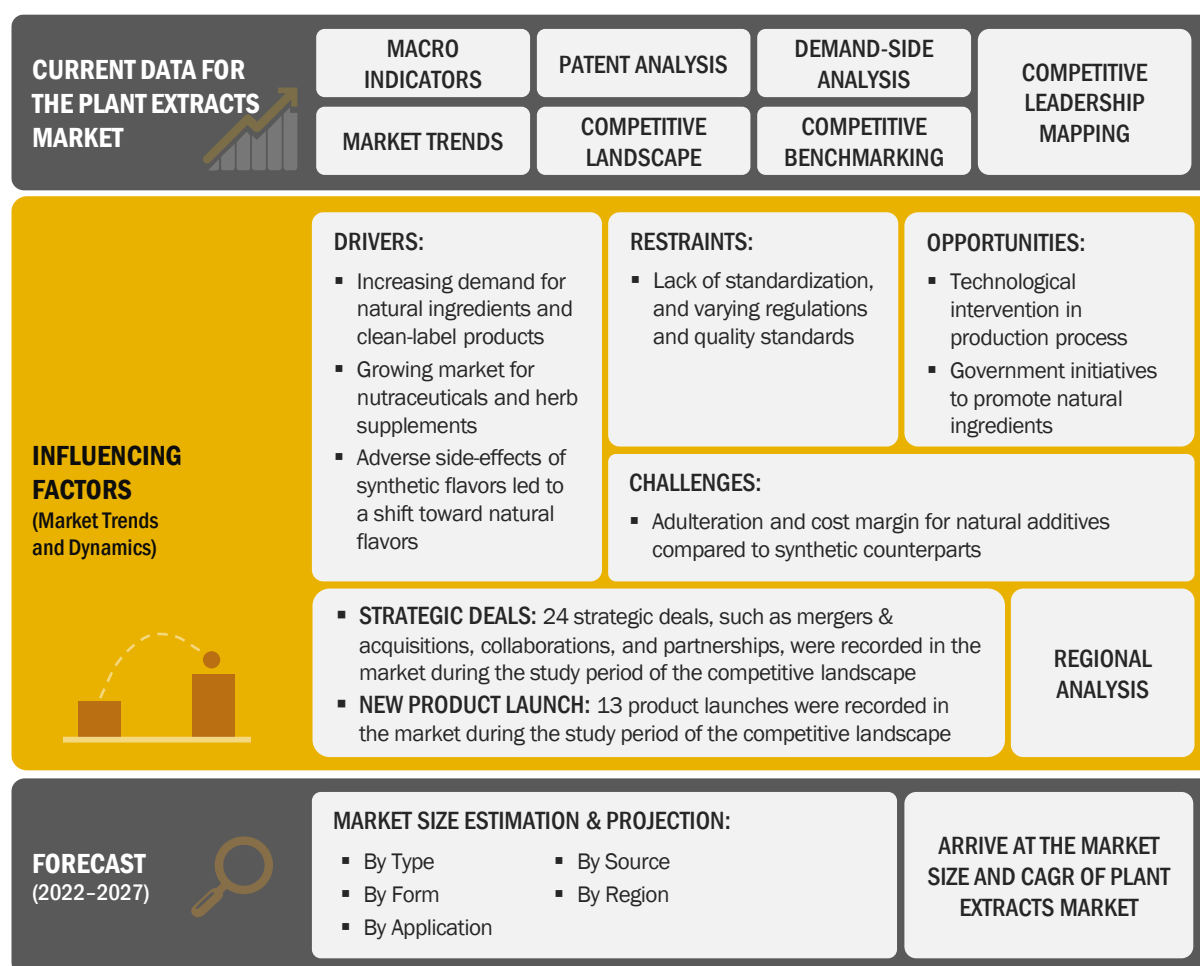
2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This research involves the extensive use of secondary sources; directories; and databases, such as the Food and Drug Administration (FDA), Herb Society of America, American Herbal Products Association, American Spice Trade Association, Essential Oils Association of India, Bloomberg, and Factiva, to identify and collect information useful for a technical, market-oriented, and commercial study of the plant extracts market. In-depth interviews were conducted with various primary respondents, such as key industry participants, Subject Matter Experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information and to assess prospects.

The following figure depicts the market research methodology applied in drafting this report on the plant extracts market.

FIGURE 2 PLANT EXTRACTS MARKET: RESEARCH DESIGN



Source: Secondary Sources, Industry Experts, Related Publications, and MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

In the secondary research process, various sources, such as the Food and Drug Administration (FDA), European Spices Association, Spices Board, European Flavors & Fragrances Association, and The International Federation of Essential Oils and Aroma Trades, were referred to identify and collect information for this study. The secondary sources also include medical journals, press releases, investor presentations of companies, whitepapers, certified publications, articles by recognized authors and regulatory bodies, trade directories, and paid databases.

Secondary research was conducted to obtain critical information about the industry's supply chain, the total pool of key players, and market classification and segmentation according to the industry trends to the bottom-most level and geographical markets. It was also used to obtain information about the key developments from a market-oriented perspective.

2.1.1.1 Key data from secondary sources

PARAMETER	SOURCE
 MARKET SIZE (Market Value)	▪ Company financials ▪ Company presentations ▪ Magazines ▪ Journals ▪ Press releases ▪ Paid databases ▪ MarketsandMarkets data repository
 COMPANY REVENUE	▪ Annual reports ▪ Company websites ▪ Public databases ▪ MarketsandMarkets data repository
 QUALITATIVE INFORMATION (Market Dynamics and Trends)	▪ Company websites ▪ Annual reports ▪ Press releases ▪ MarketsandMarkets data repository

Source: Secondary Research, Company Annual Reports, Whitepapers, Press Releases, and MarketsandMarkets Analysis

2.1.2 PRIMARY DATA

The plant extracts market comprises several stakeholders, including fruits & vegetables, and spices & herbs suppliers, ingredient suppliers, end-product manufacturers, and regulatory organizations in the supply chain. Various primary sources from both the supply and demand sides of the market were interviewed to obtain qualitative and quantitative information. Primary interviewees from the supply side include research institutions involved in R&D to introduce technology, distributors, wholesalers, importers & exporters of spices and herbs, plant extract manufacturers, and technology providers. Primary sources from the demand side include key opinion leaders, executives, vice presidents, and CEOs of food & beverage products, cosmetics, dietary supplements, and pharmaceutical manufacturing companies, through questionnaires, emails, and telephonic interviews.

2.1.2.1 Key data from primary sources

SEGMENT	MARKET DEFINITION
PLANT EXTRACTS MARKET	Plant extracts are active plant materials or secondary metabolites with high functional attributes and are majorly sourced from herbs & spices. Natural colors and phytoactives are an emerging sector.
GEOGRAPHICAL SCOPE	This includes four main regions: North America (the US, Canada, and Mexico), Europe (Germany, the UK, France, Italy, Spain, Russia, Poland, and Rest of Europe), Asia Pacific (China, India, Japan, Australia & New Zealand, Singapore, and Rest of Asia Pacific), and RoW (Brazil, Rest of South America, South Africa, Rest of Africa, and the Middle East).
TYPE	The major types covered in the study include oleoresins, essential oils, flavonoids, alkaloids, carotenoids, and other types.
FORM	The forms covered in the study include dry and liquid.
SOURCE	Based on source, the market is majorly segmented into herbs & spices, fruits & vegetables, and spices.
APPLICATION	Based on application, the market is majorly segmented into food & beverages, dietary supplements, pharmaceuticals, and cosmetics, among other applications.

2.1.2.2 Key industry insights



Innovation and new formulations have gripped the carotenoids application within the cosmetics industry. Consumers today are significantly more concerned with beauty products and a shift toward the demand for natural ingredients has resulted in an increased use of carotenoid ingredients in skin care.

- Managing Partner,
Leading Carotenoid Manufacturing Company

In terms of application, high growth is expected during the forecast period for herbal drinks. Herbal infusion beverages are gaining demand owing to the low-calorie levels, caffeine levels, and beneficial antioxidants in the drink. Ready-to-drink cans have adopted the botanical trend, offering a convenient drinking format.

- Founder and CEO,
Europe Herb-infused Beverages Company



Botanicals are gaining popularity owing to their association with sustainable credentials, natural flavors, clean-label status, and health benefits. Botanicals are associated by consumers with “premium” ingredients, adding authentic taste to the product.

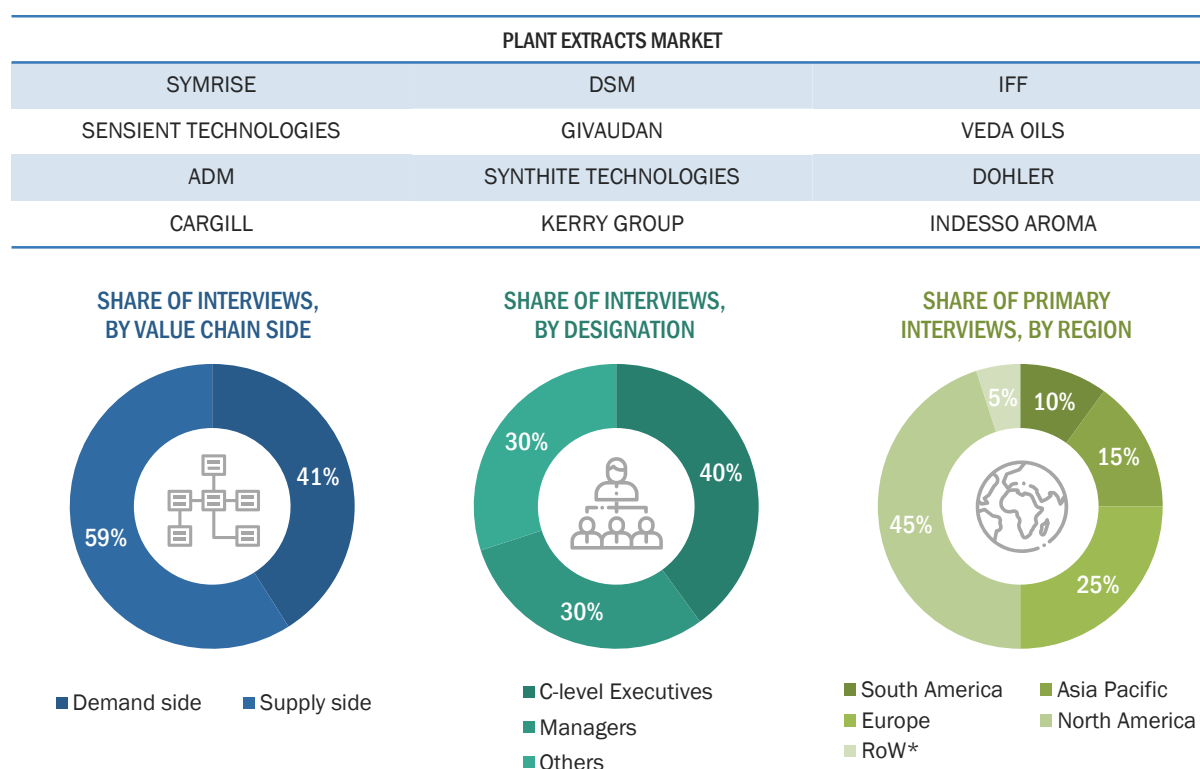
- Procurement Head,
US-based Botanicals Manufacturing Company



Source: Expert Interviews

2.1.2.3 Breakdown of primary interviews

FIGURE 3 BREAKDOWN OF PRIMARY INTERVIEWS BY COMPANY TYPE, DESIGNATION, AND REGION



*RoW includes the Middle East & Africa

2.2 MARKET SIZE ESTIMATION

Both the top-down and bottom-up approaches have been used to estimate and validate the total size of the plant extracts market. These approaches were extensively used to determine the size of the subsegments in the market. The research methodology used to estimate the market size includes the following details:

- Top-down approach:
 - The key industry and market players were identified through extensive secondary research.
 - The industry's supply chain and market size were determined through primary and secondary research.
 - All percentage share splits and breakdowns were determined using secondary sources and verified through primary sources.
 - The peer market—the essential oils and botanical extracts market—was considered to validate the market details of plant extracts.

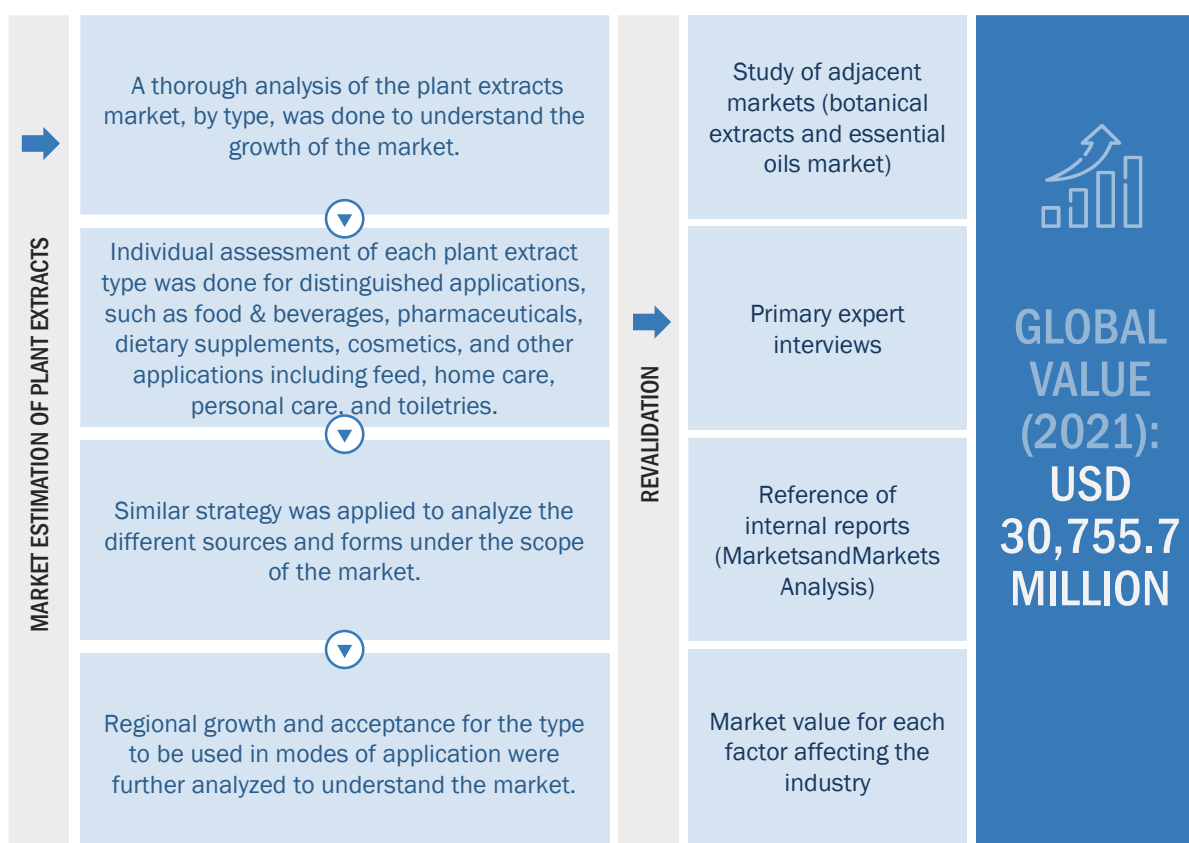
- Bottom-up approach:
 - The market size was analyzed based on the share of each type of plant extract at regional and country levels. Thus, the global market was estimated with a bottom-up approach of the type at the country level.
 - Other factors include the penetration rate of plant extracts in distinguished application sectors, such as food & beverages, dietary supplements, pharmaceuticals, cosmetics; consumer awareness; and function trends; pricing trends; the adoption rate and price factors; patents registered; and organic and inorganic growth attempts.
- All macroeconomic and microeconomic factors affecting the plant extracts market were considered while estimating the market size.
- All possible parameters that affect the market covered in this research study were accounted for, viewed in extensive detail, verified through primary research, and analyzed to obtain final quantitative and qualitative data.

The following sections (bottom-up and top-down) (supply-demand) depict the overall market size estimation process employed for this study.

FIGURE 4 PLANT EXTRACTS MARKET SIZE ESTIMATION, BY TYPE (SUPPLY-SIDE)

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Identification of the top companies, such as Symrise, Sensient Technologies, Givaudan, DSM, ADM, Cargill, and Dohler, operating in the plant extracts market	Estimation of the combined revenue of top players from the plant extracts market	Identification of the combined share of top companies in the total plant extracts market	Calculation of the total market size for the plant extracts market using the combined share of the top companies and their revenue	Revalidation of the total market size and market share estimations through primary respondents
KEY SOURCE OF INFORMATION				
Secondary sources, such as company financials, press releases, paid database (Factiva), and primary research	Secondary sources, such as company financials, press releases, paid database (Factiva), and primary research	Primary research with CEOs, VPs, marketing directors, technology, and innovation directors	MarketsandMarkets calculations	Primary research with CEOs, VPs, marketing directors, and technology and innovation directors

Source: Secondary Research, Company Annual Reports, White Papers, Press Releases, and MarketsandMarkets Analysis

FIGURE 5 PLANT EXTRACTS MARKET SIZE ESTIMATION (DEMAND-SIDE)

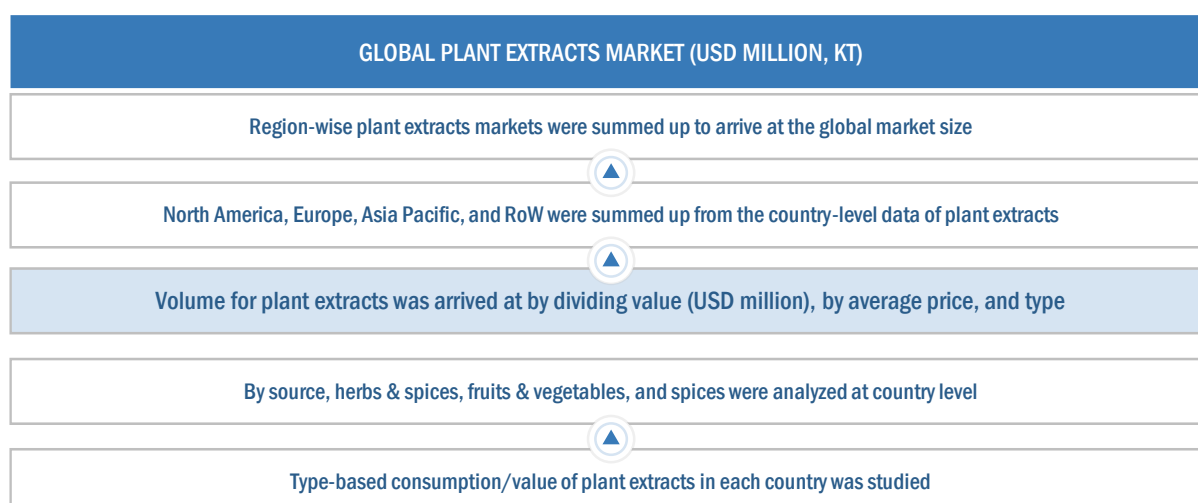
Source: Secondary Research, Company Annual Reports, White Papers, Press Releases, and MarketsandMarkets Analysis

The following figure provides an illustrative representation of the complete market size estimation process implemented in this research study for an overall estimation of the plant extracts market in a consolidated format.

2.2.1 MARKET SIZE ESTIMATION: BOTTOM-UP APPROACH

In the bottom-up approach, each country's market size for plant extracts and the types, including oleoresins, essential oils, alkaloids, carotenoids, and flavonoids, was arrived at through secondary sources, such as annual reports, investor presentations, journals, and government publications. The bottom-up procedure was also implemented on the data extracted from secondary research to validate the market segment sizes obtained.

The share of each type of plant extract as a percentage of the dietary supplements, cosmetics, food & beverages, and pharmaceuticals markets in each country was calculated from secondary sources. Country-level data for types, such as oleoresins, essential oils, flavonoids, alkaloids, and carotenoids, extracted the adoption rate of plant extracts. This was used to estimate the shares for different sources and forms in each country and across all regions. The source for each type was tracked via product mapping and studied for its penetration level to estimate the market size. Each type was studied for its commercially available form. The market size arrived at was further validated by primary respondents.

FIGURE 6 PLANT EXTRACTS MARKET SIZE ESTIMATION: BOTTOM-UP APPROACH**MARKET ESTIMATION NOTES**

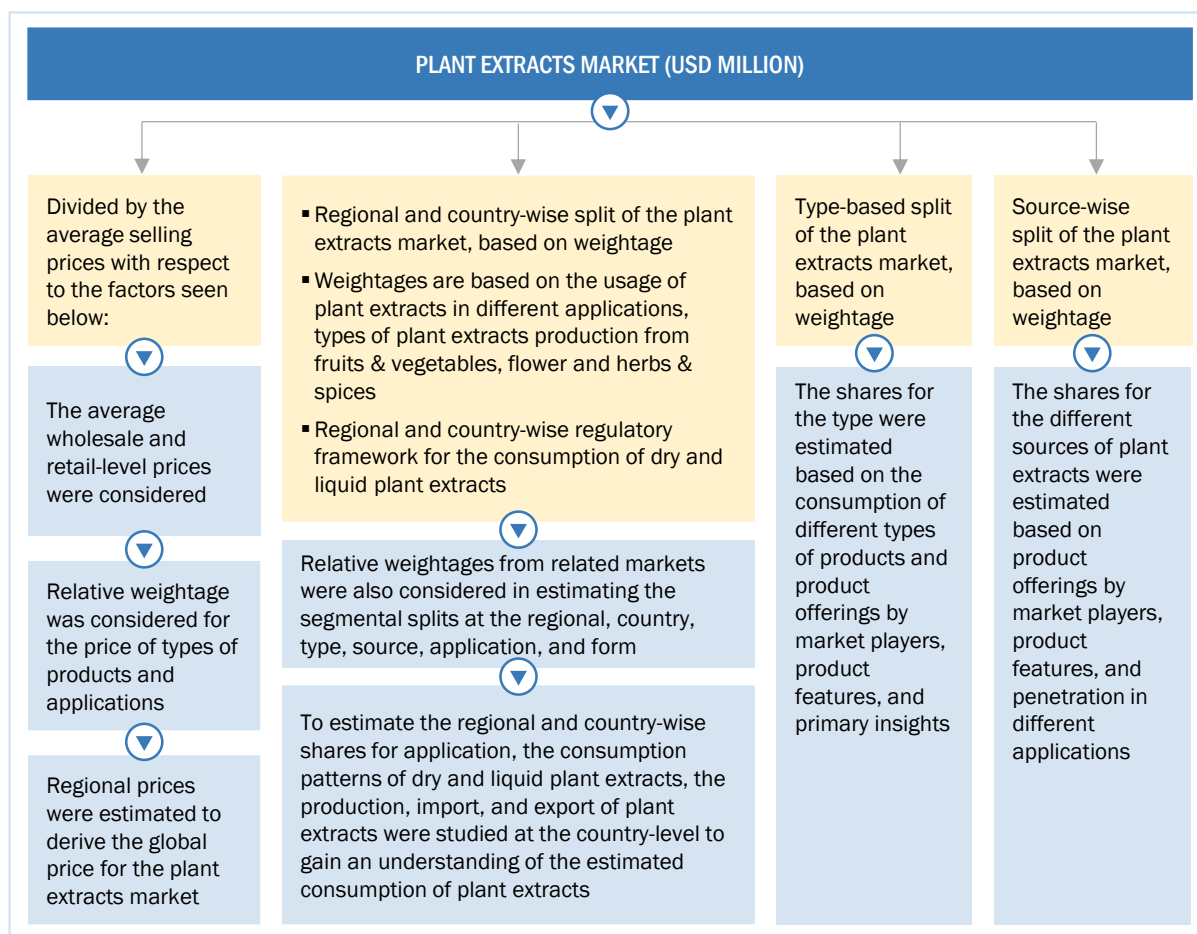
- The sales of plant extracts in each country for type was obtained from secondary sources, primaries, food & beverage associations, medical bodies, and associations and national government agencies related to nutritional supplements, cosmetics, botanical supplements, and feed.
- Prices for each type of plant extracts were obtained from secondary sources and verified through primaries.

Source: Secondary Research, Company Annual Reports, White Papers, Press Releases, and MarketsandMarkets Analysis

2.2.2 MARKET SIZE ESTIMATION: TOP-DOWN APPROACH

For the calculation of each type of specific market segment, the most appropriate immediate parent and peer market size were used for implementing the top-down procedure.

Secondary reports from the EU, Spice Board, and Food Drug and Cosmetic Act (FDCA) were considered. Further, appropriate weightage was assigned to the data derived from each parameter to arrive at the final shares for each region. The regional demand-supply trends, presence of manufacturing units, and regulatory scenario were also analyzed to validate the shares arrived at. These shares were then confirmed with primary respondents from across regions.

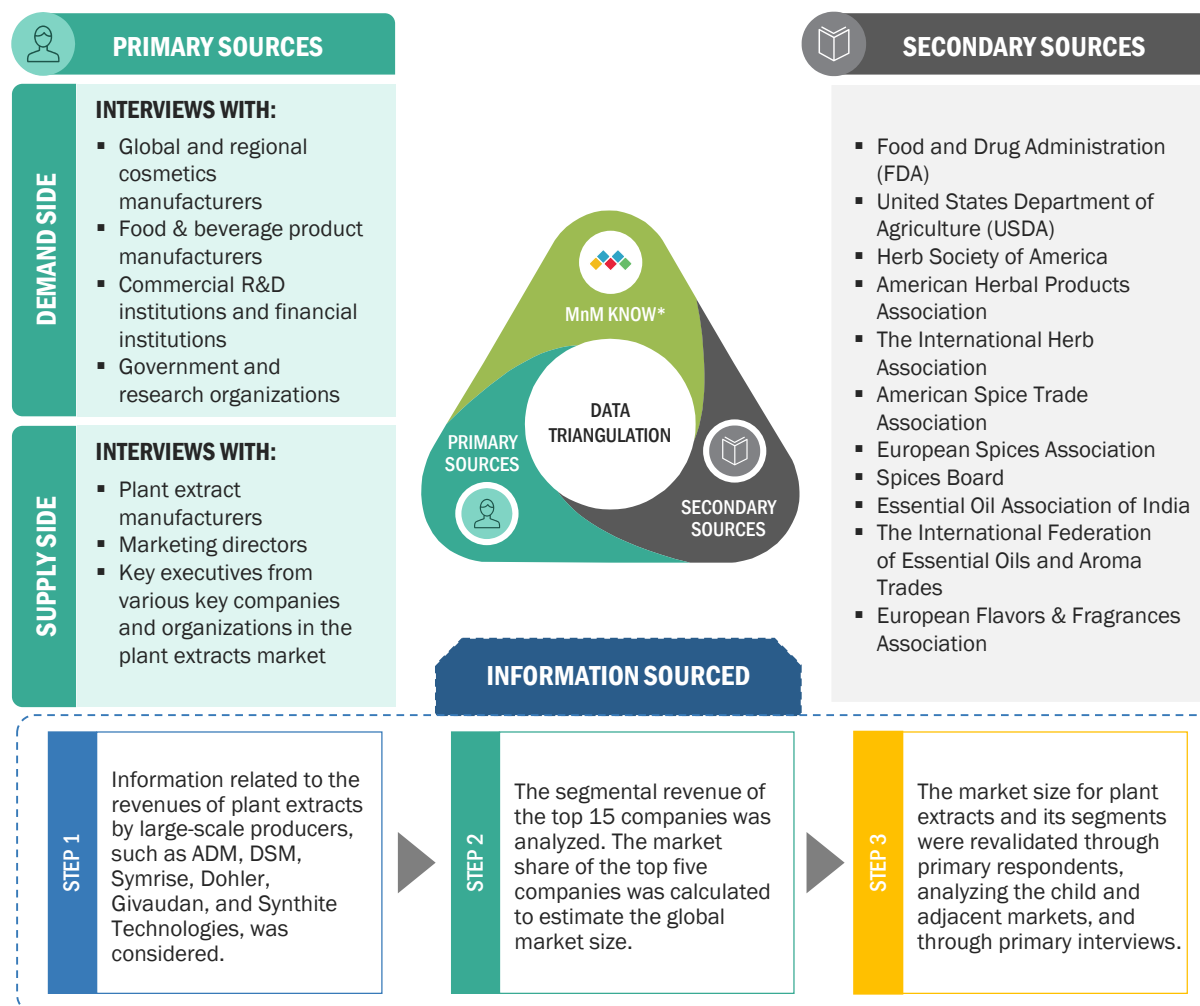
FIGURE 7 PLANT EXTRACTS MARKET SIZE ESTIMATION: TOP-DOWN APPROACH

Source: Secondary Research, Company Annual Reports, Whitepapers, Press Releases, and MarketsandMarkets Analysis

2.3 DATA TRIANGULATION

After arriving at the overall market size from the estimation process explained above, the total market was split into several segments and subsegments. The data triangulation and market breakdown procedures were employed, wherever applicable, to estimate the overall plant extracts market and arrive at the exact statistics for all segments and subsegments. The data was triangulated by studying various factors and trends from the demand and supply sides. Along with this, the market size was validated using both the top-down and bottom-up approaches.

FIGURE 8 DATA TRIANGULATION



*MnM KNOW stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledgestore," subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

Sources: Secondary Research, Company Annual Reports, White Papers, Press Releases, and MarketsandMarkets Analysis

2.4 ASSUMPTIONS

The following assumptions are made to complete the overall market engineering process of the plant extracts market:

PARAMETER	ASSUMPTION
ECONOMIC STABILITY	A positive economic climate is assumed to continue across all countries until the end of 2027.
EXCHANGE RATE	The average USD exchange rates of all foreign currencies have been considered for the respective years. Dollar fluctuations will not be significant enough to affect the forecast period to a notable extent.
POLITICAL STABILITY	Political situations are assumed to be stable throughout the projection period in regions covered in the report.
QUALITATIVE ANALYSES	The qualitative analysis conducted from the quantitative data arrived at is solely based on the understanding of the market and its trends by a team of experts involved in making this report. Regional trends and dynamics are assumed to drive the market size at regional levels.
MARKET SIZE: GROWTH	The market size estimations and growth rate projections are based on the current scenario pertaining to plant extracts. In case of limited data availability, segmental shares have been benchmarked against market trends and developments to further arrive at the final figures.
PRICING ANALYSIS	▪ The prices for distinguished products in the plant extracts market are taken at the retail level. ▪ Conversions of each unit (for oleoresins, essential oils, carotenoids, flavonoids, and alkaloids) to per tons were considered to maintain the uniformity of the data. ▪ Weighted average and a wholesale-level mark-up of 20%-30% was also considered for each type to arrive at the per unit product cost.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

2.5 RESEARCH LIMITATIONS AND ASSOCIATED RISKS

PARAMETER	RESEARCH LIMITATIONS AND ASSOCIATED RISKS
PRIMARIES, BY KEY PLAYER	Quantitative information for certain segments has been kept confidential by industry players. Hence, qualitative insights gathered during the study have been used to arrive at the market size of those subsegments.
PRIMARIES, BY REGION	In some cases, the regional market size has been derived based on weightage assigned to these markets, derived from the qualitative insights from global industry experts.
MARKET SCOPE	The quantitative information for some market segments is kept confidential by industry players. Hence, qualitative insights gathered during the study were used to arrive at the market size for certain subsegments in the Asia Pacific and RoW regions. There are a limited number of industry experts in developing markets across the African and South American regions, which provide market-related insights for subsegments. In such cases, the market size was derived based on the qualitative insights from global industry experts.

Source: Secondary Research, Company Annual Reports, White Papers, Press Releases, and MarketsandMarkets Analysis

3 EXECUTIVE SUMMARY

A plant extract is a plant substance or an active ingredient with desirable properties; this extract is removed from the plant tissues, usually by treating it with a solvent, to be used for a particular purpose. Roots, bulbs, berries, fruits, seeds, leaves, barks, and stems are among the major sources of plant extracts. Aloe vera extracts, citrus fruit extracts, herbal extracts, flower extracts, and other plant extracts have been widely used in herbal medicines, food & beverage additives, functional food & beverages, and cosmetics.

China and India are some of the leaders in producing plant extracts for various applications. The US and European countries are the major importers of plant extracts from Asian countries. Countries such as South Korea, Indonesia, Thailand, Vietnam, Australia, and New Zealand are investing extensively in the market, following the increasing demand for natural ingredients globally. However, strict regulations in developed European countries and the lack of technical specialists in developing Asian countries hamper the growth of the Asia Pacific plant extracts market.

Consumers are becoming aware of the adverse effects of consuming food & beverages processed with synthetic ingredients. Some key concerned authorities and non-profit organizations such as the FDA encourage the prohibition of using various artificial flavorings in food & beverages. The European Union (EU) has implemented stringent norms against using chemical ingredients in food products, pharmaceuticals, and dietary supplements. These norms have forced manufacturers to utilize ingredients derived from natural sources such as plant extracts. Such government regulations increase the awareness of the side-effects of synthetic flavors among customers, thus increasing the demand for natural flavors, further stimulating the plant extracts market.

As of 2021, the global plant extracts industry was valued at USD 30,755.7 million and projected to grow at a CAGR of 12.3% from 2022 to 2027 in terms of value sales. The rising consumption of nutritive and fortified food and the growth in health awareness among consumers are leading to the increased consumption of healthier diets, driving this market. Plant extracts as functional ingredients have been gaining popularity among target customers—the food & beverage industry. As a result, the rise in the demand for functional food products among health-conscious consumers eventually triggers the demand for plant extract products in the global market.

Natural plant extract-based cosmetics are becoming increasingly popular. Growing health awareness and the importance of maintaining physical appearance are fueling the global demand for plant-based natural skin care and therapeutic products. Consumers are demanding more natural alternatives for their skin, hair, and body because they believe they will be more beneficial in the long run than chemical-laden products. As a result, manufacturers of such products are increasingly incorporating plant extracts into their product lines to capitalize on the trend.

In this report, plant extracts have been studied based on type, application, form, source, and region.

TABLE 2 PLANT EXTRACTS MARKET SHARE SNAPSHOT, 2022 VS. 2027 (USD MILLION)

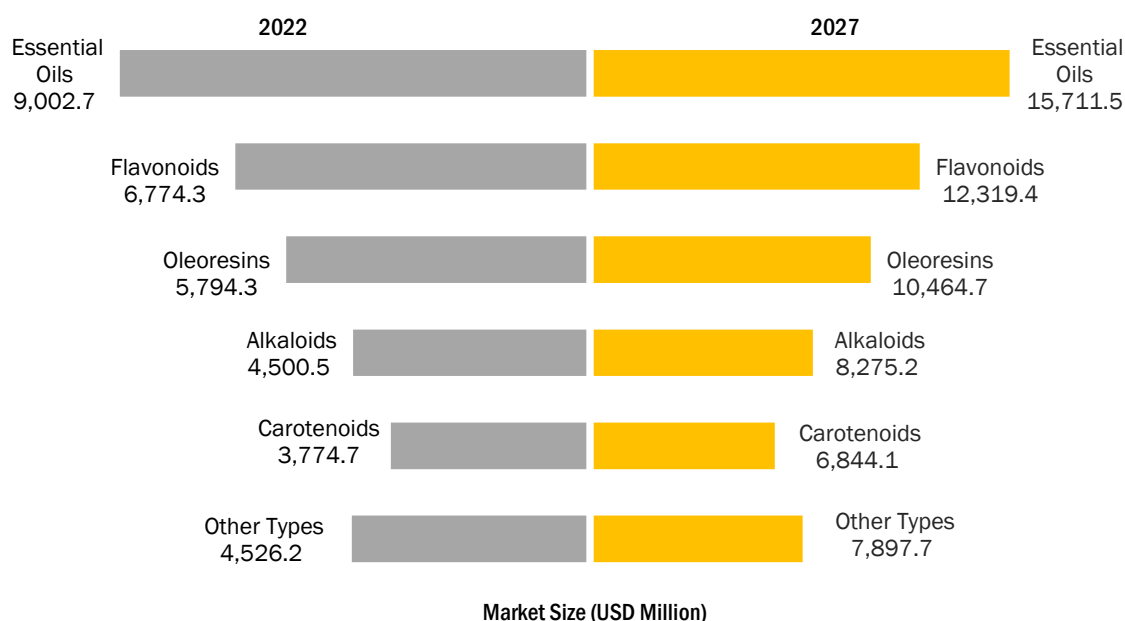
PARTICULARS	2022	2027
Global Value (USD Million)	34,372.8	61,512.6
Regional Share	- North America: 14.3% - Europe: 21.9% - Asia Pacific: 53.2% - RoW: 10.6%	- North America: 15.3% - Europe: 20.8% - Asia Pacific: 51.9% - RoW: 12.0%
Type	- Oleoresins: 16.9% - Essential Oils: 26.2% - Flavonoids: 19.7% - Alkaloids: 13.1% - Carotenoids: 11.0% - Other Types: 13.2%	- Oleoresins: 17.0% - Essential Oils: 25.5% - Flavonoids: 20.0% - Alkaloids: 13.5% - Carotenoids: 11.1% - Other Types: 12.8%
Application	- Food & Beverages: 34.0% - Pharmaceuticals: 15.2% - Dietary Supplements: 23.0% - Cosmetics: 19.2% - Other Applications: 8.7%	- Food & Beverages: 33.4% - Pharmaceuticals: 15.3% - Dietary Supplements: 23.3% - Cosmetics: 20.2% - Other Applications: 8.0%
Source	- Fruits & Vegetables: 32.0% - Herbs & Spices: 55.6% - Flowers: 12.4%	- Fruits & Vegetables: 32.2% - Herbs & Spices: 55.2% - Flowers: 12.6%
Form	- Dry: 71.8% - Liquid: 28.2%	- Dry: 72.2% - Liquid: 27.8%
Global CAGR (2022–2027)	12.3%	

Other types include anthocyanins, tannins, and lignin.

Other applications include feed, home care, personal care, and toiletries.

Rest of the World includes South America, Africa, and the Middle East.

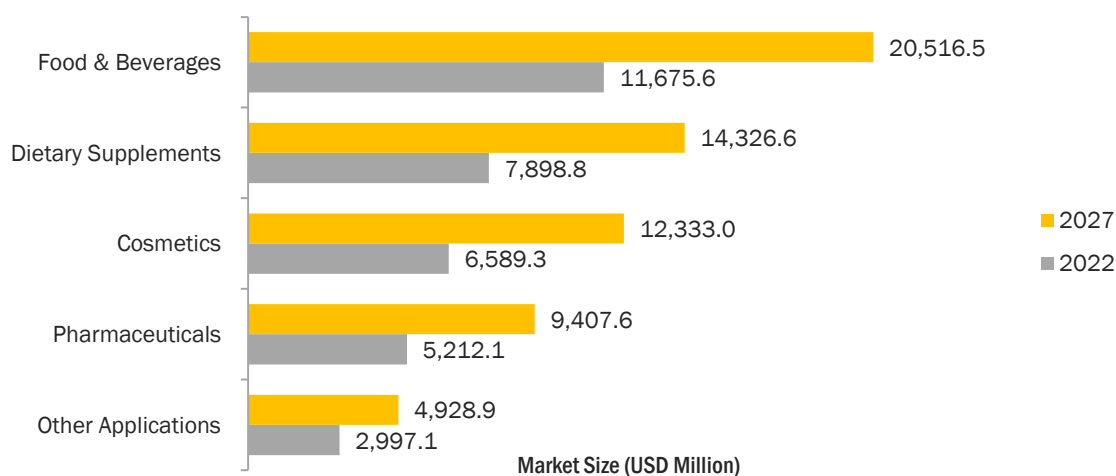
Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

FIGURE 9 PLANT EXTRACTS MARKET SIZE, BY TYPE, 2022 VS. 2027 (USD MILLION)

Other types include anthocyanins, tannins, and lignin.

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

Among types, essential oils account for a share of 26.2% in 2022. The importance of essential oils is increasing because they are mostly used in the food & beverage, cosmetics, and fragrance industries. The uses of essential oils have received increasing attention as natural additives for the shelf-life extension of food products. In Europe, sales of products that promote aromatherapy are rising. The reason is that European consumers seek more environmentally friendly and natural ways to maintain and improve their health. As a result, essential oil demand is rising in the region.

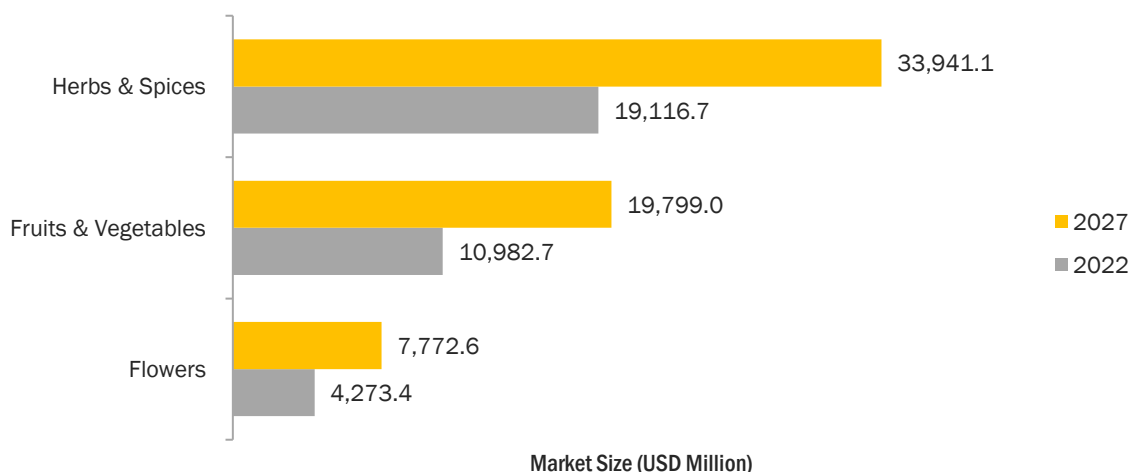
FIGURE 10 PLANT EXTRACTS MARKET SIZE, BY APPLICATION, 2022 VS. 2027 (USD MILLION)

Other applications include feed, home care, personal care, and toiletries.

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

Among applications, the food & beverages segment is estimated to account for the largest market share of 34.0% in 2022. Plant extracts are becoming increasingly important food additives due to their high concentrations of bioactive compounds. Plant and herbal extracts have long been used to enhance healthfulness, color, taste, and even the flavor of food, beverages, and supplements. Dietary supplements are projected to grow at the second highest CAGR. The surge in the consumption rate of dietary supplement products, specifically processed from plant ingredients, has been the key factor driving the growth of the segment in terms of value sales.

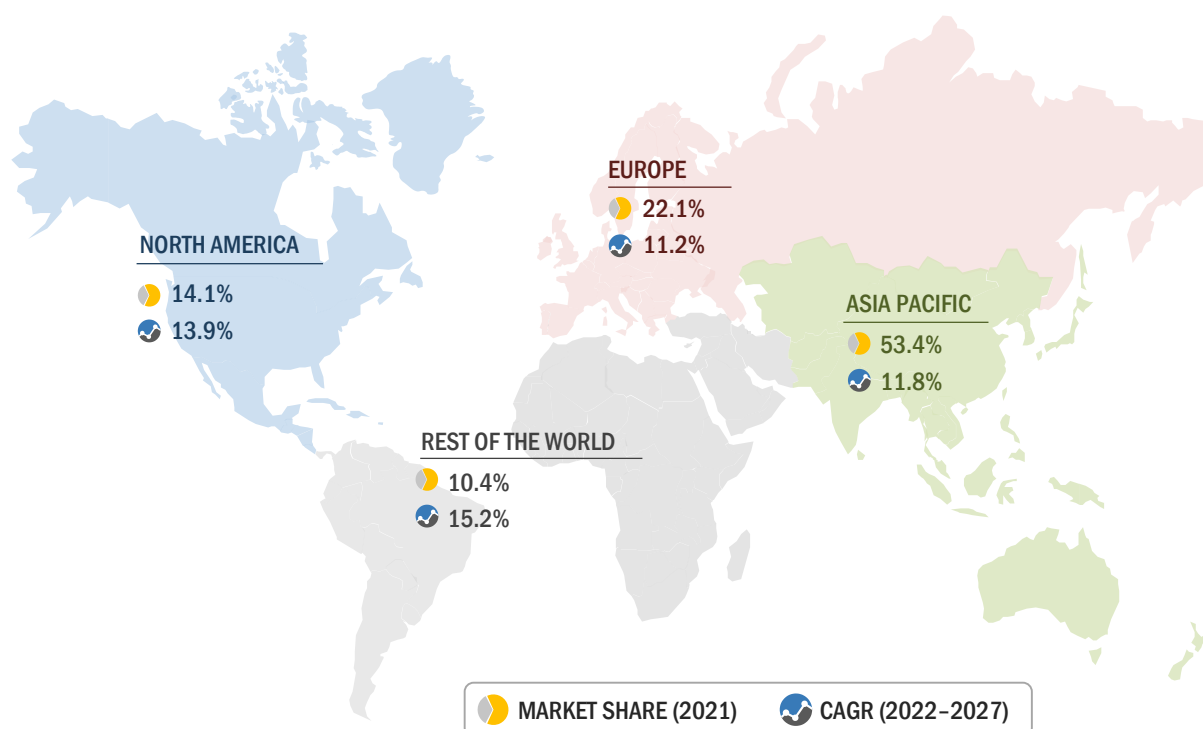
FIGURE 11 PLANT EXTRACTS MARKET SIZE, BY SOURCE, 2022 VS. 2027 (USD MILLION)



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

The herbs & spices segment is estimated to dominate the plant extracts market in 2022. The consumption of herbs & spices has been linked with several health benefits due to their medicinal properties. Due to the pandemic, spices that support immune function, such as ginger, curcuma, and garlic, were in high demand. The demand is anticipated to increase as people become more aware of the therapeutic benefits of extracts from herbs and spices. As consumers become more health conscious, they prefer natural coloring and flavoring agents to artificial ones, placing herbs and spices in high demand from the processed food and fragrance industries.

FIGURE 12 PLANT EXTRACTS MARKET SHARE AND GROWTH RATE (VALUE), BY REGION, 2021



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

The Asia Pacific market accounted for the largest share in 2021. This dominance is attributed to factors such as a surge in the consumption rate of vegan food & beverage products, the rise in the demand for functional food products, and the increase in awareness related to the health benefits associated with the consumption of clean-label food products among consumers. The rising demand for natural cosmetic products and functional food products pave the way for the market to grow in terms of value sales in Europe and North America.

Key players in this market include International Flavors & Fragrances Inc. (US), Givaudan (Switzerland), Symrise (Germany), Kerry Group Plc (Ireland), ADM (US), Synthite Industries Ltd (India), Kalsec Inc. (US), Kangcare Bioindustry Co. Ltd. (China), Carbery Group (Ireland), DSM (Netherlands), Döhler (Germany), Indesso (Indonesia), and Vidya Herbs (India).

The companies are increasing their market share using strategies such as acquisitions, expansions, partnerships, and new product launches.

In May 2022, Native Extracts Pty Ltd launched of newest blue essential oil, extracted by steam distillation of Australian rosewood. This essential oil provides a base tone of fresh forest odor, making it a one-of-a-kind product. On launching this product, the company received a huge response for sample requests. This product will create an impact on the company for its innovation and unique product profile.

In April 2022, Kerry completed the acquisition of Natreon (US), a leading supplier of branded Ayurvedic botanical ingredients based in the US. Kerry's leadership position and the ProActive Health portfolio of science-backed branded ingredients are significantly expanded as a result of the acquisition, which accelerates the company's technological growth.

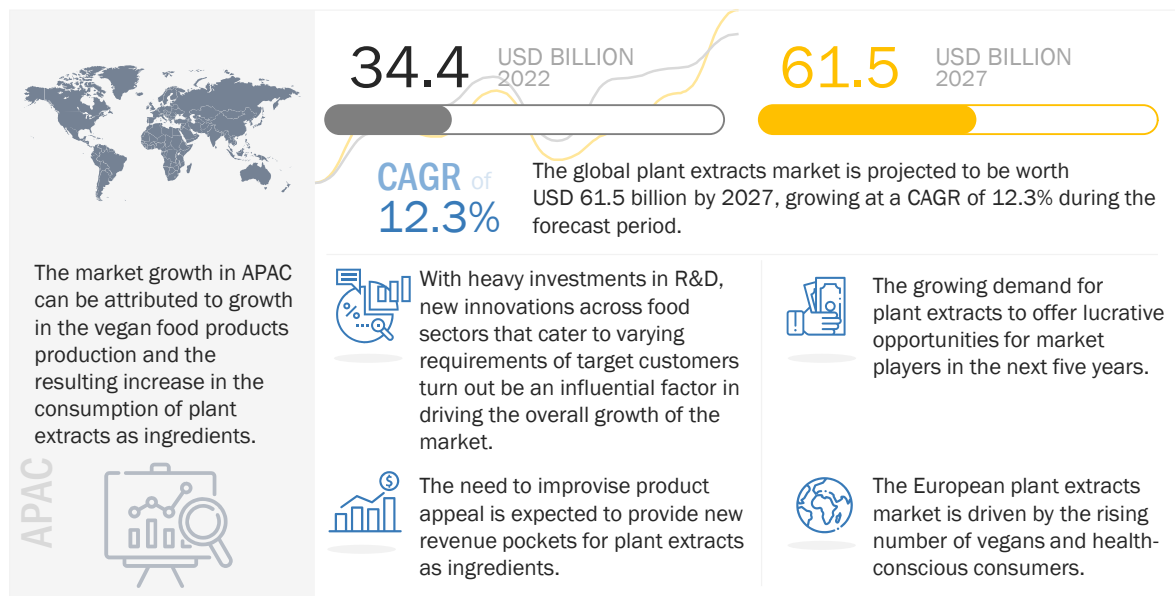
In November 2021, Givaudan opened a new digital space in China, one of the most digitalized markets, as part of its long-term strategy to expand digital capabilities. It will enable the company to capitalize on transformation opportunities while also supporting growth with local and regional customers.

In May 2021, Biospringer (France), a producer of yeast extract and yeast-derived products, announced its partnership with Indesso's savory product portfolio. This partnership focuses on innovation and the development of value-added ingredients.

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE OPPORTUNITIES FOR PLAYERS IN PLANT EXTRACTS MARKET

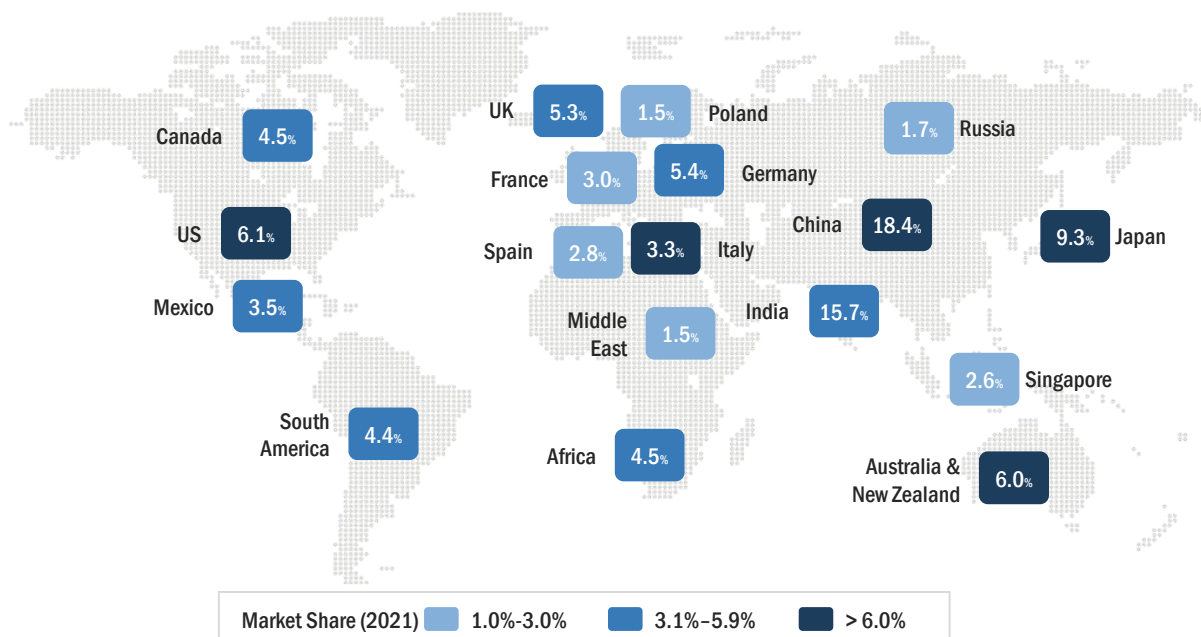
FIGURE 13 GROWING DEMAND FOR NATURAL AND PLANT-BASED INGREDIENTS



Source: Secondary Sources, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.2 PLANT EXTRACTS MARKET: MAJOR REGIONAL SUBMARKETS

FIGURE 14 ASIA PACIFIC WAS LARGEST MARKET GLOBALLY IN 2021

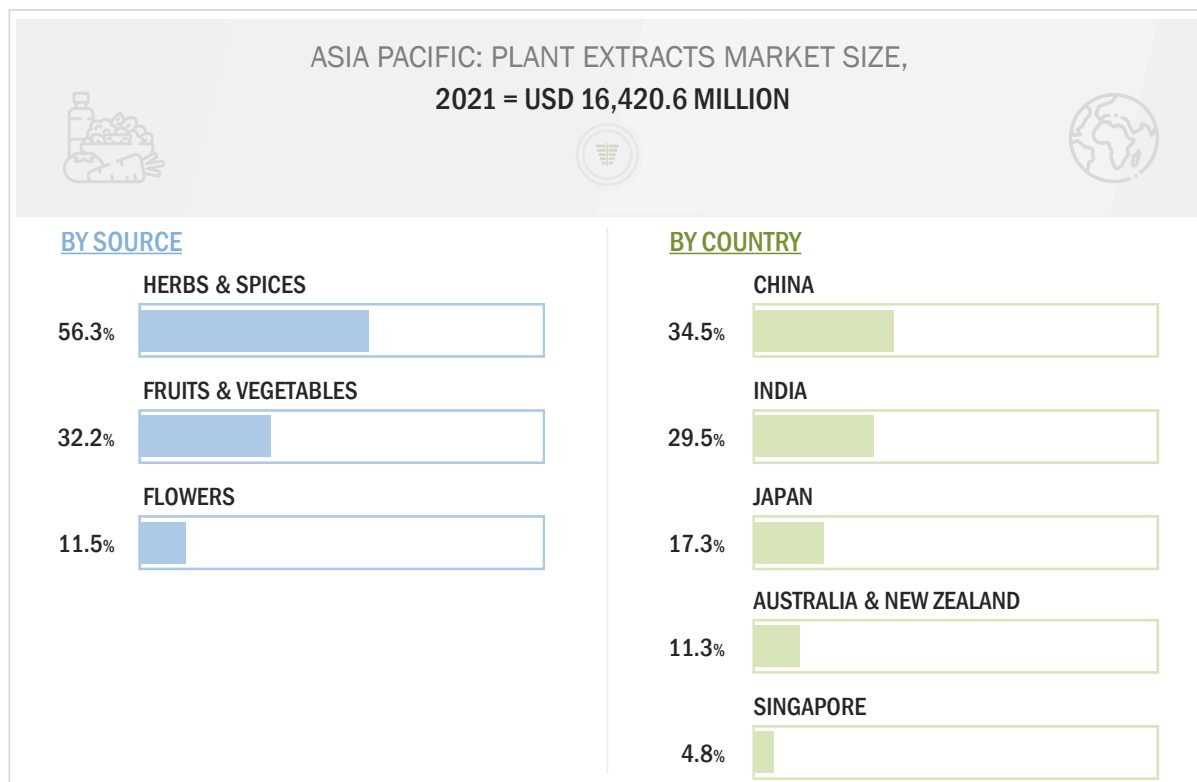


Note: The market shares have been provided only for the key regional submarkets for 2021 in terms of value.

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.3 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY SOURCE & COUNTRY

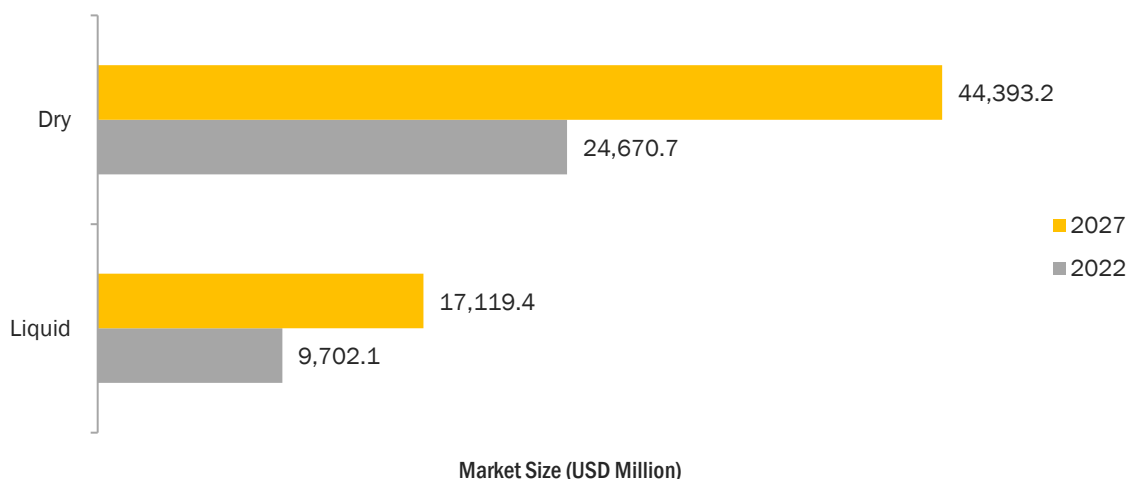
FIGURE 15 CHINA TO DOMINATE ASIA PACIFIC MARKET IN 2021



Source: Secondary Sources, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.4 PLANT EXTRACTS MARKET, BY FORM

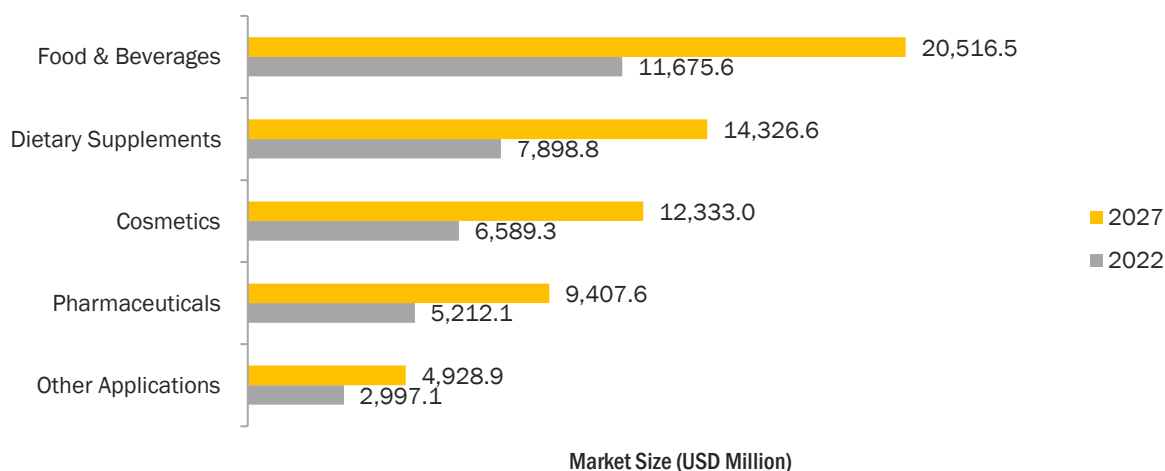
FIGURE 16 DRY SEGMENT TO DOMINATE PLANT EXTRACTS MARKET



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.5 PLANT EXTRACTS MARKET, BY APPLICATION

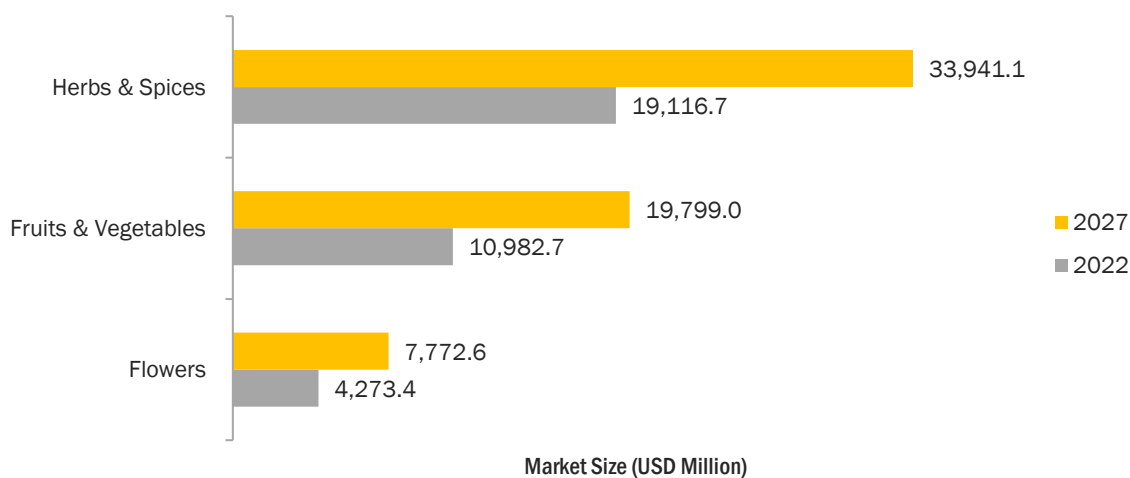
FIGURE 17 FOOD & BEVERAGES TO DOMINATE MARKET



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.6 PLANT EXTRACTS MARKET, BY SOURCE

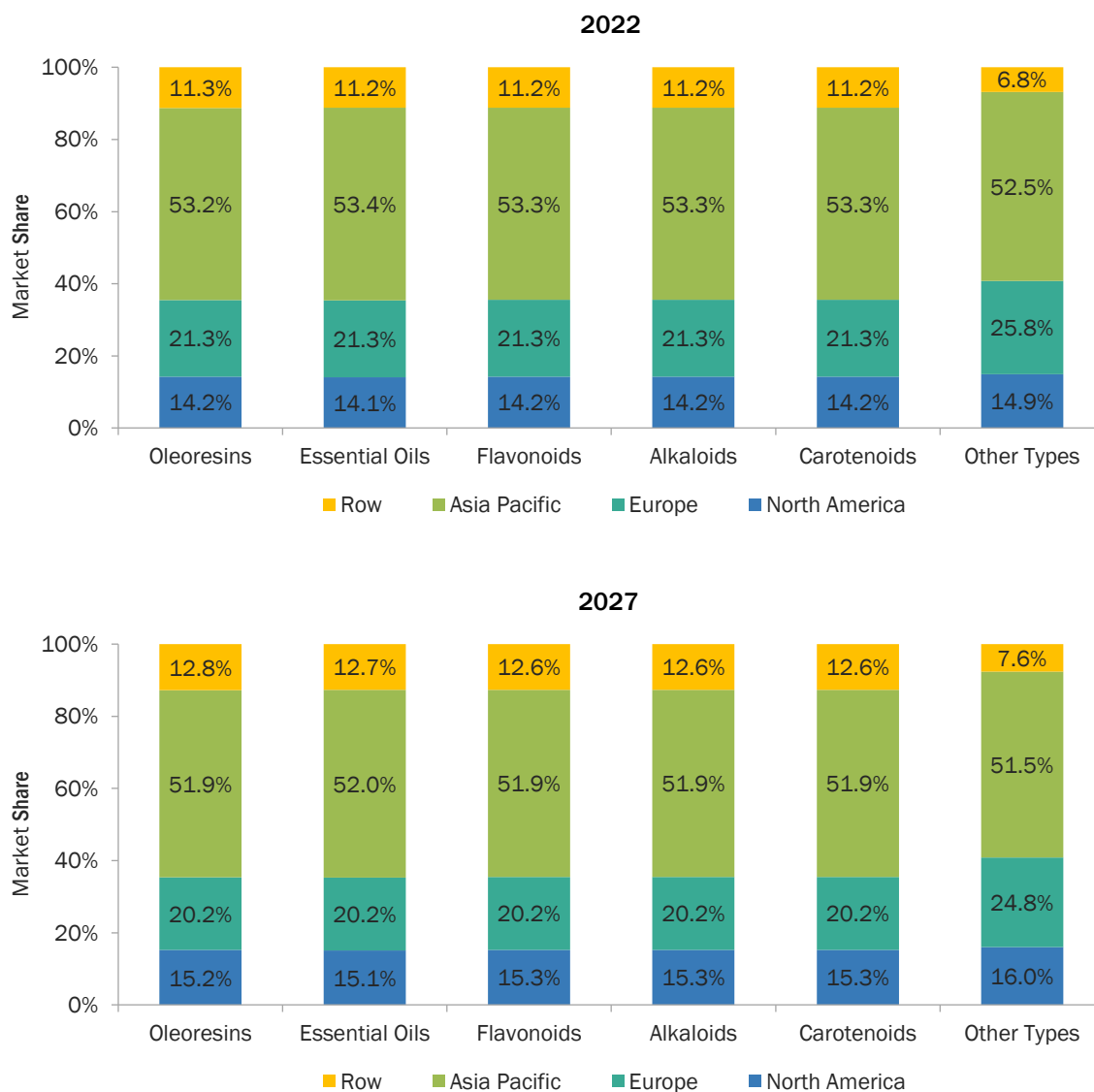
FIGURE 18 HERBS & SPICES TO DOMINATE MARKET



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

4.7 PLANT EXTRACTS MARKET, BY TYPE & REGION

FIGURE 19 ASIA PACIFIC TO DOMINATE DURING FORECAST PERIOD



Other types include anthocyanins, tannins, and lignin.

Source: Secondary Sources, Primary Interviews, Related Research Publications, Press Releases, Industry-related Journals & Books, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

Plant extracts have widespread consumer and industrial applications, which improve their scope of use, which, in turn, continues to bolster the plant extracts market. Herbs and plants are processed and consumed in different ways and forms, including ground or powdered form of a raw herb or its dried extract, plant extracts, oleoresins, and essential oils. Since plants are rich in a variety of compounds, many secondary metabolites and compounds with antioxidant properties are gaining traction. Besides, manufacturers have witnessed a growing consumer trend toward the consumption of plant-based food products. Synthetic phenolic antioxidants have been used as food additives; however, their reported negative effects on human health are a major concern. In a bid to mitigate this concern, the replacement of these synthetics with “natural,” plant-derived bioactive compounds witnesses high demand.

The plant extracts market is highly fragmented; hence, there are numerous players in the market offering a wide range of plant extracts; the major plant extract manufacturers include Givaudan (Switzerland), Sensient Technologies Corporation (US), International Flavors & Fragrances, Inc (US), Kangcare Bioindustry Co., Ltd. (China), Synthite Industries Private Ltd. (India), and PT. Indesso Aroma (Indonesia).

5.2 MACROECONOMIC INDICATORS

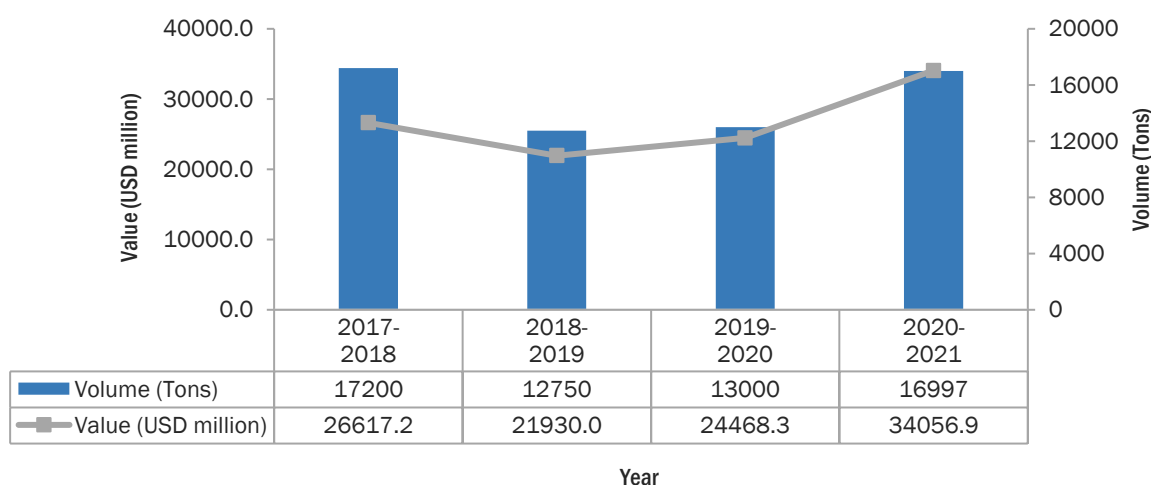
Macroeconomic variables levy short-term as well as long-term effects on socio-economic indicators of the agricultural sector, including price index, rural unemployment, and gross domestic product (GDP). Most developed countries have established agriculture as an infrastructure sector of economic development, and with the help of the abundant capacities of this sector, they promote the development of other sectors as well. The developed countries deem food security as an important parameter for political security, and hence, there is high sensitivity regarding the quality and quantity of consumption of agricultural products, especially healthy and organic food, indicating the high attention of consumers toward healthy products in the agricultural sector.

5.2.1 EXPORT OF SPICES

The Gross Domestic Product (GDP) is the primary indicator of macroeconomic performance, reflecting the overall size of an economy in an absolute value. Exports are one of the components constituting the GDP and generate foreign currency earnings in the country that produces them, thereby boosting its economic growth. Export competitiveness represents the economic growth and demand, which is regulated by trade liberalization and policies. Spices are one of the most traded agricultural commodities globally. India is the world's leading producer and exporter of spices and a significant stakeholder in spices' export trade. The use of spices in perfumery and medicine has been prevalent for long; however, some spices like black pepper have gained prominence across the Middle East and Europe in the past decade. Europe offers opportunities for companies that export spices and herbs. Europeans want a healthier lifestyle but do not have a suitable climate to grow most spices and herbs, therefore relying heavily on imports from tropical and semi-tropical countries. European importers seek high-quality, sustainably produced spices and herbs, following the production and sustainability standards of Europe. On the export front, a fluctuating demand in some of the leading spice categories as chili, turmeric, and cumin affects the year-on-year (YOY) growth in spice exports from India. As the prices rise, the demand is impacted, reflecting slow growth. As per the Directorate General of Foreign Trade (DGFT), India exported USD 4.1 billion worth of spices in FY 2021-2022, wherein USD 1.8 billion in revenue was generated from core spices such as dried chili, cumin, and turmeric, and over USD 1.2 billion export revenue from mint products, spice oils, and oleoresins. The YOY export performance declined in FY 2022; however, India's FY 2021 spice export volume increased 37% compared to FY 2020, predominantly because of the COVID-19-influenced emerging trend of demand for immunity-boosting spices. The war in Ukraine and the sanctions imposed on and by Russia and Belarus

have had a huge effect on international trade, with effects ranging from a shortage of food products, raw materials, and components to canceled trade routes, changes in demand, and price increase. Spice trade disruption leads to supply shortages and subsequent price hikes. According to the World Bank, the war disrupted the production and trade of several commodities, particularly those where Russia and Ukraine are key exporters, including energy, fertilizers, and grains. These price increases come on top of already tight commodity markets due to a solid demand recovery from the pandemic, as well as numerous pandemic-related supply constraints. Prices of spices have also increased due to local shortages and strong global demand. As Ukraine is one of the major exporters of coriander seeds, the prices have increased about 30%. Besides impacting the price of commodities, the war led to a shift in demand, and as a result, Indian exports increased as supplies from the Black Sea region were restricted. Such fluctuations and trade-driven market scenarios result in a high marginal cost for plant extracts.

FIGURE 20 INDIA: EXPORT DATA OF SPICE OILS & OLEORESINS, 2018-2021

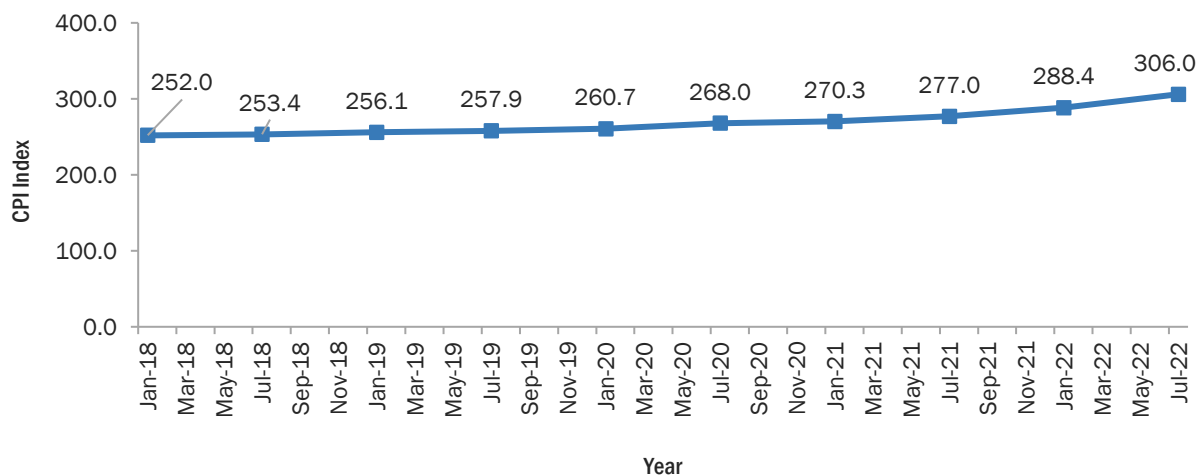


Source: DGCI&S, MoC, Indian Spices, Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

5.2.2 CONSUMER PRICE INDEX

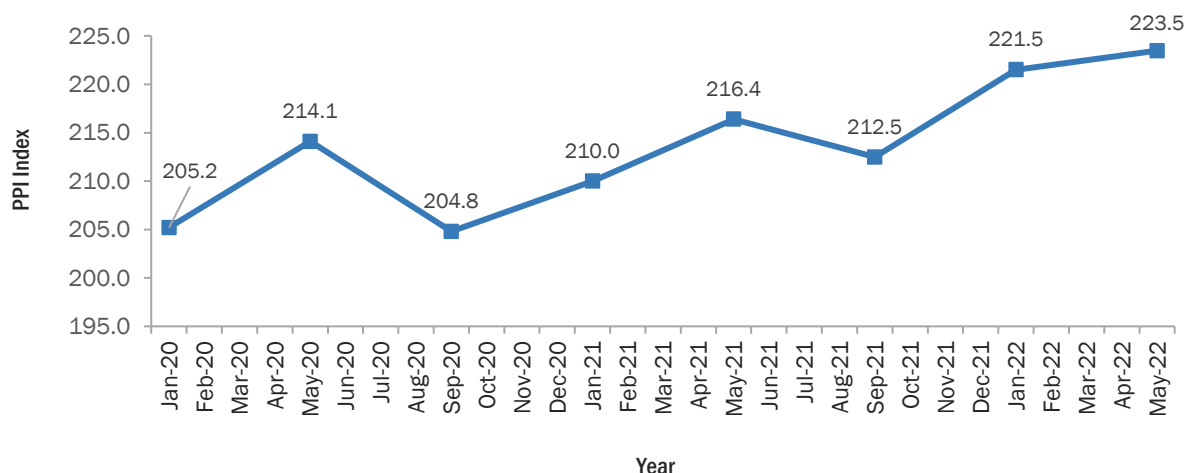
The Consumer Price Index (CPI) is a measure of inflation determined by tracking the retail prices of a good or service of constant quality and quality over time. Tracking retail prices allows the CPI to capture changes in out-of-pocket household spending over time. According to the USDA, the all-items Consumer Price Index (CPI), a measure of economy-wide inflation, increased by 1.4% from May 2022 to June 2022 before seasonal adjustment, up 9.1% from June 2021. The CPI for all food increased 1.0% from May 2022 to June 2022, and food prices were 10.4% higher than in June 2021. The percentage change in the CPI over time gives the amount of inflation over that specific period, i.e., the increase in prices of a representative basket of goods consumed. The US food price inflation accelerated in 2022. The Consumer Price Index for baked foods and cereal products increased 2% in July, its second consecutive month of such a gain, according to the Bureau of Labor Statistics of the US Department of Labor. The index for all food at home, meanwhile, rose 1.4%. As per US Congress, current food price inflation can be due to general inflationary pressure on the economy as well as food-specific factors, which may vary by the type of food. Rising food prices present challenges for many US consumers, particularly low-income ones. High CPI indicates a rise in the cost of goods sold, indicating a rise in the price of plant extracts.

FIGURE 21 US: CONSUMER PRICE INDEX (URBAN CONSUMERS) OF FOOD & BEVERAGES, 2018-2022



Source: US Bureau of Labor Statistics, Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

FIGURE 22 PURCHASER PRICE INDEX FOR FOOD (HEALTH) SUPPLEMENT STORES, 2020-2022



Source: US Bureau of Labor Statistics, Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

The Purchaser Price Index (PPI) program measures the average change over time in the selling prices received by domestic producers for their output. The prices included in the PPI are from the first commercial transaction for many products and some services. According to the US Bureau of Labor Statistics, the Producer Price Index for final demand fell 0.5% in July. Prices for final demand goods declined 1.8%, and the index for final demand services advanced 0.1%. Final demand prices moved up 9.8% for the 12 months ended in July. Wholesale inflation fell 0.5% in July, demonstrating that the price increases are slowing.

5.2.3 ORGANIC FOOD DEMAND

Consumer aspiration for health and well-being, authenticity, and clean label products is changing the food & beverages industry dynamics. New product developments, reformulations, and market strategies revolve around natural and organic food ingredients. Transparency and clean labeling are the key to product developments. Companies are introducing a greater variety of organic foods into their portfolio. The USDA has supported programs that help organic farmers respond to the growing demand for organic products.

Organic food is priced slightly higher than conventionally produced food. However, due to the rising awareness about the health benefits of these products, especially during the COVID-19 era, consumers prefer to spend extra for organically produced food. The return of 'home-made' healthy food has also contributed to consumption growth. There has been a responsible shift in consumption patterns from non-organic to organic, and has been an opportunity for many start-ups to enter the organic food space. In India, according to the APEDA, in MY 2020/2021, organic crop production reached 3.2 million metric tons (MMT), up 36% compared to MY 2019/2020.

5.3 MARKET DYNAMICS

The increasing number of applications for plant extracts across products has helped establish a strong market for these products. Plant extracts have been used in pharmaceuticals, food, and cosmetics industries across the globe. However, recently, they have gained much interest as they are relatively safe, with minimal side effects to health compared to synthetic extracts. Hence, they are widely used as antimicrobials and anti-inflammatory drugs. In cosmetics, these extracts find applications as anti-aging reagents, hair oils, mouthwash, liniments, and dental care products. Essential oils, which are one of the most important types of plant extracts, find applications in flavors & fragrances. With the surge in demand for natural ingredients and natural products due to the rise in awareness related to better dietary choices, the growth of the aging population, the increase in the trend of a healthy lifestyle, and the growth in incidences of chronic diseases, many manufacturers are investing in R&D and producing innovative extracts, which contribute to the nutritional health of consumers. With the growing vegan population preferring only plant-based products, natural plant extracts are gaining immense popularity among consumers, further expected to propel market growth. However, consumer skepticism associated with the adoption of several plant extracts in different applications and inadequate supply of raw materials, and price fluctuation could hinder the market growth to a certain extent during the forecast period.

FIGURE 23 PLANT EXTRACTS MARKET DYNAMICS

 DRIVERS	▪ Demand for essential oils from the cosmetics industry ▪ Increasing demand for natural ingredients and clean-label products ▪ Growing market for nutraceuticals and herbal supplements ▪ Adverse side-effects of synthetic flavors led to a shift toward natural flavors
 RESTRAINTS	▪ Lack of standardization, and varying regulations and quality standards ▪ Price fluctuations and inadequacy of raw materials
 OPPORTUNITIES	▪ Technological intervention in the production process ▪ Growing trend of veganism among consumers ▪ Government initiatives to promote natural ingredients
 CHALLENGES	▪ Lack of downstream processing facilities and resultant microbial contamination ▪ Adulteration and cost margin for natural additives compared to synthetic counterparts

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

5.3.1 DRIVERS

5.3.1.1 Demand for essential oils from cosmetics industry

Essential oils are widely used in cosmetics, perfumery, food and beverages, cleaning products, and home fragrances, for their relaxing, stimulating, antiseptic, bactericidal, decongestant, soothing, antispasmodic, and anti-inflammatory properties. Besides being used for fragrances, essential oils like lemon and orange possess antiseptic properties, garnering attention as promising cosmetic-active ingredients. Essential oils are also used in skin care with anti-acne, anti-aging, skin lightening, and sun protection characteristics. Skin care is the most important product category, in which several products use natural ingredients. Essential oils are mainly used as fragrances, while various other plant-based extracts are used for their active properties. Essential oils are also used in soaps, color cosmetics, and deodorants.

Companies within the ecosystem are focusing on further expanding the market for essential oils within cosmetics. Primavera Life, a producer of natural cosmetics and essential oils, uses a wide range of essential oils in its products, including lemongrass from Bhutan and Nepal, cajeput from Cambodia, and lemon verbena and myrtle from Peru. Higher-priced essential oils are usually used in the perfumery and personal care sector. According to the European Federation of Essential Oils (EFEO), fragrances, cosmetics, and aromatherapy generate about one-third of the demand for essential oils. There is an expected increase in demand from the cosmetics sector because many natural and conventional cosmetics companies use essential oils as fragrances. Some highly valued essential oils used as fragrances include eucalyptus, citrus, floral oils, tea tree, and lavender. Geraniol, citral, linalool, limonene, and citronellol are other notable fragrances popular among the masses.

Essential oils are highly versatile as a natural fragrance source in cosmetics and cosmeceuticals. The continued surge witnessed by the category is attributed to the fact that it has a wide range of natural unique, and pleasant fragrances along with being a bioactive ingredient. Essential oils and botanicals have developed a heightened portfolio over the years, primarily because of the enormous demand for natural ingredients. Essential oils demand is still under scrutiny for constituting allergens and chronic toxicity risks for consumers, necessitating the monitoring of dosage and application. The International Fragrance Association (IFRA) offers guidelines on the safe use of essential oils in cosmetic products. The concentration of essential oils consumed can also enable safe usage. The important role of essential oils in the current development of the cosmetics industry toward greener and eco-sustainable products and the constant research over mitigating the allergen and safety issues associated with essential oils improve its business expansion. For example, in France, L'Oréal has a local partnership with SICA Bio-plantes, a group of farmers, for sustainable and responsible sourcing of some of the essential oils; for example, lavender essential oil or lemon winter savory.

5.3.1.2 Increasing demand for natural ingredients and clean label products

Natural and organic products have high demand across a variety of consumer products, especially within the food & beverage industry. Consumer demand for healthy, natural ingredients is in tandem with the rise in demand for health-consciousness among consumers and sustainability quotient. European companies are launching new natural health products, while some are replacing synthetic ingredients with natural ones in product formulations. European consumers are inclined toward complementary and alternative medicines, stimulating the demand for natural ingredients. Western European countries have the most opportunities for exporters of natural ingredients for health products in developing countries. Natural ingredients used in health products have functional and active properties, a prime reason for their demand. However, there is a definite surge in demand for natural ingredients in the cosmetics industry.

Most consumers in developed countries have been looking for food products, without any inclusion of artificial additives, due to the associated long-term adverse health effects. Manufacturers are inclined toward the usage of natural flavoring agents, preservatives, and sweeteners as they are the key ingredients used to make the food tasty and to extend the shelf-life of the product. With the growing awareness of health products, consumers are increasingly looking for less processed and more natural food & beverages. This is reflected in the clean label movement spreading across the packaged food industry. Consumers are paying close attention to and examining labels and nutritional ingredients more than ever. The term 'clean label' does not have a clear or single definition and is mostly associated with the absence of artificial ingredients, GMOs, and the presence of simple ingredients that are familiar to consumers and transparency in terms of the wholesomeness of the ingredient.

Further, 'natural' and 'organic' are claims that are termed as clean label. Consumers perceive that natural ingredients have a positive impact on health, leading to an increase in the demand for naturally sourced ingredients in products for various applications, including toiletries, personal care, dietary supplements, food, and cosmetics. Ingredient labeling has resulted in manufacturers becoming increasingly cautious about the ingredients used in their products. All these factors are driving the demand for plant extracts in various industries.

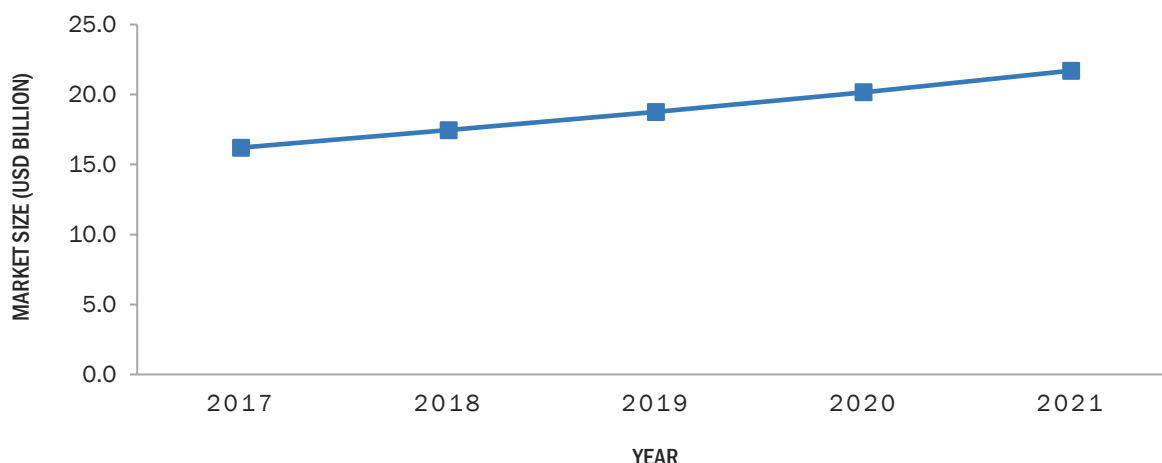
Clean label is the new norm among food & beverage products. Consumers are, more than ever, conscious of their purchased products. "Being Green" is becoming a lifestyle more than just a habit. The willingness to pay more for recognizable ingredients is the strongest in the US, highlighting the importance of clean and clear labeling in the American market. Asian consumers also put a very high value on the quality of their food and are willing to pay a premium for the best ingredients, even though average incomes are lower. Mislabeling is a challenge still affecting the industry growth; however, companies are focusing on mitigating this concern by improving product labeling and inculcating higher transparency. Demand for natural ingredients used in cosmetics and toiletries is growing, a trend expected to continue. Natural ingredients are increasingly used in conventional cosmetics and personal care products. Companies are

replacing synthetic chemicals with natural ingredients, partly because of consumer demand and because of the move toward sustainable raw materials. Growing consumer awareness of natural cosmetics and the desire of cosmetics companies to replace synthetic ingredients with natural variants are the main drivers of this growth. The top six products, in terms of opportunities for developing countries, are shea butter, coconut oil, avocado oil, frankincense essential oil, patchouli essential oil, and licorice extract.

Taking clean label to the next level, Symrise expanded its footprint of natural capabilities in North America in 2019 by manufacturing botanical and vanilla extracts and distillations in Teterboro, New Jersey. In a continuous bid, Symrise's division, Diana Food, launched the first clean label, organic and Fair-Trade acerola powder for nutritional products in 2020. In another instance, Kerry Group acquired Natreon, a US-based producer of ayurvedic botanic ingredients, in 2022. In 2021, Synergy Flavors launched an entire range of natural extracts and essences appropriate for clean-label products, including beverages, dairy products, and desserts.

5.3.1.3 Growing market for nutraceuticals and herbal supplements

The herbal supplements market is currently gaining popularity, though it goes back as a traditional solution for health and wellness. The market demand for nutraceuticals and herbal medicinal products is driven by the increasing attention to health and prevention by consumers, increased healthcare costs and search for alternative products, and greater customization of needs for different segments of the population. Herbal supplements have emerged as a popular complementary and alternative medicine/supplement to modern medicine. Besides, most of the population across the globe, especially the Millennials, are inclining toward preventative medicine following the age-old adage "prevention is better than cure." Increasing health awareness among consumers and health issues as a result of a sedentary lifestyle have triggered the shift toward nutraceuticals and herbal supplements. Post-COVID-19, companies started large-scale manufacturing of herbal concoctions, specifically targeting immunity-boosting products. Though there is limited scientific or clinical data supporting the use of herbal supplements and nutraceuticals to prevent or treat COVID-19, many consumers showed interest in natural products to improve their overall health and particularly to enhance their immune response. Forest Remedies, in 2021, launched Elderberry Immune Multi Omega 3-6-9 + Ahiflower Oil which provides immune and wellness support. Ahiflower oil is combined with pure elderberry extract, which is rich in antioxidants and vitamins, to support the immune system and overall health and wellness. Vytanutra launched into the dietary supplement market; three new formulas based on Chinese herbal lore and backed by modern research. The plant-based blends of vitamins and minerals aim to enhance normal body wellness and provide specific benefits like immune and eye health support. Germany's Dermapharm acquired herbal specialist Arkopharma in 2022 to expand its footprint in Europe's consumer health market in countries such as France, Spain, and Italy.

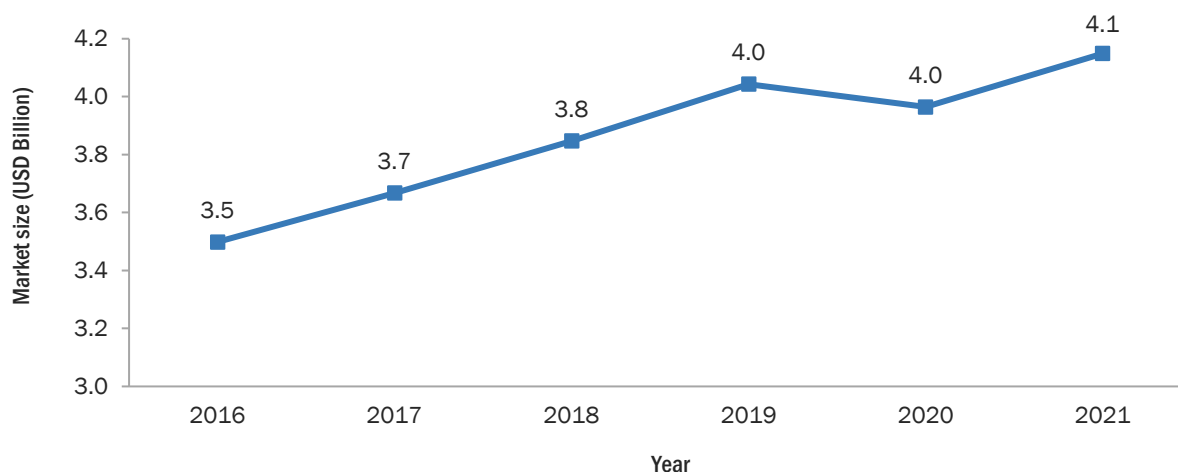
FIGURE 24 GROWING DEMAND FOR BOTANICAL DIETARY SUPPLEMENTS

Source: MarketsandMarkets Analysis

Herbal nutraceutical is used as a powerful instrument in maintaining health and acting against nutritionally induced acute and chronic diseases, thereby promoting optimal health, longevity, and quality of life. According to a report by the American Botanical Council, the demand for herbal supplements soared high in 2020. The demand was particularly high for elderberry and ashwagandha, as consumers turned to natural products marketed for such conditions as immune health and stress support. Consumer behavior has changed, leading to an increased preference for products with health-promoting attributes. Personalized nutrition is also trending, further promoting the growth of the market, especially in the beauty and personal care vertical. For instance, in 2021, Herbalife launched a new beauty-from-within daily supplement designed with lycopene, phytoene, and phytofluene from tomato extracts and carnolic acid from rosemary to support skin care.

5.3.1.4 Adverse side-effects of synthetic flavors

Food flavorings or micro ingredients are added to enhance food flavors, besides assisting in the preservation of products or help keep the product stay fresh for a prolonged period. Food flavorings are an integral part of the food & beverage industry as it confers specific sensory properties of taste and smell to processed foods. They have a complex formulation and might as well be categorized as natural, synthetic, and nature identical. Food flavors are standardized by the FDA, and flavor manufacturing companies must comply with the FDA and USDA norms for food production. Nowadays, increasing attention is being paid to the food additives used, with growing concerns related to their safety and potential health risks.

FIGURE 25 INCREASING DEMAND FOR NATURAL FLAVORS & FRAGRANCES

Source: MarketsandMarkets Analysis

Consumers are becoming increasingly aware of the adverse effects of consuming food & beverages containing synthetic ingredients. Several health organizations and food safety associations prohibit the usage of various artificial flavorings in food & beverages. For instance, on October 5th, 2018, the US Food and Drug Administration (FDA), along with the Breast Cancer Fund, the Natural Resources Defense Council, the Center for Science in the Public Interest, and the Environmental Defense Fund, had in 2018 announced a petition to ban the use of artificial flavors like benzophenone, styrene, methyl eugenol, and others. These additives caused cancer in lab animals. This regulatory intervention eventually led to the adoption of essential oils and natural extracts in the food and cosmetics industries.

According to the FDA, artificial flavor or artificial flavoring means any substance, the function of which is to impart flavor, which is not derived from a spice, fruit or fruit juice, vegetable or vegetable juice, edible yeast, herb, bark, bud, root, leaf or similar plant material, meat, fish, poultry, eggs, dairy products, or fermentation products. There is a significant rise in the number of incidences of health risks, including chest pain, dizziness, headaches, nausea, allergies, fatigue, seizures, and brain damage, since artificial flavors are extracted from petroleum and other chemicals. Problems related to health arise when food manufacturers lower their standards for profit and use low-quality chemicals and ingredients that result in health problems over time. Owing to such issues, the usage of chemical-based flavorings is decreasing, and the demand for natural extracts is growing extensively.

5.3.2 RESTRAINTS

5.3.2.1 Lack of standardization and varying regulations and quality standards

Plant extracts are a concoction of several plant compounds and are subjected to rigorous health and safety checks based on government regulations in different countries. Several studies have been conducted by regulatory bodies to assess the impact of these products on human health. Based on their results, products are classified, and Daily Intake Allowance (DIA) has been set. International bodies such as the National Food Safety and Quality Service (SENASA), Canadian Food Inspection Agency (CFIA), US Food and Drug Administration (FDA), World Health Organization (WHO), and the Committee on the Environment, Public Health and Food Safety (EU) are associated with food safety regulations. These organizations have control over the use of different chemicals and materials used in food processing, directly or indirectly.

Herbal extracts have increasing awareness and general acceptability; however, the standardization of herbal medicines is yet to be established worldwide. Standardization of herbal medicines includes prescribing a set of standards or inherent characteristics, constant parameters, and definitive qualitative and quantitative values that carry an assurance of quality, efficacy, safety, and reproducibility. It is the process of developing and agreeing upon technical standards. It is observed that there are no clear universal guidelines related to the usage of individual ingredients in end products. Regulations and guidelines vary from region to region. This has resulted in additional costs for the production for manufacturing region-specific ingredients. Hence, such varied regulations can serve as a restraint for manufacturers. Besides, quality control and standardization of medicinal plant-based products also need to be emphasized, especially with respect to the bioactive compound included. The manufacture of oleoresin also requires standardization and proper labeling. European buyers prefer a standardized good-quality product across all packaging suitable to order volumes. Concerning environmental sustainability, oleoresin export to Europe requires manufacturers to meet the UNCTAD BioTrade Principles and Criteria, and FairWild standards certification. Synthite Industries, a global leader in the oleoresins market, has Food Safety System Certification (FSSC) 2200, and International Standard for Organization (ISO) 900:2015 to assure a quality management system in place during production. The company also has ISO 14001:2015 to demonstrate an efficient environmental management system with a strict quality control system and an in-house laboratory. Such measures are a step toward mitigating the restraint posed by a lack of consistent standards and regulations.

5.3.2.2 Price fluctuations and inadequate supply of raw materials

Most herbaceous plants with functional properties are available in Asian countries. China and India are home to most of such plant species. India constitutes approximately 25,000 plants of medicinal value and is known as the 'Botanical Garden of the World.' Since these plants are majorly produced in only a few countries, the supply of such a wild variety of plants to other developed countries is limited. Various spices and herbs are available during specific seasons and are harvested annually. The continuity in the supply of raw materials is one of the major problems faced by the plant extracts market. The prices of herbs and spices may vary based on availability and season. High storage costs are involved in maintaining the stock throughout the year. Owing to seasonal availability, prices of raw materials also fluctuate a lot, which makes it difficult to standardize the prices of end products. Herb prices tend to vary widely between US terminal markets. Prices of cumin or jeera, the second-most widely consumed spice in India, shot up to a five-year high, as output has shrunk by a third. The average weighted price of cumin in wholesale markets has touched nearly INR 13,000 a quintal (100 kg), up from INR 12,522 in 2021 and INR 12,333 in 2020.

A massive disruption occurred within the supply chain during the COVID-19 outbreak. The spice industry faced severe price pressure and partial raw material shortage. German spice processors had to cope with large price increases and supply shortages in the commodity markets, and the rising dollar price situation severed. Supply bottlenecks, and an increased demand in the countries of origin and weather unpredictability cause distress at times. Lack of harvest workers, slowdown in production, and lockdown-impacted port closures have caused crop failures and shortages. Extreme weather conditions such as heavy rain in Southern Europe or enormous drought and forest fires in Eastern Europe exacerbate these failures. The spice industry in Europe and adjacent regions have a high dependency on imports. Distortions in freight and logistics have a direct impact on the spice trade. The consequences, such as the lack of availability of shipping containers, are omnipresent and lead to drastic cost increases. As the demand outstrips supply, the annual demand stands risky despite seasonal harvests and disrupting the continuity of supply restraining the market.

5.3.3 OPPORTUNITIES

5.3.3.1 Technological intervention in production processes

Technological intervention across the supply chain helps the plant extracts business to propel. Bioactive extraction is witnessing high technology adoption. Besides, “Green technologies” are being adopted to result in significant sustainable measures. For instance, polar-nonpolar-sandwiching (PNS) technology was used for the extraction of a spectrum of bioactive, including curcuminoid, polyphenols, protein, fiber, and essential oils from within the turmeric root extract. Ultrasound assisted extraction (UAE) is a green technology for bioactive extraction. Givaudan, a leading player in the flavors and fragrances industry, performs sustainable extraction through one of its innovative biomimetic technologies: Natural Deep Eutectic Solvents (NaDES). This green fractionation technology enables the extraction of an analyte from a sample according to its physical or chemical properties. Such specialized extraction technique is used to extract botanicals. Vapor Cartridge Technology LLC, in 2022, announced the patenting of an environmentally conscious method of extracting essential oils/resins from cannabis plants. An Israeli start-up, AvoMed, developed a technology that can extract a bioactive PFA (polyhydroxylated fatty alcohols) from avocado, further used as an active ingredient in topical cosmetic products to treat skin damage and inflammatory disorders.

Adulteration and authenticity are being tested using technology, promoting the consumption of herbal supplements, thereby increasing the demand for plant extracts. Herbal products sold worldwide as medicines or foods are perceived as low risk because they are considered natural and thus safe. However, the quality of these products is ineffectively regulated and controlled. DNA testing technology was used to analyze the authenticity of herbal products. The introduction of such technical expertise bolsters the demand for plant extracts.

5.3.3.2 Growing trend of veganism among consumers

Veganism and an increased shift toward vegetarianism have further propelled the demand for herbal supplements. According to the Good Food Institute, the sales of plant-based foods grew three times faster than overall food sales in 2021. According to Plant Protein's sources, 6% of US consumers are vegan. In 2022, as per a survey conducted by The Vegan Society, 17% of Brits actively reduced the amount of meat consumption, while 8% have cut back on dairy and/or eggs, 3% shunning both. 6% of the study respondents had stopped consuming fish and seafood. One in four (23%) minimized the amount of animal products. Consumers are becoming sensitive to factors such as health, sustainability, and social impact while making purchase decisions. The consumption of animal-based foods has raised environmental concerns. Some consumers also have an ethical objection to using animals in food products. The largest markets for plant-based foods include plant-based milk and plant-based meat. This trend is expected to boost the demand for various plant extracts in the food & beverage industry as plant extracts can be used to increase the flavor and nutritional value of products. Botanical extracts are a part of a number of product launches across all major segments, from beverages, to baked goods and dairy products or savory categories such as soups, sauces, and snacks. The extracts are also used in plant-based dairy and meat alternatives, providing sensory value and transparency in terms of product labeling.

An aging population and increasing consumer demand for natural supplements bolster the market for natural ingredients for health products. In the European health products market, Moringa oleifera leaf powder is mainly used in food supplements and as a powder. It is considered a superfood in the health products sector because its dense nutrient content can boost the immune system, balance hormone levels, and improve skin and bone health, as well as vision. Moringa is also increasingly used as a plant-based protein source in vegan and vegetarian products. Tipco, a Thai juice giant, launched vegetable juices and herb-infused drinks in tandem with the consumer trend of health and plant-based eating. The functional beverage comprises curcumin and finger-root. Consumers are experimenting with complementary and alternative medicines, further stimulating the demand for natural ingredients.

5.3.3.3 Government initiatives to promote natural ingredients

Awareness is being created among consumers with respect to the benefits offered by natural ingredients and extracts; governments of various countries and manufacturers have started taking initiatives and are investing in R&D. India has an indigenous supply of medicinal and herbal plants and is further promoting herbal cultivation. In 2021, the Ministry of AYUSH, Government of India, implemented a scheme supporting market-driven cultivation of prioritized medicinal plants in identified clusters/zones. The Ministry of AYUSH is supporting the cultivation of medicinal plants on farmer's land throughout the country by providing subsidies for 140 medicinal plants @30%, 50%, and 75% of the cost of its cultivation, depending on the status of availability and market demand as per operational guidelines of the Scheme. The urban population is becoming health-conscious, resulting in a spike in the alternative market for pharmaceuticals and therapies as more people are developing trust in natural products. As under National Rural Livelihoods Mission (DAY-NRLM), sustainable livelihood generation and adoption of multiple livelihoods for a household are emphasized, and low input high output lemongrass farming is being promoted as an initiative for women empowerment by the Government of India. National Congress of the Communist Party of China organized the China Intangible Cultural Heritage Expo, presenting intangible cultural heritage items related to tea, Chinese tea art, tea ceremonies, and tea processing techniques to show the herbal infusion's close connection to modern life and its function as a bridge of communication between China and the world. Besides, China tried to promote herbal cures for COVID-19.

Digitalization has taken up the world completely, an example being the automation of spice and herb processing. Digital solutions are being incorporated to cater to the rising demand globally. The robotics solutions currently used in agriculture include robot picking, harvesting, and weeding machines, precise pesticide applicators, and drones. Several European projects support developments in the area of robotics for agriculture, including a robotic herb harvester in Denmark. The international non-profit organization CABI supports the use of drones in Africa to protect crops against desert locusts. In another case, Thailand, among other countries, legalized cannabis, leading to the herb's demand within pharmaceutical supplements.

The development of new and unique plant-based flavors & fragrances favored by consumers poses new opportunity areas for plant extract manufacturers. Also, changes in lifestyles and the increase in the demand for natural flavors from consumers are observed in developing economies, leading to the growth in the demand for plant extracts. Kemin (US) offers vertically integrated, sustainably grown plant extract blends such as rosemary, green tea, spearmint, and/or acerola, under the brand NaturFORT™. NaturFORT ARGT is a unique blend of acerola, rosemary, and green tea extracts that provides optimal color and flavor shelf-life extension of meat and poultry products, while NaturFORT RS is a unique blend of rosemary and spearmint extracts that provides flavor stability in dressings and mayonnaise. The plant extract blends provide optimized oxidation protection without negative sensory impact and formulation support. Globalization has also led to greater exposure to the flavors and fragrances of various cultures. Hence, identifying new & exotic spices and herbs and their properties in accordance with the changing consumer preferences is vital for plant extract manufacturers.

5.3.4 CHALLENGES

5.3.4.1 Lack of downstream processing facilities and resultant microbial contamination

Increasing incidences of microbial contamination in spices and seasonings imported from developing countries is a major challenge, especially because of raw material shortages. The European Spice Association and the US FDA have laid down regulations to make imported spices safe to consume. The key problem of importing condiments is the difficulty of checking composition, and though it is possible to perform a test on the composition, it is not cost-effective. The lack of warehousing facilities substantiates the microbial contamination and acts as a challenge to the growth of the plant extracts market. India is the largest producer of spices globally but lacks proper warehousing facilities, ultimately leading to microbial contamination and deteriorating the quality of the product. Low investment in technology and industry infrastructure due to low-profit margins of the enterprises is a challenge. It also results in an inflow of sub-standard procedures flowing into the local supply chains. The smallholders cannot invest in appropriate infrastructure for GMP, HACCP, ISO-certified production, and manufacturing.

Heavy metal contamination in herbal medicines is a concern to humans at levels above a threshold concentration. Manufacturing of herbal medicines necessitates supreme quality, and a high level of sanitation and hygiene is necessary to avoid microbial contamination. Drying requires reducing the moisture content and preventing enzymatic and microbial activity, ensuring a good shelf-life of the product. Medicinal plants can be dried in several ways: in the open air (shaded from direct sunlight); placed in thin layers on drying frames, in wire-screened rooms, or in buildings. High hydrostatic pressure (HHP) processing is a method where the material is subjected to elevated pressures applied. HPP is a technology that improves the microbial safety and shelf-life of the product and is quite commonly applied to the food processing industries. This is a non-thermal technology that leads to the destruction of undesirable microorganisms and inactivates enzymes without affecting the bioactive constituents. Proper technology implementation and efficient handling are supposed to mitigate this challenge during the forecast period.

5.3.4.2 Adulteration and cost margins for natural additives compared to synthetic counterparts

One of the major challenges faced by the plant extracts market is the prevalence of adulterated extracts. Such adulterated products do not possess the actual properties of pure plant extracts and mislead consumers. According to the National Academy of Science, US, synthetic oils are primarily made of 95% of the chemicals that can be typically found in mineral oil and petroleum jelly, which are both by-products of petroleum production. Manufacturers of synthetic fragrances add many toxic chemicals for the fragrances to spread and dissipate into the air. Benzene derivatives and aldehydes are the most common additive chemicals apart from other known toxins, which can cause adverse health effects. The major reason for the adulteration of essential oils is their high price. For instance, high-quality, pure rose essential oil may cost approximately USD 78,000 a liter, whereas the adulterated version of the same product costs approximately USD 200–300. Since the availability of some of the plant extracts is extremely low, the rate of adulteration is significantly high. Besides, as consumers favor natural essential oils over synthetic alternatives, the price of essential oil escalates, mainly due to expensive production costs.

Synthetic substitutes for plant extracts are easily available and are used in most synthetic cosmetics as cheaper substitutes. Several synthetic drugs are available in the market, which majorly include antibiotics and anti-inflammatory drugs. Such drugs tend to cause various side-effects in comparison to herbal drugs, which act gently. Hence, several synthetic and substandard products available in the market act as substitutes for natural plant extracts and pose a challenge for plant extract manufacturers. Moreover, in most cases, it is highly difficult to recognize the synthetic additives, and thus, implying the significance of proper labeling. The FDA has established labeling requirements for fragrances in cosmetics and aromatherapy. It requires the list of ingredients under the Fair Packaging and Labeling Act (FPLA). This law cannot be used to force a company to tell “trade secrets.” Fragrance and flavor formulas are complex

mixtures of many different natural and synthetic chemical ingredients, and they are the kinds of cosmetic components that are most likely to be “trade secrets.” Information required for cosmetic essential oils or essential oil products includes the declaration of ingredients and caution statements. The FDA provides a Cosmetic Labeling Guide with full regulations for cosmetic labels and a very useful compliance guide for labeling over-the-counter (OTC) drugs. Such transparency is expected to mitigate the challenge of adulteration and promote the growth of the plant extracts market.

6 INDUSTRY TRENDS

6.1 INTRODUCTION

This chapter covers various industry trends within the plant extracts market that would help companies remain competitive in the marketplace by adopting development strategies, such as collaborations and partnerships, initiated by companies to provide fully integrated solutions.

Plant extracts are produced from various sources, such as fruits & vegetables, herbs & spices, and flowers. They are used in numerous applications in the food, beverage, cosmetics, and pharmaceutical industries. Plant extracts have been experiencing increased demand due to changes in the lifestyle of consumers and their preferences for natural and organic products. Consumers are increasingly aware of the health benefits associated with plant extracts, which provides an increasing opportunity for manufacturers to incorporate these plant extracts in various applications. These markets seem to develop and formulate new products and technologies using plant ingredients, resulting in innovation, and implementing groundbreaking products with an array of applications. These ingredients are also used in cosmetics and skin care preparation, and the market is booming due to the rising middle class, which has increased their purchasing power and increased awareness related to maintaining healthy skin.

6.2 VALUE CHAIN

This section of the report categorizes the value chain representatives for a more comprehensive understanding of the effective distribution of products to the end user and its stakeholders, resulting in the smooth function of the business and expanding the opportunities globally. This value chain plays an important role as it improves a company's efficiency, eliminates waste, increases brand recognition, and increases profits. The market value chain includes several activities, such as R&D, sourcing of raw materials, manufacturing, packaging & storage, distribution, and marketing & sales of the product, each of which is elaborated as follows:

6.2.1 RESEARCH & PRODUCT DEVELOPMENT

R&D helps in choosing the correct type of raw material. It helps understand the various properties related to each plant type and its strain. This information is useful for the extraction of different types of plant extracts for different applications like food & beverage, pharmaceuticals, and cosmetics. Research models lead to various developments and efficient extraction procedures like CO₂ extraction of plant extracts from the raw material. It also helps optimize processes, provides new and innovative solutions to the customer, and maximizes production. One such example is that of Carbery Group, where in January 2022, it announced the opening of a new R&D facility in Synergy headquarters. The expansion includes space for a nutrition laboratory, beverage application, beverage thermal processing pilot plant, flavor creation lab, and a flavor research and technology lab. This will boost the company's research and help launch new products. Similarly, Symrise, in 2019, opened an expanded innovation center in Indonesia with a total investment of approximately USD 3.4 million. With this modern facility, the company plans to develop innovative solutions for fragrances, cosmetics, food, and beverages.

6.2.2 RAW MATERIAL SOURCING

Plantation cultivation and vertical integration are gaining popularity, providing firms with a consistent supply and greater quality control. This trend strengthens the value chain, but it also risks taking production tools away from poor people, limiting their participation in the value chain. Several companies, including Symrise, have helped locals improve their production techniques by promoting better harvesting and cultivation. This improves the quality of raw materials while benefiting local producers and ecosystems. Companies are increasingly focused on sourcing raw materials in a way that minimizes negative environmental impacts, i.e., in a sustainable manner. They are implementing various strategies to ensure sustainability in response to consumer demand, which is a primary driver of the sustainability imperative. In 2021, Symrise, for example, established a Responsible Sourcing Steering Committee (RSSC) to implement the company's due diligence. The decision-making body establishes the procedures for evaluating raw material suppliers and implementing appropriate measures, as well as for incorporating the assessment criteria into business processes. The company also intends to increase the number of suppliers who set their environmental goals, to achieve 100% responsible sourcing of strategic agricultural raw materials by 2025. Givaudan, a major ingredient manufacturer and supplier, also aspires to become B Corp certified. B Corps are organizations that meet the highest standards of social and environmental performance, public transparency, and legal accountability. It also intends to source all materials and services in a manner that protects both people and the environment by 2030. It has strengthened and expanded its responsible sourcing program into the Sourcing4Good umbrella program.

In the plant extracts market, the global demand for spices is increasing, with more people realizing the health benefits associated. Spice raw material trade is fragmented, with smallholders producing most spices. The sourcing of raw materials is always vulnerable to challenges such as profitability due to fluctuating raw material prices, human rights risks, and the effects of climate change.

6.2.3 MANUFACTURING

In the production segment, the processing of various plant sources occurs to form extracts. A preliminary test to find both primary and secondary metabolites in an extract is called phytochemical screening. To detect, a number of qualitative analyses are used, including the tests of Dragendorff and Mayer for alkaloids and Shinoda for flavonoids. There are several methods for extracting plant extracts, some of which include maceration, infusion, percolation, extraction with supercritical CO₂, and Microwave Assisted Extraction. Plant extracts are separated into different fractions through the process of fractionation. Techniques for fractionation can be categorized as either physical or chemical. The fractional distillation, fractional crystallization, fractional liberation, sublimation, and separation funnel methods are physical techniques for separating compounds from mixtures. Contrarily, chemical methods purify substances using the appropriate reagents based on the type of functional groups they possess.

Plant extracts need to follow stringent regulatory guidelines and must comply with the same. Before any products are sent out for distribution, they must undergo tests at state-accredited labs, which screen the extracts both for quality and for the presence of contaminants. Sustainability has also made its way into the production aspects. For instance, Sensient Technologies Corporation's New Zealand manufacturing facility gets 100% of its electrical energy from renewable sources, helping to offset over 220 metric tons of CO₂ emissions each year. Furthermore, by 2020, all three UK manufacturing facilities were using 100% renewable electricity.

6.2.4 PACKAGING, STORAGE, AND DISTRIBUTION

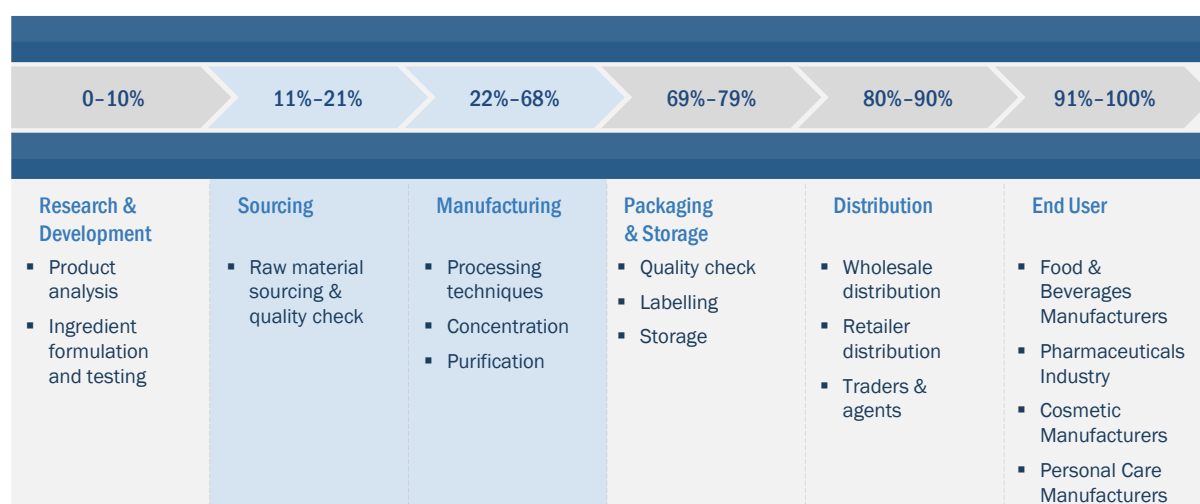
Production is then followed by packaging, storage, and distribution. The shelf-life of different plant extracts varies, and they should be stored separately. Recommended storage areas should be well-ventilated and designed to prevent contamination by insects, rodents, and other pests, with special attention paid to cleanliness and upkeep.

Distribution is possible through both online and offline channels. The rapid growth of the global B2B e-commerce channel opens a plethora of opportunities for both business buyers and sellers. The use of the Internet to facilitate business transactions promises benefits such as increased buyer and seller access. The business-to-business market is still in its infancy and is expected to evolve over time on the Internet.

6.2.5 END USERS

The end users for plant extracts are food & beverage, pharmaceuticals, dietary supplements, cosmetics, and personal care products manufacturers. The end product enters the market to cater to different consumer needs, but before the sales, the product must be authorized by the respective regulatory authority to obtain certificates for sales & marketing. To get approval, the manufacturing companies must submit required data regarding the safety and efficacy of the product in a set format. The regulatory authority is constantly reviewing and framing laws to enhance the entire process as per the scope and function of the various application industries.

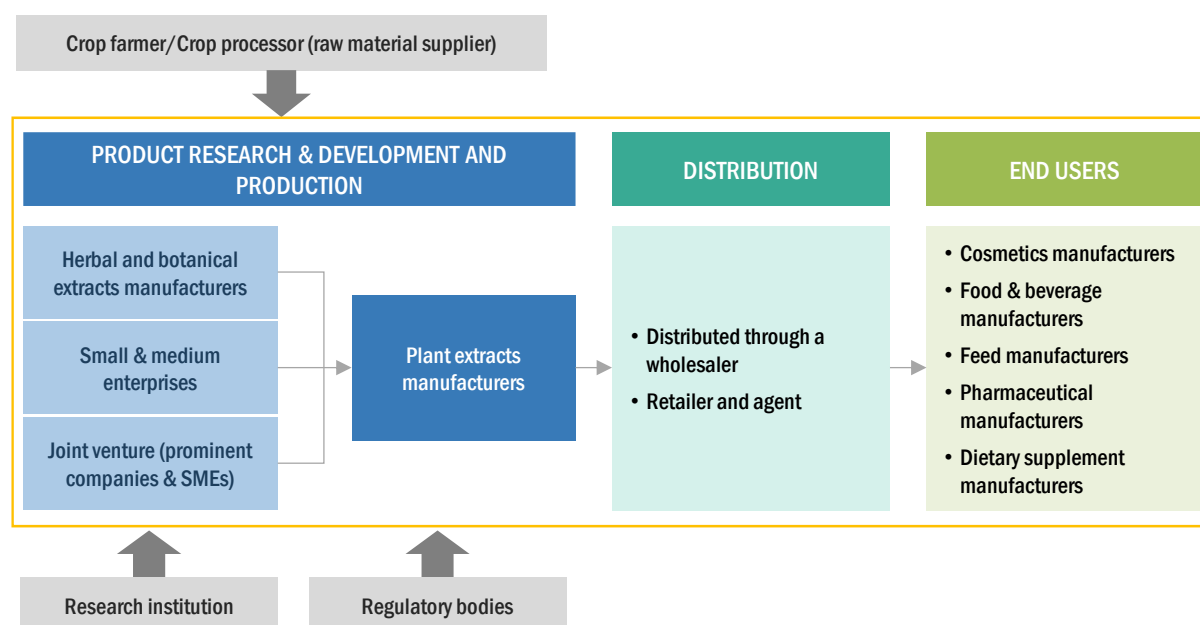
FIGURE 26 PLANT EXTRACTS MARKET: VALUE CHAIN



Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.3 SUPPLY CHAIN ANALYSIS

The supply chain plays an important role in the global plant extracts market. Due to the rapid increase in the demand for organic, plant-based products, the demand for plant extracts is increasing; thus, the supply chain must be efficient. The major participants of the supply chain are suppliers, regulatory bodies, distributors, and end users. To establish a robust supply chain system, various resources for infrastructure and machinery are required, along with capital for running costs of continuous operations. The availability of raw materials and fluctuating prices are factors affecting the market. R&D is a crucial part of the supply chain. Due to the regulations and a rapidly growing industry, there are still many complexities in the supply chain like, in countries which do not have established plant extracts industries, where the industry is fragmented, producers lack the understanding of market access requirements and standards, as well as the technical skills required to implement them and develop extracts suitable for the international market. Furthermore, fluctuations in quality make standardization difficult and reduce extracts' suitability for international markets.

FIGURE 27 PLANT EXTRACTS MARKET: SUPPLY CHAIN

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.4 MARKET MAP AND ECOSYSTEM OF PLANT EXTRACTS

The market map of the ecosystem is categorized into demand-side and supply-side companies.

6.4.1 DEMAND SIDE

The demand-side of the ecosystem includes food & beverage, dietary supplement, pharmaceutical companies, skin care, and cosmetics manufacturers, such as Cronos Group Inc. (the company offers a range of hemp-derived CBD-based skincare products including body oil, lotion, and cream) and Nestlé Health Science partnered with Valbiotis (France) to develop TOTUM-63, a patented combination of five plant extract designed to reduce the risk of developing type 2 diabetes. The partnership aimed to commercialize TOTUM-63, while Danone launched the Volvic Essence range in France and Germany in 2018. It is natural mineral water infused with fruit and herbal extracts. It is a natural and organic choice for consumers seeking healthy hydration because it contains no sugar or additives. The global demand for plant extract products is enormous as these companies continuously invest in developing new products of high quality and with a longer shelf-life.

6.4.2 SUPPLY SIDE

The supply side covers plant extract manufacturers, regulatory bodies, certification providers, start-ups, and wholesalers & distributors. The key manufacturers and suppliers in this market are Givaudan (Switzerland), Symrise (Germany), Synthite Industries Ltd (India), and Indesso (Indonesia). The regulatory bodies include the European Medical Cannabis Association, The Bureau of Food and Drugs (BFAD), The Food and Agriculture Organization, and National Agency for Food Herb Society of America, the European Flavour Association, and Drug Administration and Control (NAFDAC).

FIGURE 28 PLANT EXTRACTS MARKET: ECOSYSTEM MAP

Source: Secondary Literature and MarketsandMarkets Analysis

6.4.3 ECOSYSTEM MAP

TABLE 3 PLANT EXTRACTS MARKET: ECOSYSTEM

COMPANY	ROLE IN THE ECOSYSTEM
Vidya Herbs (India)	Manufacturer
Symrise (Germany)	Manufacturer
Sensient Technologies Corporation (US)	Manufacturer
Givaudan (Switzerland)	Manufacturer
ADM (US)	Manufacturer
Kerry Group Plc (Ireland)	Manufacturer
Arjuna Natural (India)	Manufacturer
Synthite Industries Ltd (India)	Manufacturer
Herbochem (India)	Manufacturer

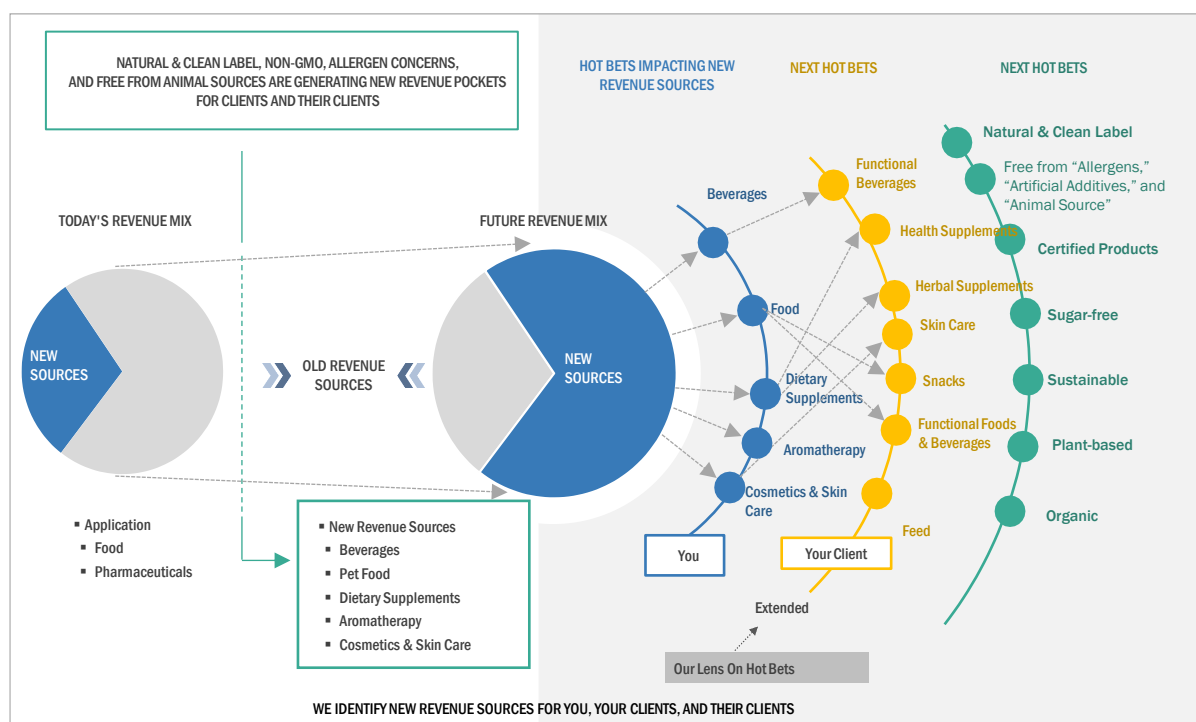
Indesso (Indonesia)	Manufacturer
Rutland Biodynamics (US)	Manufacturer
Augustus Oils Ltd (UK)	Manufacturer
Green Plant Extract (Spain)	Manufacturer
Prinova Europe (UK)	Manufacturer
Freedom Foods Group Limited (UK)	Food & beverages (End User)
Frías Nutrición (Spain)	Food & beverages (End User)
Danone SA (Spain)	Food & beverages (End User)
Unilever plc (UK)	Food & beverages (End User)
Brisan Group (US)	Food & beverages (End User)
Nestlé (Switzerland)	Food & beverages (End User)
PepsiCo (US)	Food & beverages (End User)
GoodLifeNutrition (India)	Dietary Supplements (End User)
Biogena (Austria)	Dietary Supplements (End User)
Gaia Herbs (US)	Dietary Supplements (End User)
The Boots Company Plc (UK)	Cosmetic manufacturer (End User)
Josie Maran Cosmetics LLC (US)	Cosmetic manufacturer (End User)
Aura Herbal Pvt Ltd (India)	Cosmetic manufacturer (End User)
Parchem Fine & Specialty Chemicals (US)	Distributor
DYNADIS SARL (France)	Distributor
Natac (Spain)	Distributor

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

6.5 TRENDS/DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

6.5.1 REVENUE SHIFT AND NEW REVENUE POCKETS IN PLANT EXTRACTS MARKET

The diagram illustrates some of the key influential entities or factors that would help drive the demand for plant extracts for target customers as well as to its subsequent customers. From the client's perspective, functional food & beverage, health supplement, skin care, and personal care products would help generate demand from the client's side. For its subsequent client, some of the key mandatory labels such as natural and clean labels, certified product suppliers, and organic would help generate sales growth.

FIGURE 29 REVENUE SHIFT IMPACTING PLANT EXTRACTS MARKET

Source: Secondary Literature and MarketsandMarkets Analysis

The use of plant extracts in beverages is frequently associated with their specific health properties, such as the ability to energize, boost the immune system, improve skin health, increase focus, and provide relaxation. Cosmetics & skin care products are also increasing the use of plant extracts in their products due to the rising demand for chemical-free products. Moreover, with the rising trend of healthy eating and a healthy lifestyle, the demand for herbal dietary supplements has increased tremendously. These factors would provide growth opportunities for plant extract manufacturers due to increased demand from end users like dietary supplements, food & beverage, and pharmaceuticals. Moreover, as the demand for clean-label and natural alternatives increases, the market for plant extract will see a further increase.

6.6 TECHNOLOGY ANALYSIS

6.6.1 GREEN TECHNOLOGIES

More environmentally friendly and selective extraction techniques are in high demand. Safer extraction methods protect the environment and its natural resources while limiting the negative impact of human involvement. This has heightened the interest in supercritical fluid extraction (SFE), microwave-assisted extraction (MAE), and ultrasound-assisted extraction (UAE), all of which have been used to improve the extraction of plant extracts compounds from various sources. Supercritical fluid extraction is a green natural product extraction technology (SFE).

Supercritical fluid extraction (SFE) is the process of separating one component (the extractant) from another (the matrix) by using supercritical fluids as the extracting solvent. SFE can be used as a sample preparation step for analytical purposes, or on a larger scale to remove unwanted material from a product (e.g., decaffeination) or to collect the desired outcome (e.g., essential oils). Supercritical carbon dioxide extraction conditions are above the critical temperature of 31°C and critical pressure of 74 bar. SFE is primarily used on a large scale for decaffeination and the production of hop extracts. Other industrial applications, such as the extraction of aroma and oil from fruits and spices, are gaining popularity.

Ultrasound and microwave-assisted extraction provide several benefits, including efficient cell wall disruption. Microwave-assisted extraction (MAE) uses microwave radiation to heat the sample-solvent mixture, increasing the solvent's accessibility in the sample and speeding up the partitioning process. Ultrasound assisted extraction (UAE) generates local hotspots at the macroscopic scale with high shear stress and temperature by producing cavitation bubbles that burst at the surface of the plant sample matrix, inducing the destruction of plant cell walls, and facilitating the mass transfer of active compounds into the solution. In comparison to traditional methods, ultrasound and microwave-assisted extraction not only provide efficient cell wall disruption but also use less solvent and take less time.

PHWE, also known as Pressurized Solvent Extraction (PSE) or Accelerated Solvent Extraction (ASE), uses high temperatures and pressures to improve the extraction of phytochemicals. The high pressure is used to maintain the solvent in the liquid phase even when heated above its boiling point. The process needs only a small amount of solvent but requires labor-intensive washing and filtration after the extraction step. Extraction efficiency can be improved by selecting the appropriate solvent or mixture, pressure (usually 3–20 MPa), temperature (usually 100–200 °C), and extraction time. High temperature leads to an increase in the conductivity of the water and a decrease in surface tension, leading to a rise in the propagation of phytochemical molecules through the water. For different classes of compounds present in a variety of matrices, including environmental, food, and botanical samples, pressurized hot water extraction (PHWE) has emerged as a popular green extraction technique. Using water alone as the solvent is one of the greenest and cleanest technologies. Water can easily change its physico-chemical characteristics, such as its self-ionization, dielectric constant, viscosity, diffusivity, density, and surface tension, in a pressurized hot water extraction (PHWE), enhancing its solvent properties. Additionally, PHWE has an environmentally friendly process, a quicker extraction time, and lower extracting agent costs.

6.6.2 DRYING TECHNOLOGY

For a longer shelf-life, botanical extracts, like aqueous extracts, are dried. The optimized and well-controlled drying process can result in a high-quality product with minimal active ingredient losses. Some emerging advanced drying technologies include freeze drying, microwave vacuum drying, and infrared drying.

Freeze-drying food involves lowering the temperature of the product to below freezing and then extracting the water in the form of vapor using a high-pressure vacuum. The vapor condenses on a condenser, freezes, and is removed. The key advantages of freeze drying over conventional drying technologies include the retention of morphological structures as well as biological and chemical properties, as well as the high recovery of volatile and heat-sensitive components. Because of the high energy consumption and operating and maintenance costs, freeze drying remains one of the most expensive drying processes. Today, freeze-drying is used in a variety of common food products such as fruits and vegetables. Since their volume and weight can be greatly reduced, freeze-dried extracts are simple to store and transport. Additionally, by using this process to concentrate the product, their potency compared to their natural form is increased. Moreover, there is a high retention of flavor and aroma. These advantages make freeze drying a popular choice.

Microwave vacuum drying is a method of dehydration that combines microwave radiation energy and vacuum. Microwave radiation can rapidly heat samples by converting electromagnetic energy to thermal energy. It is more cost-effective than freeze drying because the drying process is faster, allowing for a higher throughput for the same plant dimensions. Microwave energy can significantly improve the drying of aromatic and medicinal plants, conserve energy, and maintain product quality; thus, it is increasingly used by manufacturers.

6.6.3 CELLULAR EXTRACTION

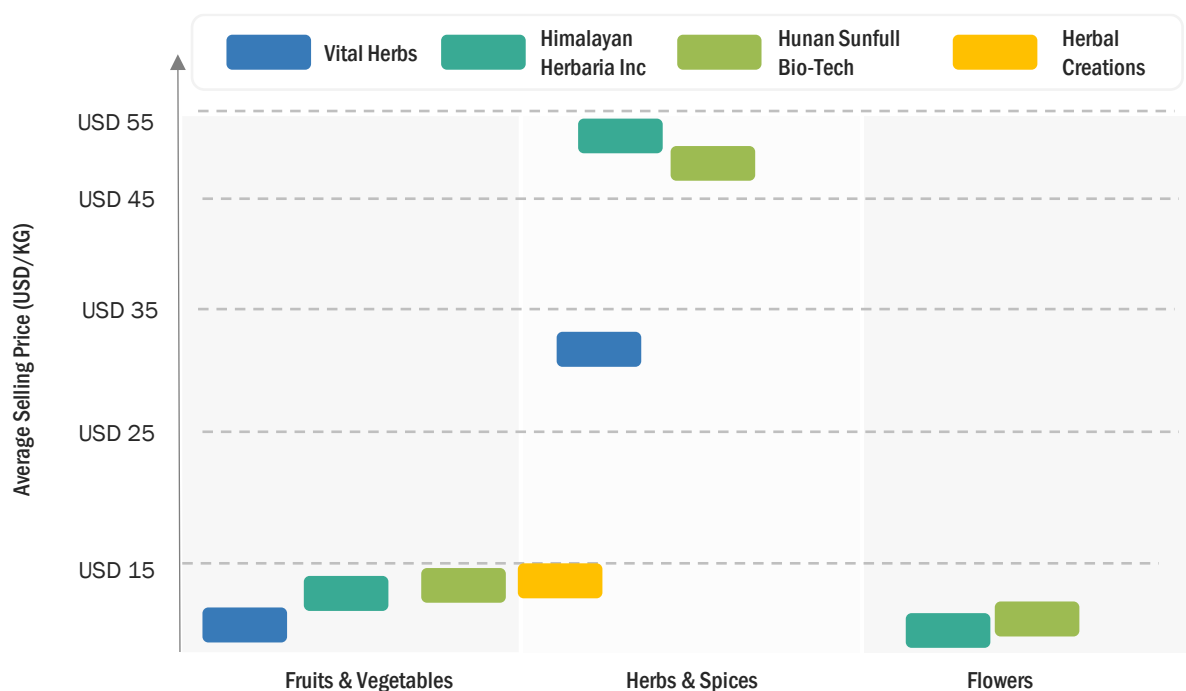
Cellular extraction is a method of incorporating water-soluble plant extracts (the entourage) into a liquid matrix while preserving the plant matter. To achieve an equilibrium between a plant cell and the extractant environment, cellular extraction is performed in a pressurized, airless environment with increased osmolarity. Cellular extraction provides a complete entourage of phytoactive nutrients. It also ensures that the molecule's cellular integrity and stability are not compromised. It effectively delivers the plant's full profile of water-soluble or oil-soluble phyto-compounds for long periods, making them as bioavailable as they are in the plant itself.

6.7 PRICING ANALYSIS

In the global plant extracts market, the product based on source is divided into fruits & vegetables, herbs & spices, and flowers. This section provides insights into the pricing of various sources offered in the market based on the region.

6.7.1 SELLING PRICE CHARGED BY KEY PLAYERS IN TERMS OF PRODUCT SOURCE

FIGURE 30 SELLING PRICE OF KEY PLAYERS FOR PLANT EXTRACTS' PRODUCT SOURCE



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

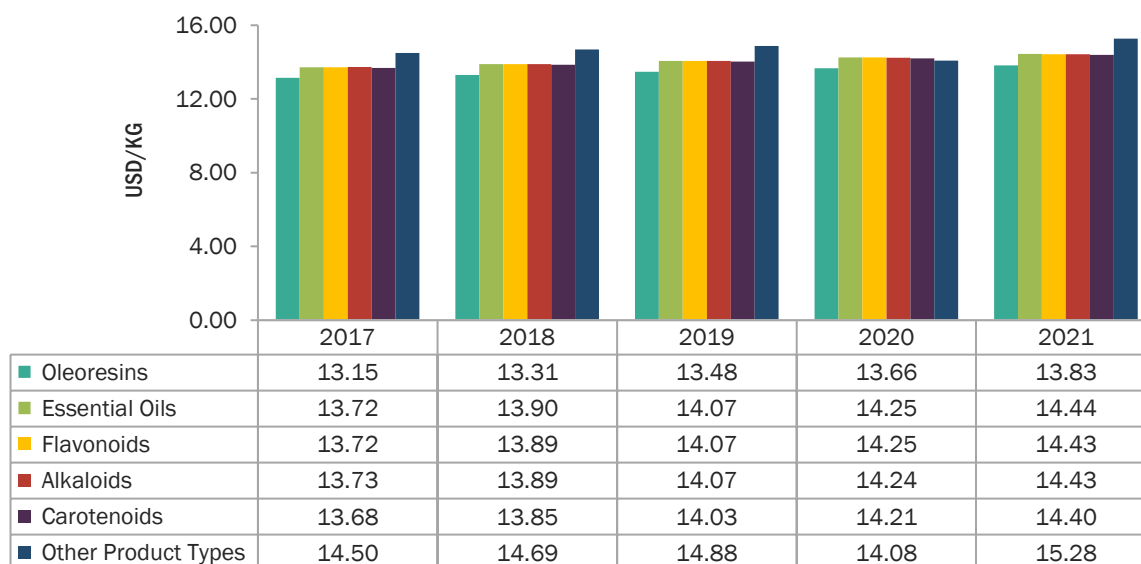
TABLE 4 SELLING PRICE OF KEY PLAYERS FOR PRODUCT FORM (USD/KG)

PRODUCT FORM	VITAL HERBS	HIMALAYAN HERBARIA INC.	HUNAN SUNFULL BIO-TECH	HERBAL CREATIONS
Fruits & Vegetables	6.26	8.13	9.00	15.02
Herbs & Spices	31.30	56.3	50.1	-
Flowers	-	1.1	1.00	-

Source: Investor Presentations, Expert Interviews, Industry Journals, Magazines, and MarketsandMarkets Analysis

A range of factors influences the prices, including the cultivation environment, cultivars, yield, facility scale, and processing. The pricing in the plant extracts domain greatly depends on the raw material supplier, extraction process, and season. Thus, there are variations in the prices for different manufacturers.

FIGURE 31 AVERAGE SELLING PRICE IN KEY REGIONS, BY PRODUCT TYPE, 2017–2021 (USD/KG)



Source: Company Publications and MarketsandMarkets Analysis

The key product types gaining traction include essential oils, oleoresins, flavonoids, alkaloids, and carotenoids. The prices of essential oil vary depending on the extraction procedure and purity. Pure essential oil is expensive compared to essential oil blended in a carrier oil, as specific farming conditions must be maintained to cultivate a plant that will be used as a raw material for the extraction, which is only possible at a higher cost. A significant amount of the raw plant material is required to create just one drop of essential oil because essential oils are highly concentrated plant constituents. These factors combined with testing, packaging, storage, distribution, and marketing add to the cost of the final product.

TABLE 5 OLEORESINS: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

Region	2017	2018	2019	2020	2021
North America	13.26	13.42	13.59	13.76	13.94
Europe	14.23	14.41	14.60	14.79	14.98
Asia Pacific	12.75	12.91	13.07	13.23	13.40
Rest of the World	12.35	12.51	12.67	12.84	13.01

Source: Company Publications and MarketsandMarkets Analysis

TABLE 6 ESSENTIAL OILS: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

Region	2017	2018	2019	2020	2021
North America	14.08	14.26	14.44	14.62	14.81
Europe	14.67	14.86	15.05	15.25	15.45
Asia Pacific	13.41	13.57	13.74	13.91	14.09
Rest of the World	12.73	12.89	13.06	13.24	13.41

Source: Company Publications and MarketsandMarkets Analysis

TABLE 7 FLAVONOIDS: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

Region	2017	2018	2019	2020	2021
North America	14.37	14.55	14.73	14.92	15.11
Europe	14.37	14.56	14.75	14.94	15.13
Asia Pacific	13.27	13.44	13.60	13.77	13.95
Rest of the World	12.85	13.02	13.19	13.37	13.55

Source: Company Publications and MarketsandMarkets Analysis

TABLE 8 ALKALOIDS: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

Region	2017	2018	2019	2020	2021
North America	14.37	14.55	14.73	14.92	15.11
Europe	14.37	14.56	14.75	14.94	15.13
Asia Pacific	13.27	13.44	13.60	13.77	13.95
Rest of the World	12.85	13.02	13.19	13.37	13.55

Source: Company Publications and MarketsandMarkets Analysis

TABLE 9 CAROTENOIDS: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

Region	2017	2018	2019	2020	2021
North America	14.22	14.40	14.58	14.77	14.96
Europe	14.37	14.56	14.75	14.94	15.13
Asia Pacific	13.27	13.44	13.60	13.77	13.95
Rest of the World	12.85	13.02	13.19	13.37	13.55

Source: Company Publications and MarketsandMarkets Analysis

TABLE 10 OTHER PRODUCT TYPES: AVERAGE SELLING PRICE, BY REGION, 2017–2021 (USD/KG)

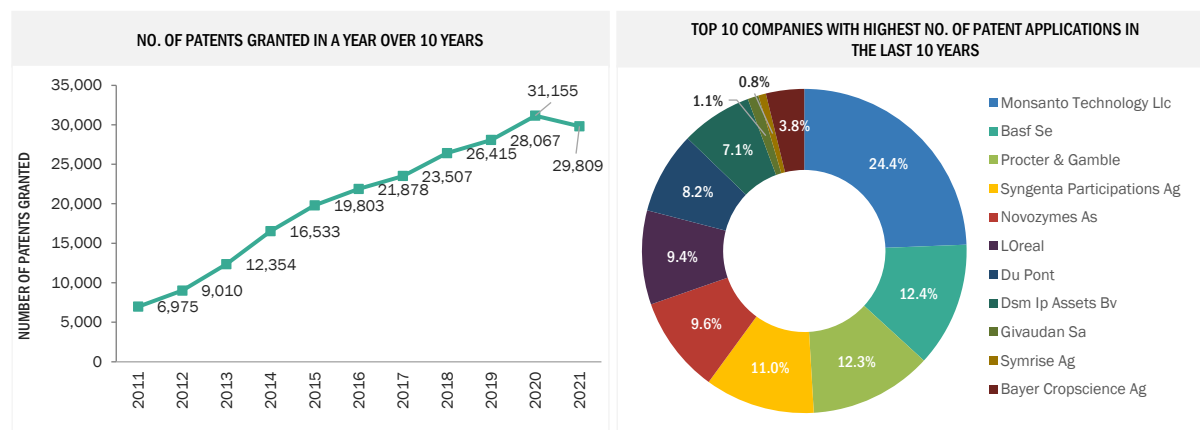
Region	2017	2018	2019	2020	2021
North America	14.38	14.56	14.74	14.93	15.12
Europe	16.02	16.24	16.45	16.67	16.90
Asia Pacific	14.79	14.98	15.18	15.37	15.57
Rest of the World	12.82	12.99	13.16	13.34	13.52

Source: Company Publications and MarketsandMarkets Analysis

6.8 PLANT EXTRACTS MARKET: PATENT ANALYSIS

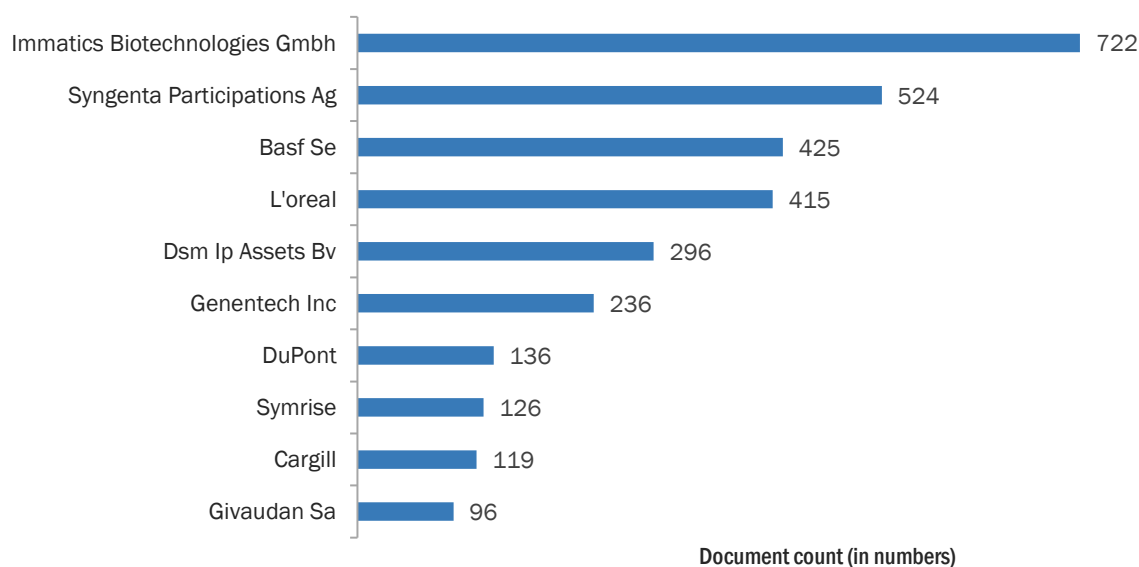
The plant extracts market has witnessed exponential growth in the last few years. With the rise in demand and increase in the number of R&D activities pertaining to plant extracts, companies are testing new compositions to develop products with enhanced features. For the patent analysis conducted for this study, patents were captured from Lens.org, where more than 242,557 patents were granted related to the plant extracts market between 2019 and 2022.

FIGURE 32 NUMBER OF PATENTS GRANTED FOR PLANT EXTRACTS, 2011–2021

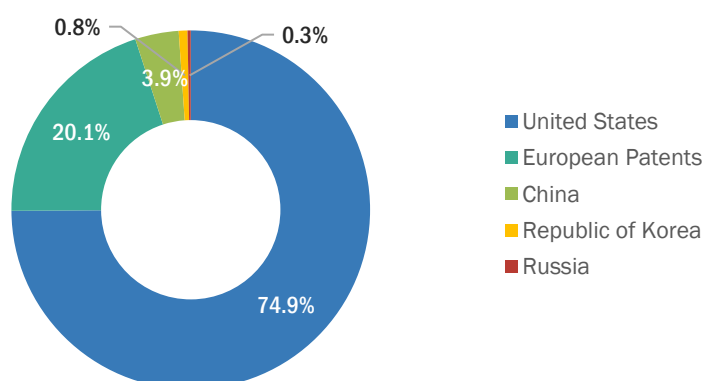


Source: Lens.org

FIGURE 33 TOP PATENT APPLICANTS FOR PLANT EXTRACTS, 2019–2022



Source: Lens.org

FIGURE 34 REGIONAL ANALYSIS OF PATENTS GRANTED FOR PLANT EXTRACTS, 2019-2022

Source: Lens.org

6.8.1 LIST OF MAJOR PATENTS

This section showcases product innovations through an overview of the patents.

TABLE 11 LIST OF FEW PATENTS IN PLANT EXTRACTS MARKET, 2019–2022

MONTH & YEAR	TITLE	PUBLICATION NUMBER	INVENTOR	DESCRIPTION
July 2022	Multipurpose mosquito repellent composition and method of manufacturing the same	US 11395496 B2	Loc Thanh	A multipurpose mosquito repellent composition is created by combining a foundation mixture of several aromatic and medicinal oils with an anti-insect extract or essential oil and at least two odorless and anti-insect effect extending extracts or essential oils.
March 2022	Plant extract inhibiting tyrosinase activity and application of plant extract	CN 110433114 A	Hunan Univ of Sci & Eng	The invention discloses a plant extract inhibiting tyrosinase activity and application of the plant extract. The plant extract is prepared from a ginkgo biloba extract, a pomegranate rind extract, a pyracantha fortuneana extract, a radix angelicae dahuricae extract, and a small convolvulus extract.
May 2020	Skincare plant extract, extracting device and extracting process	CN 111151024 A	Liang Chen	The invention relates to a skin care plant extract, an extracting device, and an extracting process. The skin care plant extract comprises the following raw materials in parts by weight: 25 parts of ginseng, 15 parts of edible tulip, 10 parts of rhizoma corydalis, 7 parts of radix bupleuri, and 7 parts of safflower carthamus.

March 2020	Plant extract composition and preparation method and application of plant extract composition in cosmetics	CN 110897957 A	Jiangsu Ruiting Biotechnology Co Ltd	The invention discloses a plant extract composition and a preparation method and application of the plant extract composition in cosmetics.
January 2020	Plant Extracts	SE 542639 C2	Oriflame Cosmetics Ag	Cosmetic formulations comprising plant extracts from Glycyrrhiza uralensis. The present invention provides a plant extract obtained from the Glycyrrhiza uralensis.
November 2019	Plant Extracts for the Treatment and Prevention of Infections	CA 2823086 C	Kamedis Ltd, Bio Fd&c Co Ltd	A composition comprising a Sanguisorbae officinalis plant extract and at least one additional plant extract selected from the group consisting of an Ailanthus altissima plant extract.

Source: Lens.org

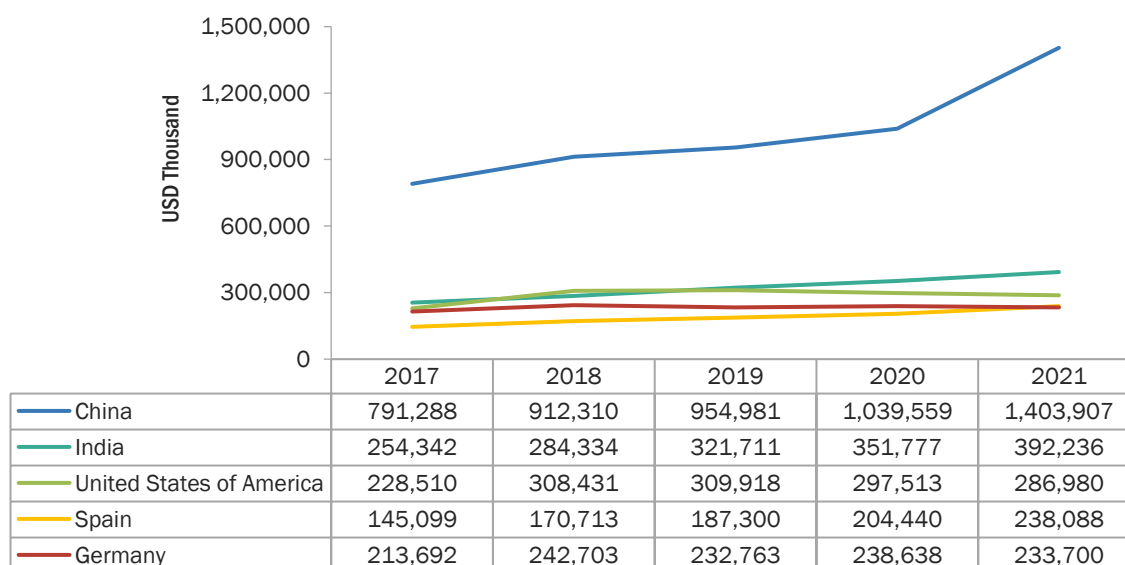
6.9 TRADE ANALYSIS: PLANT EXTRACTS MARKET

6.9.1 EXPORT SCENARIO OF VEGETABLE SAPS AND EXTRACTS

Major exporting countries include China, India, the US, Spain, and Germany. The global demand for plant extracts is increasing, and the pandemic has induced a positive demand shift. Increased global demand for plant extracts can translate into better economic prospects for the growing middle class, as well as increased demand for chemical-free, organic, and sustainable products. The growing consumer awareness of the importance of a healthy lifestyle and food safety has also contributed to the growing demand for plant extracts, which are then used in the production of food & beverages, cosmetics, pharmaceuticals, and dietary supplements.

The Government has taken several measures to promote India's exports. For example, under the Department of Commerce's Market Access Initiative (MAI) Scheme, EPCs/Trade Bodies are provided financial assistance for participation and organization of Trade Fairs, Buyer Seller Meets (BSMs), research & product development, and market studies. Furthermore, the Ministry of AYUSH is assisting the industry in the establishment of quality standards through its quality certification programs such as AYUSH mark and Premium mark. Finally, the Ministry of AYUSH signed MoUs with 25 countries to promote traditional medicine and homeopathy, including Nepal, Bangladesh, WHO Geneva, Germany (joint declaration), Brazil, and Zimbabwe, which will benefit from exports in the long run.

FIGURE 35 VEGETABLE SAPS AND EXTRACTS EXPORTS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 130219 Vegetable saps and extracts (excluding licorice, hops, and opium)

Source: ITC Trade Map

TABLE 12 EXPORT DATA OF VEGETABLE SAPS AND EXTRACTS FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE EXPORTED (USD THOUSAND)	VOLUME EXPORTED (TONS)
China	1,403,907	37,234
India	392,236	14,220
United States of America	286,980	9,067
Spain	238,088	10,491
Germany	233,700	5,246
Italy	126,447	11,750
Vietnam	99,792	13,455
Republic of Korea	81,905	2,155
Switzerland	71,557	1,625
Netherlands	65,528	1,973
Belgium	42,382	1,528
Canada	34,605	1,703
Japan	33,308	474
Brazil	33,213	6,307
Singapore	28,256	530

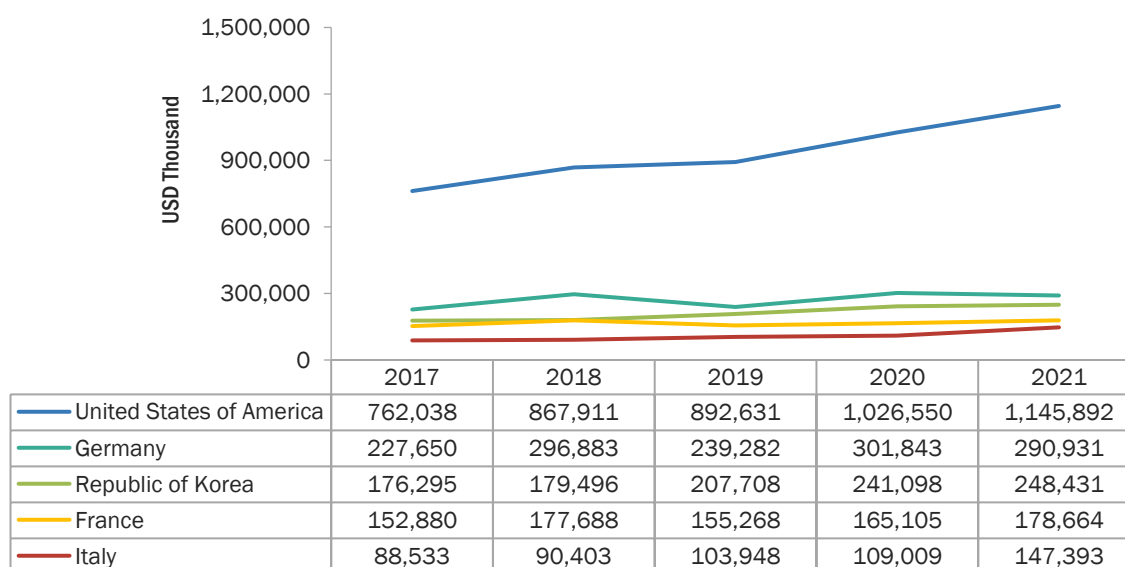
Note: Commodity code for the above data is 130219 Vegetable saps and extracts (excluding licorice, hops, and opium)

Source: ITC Trade Map

6.9.2 IMPORT SCENARIO OF VEGETABLE SAPS AND EXTRACTS

The import of vegetable saps and extract is observed in countries such as the US, Germany, the Republic of Korea, France, and Italy. Organic and natural food products have become priorities for food processors and consumers, owing to the food quality, safety, sustainability, and social acceptance. Consumers in the US and other developed countries are now conscious about their purchasing habits as food labeling is often checked for ingredients' origin, impact of supply chain over environment, and manufacturing practices. Such factors are driving the market for plant extracts in the US. In Germany, drugs derived from natural sources have gained popularity. This means that raw materials for these products, such as fruit & vegetable extracts, medicinal plants and extracts from their leaves, flowers, roots, and seeds, are in high demand. Furthermore, due to concerns such as animal welfare and environmental sustainability, Germany's vegan and the vegetarian population is growing. Following a surge in other categories like meat, the vegan and vegetarian trend is blowing up in chocolate and sugar confectionery, prompting manufacturers to switch to plant-based ingredients, thus increasing the demand for plant extracts, which can be used as additives, flavoring, and coloring agents.

FIGURE 36 VEGETABLE SAPS AND EXTRACTS IMPORTS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 130219 Vegetable saps and extracts (excluding licorice, hops, and opium)

Source: ITC Trade Map

TABLE 13 IMPORT DATA OF VEGETABLE SAPS AND EXTRACTS FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE IMPORTED (USD THOUSAND)	VOLUME IMPORTED (TONS)
United States of America	1,145,892	43,482
Germany	290,931	7,536
Republic of Korea	248,431	8,635
France	178,664	17,720
Italy	147,393	7,653
China	132,862	8,324
Japan	129,028	4,896

Spain	121,553	8,268
Canada	95,792	3,513
Netherlands	93,021	1,3726
Australia	79,075	2,403
United Kingdom	75,767	3,660
Austria	61,375	1,393
Belgium	56,758	4,058
Taipei, Chinese	51,298	853

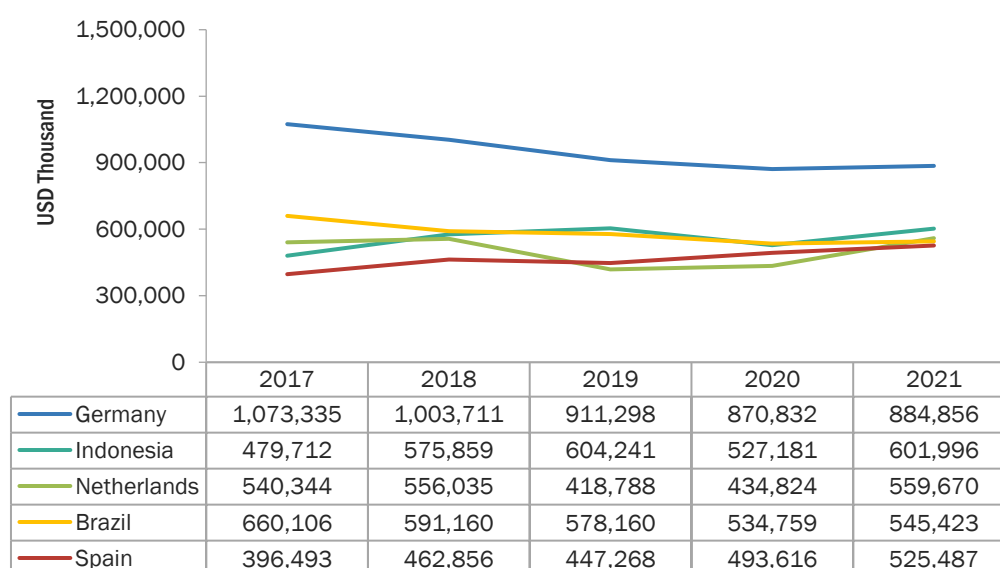
Note: Commodity code for the above data is 130219 Vegetable saps and extracts (excluding licorice, hops, and opium)

Source: ITC Trade Map

6.9.3 EXPORT SCENARIO OF EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ

Major exporting countries include Germany, Indonesia, the Netherlands, Spain, and Brazil. Brazil has continued to export since the COVID-19 pandemic began. The depreciation of the Brazilian currency aided exporters as well. However, a drop in coffee production will result in a drop in global exports, which could cause local supply and demand imbalances and drive up consumer prices. Vietnam, on the other hand, is a major coffee exporter, but it needs to invest in value-added coffee industries if it wants to move up the global coffee supply chain and make the industry more sustainable. This opens up numerous opportunities for foreign companies, particularly those with expertise in the production of processed coffee products. The signing of the EU-Vietnam Free Trade Agreement opens up new opportunities for Vietnamese brands to export processed coffee to the EU.

FIGURE 37 EXPORTS: EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 2101 Extracts, essences, and concentrates, of coffee, tea, or maté and preparations with a basis of these products or with a basis of coffee, tea, or maté; roasted chicory and other roasted coffee substitutes, and extracts, essences, and concentrates thereof

Source: ITC Trade Map

TABLE 14 EXPORT DATA OF EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE EXPORTED (USD THOUSAND)	VOLUME EXPORTED (TONS)
Germany	884,856	95,127
Indonesia	601,996	197,506
Netherlands	559,670	37,145
Brazil	545,423	99,092
Spain	525,487	159,156
Malaysia	517,980	107,474
Vietnam	465,125	0
India	401,636	88,507
United States of America	390,447	68,897
United Kingdom	327,656	34,360
Colombia	302,657	24,057
Poland	297,881	53,283
China	287,555	25,863
France	261,584	30,668
Russian Federation	248,056	52,985

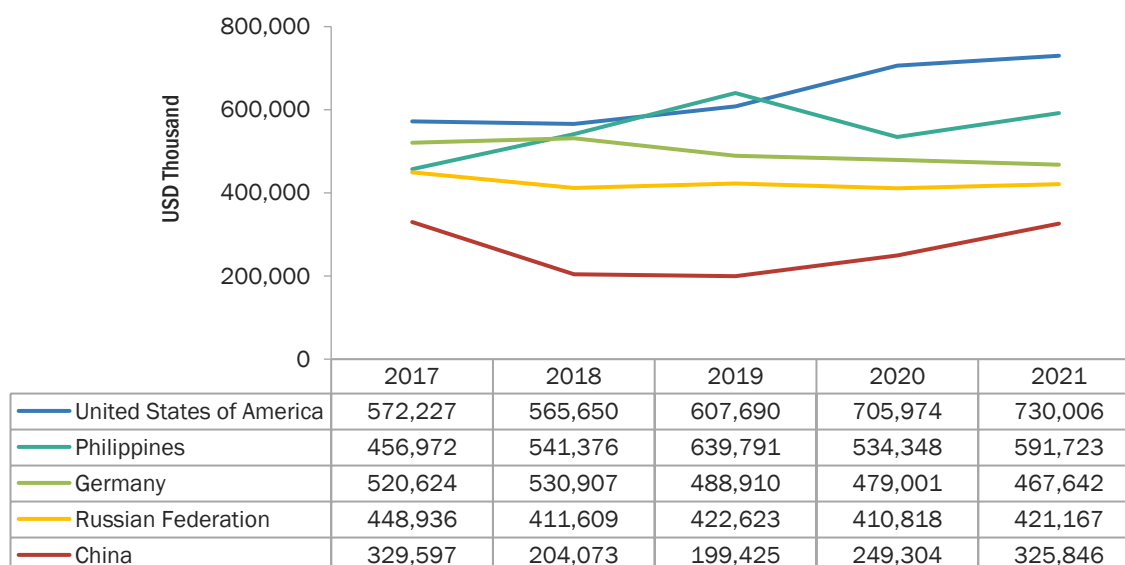
Note: Commodity code for the above data is 2101 Extracts, essences, and concentrates, of coffee, tea, or maté and preparations with a basis of these products or with a basis of coffee, tea, or maté; roasted chicory and other roasted coffee substitutes, and extracts, essences, and concentrates thereof

Source: ITC Trade Map

6.9.4 IMPORT SCENARIO OF EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ

The US is the top importer of coffee and tea extracts, essences, and concentrates. Coffee and tea consumption is increasing in the US, and the driving forces behind the top coffee are quality and convenience. Consumers seek instant coffee because they want speed, savings, and convenience without sacrificing the flavor. In terms of tea, premium and herbal tea are making inroads. Furthermore, coffee and tea-flavored food & beverages are becoming more popular in the country. Coffee is also a great flavor option for the fitness and nutrition categories. These factors are compelling manufacturers to experiment with new flavor combinations, which is driven by the increased consumer demand.

FIGURE 38 IMPORTS: EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 2101 Extracts, essences, and concentrates, of coffee, tea, or maté and preparations with a basis of these products or with a basis of coffee, tea, or maté; roasted chicory and other roasted coffee substitutes, and extracts, essences, and concentrates thereof

Source: ITC Trade Map

TABLE 15 IMPORT DATA OF EXTRACTS, ESSENCES, AND CONCENTRATES OF COFFEE, TEA, OR MATÉ FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE IMPORTED (USD THOUSAND)	VOLUME IMPORTED (TONS)
United States of America	730,006	137,508
Philippines	591,723	208,788
Germany	467,642	70,559
Russian Federation	421,167	65,587
China	325,846	45,962
France	324,744	34,953
Netherlands	284,289	30,774
Poland	282,430	36,725
United Kingdom	253,343	32,555
Japan	213,716	28,202
Canada	211,324	45,090
Australia	193,845	15,148
Turkey	166,025	26,849
Republic of Korea	165,861	15,221
Czech Republic	145,261	20,143

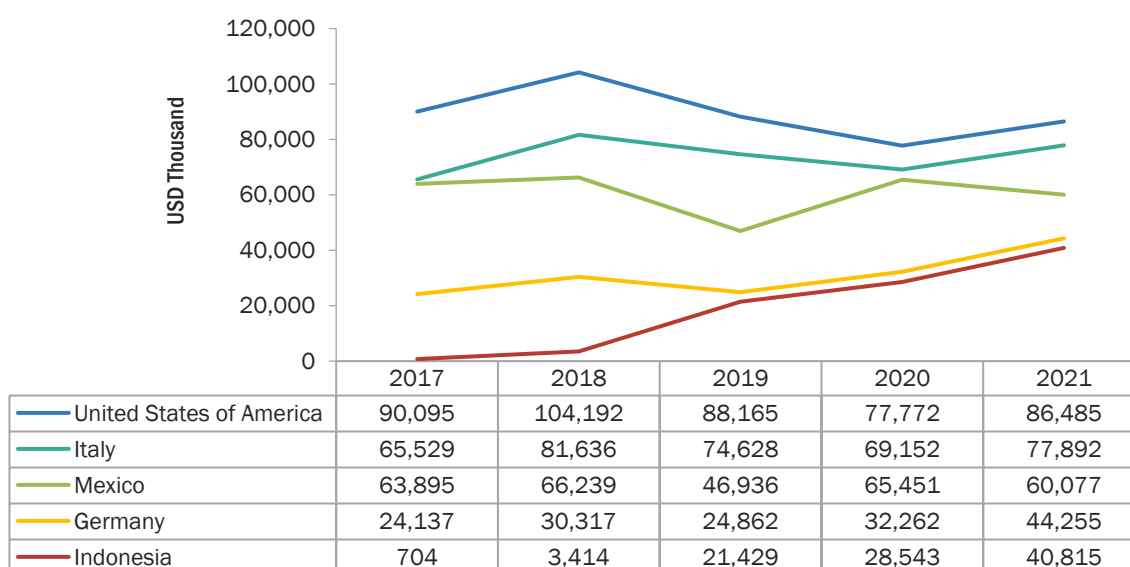
Note: Commodity code for the above data is 2101 Extracts, essences, and concentrates, of coffee, tea, or maté and preparations with a basis of these products or with a basis of coffee, tea, or maté; roasted chicory and other roasted coffee substitutes, and extracts, essences, and concentrates thereof

Source: ITC Trade Map

6.9.5 EXPORT SCENARIO: ESSENTIAL OILS FROM CITRUS FRUITS

Major exporting countries include the US, Italy, Mexico, Germany, and Indonesia. The rising preference of consumers for natural essential oil ingredients such as oranges and the rise in purchasing power parity in developing countries drive the demand for essential oils in applications such as aromatherapy, health care, and food & beverages. Moreover, a rise in awareness about functionalities, health benefits, and applications of essential oils will drive the growth of this market. Indonesia has emerged as the market leader in essential oils. Essential oil exports from Indonesia are primarily used in fragrance, medicine, and cooking. The perfume industry is a major consumer of essential oils. Indesso (Indonesia) is one of Indonesia's successful exporters, marketing essential oils, natural extracts, and aromatic chemicals to global markets. The company exports essential oil to over 20 countries, including developed nations such as the US, Switzerland, France, Canada, the UK, and Japan.

FIGURE 39 EXPORTS: ESSENTIAL OILS OF CITRUS FRUITS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 330119 Essential oils of citrus fruit, whether or not terpeneless, incl. concretes and absolutes (excluding those of sweet and bitter orange and lemon)

Source: ITC Trade Map

TABLE 16 EXPORT DATA OF ESSENTIAL OILS OF CITRUS FRUIT FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE EXPORTED (USD THOUSAND)	VOLUME EXPORTED (TONS)
United States of America	86,485	2,549
Italy	77,892	1,177
Mexico	60,077	1,786
Germany	44,255	757
Indonesia	40,815	1,340
United Kingdom	34,548	888
France	24,339	478
Netherlands	14,464	421
Brazil	12,416	317

Canada	10,854	391
Spain	9,085	399
India	9,052	361
South Africa	8,166	1,112
Egypt	7,392	130
Israel	4,477	0

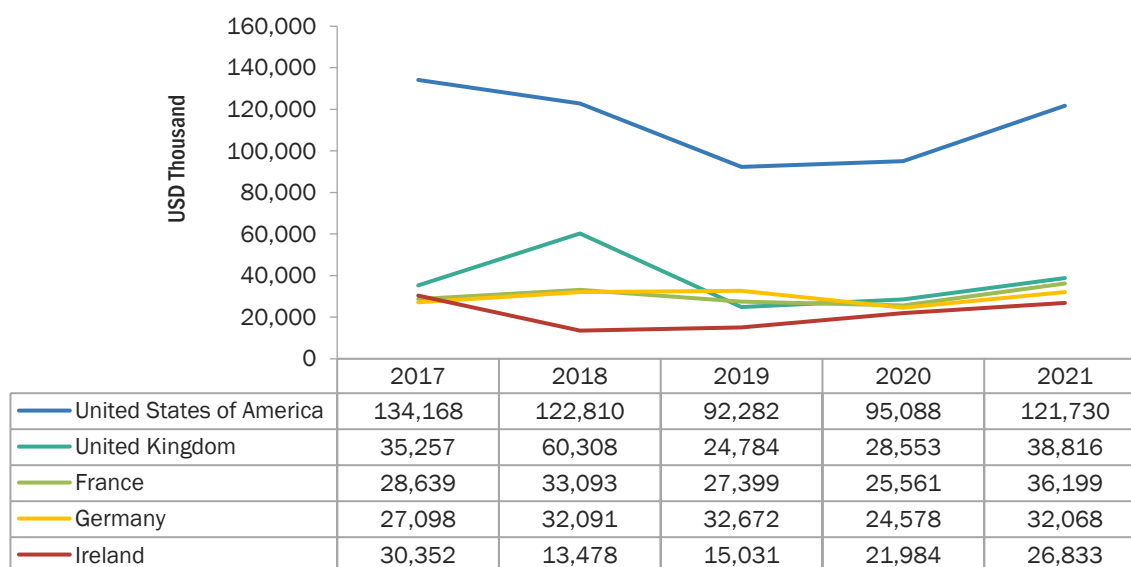
Note: Commodity code for the above data is 330119 Essential oils of citrus fruit, whether or not terpeneless, incl. concretes and absolutes (excluding those of sweet and bitter orange and lemon)

Source: ITC Trade Map

6.9.6 IMPORT SCENARIO: ESSENTIAL OILS FROM CITRUS FRUITS

The US is the largest country in terms of the import and export of essential oils globally. The consumption of essential oil products in the US is high. The market for essential oils is witnessing fast growth due to the increasing applications of essential oils in various end-use applications such as aromatherapy, health care, and food & beverages. The UK is one of the key consumers of essential oils. This market is growing; essential oil products are increasingly used by consumers due to their health benefits. The rising adoption of a healthier lifestyle by consumers in the UK and the increase in the usage of essential oils in food products & consumer health products have impacted the growth of essential oils in various applications in this country.

FIGURE 40 IMPORTS: ESSENTIAL OILS OF CITRUS FRUITS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 330119 Essential oils of citrus fruit, whether or not terpeneless, incl. concretes and absolutes (excluding those of sweet and bitter orange and lemon)

Source: ITC Trade Map

TABLE 17 **IMPORT DATA FOR ESSENTIAL OILS OF CITRUS FRUITS FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)**

COUNTRY	VALUE IMPORTED (USD THOUSAND)	VOLUME IMPORTED (TONS)
United States of America	121,730	6,089
United Kingdom	38,816	1,176
France	36,199	738
Germany	32,068	687
Ireland	26,833	893
China	26,317	362
Saudi Arabia	18,787	306
Japan	18,668	222
Singapore	18,378	396
Netherlands	17,469	463
Mexico	17,024	538
Indonesia	13,161	510
Switzerland	10,239	141
Canada	10,072	651
Italy	9394	280

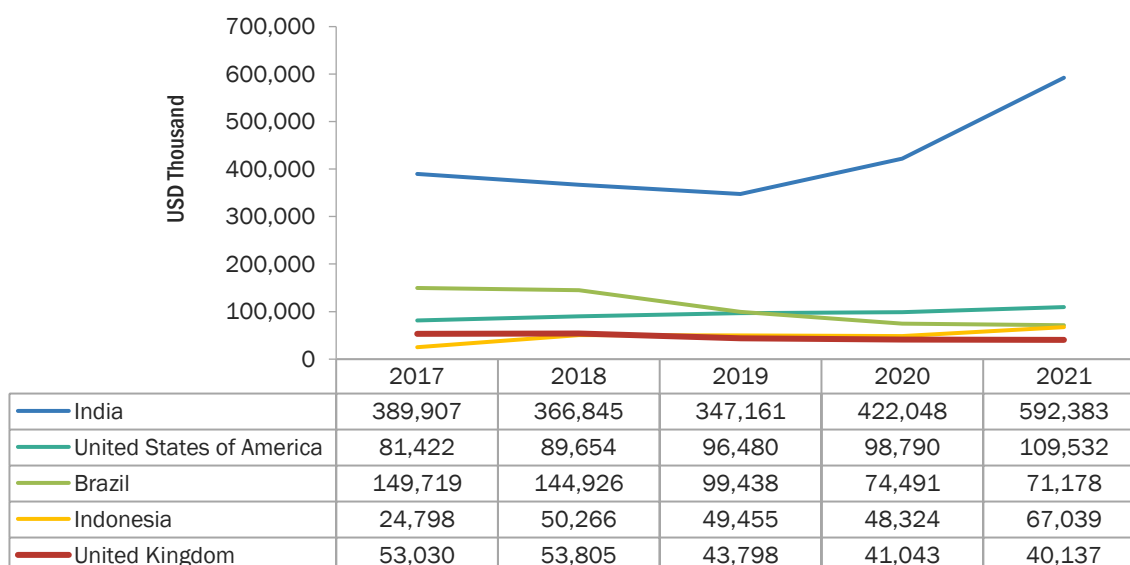
Note: Commodity code for the above data is 330119 Essential oils of citrus fruit, whether or not terpeneless, incl. concretes and absolutes (excluding those of sweet and bitter orange and lemon)

Source: ITC Trade Map

6.9.7 EXPORT SCENARIO: EXTRACTED OLEORESINS

Major exporting countries include India, the US, Brazil, the UK, and Indonesia. Spices can be replaced with oleoresins without losing any of their flavor or aroma qualities. India, China, and Sri Lanka are the top 3 countries that export paprika oleoresin. Oleoresin exports from India, which are spice extracts that give food products color and flavor, increased significantly in 2021 as the pandemic increased the demand for processed foods and snacks. Among Indian spices, spice oil and oleoresins rank fourth in terms of export revenue. Additionally, oleoresin demand is anticipated to increase globally due to rising demand for natural products and rising processed food consumption.

FIGURE 41 EXPORTS: EXTRACTED OLEORESINS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 330190 Extracted oleoresins; concentrates of essential oils in fats, fixed oils, waxes and the like, obtained by enfleurage or maceration; terpenic by-products of the deterpenation of essential oils; aromatic aqueous distillates and aqueous solutions of essential oils

Source: ITC Trade Map

TABLE 18 EXPORT DATA OF EXTRACTED OLEORESINS FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE EXPORTED (USD THOUSAND)	VOLUME EXPORTED (TONS)
India	592,383	20,616
United States of America	109,532	8,167
Brazil	71,178	19,085
Indonesia	67,039	2,666
Sri Lanka	52,085	1,198
United Kingdom	40,137	3,155
Germany	35,918	2,128
France	33,914	1,752
Netherlands	33,412	1,823
Spain	31,928	1,720
China	21,794	1,106
Thailand	21,745	355
Austria	20,213	3,033
Australia	19,852	0
Switzerland	12,138	372

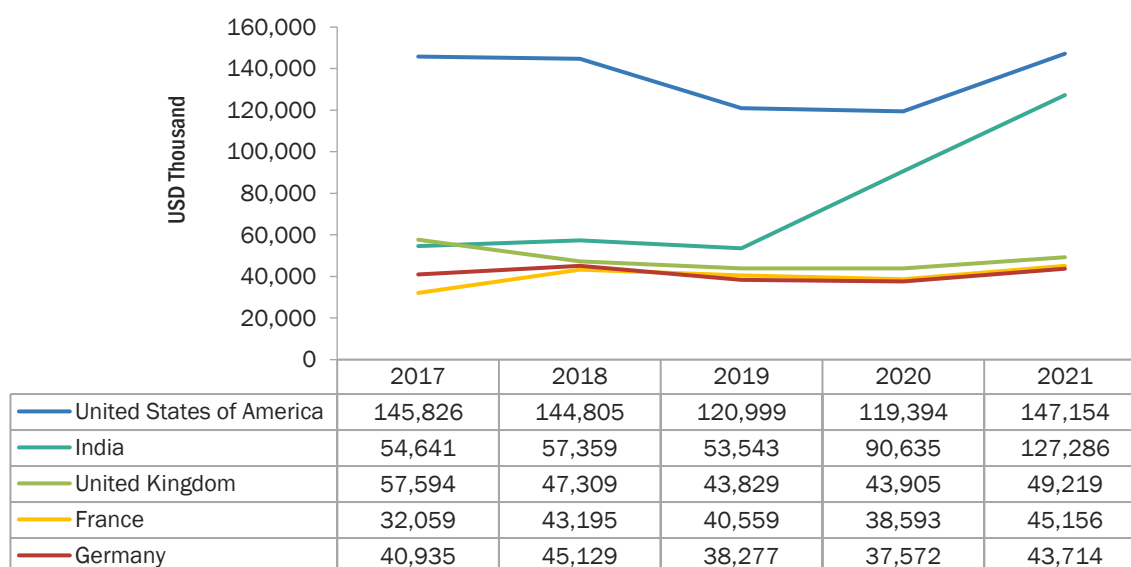
Note: Commodity code for the above data is 330190 Extracted oleoresins; concentrates of essential oils in fats, fixed oils, waxes and the like, obtained by enfleurage or maceration; terpenic by-products of the deterpenation of essential oils; aromatic aqueous distillates and aqueous solutions of essential oils

Source: ITC Trade Map

6.9.8 IMPORT SCENARIO: EXTRACTED OLEORESINS

There is a growing market for healthier food and beverage products that contain natural ingredients in Europe, which has a strong food processing industry. Additionally, consumers in the region are paying more attention to the ingredients in the food and beverage products they purchase. In Europe, there is a growing market for natural plant flavors and colors. For producers of oleoresins in developing nations, this is an opportunity.

FIGURE 42 IMPORTS: EXTRACTED OLEORESINS, BY KEY COUNTRY, 2017–2021 (USD THOUSAND)



Note: Commodity code for the above data is 330190 Extracted oleoresins; concentrates of essential oils in fats, fixed oils, waxes and the like, obtained by enfleurage or maceration; terpenic by-products of the deterpenation of essential oils; aromatic aqueous distillates and aqueous solutions of essential oils

Source: ITC Trade Map

TABLE 19 IMPORT DATA FOR EXTRACTED OLEORESINS FOR KEY COUNTRIES, 2021 (VALUE AND VOLUME)

COUNTRY	VALUE IMPORTED (USD THOUSAND)	VOLUME IMPORTED (TONS)
United States of America	147,154	8,914
India	127,286	4,955
United Kingdom	49,219	3,459
France	45,156	3,636
Germany	43,714	3,995
Spain	43,043	2,458
China	38,969	2,427
Indonesia	33,585	1,833
Thailand	31,484	619
United Arab Emirates	29,289	0
Korea, Republic of	28,522	963
Netherlands	26,990	2,206

Saudi Arabia	25,664	1,654
Mexico	24,738	1,303
Singapore	19,218	0

Note: Commodity code for the above data is 330190 Extracted oleoresins; concentrates of essential oils in fats, fixed oils, waxes and the like, obtained by enfleurage or maceration; terpenic by-products of the deterpenation of essential oils; aromatic aqueous distillates and aqueous solutions of essential oils

Source: ITC Trade Map

6.10 CASE STUDIES: PLANT EXTRACTS MARKET

6.10.1 ARJUNA NATURAL: NATURAL SOLUTION FOR FRUIT JUICE PRESERVATION

Problem Statement	A women's health food brand aspired to develop the first one-of-a-kind solution for healthy living for women. It desired to provide customers with 100% healthy fruit juice. This translates to the use of strong preservatives to achieve the required shelf-life, which contradicts the brand philosophy and approach in search of a 100% natural solution.
Solutions Offered	The client's team collaborated closely with Arjuna's scientific research team. Fruit juice samples were examined to determine the causes of spoilage. Acetic acid bacteria, yeast, and Rhizopus sp. were the microbes identified in this case. They wanted a shelf-life of at least 6 months. The search for viable solutions began to combat microbes. A potential winner was identified and formulated through careful selection and experimentation of various natural extracts. The client was presented with Extenfo for fruit juice (AXFC 105).
Outcome	The brand evaluated the shelf-life of its product using the new natural preservatives. The product had a shelf-life of 9 months and no sensory fouling or sedimentation at an applied dosage of 750 PPM. This allowed them to make the label claim 'Natural Flavor,' claiming that it was a 100% natural product.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.10.2 SYMRISE: HALAL VANILLA

Problem Statement	Natural and halal-certified raw materials with authentic flavor profiles are in high demand. Muslim consumers want high-quality vanilla in their products, with the same complex and rich flavor profile as traditional vanilla extracts.
Solutions Offered	In Madagascar, Symrise chose a fully backward integrated sourcing model. Direct collaboration with farmers ensures quality and supply. Furthermore, its quality control lab in Madagascar enabled it to consistently check the quality of vanilla beans. The company improved its processing even further, such as improving the extraction method to preserve the volatile and sensitive flavor components of vanilla beans in extracts. This enabled it to produce halal extracts without the need for a heating step.
Outcome	Symrise improved the extraction method, allowing for the production of vanilla ingredients without the use of alcohol, ensuring the full and authentic flavor profile of vanilla. It was able to use the best and most responsibly sourced vanilla beans in Madagascar due to its backward integrated approach.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.11 KEY CONFERENCES AND EVENTS, 2022–2023

This section overviews the key conferences and events occurring in different quarters during 2022–2023.

TABLE 20 KEY CONFERENCES AND EVENTS IN PLANT EXTRACTS MARKET

YEAR/ QUARTER	NORTH AMERICA	EUROPE	ASIA PACIFIC	REST OF THE WORLD
Q1'22	-	[March] EOHUB Meeting In UK	[March] 6th Edition of Global Congress on Plant Biology and Biotechnology	[March] CannX 2022: The 6th International Medical Cannabis Conference
	-	[February] The RCA Conference	-	-
Q2'22	[July] NBJ Summit	[May] 2nd Annual European Cannabis Conference	[May] Essential Oil Association of India International Congress and Expo 2022	[May] Globoil
	[June] CannabisWiki Conference + Expo	[May] Vitafoods	[May] International Congress and Expo, 2022	-
	-	[June] The International Federation of Aromatherapists (IFA)	[May] International Congress & Expo 2022	-
Q3'22	[July] 2022 IFT	[July] ICBC B2B	-	[September] WPE-WHPE Exhibition
	[September] doTERRA Connection 2022	[September] 52nd International Symposium on Essential Oils	-	[September] 2022 FI Asia
	[July] KIND Cannabis Week	[August] 70th International Congress and Annual Meetings of the Society for Medicinal Plant and Natural Product Research	-	-
Q4'22	[October] SupplySide West - Las Vegas	[November] FI Europe - Paris	[November] CPHI India	[November] Gulfood 2022
	[September] The International Federation of Essential Oils and Aroma Trades	[December] International Conference on Chemistry of Essential Oils	[December] Natural Extracts 2022	-
Q1'23	[March] EXPO WEST	[January] Future cannabis strategies Europe 2023	[January] International Spice Conference 2023	[February] Gulfood 2023
	-	[March] International Conference on Essential Oils	-	-

		[March] International Conference on Essential Oils	-	-
Q2'23		[April] 26th Euro-Global Summit on Food and Beverages , Advanced Technological Developments in Food Exploration	[May] Traditional Medicine, Ethnomedicine And Natural Therapies	-
		[June] International Conference on Antimicrobial Plant Extracts, Phytochemicals and Plant Products	-	-
Q3'23	-	-	-	-
Q4'23		[October] The International Federation of Essential Oils and Aroma Trades (IFEAT) conference		
		[November] International Conference on Biostimulants in Plants Science, Seaweed and Plant Extracts		

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.12 TARIFF AND REGULATORY LANDSCAPE

6.12.1 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

There are different regulatory bodies to govern the plant extracts market worldwide. This section includes a region-wise list of key regulatory bodies, government agencies, industry associations, and non-profit organizations relevant to the market.

TABLE 21 NORTH AMERICA: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	DESCRIPTION
US	Food and Drug Administration	Government Regulatory	The Food and Drug Administration is responsible for protecting public health by ensuring the safety, efficacy, and security of human and veterinary drugs, biological products, and medical devices and ensuring the safety of the country's food supply, cosmetics, and products that emit radiation.
US	US Department of Agriculture	Government Regulatory	It is the federal executive department responsible for developing and executing federal laws related to farming, forestry, rural economic development, and food. It aims to meet the needs of commercial farming and livestock food production, promotes agricultural trade and production, works to assure food safety, protects natural resources, fosters rural communities, and works to end hunger in the US and internationally.
US	Food and Agriculture Organization	Government Regulatory	The Food and Agriculture Organization of the United Nations is a specialized agency that leads international efforts to defeat hunger and improve nutrition and food security.
US	Herb Society of America	Government Regulatory	The Herb Society of America is dedicated to promoting the knowledge of herbs through educational programs, research, and sharing the experience of its members with the community.
US	American Herbal Products Association	Government Regulatory	The AHPA advocates the US Congress; state governments; FDA; FTC; USDA; FWS; NIH; NCCIH; US Pharmacopeia; and others for effective laws and regulations that promote the responsible commerce of herbal products.
US	American Spice Trade Association	Government Regulatory	The American Spice Trade Association is an American trade association for companies importing, processing, and distributing food spices.
Canada	Health Canada	Government Regulatory	Health Canada regulates natural health products (NHPs) so that Canadians can have confidence that the products they use are safe, effective, and of high quality.
Canada	Natural and Non-prescription Health Products Directorate	Government Regulatory	The Natural and Non-prescription Health Products Directorate is the Canadian regulating authority for natural health products and non-prescription drugs.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 22 EUROPE: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	DESCRIPTION
EU	Committee on Herbal Medicinal Products	Government Regulatory Body	The Committee on Herbal Medicinal Products (HMPC) is the European Medicines Agency's (EMA) committee responsible for compiling and assessing scientific data on herbal substances, preparations, and combinations, to support the harmonization of the European market.
EU	European Spice Association	Government Regulatory Body	European Spice Association, ESA, is a non-profit association representing the interests of its members in all matters pertaining to the processing, packing, quality assurance, and food safety and/or marketing of herbs, spices, and spice products (e.g., seasonings).
EU	European Flavors & Fragrances Association	Government Regulatory Body	EFFA is a European non-profit trade association for the flavor industry. The organization originally focused on flavors and fragrances but decided to focus solely on flavors in 2009.
UK	Medicines and Healthcare products Regulatory Agency	Government Regulatory Body	The Medicines and Healthcare products Regulatory Agency is an executive agency of the Department of Health and Social Care in the United Kingdom which is responsible for ensuring that medicines and medical devices work and are acceptably safe.
Germany	The International Federation of Essential Oils and Aroma Trades	Government Regulatory Body	IFEAT – The International Federation of Essential Oils and Aroma Trades – is a trade association founded in 1977. It represents the interests of companies involved in the production, processing, trading, and manufacturing of the many thousands of ingredients used in flavors, fragrances, and aromatherapy.
France	National Agency for the Safety of Medicines and Health Products (ANSM)	Government Regulatory Body	It ensures the safety of health products and promotes access to therapeutic innovation, on behalf of the French Government.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 23 ASIA PACIFIC: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	DESCRIPTION
Australia	Australian Government's Therapeutic Goods Administration (TGA)	Government Agency	The Australian Government's Therapeutic Goods Administration (TGA) is responsible for ensuring the therapeutic goods available in Australia are safe and fit for their intended purpose.
India	Essential Oil Association of India	Government Agency	The Essential Oil Association of India (EOAI) was founded in 1956 as a premier organization of essential oil scientists, researchers, industrialists, traders, manufacturers, exporters, importers, and farmers.
India	Spices Board	Government Agency	The Spices Board (Ministry of Commerce and Industry, Government of India) is the primary organization responsible for the development and international promotion of Indian spices. The Board serves as an international link between Indian exporters and foreign importers.
India	Ministry of AYUSH	Government Agency	The Ministry of AYUSH, a ministry of the Government of India, is responsible for developing education, research, and propagation of traditional medicine systems.
Japan	Pharmaceuticals and Medical Devices Agency	Government Agency	The Pharmaceuticals and Medical Devices Agency (PMDA) is a Japanese regulatory agency that collaborates with the Ministry of Health, Labor, and Welfare. Its responsibility is to safeguard public health by ensuring the safety, efficacy, and quality of pharmaceuticals and medical devices.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 24 REST OF THE WORLD: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
South Africa	South African Health Products Regulatory Authority (SAHPRA)	Government Agency	SAHPRA is a National Department of Health body established by the South African government to ensure the health and well-being of humans and animal health.
Colombia	The Colombian Agricultural Institute	Association	The ICA is the entity responsible for ensuring the quality of agricultural inputs, seeds, and the protection of plant health in Colombia. This office issues multiple permits that are needed to cultivate any vegetable product, including those containing or deriving from cannabis.

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.12.2 REGULATORY FRAMEWORK

6.12.2.1 North America

6.12.2.1.1 United States

Unlike many regulatory systems, the US has no 'herbal' or 'traditional' medicine category. Four overarching principles help define the US regulations:

- Route of administration
- Form or formulation
- Safety
- Intended use—what the product is intended to do

For example, a topically applied botanical product can never be a food or dietary supplement; however, depending on its safety and intended use, it might be a drug, cosmetic, or device. Plants used in vaccines or as substrates for monoclonal antibodies are 'biologics,' which are regulated under the US Public Health Service Act.

Foods, drugs, cosmetics, and devices are regulated under the Federal Food, Drug and Cosmetic Act (FDCA). Regardless of an ingredient's current market channel, if its labeling shows that it is intended to affect the structure or function of the body and/or 'treat,' 'prevent,' or 'mitigate' a medical condition, the US regulates it as a 'drug.'

Botanical ingredients are often marketed as foods. A botanical can be a conventional food (vegetables and spices), a dietary supplement, or a 'food for special dietary use.' As botanicals are explicitly named in the Dietary Supplement Health and Education Act (DSHEA) as ingredients that can be used in finished dietary supplements, it was commonly thought that botanicals could not be drugs. A dietary supplement is 'a product (other than tobacco) that is intended to supplement the diet for other general food products containing the same set of nutrition. DSHEA also specifies the formulations for dietary supplements, which differ from those of conventional foods.

In the policy guidance, FDA provides a regulatory definition for 'botanicals' as 'products that include plant materials, algae, macroscopic fungi, and combinations.' However, it also describes what products do not fall under this definition, which are highly purified ingredients derived from plants, such as Taxol, products of fermentation, or homeopathic drugs. These exceptions are addressed under different regulatory approaches in the US.

6.12.2.1.2 Canada

Canadian law does not explicitly define 'botanicals.' Similar to the US, Canada allows botanical ingredients and finished products to be marketed in a variety of domestic channels. Botanicals can be marketed in Canada as foods, food additives, cosmetics, and devices. Botanical ingredients, such as acacia gum, agar, carrageenan, and pectin, are included in Health Canada's 'List of Permitted Emulsifying, Gelling, Stabilizing, or Thickening Agents (Lists of Permitted Food Additives).'

When used in cosmetics, Canadian regulations require that botanical ingredients be listed according to genus and species or the complete International Nomenclature of Cosmetic Ingredients (INCI) name.

Unique to Canada is a category called Natural Health Products (NHPs). By regulation, NHPs are defined as products that may contain: "A plant or a plant material, an alga, a bacterium, a fungus, or a nonhuman animal material, extracts or isolates, and vitamins and its synthetic duplicate." NHPs include homeopathic medicines or herbal medicines of traditional medicine practices that are intended for use in diagnosis,

treatment, mitigation, or prevention of a disease, disorder, or abnormal physical state or its symptoms. The intended use of NHPs is to 'restore or correct organic functions or modify organic functions in humans, in a manner that maintains or promotes health.' NHPs are regulated by the Nonprescription Natural Health Products Directorate (NNHPD). As such, they are not foods, fortified foods, prescription products, or drugs administered by puncturing the skin or any substance that is regulated under the Tobacco or the Controlled Drugs and Substances Acts of Canada. NHPs are sold 'Direct To Consumer' (DTC), and therefore, they are restricted to oral, topical, or sublingual routes of administration. Canada regulates NHPs based on their intended use that includes the health condition (serious, major, minor); health effect (diagnoses, treatment, cure, risk reduction, prevention, general health maintenance, support t); or general health claims (relate to modifying organic functions in a manner that maintains or promotes health (nutrient structure-function and quality of life claims). Health Canada has not yet provided explicit guidance related to the development pathway of botanicals as prescription drugs.

6.12.2.2 Europe

The Committee on Herbal Medicinal Products (HMPC) is the European Medicines Agency's (EMA) committee responsible for compiling and assessing scientific data on herbal substances, preparations, and combinations to support the harmonization of the European market.

The Committee on Herbal Medicinal Products (HMPC) issues scientific opinions on herbal substances and preparations, along with information on recommended uses and safe conditions, on behalf of the European Medicines Agency (EMA). This gives companies and national competent authorities a clear reference point when preparing or assessing an application for marketing authorization or registration of herbal medicinal products in European Union (EU) Member States.

Guideline on the declaration of herbal substances and herbal preparations in herbal medicinal products/traditional herbal medicinal products.

- Herbal preparations are diverse in character, ranging from simply processed, comminuted plant material to complex processed preparations, such as refined extracts. The declaration of an herbal preparation should cover the name of the herbal substance and the definition of the herbal preparation, including the physical state, a ratio of herbal substance to genuine herbal preparation (DER genuine, also named native DER), and extraction solvent(s), if appropriate. The name of the herbal substance is the scientific Latin name of the plant species according to the binomial system (genus, species, variety, and author) with the Latin term of the plant part, followed by the common name of the monograph of the European Pharmacopoeia, if available, or else of the Pharmacopoeia of a Member State, if available, or else the common name of the herbal substance (in brackets). In those special cases, where different Latin plant species apply to the same herbal substance, the list of Latin names could be shortened to the genus name followed by the word 'species,' for instance, "Crataegus" species. This option is only applicable in cases where no restrictions concerning the species used are known from the quality documentation. In special cases, where necessary, only the scientific Latin name of the plant species may be used together with the common term for the plant part. In addition, the declaration of herbal preparations needs to reflect the different extract types (type of herbal preparation) as described in the European Pharmacopoeia.
- 'Standardized herbal preparations:' are adjusted within an acceptable tolerance to a given content of constituents with known therapeutic activity; standardization is achieved by adding excipients for adjustment to herbal preparations or by blending batches of herbal preparations/herbal substances used in the manufacturing process. For such preparations, the name and content of the constituent (s) with known therapeutic activity should be stated. The equivalent quantity of genuine herbal preparation should be given (as a range, if applicable).

- 'Quantified herbal preparations:' are adjusted to a defined range of constituents (active markers); adjustments are made by blending batches of herbal preparations/herbal substances used in the manufacturing process. For such preparations, the name and content of the active markers should be stated in a range. The equivalent quantity of the genuine herbal preparation should be stated, quoting either the corresponding amount of herbal substance (given as a range) or the DER genuine.
- 'Other herbal preparations:' are essentially defined by their production process and their specifications. For such preparations, the name and content of the analytical marker(s) should not be stated. The quantity of the genuine herbal preparation should be stated, quoting either the corresponding amount of herbal substance (given as a range) or the DER genuine. When solvent(s) are used in the manufacturing of an herbal preparation (extraction solvent), the name and composition of the solvent used in the first extraction step should be included in the declaration of the herbal medicinal product. If purification procedures are used in the manufacture of herbal preparation, the word 'refined' should be added to the name of the herbal preparation, where applicable.

6.12.2.3 Asia Pacific

6.12.2.3.1 China

As China starts to emerge and recover from COVID-19, consumers are now increasingly relying on the consumption of plant-based food products. Hence, any food manufacturer strategizing on coming up with plant-based food products should adhere to the following laws:

- One must confirm all plant-based ingredients are lawful to use in food under China's food laws. Approval from the National Health Commission (NHC) is required if the ingredient is considered 'new,' meaning there is no history of consumption by the Chinese population. In addition to the plant-based ingredients, one must not overlook the other ingredients and food additives used in the food, as they should have proper status under the Chinese requirements. For instance, it should be confirmed that the use levels of all food additives in the finished plant-based food comply with any applicable limits outlined in China's food additive standard.
- A product standard for the plant-based ingredient or food should be identified. China's food regulatory system is based on standards, which detail the safety and quality requirements of the food. Hence, identifying a proper product standard is a crucial component of compliance. For instance, when a plant-based food is imported, upon its arrival at the port, customs officials will examine the food in accordance with the applicable Chinese standard(s). Accordingly, identifying the correct standard(s) for the product before exporting it to China is critical.
- One must ensure proper labeling is used to promote plant-based foods. This would include an appropriate product name and other statements or claims that may be put on the package. For example, whether the qualifying terms 'artificial,' 'imitated,' 'plant-based,' and 'veggie' may be used, and whether the plant-based food, per se, can refer to the terms 'milk,' 'sausage,' 'burger,' and 'egg,' which are traditionally derived from animals.

6.12.2.3.2 India

Phytopharmaceuticals are a term used to describe plant-based pharmaceuticals in India. They comprise all plant-based medicines, such as Ayurvedic, Siddha, and Unani (ASU) drugs. The Central Drugs Standards Control Organization has control over phytopharmaceuticals (CDSCO). The CDSCO has published a draft regulation that defines and lays out guidelines for phytopharmaceuticals as an amendment to the Drugs & Cosmetic Rules 1945. Rule 2 of the Drugs and Cosmetics Rules, 1945 states that a "Phytopharmaceutical Drug" is defined as a purified and standardized fraction with a defined minimum four bio-active or phytochemical compounds (qualitatively and quantitatively assessed) of an extract of a medicinal plant or

its part, for internal or external use by people or animals for the diagnosis, treatment, mitigation, or prevention of any disease or disorder, but does not include administration by parenteral means.

Standards for Health supplements and Nutraceuticals are specified under Food Safety and Standards. These regulations cover eight categories of Functional foods, namely, Health Supplements, Nutraceuticals, Food for Special Dietary Use, Food for Special Medical Purpose, Specialty food containing plants or botanicals, Foods containing Probiotics, Foods containing Prebiotics and Novel Foods.

The Department of Biotechnology supports multidisciplinary research across the nation that aims to create new products and methods using medicinal and aromatic plants. The DBT has taken the initiative to develop the Northeast Region's Phytopharmaceuticals Mission Program using cutting-edge scientific tools in accordance with DCGI-CDSCO protocols. In order to launch a phytopharmaceutical mission in the Northeast region with the goal of developing phytopharmaceuticals from medicinal plants in the region in accordance with DCGI-CDSCO (Ministry of Health & Family Welfare, Government of India) protocols, the Department has organized a series of consultation meetings with participation from all stakeholders from various Ministries/Departments as well as representatives from academia and industry.

6.12.2.3.3 Australia

In Australia, herbal extracts can be approved for use as ingredients in complementary medicines or as complementary medicines themselves. These medicines may be either listed (lower risk) or registered (higher risk) on the Australian Register of Therapeutic Goods. The eligibility of specific ingredients, including herbal extracts, for use in listed medicines, is determined by TGA. Herbal ingredients used in registered medicines are evaluated as part of the application for approval of the medicine. Herbal ingredients differ from pharmaceutical ingredients in that they are complex multi-component substances. Natural variation is often seen in extracts of herbal ingredients, and fluctuations in the availability of extracts may result in differences in the extraction profiles of individual raw material batches of the same herbal ingredient.

The Commonwealth of Australia, New Zealand, and Australian states and territories have collaborated to create and implement uniform food standards under the Australia and New Zealand food regulation system.

The Food Standards Australia New Zealand Act of 1991 governs the statutory organization known as Food Standards Australia New Zealand. The Act gives governments, business, and the public a focal point for working together to create and uphold consistent food regulations in Australia and New Zealand.

Foods (Standard 1.5.1) This Standard controls the distribution of novel foods and novel food components. These foods cannot be sold under this Standard unless they are included in Schedule 25 of the Code. The specific permission may impose requirements pertaining to things like the requirement for preparation or cooking instructions, warning statements or other advice, or the requirement to meet specific composition or purity requirements. Before adding the novel food to Schedule 25, FSANZ will evaluate its safety for human consumption. The Authority's safety assessment guidelines will be followed in performing the safety assessment.

Novel foods can also be found in other foods as ingredients. Plants, animals, and their parts, plant or animal extracts, herbs, including extracts, dietary macro-components, single chemical entities, microorganisms, including probiotics, and foods produced from new sources or by a process never before used to make food are just a few examples of novel food categories.

6.12.2.4 South America

6.12.2.4.1 Brazil

The Brazilian health legislation, which provides for herbal medicines registration, has recently been updated and published as RDC 14/10, in line with the scientific and technological developments, enabling the expansion of the population's access to these medicines (Brasil, 2010b).

In Brazil, herbal medicines can be registered only by pharmaceutical industries previously authorized by the Brazilian Health Surveillance Agency to manufacture these products. The company must also confirm that it is able to manufacture pharmaceuticals in accordance with the Good Manufacturing Practices and Control protocol (BPFC). To achieve this certification, the company must follow internationally standardized norms, internalized in Brazil by the newly published RDC 17/10 (Brasil, 2010a). For herbal medicine production, industries must prove they have employed a qualified pharmacist who will be the technically responsible officer for the product.

6.12.2.4.2 Argentina

Herboristerias may be sold as plant medicines, but not as mixtures. Plant drug mixtures are regulated (Law No. 16.463). A Ministry of Health regulation from 1993 established the requirement for herbal medicines to be registered. Crude extracts, extracts or fractions of complex chemical compositions, and pure active principles are all under strict control thanks to the Argentinian National Pharmacopeia. There are about 889 monographs in Argentina. 33 describe extracts or fractions, while 56 describe only crude drugs.

However, there is a lack of control over the raw materials, the wild plants, the scientific standards for plant collection, and the drying, preserving, or grinding processes.

6.13 PORTER'S FIVE FORCES ANALYSIS

Porter's five forces analysis is an important business analysis model that helps better understand the market dynamics and current positioning of the market stakeholders, which are buyers, suppliers, and major players, in the value chain. It also helps understand the threat posed by substitutes and new entrants, the two major factors that can influence the market drastically over time. The five forces model measures competitive intensity, attractiveness, and profitability.

The competitive environment prevailing in the global plant extracts market can be analyzed with the help of Porter's five forces analysis. Porter's five forces model provides a framework to analyze the implications of market trends and consumer demand through a deeper look at each of the five factors of the plant extracts industry. The analysis below enables an improved understanding of market dynamics in the plant extracts market. The factors that affect the existing business environment, along with the demand and supply in the market, are analyzed.

TABLE 25 PLANT EXTRACTS MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Degree of Competition	High
Bargaining Power of Suppliers	High
Bargaining Power of Buyers	Low
Threat of Substitutes	High
Threat of New Entrants	Moderate

Source: Secondary Literature and MarketsandMarkets Analysis

6.13.1 DEGREE OF COMPETITION

The degree of competition in an industry depends on the number of players and the strategies organizations are introducing such as price discounting, introducing new products, advertising campaigns, and service improvements. For instance, Vidya Herbs launched a new green coffee extract, CGA-7, that has high-purity chlorogenic acid and helps in weight management. The global plant extracts market is flourishing because of the global shift toward plant-based diets as they are a healthier and sustainable option. The consumption of pulses is increasing since they are a good source of vitamins, minerals, phenolic compounds, antioxidants, and alkaloids.

There are several major companies operating in the plant extracts segment that dominate their respective markets. Thus, companies are competing to gain a significant market share. Plant extract manufacturers invest a considerably large proportion of their revenues in R&D activities. There are new extraction technologies developed to produce high-quality plant extracts efficiently. The global market players have been setting up facilities to develop cost-effective solutions to cater to the requirements of the emerging economies. This leads to a high level of rivalry between market players.

Hence, the degree of competition in the market is projected to be high.

6.13.2 BARGAINING POWER OF SUPPLIERS

The suppliers in this market are farmers and agriculture producers who supply raw materials to the plant extracts manufacturing companies. Manufacturing companies have multiple options for sourcing raw materials. The raw materials used are fruits, vegetables, herbs, and spices. Various leading players have a well-established partnerships with local suppliers for constant, hassle-free procurement of raw materials, giving these companies an added advantage. However, inconsistent production, seasonal availability, and because very few countries are cultivating the required crops, specifically in the case of herbs & spices, cause disruption in the continuity of the supply of raw materials, which is one of the major problems faced by the plant extracts market.

Thus, the bargaining power of the suppliers is projected to be high.

6.13.3 BARGAINING POWER OF BUYERS

The bargaining power of buyers determines the bargaining power of target customers and their effects on pricing and quality. Plant extracts are used as an ingredient to be used in further applications, so it is not directly sold to consumers. This market has several buyers, from various food sectors, such as food & beverages manufacturers, cosmetic manufacturers, and the pharmaceuticals industry. Furthermore, sellers may find different ways like customization of extracts as per the buyers for increasing the cost that buyers incur when switching from one seller to another. Moreover, quality, affordability, and availability will be an issue while switching the supplier. Since switching cost for buyers is high, the bargaining power is low.

Hence, the bargaining power of buyers is projected to be low.

6.13.4 THREAT OF SUBSTITUTES

The threat of substitutes is identifying substitutes for products or services that can fulfill the same purpose as products of the industry of the considered industry. As growing trends toward plant-based, clean-label, vegan, and sustainable diets are increasing, customers are switching to plant extracts to fulfill their dietary, personal care, and homecare needs. However, synthetic flavors and colors are major substitutes for natural extracts as they are less expensive, such as synthetic vanilla.

Therefore, the threat of substitutes is projected to be high

6.13.5 THREAT OF NEW ENTRANTS

The plant extracts market is competitive, with key and local players at the regional levels operating in the market. Due to the increasing consumer demand for clean plant-based diet and the increase in awareness related to plant-based products, the demand for new and innovative products would also increase. Thus, the market has ample scope for growth opportunities. The threat remains moderate because the sector requires high capital investment. Also, established players have expensive and large production facilities that produce high-grade, quality products.

Hence, it can be concluded that the threat of new entrants will be moderate.

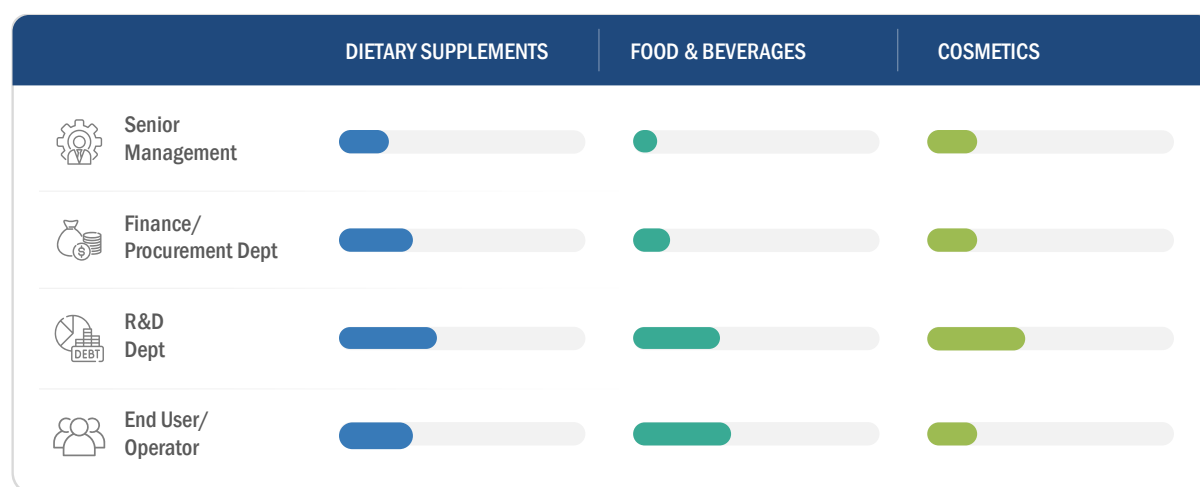
6.14 KEY STAKEHOLDERS AND BUYING CRITERIA

This section elaborates on Porter's five force model. It elaborates on the role of the buyer and the customers and gives a detailed insight into the buying cycle. The first section identifies the key stakeholders and their extent of influence in decision-making from the top applications of the plant extracts market.

6.14.1 KEY STAKEHOLDERS IN BUYING PROCESS

The table below summarizes the involvement of different stakeholders in the buying process for the top applications of the plant extracts market. The applications are selected based on the market share during the forecast period. The R&D department has an influence on both pharmaceuticals and cosmetics.

FIGURE 43 INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP APPLICATIONS



Source: Secondary Research, Primary Interviews, and MarketsandMarkets Analysis

TABLE 26 INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP APPLICATIONS (%)

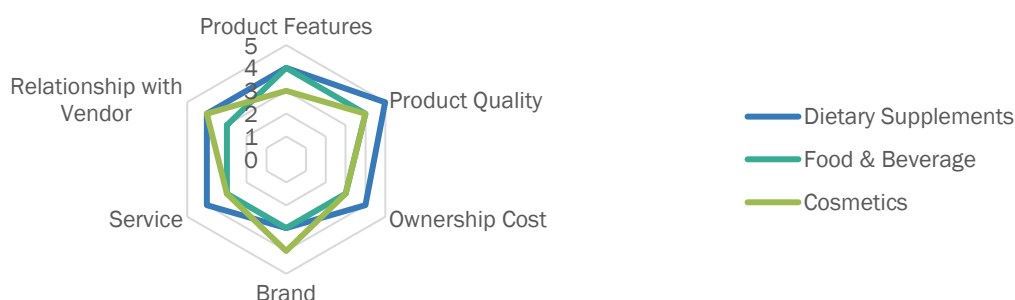
Stakeholders	Dietary Supplements	Food & Beverages	Cosmetics
Senior Management	20%	10%	20%
Finance/Procurement Dept	30%	15%	20%
R&D Dept	40%	35%	40%
End-user Operator	30%	40%	20%
Total	100%	100%	100%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

6.14.2 BUYING CRITERIA

This section identifies the key criteria for selecting suppliers or vendors and their weightage for the top three applications. The key buying criteria for plant extracts varies according to the applications for which it is being utilized. For instance, the product feature of dietary supplements is high since they must pass various quality checks and regulations before they are used for manufacturing supplements.

FIGURE 44 KEY CRITERIA FOR SELECTING SUPPLIER/VENDOR



Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 27 KEY CRITERIA FOR SELECTING SUPPLIER/VENDOR

Buying Criteria	Dietary Supplements	Food & Beverages	Cosmetics
Product Features	4	4	3
Product Quality	5	4	4
Ownership Cost	4	3	3
Brand	3	3	4
Service	4	3	3
Relationship with Vendor	4	3	4

Note: Respondents rated each criterion on a scale of 1–5 where 1 indicates Least Important and 5 indicates Most Important or Critical.

7 PLANT EXTRACTS MARKET, BY TYPE

KEY FINDINGS

- The shift in preferences of consumers from synthetic products to natural products in various applications is projected to drive the growth of the market during the forecast period.
- In 2021, essential oils accounted for a 25.6% share in the plant extracts market and are projected to grow at a CAGR of 11.8% during the forecast period. Due to the health advantages offered by essential oils, such as remedies for pain management, and insomnia, they are used in aromatherapy products and treatments.
- The flavonoids segment is projected to witness the second-fastest growth globally, at a CAGR of 12.7%, and reach USD 12,329.5 million by 2027.
- The rising popularity of pharmaceutical products has increased the use of flavonoids. This is due to their anti-oxidative, anti-inflammatory, anti-mutagenic, and anti-carcinogenic properties.
- The alkaloids segment is growing at the highest CAGR of 13.0% due to their application as local anesthesia.

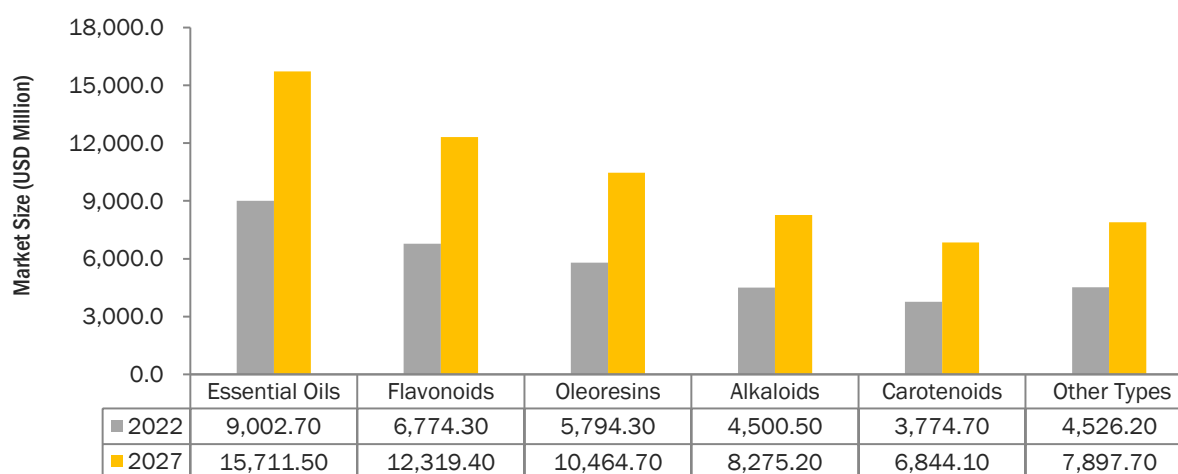
7.1 INTRODUCTION

Plant extracts that are derived from different parts of plants are used in various food and non-food applications due to their aromatic, antimicrobial, and medicinal properties. The different types of plant extracts include essential oils, oleoresins, flavonoids, alkaloids, carotenoids, and others. Regular consumption of these extracts reduces the risk of several chronic diseases, such as cardiovascular diseases, cancer, diabetes, high blood pressure, rickets, and osteoporosis. Most extracts are produced through various distillation processes. These extracts are available in various forms, including powder, capsules, and liquid.

Various types of plant extracts are used as key ingredients in food & beverage products and dietary supplements due to their health and nutritional benefits, which cater to nutritional deficiencies, provide required nutrients in the body, and boost the immunity and overall well-being of consumers.

According to the National Library of Medicine publication in 2021, Plant extracts and bioactive dietary components have the potential to modulate risk factors and manage symptoms for a wide range of common childhood disorders, including memory impairment, respiratory illnesses, gastrointestinal disorders, metabolic derangements, and oral cavity pathologies. With the growing health awareness, the increasing prevalence of chronic diseases, and the need for nutrition enhancements among consumers, plant extracts find applications in various end-user industrial applications. The demand for cosmetics products made of plant-derived ingredients also remains high as these ingredients enhance the product value due to their health benefits for the skin. Natural extracts, specifically plant extracts, are currently gaining popularity in the cosmetics industry. The use of antioxidants in topical formulations can help to reduce oxidative stress in the skin, which has been linked to aging. Owing to its increasing applications across industries, the growth of the plant extracts market is projected to remain high during the forecast period.

FIGURE 45 PLANT EXTRACTS MARKET, BY TYPE, 2022 VS. 2027 (USD MILLION)



Source: Related Research Publications, Government Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 28 PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	2,816.3	3,192.4	3,569.8	4,151.3	4,631.1	5,178.2	13.0%
Essential Oils	4,424.4	4,984.3	5,539.4	6,402.0	7,098.1	7,888.0	12.3%
Flavonoids	3,285.6	3,725.7	4,167.8	4,848.4	5,410.8	6,052.3	13.0%
Alkaloids	2,141.0	2,435.6	2,733.4	3,190.0	3,571.6	4,007.9	13.4%
Carotenoids	1,846.2	2,090.6	2,335.4	2,712.9	3,023.4	3,377.1	12.8%
Other Types	2,369.4	2,673.2	2,975.0	3,442.8	3,821.9	4,252.1	12.4%
Total	16,882.9	19,101.8	21,320.7	24,747.3	27,557.0	30,755.7	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 29 PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	5,797.9	6,507.5	7,321.9	8,259.0	9,311.9	10,471.9	12.6%
Essential Oils	8,777.8	9,791.6	10,949.6	12,275.4	13,756.1	15,346.4	11.8%
Flavonoids	6,779.1	7,611.6	8,567.4	9,667.5	10,904.1	12,329.5	12.7%
Alkaloids	4,503.7	5,073.0	5,728.5	6,484.9	7,338.0	8,281.9	13.0%
Carotenoids	3,777.3	4,235.2	4,760.4	5,364.1	6,041.7	6,849.7	12.6%
Other Types	4,736.9	5,289.4	5,920.5	6,643.1	7,450.3	8,233.2	11.7%
Total	34,372.8	38,508.4	43,248.2	48,694.1	54,802.0	61,512.6	12.3%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 30 PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	218.0	244.1	269.6	309.7	341.3	376.9	11.6%
Essential Oils	326.9	363.9	399.5	456.0	499.4	548.1	10.9%
Flavonoids	244.5	273.9	302.7	347.8	383.3	423.3	11.6%
Alkaloids	159.4	179.1	198.5	228.8	253.0	280.3	12.0%
Carotenoids	137.6	153.9	169.8	194.8	214.4	236.5	11.4%
Other Types	161.2	179.7	197.6	225.8	247.6	272.0	11.0%
Total	1,247.6	1,394.6	1,537.7	1,763.0	1,938.9	2,137.2	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 31 PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	417.1	462.7	514.5	573.5	640.8	716.5	11.4%
Essential Oils	602.9	664.7	734.6	813.8	903.8	1,001.3	10.7%
Flavonoids	468.7	520.0	578.4	644.8	720.7	809.5	11.5%
Alkaloids	311.3	346.6	386.7	432.5	485.0	543.9	11.8%
Carotenoids	261.5	289.7	321.8	358.3	399.9	450.2	11.5%
Other Types	299.5	330.5	365.6	405.3	450.5	498.5	10.7%
Total	2,361.0	2,614.3	2,901.6	3,228.3	3,600.7	4,019.7	11.2%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.2 OLEORESINS

7.2.1 OLEORESINS PRESERVE ROBUST FLAVOR AND AROMA OF SPICES

Spices are used for flavor, color, aroma, and preservation of food or beverages. They may be derived from several parts of plants: bark, buds, flowers, fruits, leaves, rhizomes, roots, seeds, stigmas, and styles or the entire plant tops. Spices are concentrated sources of antioxidants.

Oleoresins are concentrated herbal and spice extracts. They are semi-solid resin-like extracts of essential or fatty oils. They have pungency and aroma, which add to the flavor of food. They are obtained through a solvent extraction process from an herb or spice that is cleaned and ground to the desired meshing size. Solvents such as hexane, acetone, ethylene dichloride, and alcohol can all be used for extraction. The solvents are percolated through ground herbs or spices during extraction. The excess solvent is then removed from the extract, but the extraction process depends on the type of spice or herb. Microorganisms and enzymes are also absent in oleoresins. Oleoresins can be flavor-standardized and have a long shelf-life. They dissolve in fats and oils, and some oleoresins contain important flavor notes that essential oils lack. Oleoresins can be used in various applications, such as in flavor, perfume, and cosmetics industries, due to their properties.

Black pepper oleoresins have both flavor and antioxidant properties; they are used in food and health products. They have antimicrobial properties and hence can also be used as a colorant and a food preservative. Black pepper oleoresins are primarily produced in India. Paprika oleoresin is primarily used as a flavoring and coloring agent. Capsicum oleoresins have a stronger flavor than paprika oleoresins and are mostly used to flavor foods. Capsicum oleoresins are manufactured in India, Africa, and China. India, Spain, Morocco, the US, Mexico, and South Africa produce paprika oleoresins. Orange juice, cheese, sauces, confectionery, and processed meat products are all colored with paprika oleoresins. Turmeric oleoresins contain coloring matter, volatile oil, fatty oil, and bitter principles, as well as the coloring components curcumin and curcuminoids. Its color is typically orange-red but varies depending on the pH. Turmeric oleoresin is primarily manufactured in India and is used in fermented foods, sweets, bakery products, dairy products, and snacks. Ginger oleoresin can be found in gingerbread, ginger ale, sweets, curry powders, soft drinks, sauces, and alcoholic beverages. The application of these oleoresins has benefitted food & beverage manufacturers to produce products with innovative products. Most companies are setting up their manufacturing plants in India, Indonesia, and Malaysia, due to the high availability of raw materials to produce oleoresins. Thus, Asian countries are projected to account for a larger share in spice production compared to the North American and European regions.

There are several advancements in oleoresin extraction. Some of them include:

Microencapsulation: Oleoresins, like other natural substances, oxidize when exposed to light, heat, and oxygen. However, oleoresins can be microencapsulated by spray drying. This process protects and preserves their beneficial compounds from deterioration. It also turns the spice into a flowable powder, allowing for the controlled release of active ingredients. Furthermore, microcapsule powders can be used as a natural food colorant. Givaudan, one of the most well-known names in the encapsulated flavors and fragrances industry, recently developed the “PrimeLock+” technology, which mimics animal fat cells using encapsulated coconut oil. This vegan-friendly solution offers encapsulated flavors to reduce high-fat content while enhancing the juiciness of plant-based products to help consumers achieve the feel and flavor of the meat.

Green credentials: There is a growing interest in greener extraction methods that also improve oleoresin quality. For example, supercritical fluids, such as supercritical carbon dioxide, have the properties of both a gas and a liquid. This property is useful in the extraction of high-value compounds, fatty acids, and essential oils. Furthermore, the option provides speed, high efficiency, selectivity, and sustainability, and it is expected to grow.

Flavor preservation: Oleoresins have advantages over essential oils in terms of flavor preservation, despite being less concentrated. Extraction methods and the heavy oils in oleoresins aid in the preservation of low flavor notes that would otherwise be lost during steam distillation. Because of this, oleoresins are the closest liquid extractive to the true essence of the source product.

The increasing demand for authentic cuisines globally is one of the foremost factors driving the consumption of oleoresins. The growing fondness for enjoying various flavors in foods and snacks is expected to prompt manufacturers to produce high-quality, appealing, and reliable products that can maintain consistent standards globally. Oleoresins are produced to preserve the spice's robust flavor and aroma and are widely used in the natural food color and flavor market. The growing demand for natural and clean label products, as well as strict regulations prohibiting the use of artificial colors and flavors in food products, are driving the oleoresin market demand. Furthermore, the additional health benefits of specific natural food colors and flavors, as well as advanced technological innovations, have fueled the global growth of the natural food colors and flavors market.

North America has a high demand for processed foods, which inadvertently creates a demand for oleoresins. Plant-derived medicines have long been used in Asian countries for a variety of purposes. The oleoresins market has gained commercial traction globally in the last 10 to 15 years, owing to an increase in diseases caused by synthetic and chemical-based products such as irritable bowel syndrome (IBS), attention deficit hyperactivity disorder (ADHD), or autism spectrum disorder. Furthermore, significant investment in plant-derived nutritional and health products is expected to present promising opportunities for Asia Pacific.

TABLE 32 OLEORESINS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	373.3	428.1	484.4	570.0	643.4	727.9	14.3%
Europe	638.2	715.7	791.8	910.9	1,005.4	1,112.3	11.8%
Asia Pacific	1,533.9	1,731.8	1,928.9	2,234.3	2,482.7	2,765.0	12.5%
RoW	270.9	316.7	364.7	436.1	499.7	573.1	16.2%
Total	2,816.3	3,192.4	3,569.8	4,151.3	4,631.1	5,178.2	13.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 33 OLEORESINS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	824.6	936.5	1,066.1	1,216.8	1,388.2	1,593.3	14.1%
Europe	1,232.1	1,368.2	1,523.1	1,699.7	1,896.1	2,110.0	11.4%
Asia Pacific	3,083.7	3,447.5	3,863.7	4,341.1	4,875.2	5,426.1	12.0%
RoW	657.5	755.3	869.0	1,001.4	1,152.4	1,342.4	15.3%
Total	5,797.9	6,507.5	7,321.9	8,259.0	9,311.9	10,471.9	12.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 34 OLEORESINS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	28.5	32.3	36.1	41.9	46.7	52.2	12.9%
Europe	45.4	50.3	54.9	62.4	68.0	74.2	10.3%
Asia Pacific	121.8	135.8	149.5	171.0	187.6	206.4	11.1%
RoW	22.2	25.6	29.2	34.4	38.9	44.0	14.7%
Total	218.0	244.1	269.6	309.7	341.3	376.9	11.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 35 OLEORESINS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	58.5	65.6	73.8	83.2	94.1	107.9	13.0%
Europe	81.2	89.1	98.0	108.0	119.3	132.0	10.2%
Asia Pacific	227.5	251.4	278.4	309.1	344.0	383.2	11.0%
RoW	49.9	56.6	64.3	73.2	83.4	93.3	13.3%
Total	417.1	462.7	514.5	573.5	640.8	716.5	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.3 ESSENTIAL OILS

7.3.1 RISING TREND OF AROMATHERAPY IN DEVELOPING COUNTRIES

Essential oils are also called volatile odoriferous oils. They are concentrated liquids of complex mixtures of volatile compounds extracted from several plant organs: for example, leaves, peels, barks, flowers, buds, and seeds. A range of essential oils has been found to have various degrees of antimicrobial activity and are believed to have antiviral, nematocidal, antifungal, insecticidal, and antioxidant properties. It is extracted from over 300 different types of botanical species. The product types considered in this report include orange oil, lemon oil, lime oil, peppermint oil, cornmint oil, citronella oil, spearmint oil, geranium oil, clove leaf oil, eucalyptus oil, jasmine oil, tea tree oil, rosemary oil, lavender oil, and others. Orange oil is estimated to be the most dominant essential oil, owing to its varied applications. It can be used in the food & beverage industry as a flavoring ingredient, in home care and toiletries applications as a fragrance, and due to its numerous medicinal benefits, it is useful in aromatherapy and health applications.

Some essential oils have been used as medicines. Numerous efforts are being made to explore essential oils usage in the treatment of various infectious diseases that are supernumerary to pharmaceutical remedies. Previously, essential oils were used only for the treatment of various infectious diseases globally. Essential oils are used in aromatherapy, which is a complementary therapy using essential oils and other aromatic plant compounds. This therapy does not provide a cure for diseases, rashes, or illnesses, but it can support the conventional treatment of various conditions.

The importance of essential oils is increasing because they are mostly used in the food & beverage, cosmetics, and fragrance industries. The uses of essential oils have received increasing attention as natural additives for the shelf-life extension of food products. Distillation is the most practiced method of extraction of essential oils. However, citrus oils are extracted using the cold press method. The carbon dioxide method is a relatively new kind of extraction. Essential oils can be incorporated into packaging, in which they can provide multi-functions termed as 'active or smart packaging.' Those essential oils are able to modify the matrix of packaging materials, thereby rendering improved properties. For example, chitosan films containing Eucalyptus globulus essential oil were developed for the packaging of sliced sausages that have a high potential to reduce antimicrobial activity and control food-borne contamination in food systems. Chitosan-based coatings containing lemon essential oil were effective in delaying the ripening process in strawberries due to their reduced respiratory rate.

The growth markets for the uses of essential oils are expected to gain higher demand in rapidly developing and less developed countries such as Cambodia, Bangladesh, and Myanmar. In the long-term, the amount of essential oils consumed in these growing markets is predicted to exceed the current consumption of industrialized countries. The demand is predicted to be the greatest for low-value fragrance oils used in soaps, detergents, and related products with smaller growth in the flavor oils, for non-alcoholic beverages (citrus and spice oils), and oral care products (mint oils).

TABLE 36 ESSENTIAL OILS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	595.2	678.6	763.2	892.6	1,001.6	1,126.3	13.6%
Europe	1,024.4	1,142.3	1,256.6	1,437.4	1,577.5	1,735.1	11.1%
Asia Pacific	2,409.5	2,704.2	2,994.0	3,447.1	3,807.5	4,215.2	11.8%
RoW	395.2	459.2	525.6	624.8	711.6	811.3	15.5%
Total	4,424.4	4,984.3	5,539.4	6,402.0	7,098.1	7,888.0	12.3%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 37 ESSENTIAL OILS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,268.4	1,431.8	1,620.3	1,838.2	2,084.5	2,372.0	13.3%
Europe	1,911.2	2,110.2	2,335.6	2,591.7	2,874.6	3,155.5	10.5%
Asia Pacific	4,673.0	5,193.1	5,785.3	6,461.4	7,213.6	7,988.5	11.3%
RoW	925.2	1,056.6	1,208.4	1,384.2	1,583.5	1,830.5	14.6%
Total	8,777.8	9,791.6	10,949.6	12,275.4	13,756.1	15,346.4	11.8%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 38 ESSENTIAL OILS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	42.8	48.2	53.5	61.8	68.5	76.1	12.2%
Europe	70.7	77.9	84.6	95.5	103.5	112.3	9.7%
Asia Pacific	181.9	201.7	220.6	250.9	273.7	299.2	10.5%
RoW	31.5	36.1	40.8	47.8	53.8	60.5	14.0%
Total	326.9	363.9	399.5	456.0	499.4	548.1	10.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 39 ESSENTIAL OILS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	84.7	94.4	105.6	118.4	133.0	152.0	12.4%
Europe	122.2	133.3	145.7	159.7	175.4	192.8	9.5%
Asia Pacific	327.9	360.1	396.5	437.5	484.1	532.1	10.2%
RoW	68.1	76.8	86.8	98.2	111.2	124.4	12.8%
Total	602.9	664.7	734.6	813.8	903.8	1,001.3	10.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.4 FLAVONOIDS

7.4.1 ANTI-OXIDATIVE AND ANTI-INFLAMMATORY PROPERTIES

Flavonoids are naturally occurring substances with variable phenolic structures that can be found in fruits, vegetables, grains, bark, roots, stems, flowers, tea, and wine. These natural products are well known for their health benefits, and efforts are being made to isolate flavonoids. Flavonoids are now widely recognized as an essential component in a wide range of nutraceutical, pharmaceutical, medicinal, and cosmetic applications. This is due to their ability to modulate key cellular enzyme functions as well as their anti-oxidative, anti-inflammatory, anti-mutagenic, and anti-carcinogenic properties. Flavonoids are found in many plant-based food and beverages, including tea and wine. Numerous studies have demonstrated the advantages of these phytonutrients. The rising popularity of pharmaceutical products has increased the use of flavonoids.

Flavonoids are small organic compounds that are normally absorbed by the human body over long periods, making them one of the safest non-immunogenic drugs. Flavonoid is a type of natural polyphenol compound found in fruits with a variety of structures, including flavonols, flavones, flavanones, flavanols, anthocyanins, isoflavones, and chalcones. It is now used in supplements, cosmetics, and medicines due to its potential health benefits, and researchers are increasingly interested in its role in atherosclerosis. Due to the rising demand for herbal medicines for various diseases, the demand for flavonoids is increasing in the market.

Flavonoids are widely used in cosmetics due to their beneficial effect on keeping the skin healthy and youthful. Creams and cosmetic products containing these compounds smoothen and moisturize the skin, prevent wrinkles, aid in collagen synthesis, have anti-edema properties, and provide UV protection. The market is driven by the rising consumer demand for nutraceuticals, which offers opportunities for plant flavonoids.

In January 2022, Biosyntia, a Danish biotech company, and Lantana Bio, a French biotech company, partnered to accelerate the development of sustainable flavonoids for use in dietary supplements, food and drink fortification, and personal care. The two companies will collaborate to develop technology that will allow for the easy extraction of plant extracts. Due to extraction issues, several flavonoids are not available in the consumer market or are expensive. Advanced techniques and strategies are constantly being developed to overcome the limitations of traditional extraction methods for these compounds. These include ultrasound-assisted extraction (UAE), supercritical fluid extraction (SFE), microwave-assisted extraction (MAE), solid-phase extraction (SPE), enzyme-assisted extraction, pressurized liquid extraction (PLE or accelerated solvent extraction—ASE), and extraction assisted by pulsed electric field (PEF). These modern techniques are effective at extracting flavonoids from a wide range of sources. More importantly, they allow for the reduction of organic solvents and their replacement with alternative “green” solvents while achieving high yields in short extraction times. As a result, these methods can be classified as “green extraction.” The advantages of such techniques have been immense in the production of extracts with good quality and yield, lower consumption of solvent, energy, and shorter extraction time.

TABLE 40 FLAVONOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	435.4	499.7	565.7	666.0	752.2	851.6	14.4%
Europe	745.3	836.2	925.4	1,065.1	1,176.0	1,301.4	11.8%
Asia Pacific	1,791.5	2,023.4	2,254.5	2,612.3	2,903.8	3,235.2	12.5%
RoW	313.4	366.5	422.1	505.0	578.8	664.2	16.2%
Total	3,285.6	3,725.7	4,167.8	4,848.4	5,410.8	6,052.3	13.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 41 FLAVONOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	965.3	1,096.9	1,249.5	1,426.9	1,628.8	1,880.4	14.3%
Europe	1,442.2	1,602.0	1,784.0	1,991.7	2,222.5	2,486.8	11.5%
Asia Pacific	3,609.4	4,036.7	4,525.7	5,086.7	5,714.7	6,395.1	12.1%
RoW	762.3	876.0	1,008.2	1,162.3	1,338.0	1,567.2	15.5%
Total	6,779.1	7,611.6	8,567.4	9,667.5	10,904.1	12,329.5	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 42 FLAVONOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	30.7	34.8	38.9	45.2	50.4	56.4	12.9%
Europe	52.5	58.2	63.6	72.2	78.7	86.0	10.4%
Asia Pacific	136.6	152.5	167.8	192.0	210.8	232.0	11.2%
RoW	24.7	28.5	32.4	38.3	43.3	49.0	14.7%
Total	244.5	273.9	302.7	347.8	383.3	423.3	11.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 43 FLAVONOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	63.1	70.9	79.8	90.1	101.9	117.5	13.2%
Europe	94.1	103.3	113.6	125.3	138.5	154.1	10.4%
Asia Pacific	255.8	282.7	313.2	347.9	387.3	433.8	11.1%
RoW	55.6	63.1	71.7	81.6	93.0	104.1	13.4%
Total	468.7	520.0	578.4	644.8	720.7	809.5	11.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.5 ALKALOIDS

7.5.1 ANTI-CANCEROUS PROPERTIES

Many plant species, primarily flowering plants but also some animals, produce alkaloids. Plants produce and store a variety of organic compounds known as secondary metabolites, such as amino acids, proteins, carbohydrates, fats, and alkaloids.

Alkaloids in plants are typically found in tissues as an aqueous solution. The extraction method is commonly used to isolate them. Special extraction methods for commercially useful alkaloids were developed. In general, alkaloid mixtures should be dissolved in some solvent with reagents. Alkaloids can be recovered from a solution using the extraction method. The alkaloid can then be separated from the mixture and obtained in pure form. Certain solvents must be used to obtain crystalline alkaloids. Chromatography is another method. It makes use of differences in the degrees of adsorption of various alkaloids in a solvent system on solid materials like silica or alumina. Many alkaloids have local anesthetic properties, but their clinical application is limited. Morphine is one of the most well-known alkaloids that has been and continues to be used for medical purposes. This alkaloid is a potent narcotic used to relieve pain, but its utility is limited due to its addictive properties. Codeine, a morphine ether derivative, has excellent analgesic activity and is relatively non-addictive. These alkaloids stimulate the respiratory or cardiac systems. Atropine is used as a medication in many clinical applications. For example, atropine injections are used to treat bradycardia (low heart rate). Tubocurarine is an alkaloid that is found in poison curare and is used as a muscle relaxant in surgery.

Numerous alkaloids isolated from medicinal plants and herbs have antiproliferative and anticancer effects on a variety of cancers in vitro and in vivo. As anticancer drugs, vinblastine, vinorelbine, vincristine, and vindesine have already been developed. Berberine is an isoquinoline alkaloid widely distributed in natural herbs. It has anticancer potential by interfering with the multiple aspects of tumorigenesis and tumor progression. However, most of these alkaloids are poorly soluble in water, have low bioavailability, and are difficult to reach the cancer site. Structural changes, changing the drug delivery system, and the advancement of nanotechnology will resolve these issues. Nanotechnology is one of the input novel drug delivery methods under investigation, with nanoformulation expected to have a wide range of advantages over conventional plant constituent preparations. A combination of herbal medicine and nanotechnology can be an important tool for the advancement of herbal medicine research by increasing bioavailability and decreasing toxicities.

TABLE 44 ALKALOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	283.7	326.7	371.0	438.2	496.5	563.9	14.7%
Europe	485.7	546.6	606.9	700.8	776.2	861.8	12.2%
Asia Pacific	1,167.4	1,322.8	1,478.6	1,718.8	1,916.7	2,142.4	12.9%
RoW	204.2	239.6	276.9	332.3	382.1	439.8	16.6%
Total	2,141.0	2,435.6	2,733.4	3,190.0	3,571.6	4,007.9	13.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 45 ALKALOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	641.3	731.1	835.5	957.2	1,096.1	1,263.1	14.5%
Europe	958.1	1,067.7	1,192.8	1,336.0	1,495.7	1,670.4	11.8%
Asia Pacific	2,397.9	2,690.4	3,026.0	3,412.1	3,845.7	4,295.7	12.4%
RoW	506.4	583.9	674.2	779.6	900.4	1,052.7	15.8%
Total	4,503.7	5,073.0	5,728.5	6,484.9	7,338.0	8,281.9	13.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 46 ALKALOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	20.0	22.7	25.5	29.7	33.3	37.3	13.3%
Europe	34.2	38.0	41.7	47.5	52.0	56.9	10.7%
Asia Pacific	89.0	99.7	110.0	126.4	139.2	153.6	11.5%
RoW	16.1	18.6	21.3	25.2	28.6	32.5	15.1%
Total	159.4	179.1	198.5	228.8	253.0	280.3	12.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 47 ALKALOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	42.0	47.3	53.4	60.4	68.6	78.9	13.5%
Europe	62.5	68.8	76.0	84.0	93.2	103.5	10.6%
Asia Pacific	169.9	188.4	209.4	233.3	260.7	291.4	11.4%
RoW	36.9	42.0	47.9	54.7	62.6	70.0	13.7%
Total	311.3	346.6	386.7	432.5	485.0	543.9	11.8%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.6 CAROTENOIDS

7.6.1 INCREASING INCLINATION OF CONSUMERS TOWARD NATURAL PRODUCTS

Carotenoids are a type of natural pigment that is widely distributed in nature and is produced by plants, certain bacteria, fungi, and algae. These molecules are divided into two groups: carotenes, which are pure hydrocarbons (α -/ β -carotenes and lycopene), and xanthophylls, which are oxygenated derivatives (lutein, zeaxanthin, astaxanthin, fucoxanthin, and cryptoxanthin). These pigments enhance the nutritional value of foods and are used for coloring. These factors have increased the market size of carotenoids. Carotenes have been approved by the European Commission as Group II food additives under the E160 code and are used as an orange-red pigment in non-alcoholic beverages, cheese, pastry, and ice cream. Furthermore, both pigments are vitamin A (retinal, retinol, and retinoic acid) precursors.

One of the most popular strategies is new product development or product innovation for market growth. In August 2018, BASF introduced Lucantin NXT, a new carotenoid product. The product is designed to have a long shelf-life, improved stability, and high homogeneity. GacLife recently launched five new health beverages containing carotenoids, which provide powerful antioxidants to protect human health. Some of the key players are focusing on collaborations to expand their product portfolio. ExcelVite, for example, announced a collaboration with USP in April 2019. The company intends to create a carotenoid monograph to meet its commitment to achieving the highest ingredient standards.

Epidemiological studies have shown a correlation between a high carotenoid intake in the diet with a reduced risk of breast, cervical, ovarian, colorectal cancers, and cardiovascular & eye diseases. Cancer prevention by dietary carotenoids involves various mechanisms, including effects on gap junctional intercellular communication, growth factor signaling, cell cycle progression, differentiation-related proteins, retinoid-like receptors, antioxidant response element, nuclear receptors, AP-1 transcriptional complex, and inflammatory cytokines. Carotenoids have low bioavailability and bio-accessibility due to poor water solubility, degradation in the presence of environmental stresses such as oxygen, light, and high heat, and rapid degradation during digestion. Nanoencapsulation technologies, such as wall or encapsulation materials, are increasingly used to improve the functionality of food products. Nanoencapsulation is a versatile process used to protect, entrap, and deliver food bioactive products such as carotenoids from a variety of environmental conditions to extend the shelf-life and provide controlled release. Carotenoids can be nano-encapsulated to improve their functional properties. Carotenoids in stable nanoencapsulation could be used in food products. Owing to bolstering demand for its nutritional benefits, there is an increase in carotenoid demand in the market.

TABLE 48 CAROTENOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	244.7	280.4	317.0	372.7	420.3	475.2	14.2%
Europe	418.8	469.2	518.5	596.0	657.1	726.2	11.6%
Asia Pacific	1,006.7	1,135.4	1,263.3	1,461.7	1,622.6	1,805.2	12.4%
RoW	176.1	205.6	236.5	282.6	323.4	370.6	16.0%
Total	1,846.2	2,090.6	2,335.4	2,712.9	3,023.4	3,377.1	12.8%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 49 CAROTENOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	537.9	610.3	694.3	791.7	902.5	1,044.7	14.2%
Europe	803.6	891.4	991.3	1,105.1	1,231.5	1,381.6	11.4%
Asia Pacific	2,011.2	2,246.1	2,514.6	2,822.4	3,166.4	3,552.8	12.1%
RoW	424.7	487.4	560.2	644.9	741.4	870.7	15.4%
Total	3,777.3	4,235.2	4,760.4	5,364.1	6,041.7	6,849.7	12.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 50 CAROTENOIDS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	17.4	19.7	22.0	25.6	28.5	31.8	12.8%
Europe	29.5	32.6	35.6	40.4	44.0	48.0	10.2%
Asia Pacific	76.8	85.5	94.0	107.5	117.8	129.4	11.0%
RoW	13.9	16.0	18.2	21.4	24.2	27.4	14.5%
Total	137.6	153.9	169.8	194.8	214.4	236.5	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 51 CAROTENOIDS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	35.5	39.9	44.8	50.5	57.0	65.9	13.2%
Europe	52.5	57.5	63.1	69.5	76.7	85.6	10.3%
Asia Pacific	142.5	157.3	174.0	193.0	214.6	241.0	11.1%
RoW	31.0	35.1	39.8	45.3	51.6	57.7	13.2%
Total	261.5	289.7	321.8	358.3	399.9	450.2	11.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

7.7 OTHER TYPES

Other types include anthocyanins, tannins, saponins, and lignin. Anthocyanins are naturally occurring phytochemicals that have attracted the interest of consumers and the food industry due to their diverse biological properties and technological applications. Nonetheless, traditional extraction techniques based on thermal technologies can jeopardize anthocyanin recovery and stability, reducing global yield and/or limiting their application in food systems. Natural antioxidants and antimicrobials, anthocyanins, have been used as food preservatives. Anthocyanins have protective and promoting effects on human health by preventing and mitigating the onset and/or progression of neurodegenerative diseases, metabolic syndromes, cancers, and other diseases due to their diverse biological properties. One of the major factors driving the growth of the anthocyanins market is the increase in demand for natural food colors in the food & beverage sector. The increased demand for the product due to its superior properties, such as microcirculation improvement, antimicrobial, anti-allergic, and anti-inflammatory properties, as well as its expanding application scope in pharmaceuticals and specialty drugs, is propelling the anthocyanins market. With numerous global players such as Symrise, Oterra, Kalsec, Sensient, Roha, Naturex, and ADM, the anthocyanin ingredients market is highly competitive and fragmented. Kalsec manufactures red radish variety of anthocyanin in Europe for food, beverage, and personal care customers in Germany, Italy, and the UK, among others. Roha is the leading manufacturer and supplier of anthocyanins extracted from grapes for beverage and confectionery applications in the South Asian region.

Anthocyanins are highly unstable compounds that are easily affected by temperature, oxygen, enzymes, light (UV-visible), pH, ascorbic acid, and other substances (e.g., co-pigments and metals). As a result, during the extraction process, anthocyanins may be degraded by one or more factors. As a result, the selection of extraction technology and extraction conditions are critical steps in the recovery and purification of anthocyanins for food applications.

Tannins (also known as tannic acid) are water-soluble polyphenols found in a variety of plant foods. Wattle, trefoil, sainfoin, quebracho, mangroves, eucalyptus, and canaigre are examples of tannin-containing plants. These plants are easy to obtain, and the processes for producing tannins are also simple, accelerating market growth. Tannins have antimicrobial activities. Their antimicrobial properties appear to be related to the hydrolysis of the ester linkage between gallic acid and polyols that occur after many edible fruits ripen. Tannins in these fruits act as a natural defense against microbial infections. Tannic acid's antimicrobial properties can also be used in food processing to extend the shelf-life of certain foods, such as catfish fillets. They have also been shown to have other physiological effects, such as accelerating blood clotting, lowering blood pressure, lowering serum lipid levels, causing liver necrosis, and modulating immune responses. Due to the physiological effects of tannins, their demand is increasing.

Lignin is an aromatic compound that is found in abundance as a renewable resource because it is a major component of plant cell walls. They are generally obtained as a by-product from the black liquor of chemical pulping processes. Lignins are currently used to produce energy; efforts are being made to find new applications, as they are relatively cheap. Since these readily available plant-derived materials are biocompatible, research on their potential use in biomedical applications has increased. In many cases, the application of lignins is based on their antioxidant capacity, antiviral, and antitumor effects.

TABLE 52 OTHER TYPES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	313.0	357.7	403.3	472.8	531.7	599.3	13.9%
Europe	621.3	694.7	766.3	878.9	967.1	1,066.5	11.4%
Asia Pacific	1,275.4	1,435.0	1,592.6	1,838.0	2,034.8	2,257.6	12.1%
RoW	159.7	185.7	212.7	253.0	288.3	328.8	15.5%
Total	2,369.4	2,673.2	2,975.0	3,442.8	3,821.9	4,252.1	12.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 53 OTHER TYPES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	676.3	765.0	867.4	986.0	1,120.2	1,267.3	13.4%
Europe	1,177.6	1,303.4	1,446.1	1,608.4	1,788.1	1,984.1	11.0%
Asia Pacific	2,508.0	2,792.7	3,117.2	3,487.8	3,900.7	4,252.1	11.1%
RoW	375.1	428.3	489.8	560.9	641.4	729.7	14.2%
Total	4,736.9	5,289.4	5,920.5	6,643.1	7,450.3	8,233.2	11.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 54 OTHER TYPES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	22.0	24.9	27.7	32.1	35.6	39.6	12.5%
Europe	39.3	43.4	47.2	53.4	58.0	63.1	9.9%
Asia Pacific	87.3	97.0	106.3	121.1	132.4	144.9	10.7%
RoW	12.6	14.5	16.4	19.2	21.6	24.3	14.0%
Total	161.2	179.7	197.6	225.8	247.6	272.0	11.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 55 OTHER TYPES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	44.2	49.4	55.4	62.2	70.0	78.5	12.2%
Europe	68.8	75.2	82.4	90.5	99.6	108.5	9.5%
Asia Pacific	159.1	175.0	192.9	213.2	236.2	261.5	10.4%
RoW	27.4	30.9	34.9	39.5	44.7	50.0	12.8%
Total	299.5	330.5	365.6	405.3	450.5	498.5	10.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8 PLANT EXTRACTS MARKET, BY APPLICATION

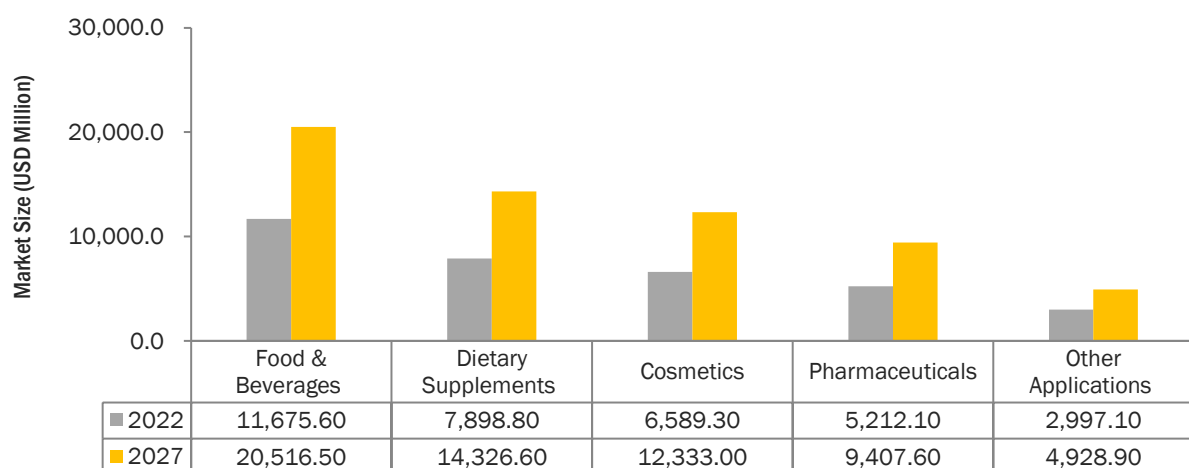
KEY FINDINGS

- The food & beverages segment dominated the plant extracts market in 2021. It is projected to reach USD 20,516.5 million by 2027.
- There is an increasing trend of veganism globally, which has resulted in increased consumption of plant-based food and also increased the usage of plant extracts in various food & beverage applications.
- The pharmaceuticals segment is projected to reach USD 9,407.6 million by 2027, in terms of value, due to the increased requirements for various plant extracts in the production of different medicines and the increased shift toward natural medicines.
- The cosmetics segment is expected to grow at the highest CAGR of 13.4% during the forecast period due to the rising demand for natural cosmetic products to avoid any side-effects caused by chemical-based cosmetics.

8.1 INTRODUCTION

Plant extracts are gaining widespread acceptance across industries, such as pharmaceuticals, dietary supplements, food & beverages, and cosmetics. This acceptance is attributed to the expansion of these industries in various countries and the rising consumption of these products globally. As plant extracts are completely natural, their 'clean label' allows greater preference by consumers who are slowly shifting toward natural, healthy, and organic products. The rising consumer demand for natural food ingredients is expected to propel the demand for various plant extracts across all applications. In the pharmaceuticals and dietary supplements industries, the demand for plant extracts is on the rise. This growth can be attributed to the increasing awareness of people related to bacterial resistance toward allopathic medicines. Due to the increasing incidents of illnesses due to stressful and busy lifestyles, consumers are demanding functional food and supplements for regular consumption. Some of these supplements include phytomedicines and herbal extracts, which are composed of naturally occurring components; they are scientifically demonstrated to promote positive effects on the target functions beyond basic nutrition. Other factors, such as healthy aging and the threat of antimicrobial resistance, also contribute to this significantly. Likewise, within the food & beverage industry, there is a growing trend of consuming natural ingredients with additional functional properties, which has been boosting the demand for plant extracts. The use of plant extracts in beverages is frequently associated with their specific health properties, such as the ability to energize, boost the immune system, improve skin health, increase focus, and provide relaxation.

FIGURE 46 PLANT EXTRACTS MARKET, BY APPLICATION, 2022 VS. 2027 (USD MILLION)



Source: Secondary Research, Primary Interviews, Related Research Publications, Industry Journals, Press Releases, and MarketsandMarkets Analysis

TABLE 56 PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (USD MILLION)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	5,840.3	6,587.7	7,330.6	8,483.0	9,417.4	10,478.7	12.4%
Pharmaceuticals	2,533.7	2,871.6	3,210.7	3,733.1	4,164.2	4,655.5	12.9%
Dietary Supplements	3,817.4	4,330.8	4,846.9	5,641.1	6,298.5	7,048.5	13.0%
Cosmetics	3,045.5	3,481.1	3,925.2	4,602.5	5,177.2	5,836.8	13.9%
Other Applications	1,646.1	1,830.6	2,007.3	2,287.6	2,499.7	2,736.1	10.7%
Total	16,882.9	19,101.8	21,320.7	24,747.3	27,557.0	30,755.7	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 57 PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	11,675.6	13,040.7	14,601.5	16,390.4	18,390.6	20,516.5	11.9%
Pharmaceuticals	5,212.1	5,849.3	6,580.6	7,422.0	8,367.5	9,407.6	12.5%
Dietary Supplements	7,898.8	8,873.1	9,992.2	11,280.9	12,730.3	14,326.6	12.6%
Cosmetics	6,589.3	7,456.6	8,458.7	9,619.5	10,934.5	12,333.0	13.4%
Other Applications	2,997.1	3,288.8	3,615.2	3,981.3	4,379.1	4,928.9	10.5%
Total	34,372.8	38,508.4	43,248.2	48,694.1	54,802.0	61,512.6	12.3%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 58 PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (KT)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	430.7	480.2	528.1	603.9	662.4	728.3	11.1%
Pharmaceuticals	189.8	208.7	226.4	255.2	275.9	298.9	9.5%
Dietary Supplements	282.1	316.2	349.6	401.9	443.2	489.8	11.7%
Cosmetics	226.2	255.3	284.3	329.2	365.6	407.0	12.5%
Other Applications	118.8	134.1	149.3	172.8	191.8	213.3	12.4%
Total	1,247.6	1,394.6	1,537.7	1,763.0	1,938.9	2,137.2	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 59 PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (KT)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	802.4	886.2	981.0	1,088.6	1,211.0	1,351.6	11.0%
Pharmaceuticals	324.4	352.8	384.5	419.9	459.5	512.5	9.6%
Dietary Supplements	542.6	602.4	670.4	747.9	836.5	936.4	11.5%
Cosmetics	454.0	507.6	569.0	639.2	719.9	804.9	12.1%
Other Applications	237.6	265.3	296.7	332.7	373.8	414.4	11.8%
Total	2,361.0	2,614.3	2,901.6	3,228.3	3,600.7	4,019.7	11.2%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8.2 PHARMACEUTICALS

8.2.1 INCREASED DEMAND FOR PLANT-BASED MEDICINES

Plant extracts have served as an important source of bioactive compounds for various drug discovery programs, and several important drugs have been isolated and identified from plants. Beyond their direct use in traditional medicine, medicinal plants are used on an industrial scale to produce total extracts and tinctures, purified extracts, and chemical products. Regulatory approval of some of the more widely known medicinal herb extracts has been made easier than that of new chemical entities. Some phytopharmaceutical companies are focusing their efforts on single molecules or plant extracts with the aim of simplifying the regulatory process by avoiding complex mixtures and combinations.

Phytochemical and phytopharmacological sciences have determined the biological active constituents of each medicinal plant, which include tannins, flavonoids, and terpenoids. These active constituents are highly water soluble but have low absorption due to the difficulty of crossing the lipid membrane and their large molecular size, resulting in low bioavailability and efficacy. As a result, vehicles must be encapsulated with plant extract drugs to improve drug solubility, reduce degradation, minimize toxicity, and increase drug active absorption, resulting in high bioavailability. Nanotechnology is viewed as a drug delivery system that can not only improve the bioavailability and effectiveness of herbs but also reintroduce other components that were discarded due to poor characteristics.

The usage of oleoresin has been rapidly increasing in the pharmaceutical & nutraceuticals segment for the development of health supplements. Plant-based oleoresin has its uses in health maintenance and promotion and in dietary and other nutraceutical supplements. Alkaloids are well known therapeutically as aesthetics, cardioprotective agents, and anti-inflammatory agents. Morphine, strychnine, quinine, ephedrine, and nicotine are examples of well-known alkaloids used in clinical settings. Dietary carotenoids help to prevent a variety of diseases, including certain cancers, cardiovascular diseases, and eye diseases. Carotenoids are frequently used in nutritional supplements, skin care cosmetics, pharmaceutical products, and as a natural food color because of their bioactivities. The usage of different types of plant extracts for pharmaceuticals is driving the growth of the market.

Established players are adopting new product launches and acquisitions to cater to the demand from customers. In May 2022, Symrise released a new line of Aronia health actives. The collection includes an Aronia extract and an Aronia juice powder that are both high in polyphenols and anthocyanins. Aronia, often known as black chokeberry, is rich in antioxidants like polyphenols, particularly anthocyanins. This product line adds to the Diana Food™ portfolio of health actives, which is part of the Taste, Nutrition, and Health division. In January 2020, ADM acquired Yerbalatina Phytoactives, a Brazilian manufacturer of plant-based extracts and ingredients. To stay ahead of health and wellness trends, ADM is expanding its

portfolio of ingredients derived from natural sources that promote health. This acquisition expands ADM's already substantial presence in Brazil.

TABLE 60 PHARMACEUTICALS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	336.8	386.4	437.3	514.6	581.0	657.5	14.3%
Europe	584.2	655.3	725.1	834.5	921.3	1,019.4	11.8%
Asia Pacific	1,391.6	1,571.4	1,750.5	2,027.9	2,253.8	2,510.5	12.5%
RoW	221.1	258.6	297.8	356.1	408.1	468.2	16.2%
Total	2,533.7	2,871.6	3,210.7	3,733.1	4,164.2	4,655.5	12.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 61 PHARMACEUTICALS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	745.1	846.3	963.7	1,100.2	1,255.4	1,441.4	14.1%
Europe	1,129.6	1,254.6	1,397.0	1,559.5	1,740.0	1,937.1	11.4%
Asia Pacific	2,800.3	3,131.1	3,509.6	3,943.9	4,430.0	4,931.6	12.0%
RoW	537.2	617.2	710.2	818.5	942.0	1,097.5	15.4%
Total	5,212.1	5,849.3	6,580.6	7,422.0	8,367.5	9,407.6	12.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 62 PHARMACEUTICALS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	24.7	27.5	30.2	34.5	37.8	41.4	10.9%
Europe	40.9	44.7	48.1	53.8	57.7	62.0	8.7%
Asia Pacific	104.5	114.3	123.3	138.2	148.6	160.1	8.9%
RoW	19.7	22.3	24.8	28.7	31.9	35.4	12.4%
Total	189.8	208.7	226.4	255.2	275.9	298.9	9.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 63 PHARMACEUTICALS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	45.6	50.2	55.4	61.3	68.0	77.9	11.3%
Europe	66.8	72.1	78.0	84.6	91.9	101.0	8.6%
Asia Pacific	172.8	186.8	202.5	219.9	239.3	266.1	9.0%
RoW	39.3	43.6	48.6	54.1	60.3	67.5	11.4%
Total	324.4	352.8	384.5	419.9	459.5	512.5	9.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8.3 DIETARY SUPPLEMENTS

8.3.1 BENEFITS SUCH AS BOOSTING IMMUNITY RELATED TO HERBAL SUPPLEMENTS

According to the FDA, a dietary supplement is a product intended for ingestion that, among other requirements, contains a “dietary ingredient” intended to supplement the diet. Increasing awareness regarding nutrition and healthy lifestyles is driving the market for dietary supplements. It is widely known that having a healthy immune system can help reduce the risk or severity of diseases and infections. When combined with proper nutrition, dietary supplements can enhance the body’s natural defenses and immunity. As people become more conscious of holistic well-being, the demand for immunity-boosting products is expected to increase in the coming years.

With the increase in the trend of healthy eating and healthy lifestyles, the demand for herbal dietary supplements has increased tremendously. Herbal dietary supplements are products made from plants, plant parts, or plant extracts. They contain one or more ingredients meant to supplement the diet. Herbal dietary supplements improve health, and these ‘natural’ remedies are both effective and free from the side-effects that may occur with other medications. In June 2021, Rhuleave-K was introduced by Arjuna Natural as a revolutionary pain management solution. It is a one-of-a-kind product made of extracts of curcuma longa and boswellia serrata. It helps to get relief from pain without any side-effects. All these factors are increasing the consumption of herbal dietary supplements. As the market is growing, companies are adopting partnerships and mergers & acquisitions to acquire a larger share of the market. In August 2020, IFF Health, a part of International Flavors & Fragrances Inc., and DolCas Biotech partnered on the sales and marketing of Curcugen, a next-gen, all-natural turmeric extract for both dietary supplement and food & beverage industries. It aims to increase the interest in curcumin and expand the activities in the market for nutraceuticals. In September 2020, the company announced the acquisition of Core Botanica, one of the leading manufacturers of customized mint leaf products in North America. The acquisition expands the group’s portfolio of raw materials and botanical ingredients, servicing the food and dietary supplement industries, along with the tea and beverage markets.

TABLE 64 DIETARY SUPPLEMENTS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	507.5	582.7	660.1	777.6	878.8	995.5	14.4%
Europe	880.1	988.2	1,094.7	1,261.0	1,393.5	1,543.4	11.9%
Asia Pacific	2,096.6	2,369.9	2,642.6	3,064.3	3,408.9	3,800.9	12.6%
RoW	333.2	389.9	449.5	538.2	617.3	708.8	16.3%
Total	3,817.4	4,330.8	4,846.9	5,641.1	6,298.5	7,048.5	13.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 65 DIETARY SUPPLEMENTS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,129.1	1,283.8	1,463.4	1,672.2	1,910.0	2,195.0	14.2%
Europe	1,711.8	1,903.2	2,121.3	2,370.2	2,647.3	2,949.9	11.5%
Asia Pacific	4,243.8	4,749.7	5,329.2	5,994.4	6,739.8	7,510.2	12.1%
RoW	814.1	936.3	1,078.4	1,244.0	1,433.2	1,671.4	15.5%
Total	7,898.8	8,873.1	9,992.2	11,280.9	12,730.3	14,326.6	12.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 66 DIETARY SUPPLEMENTS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	36.5	41.4	46.3	53.9	60.1	67.2	13.0%
Europe	60.8	67.4	73.7	83.8	91.4	99.9	10.5%
Asia Pacific	158.3	176.8	194.7	223.0	245.0	269.8	11.3%
RoW	26.5	30.6	34.9	41.2	46.6	52.9	14.8%
Total	282.1	316.2	349.6	401.9	443.2	489.8	11.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 67 DIETARY SUPPLEMENTS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	75.4	84.7	95.4	107.7	121.9	140.0	13.2%
Europe	109.5	120.3	132.4	146.1	161.7	179.1	10.3%
Asia Pacific	297.7	329.3	365.2	405.8	452.2	504.4	11.1%
RoW	60.0	68.1	77.5	88.2	100.7	112.9	13.5%
Total	542.6	602.4	670.4	747.9	836.5	936.4	11.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8.4 FOOD & BEVERAGES

8.4.1 RISING VEGAN POPULATION DEMANDING PLANT-BASED FOODS

Plant extracts are becoming increasingly important food additives due to their high concentrations of bioactive compounds such as polyphenols and carotenoids, which have antimicrobial and antioxidant activity, particularly against low-density lipoprotein (LDL) and deoxyribonucleic acid (DNA) oxidative changes. Consumers are looking for products with natural ingredients and added functional properties that support a healthy lifestyle. Plant extracts play an important role in this case, providing a variety of functional benefits while also being natural. Plant extracts are thus no longer restricted to the more specialized nutraceutical market but have spread to the broader food segment. They are used in beverages, dairy products, meat products, bakery, and confectionery products to provide health benefits. Plant and herbal extracts have long been used to enhance healthfulness, color, taste, and even flavor in food, beverages, and supplements. Plant extracts are increasingly becoming important additives in the food industry due to their antimicrobial and antioxidant activities that delay the development of off-flavors and improve the shelf-life and color stability of food products. Owing to their natural origin, they are excellent candidates to replace synthetic compounds, which are generally considered to have toxicological and carcinogenic effects. However, the efficient extraction of these compounds from their natural sources, along with the determination of their activity in the commercialized products, have been challenges for researchers and food chain contributors to develop products with positive effects on human health. Spices that are also extracted from plants play a significant role in the food cooked and consumed worldwide. Spices come with notable nutritional values and provide some major health benefits.

Plant-based extract beverages may improve antioxidant status and reduce oxidative stress in humans when consumed as part of a balanced diet. Furthermore, unlike coffee and tea, many commonly consumed herbal beverages do not contain detectable caffeine levels. Herbal beverages are classified as natural health products by Health Canada (NHPs). However, moderate consumption (2-3 cups/day) of selected herbal teas such as citrus peel, lemon balm, ginger, orange peel, and rosehip are recommended by Health Canada during pregnancy and breastfeeding.

New technologies are being introduced in the plant extracts market to increase the efficiency of product development. In February 2021, Givaudan Taste & Wellbeing introduced Advanced Tools for Modelling (ATOM), which uses artificial intelligence to optimize food and flavor formulation to generate new options and insights based on consumer preferences. The tools investigate ingredient synergies and identify flavor drivers that are both positive and negative. This will support the company's strategy and intent to lead the way in digitalization; these pioneering tools open new avenues for creative development and have the potential to shorten the time to market for new products.

There is an increasing trend of veganism globally, which has resulted in increased consumption of plant-based food. In an NSF global survey conducted in 2021, 88% of food industry practitioners predicted an increase in demand for plant-based products. 74% believe consumers choose plant-based foods for a healthier lifestyle, while 60% believe it is more environmentally friendly. The demand for beverages extracted from plants has recently increased with the rising population of health-conscious people. Herbal beverages are now consumed as part of a balanced diet, may improve the antioxidant status, and enhance the overall health status. Herbal teas or beverages are rich sources of natural bioactive compounds such as carotenoids, phenolic acids, flavonoids, coumarins, alkaloids, polyacetylenes, saponins, and terpenoids. All these above applications in the food & beverage industry are propelling the demand for various plant extracts.

TABLE 68 FOOD & BEVERAGES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	776.4	886.4	998.4	1,169.4	1,314.0	1,479.9	13.8%
Europe	1,346.5	1,503.3	1,655.6	1,896.2	2,083.5	2,294.5	11.2%
Asia Pacific	3,207.6	3,604.9	3,996.8	4,608.1	5,097.0	5,650.6	12.0%
RoW	509.8	593.2	679.9	809.3	922.9	1,053.7	15.6%
Total	5,840.3	6,587.7	7,330.6	8,483.0	9,417.4	10,478.7	12.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 69 FOOD & BEVERAGES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,669.0	1,886.9	2,138.4	2,429.6	2,759.3	3,143.4	13.5%
Europe	2,530.3	2,797.1	3,099.8	3,443.8	3,824.4	4,224.5	10.8%
Asia Pacific	6,272.9	6,980.7	7,787.5	8,709.5	9,736.5	10,755.1	11.4%
RoW	1,203.3	1,376.0	1,575.8	1,807.5	2,070.5	2,393.6	14.7%
Total	11,675.6	13,040.7	14,601.5	16,390.4	18,390.6	20,516.5	11.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 70 FOOD & BEVERAGES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	55.8	62.9	70.0	81.0	89.9	99.9	12.4%
Europe	93.5	103.1	112.1	126.8	137.6	149.7	9.9%
Asia Pacific	239.8	266.3	291.8	332.4	363.2	397.8	10.7%
RoW	41.6	47.8	54.2	63.7	71.7	80.8	14.2%
Total	430.7	480.2	528.1	603.9	662.4	728.3	11.1%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 71 FOOD & BEVERAGES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	111.4	124.5	139.4	156.5	176.1	201.3	12.6%
Europe	163.2	178.3	195.2	214.3	235.9	260.6	9.8%
Asia Pacific	436.7	480.4	529.8	585.7	649.1	721.9	10.6%
RoW	91.2	103.1	116.6	132.1	150.0	167.7	13.0%
Total	802.4	886.2	981.0	1,088.6	1,211.0	1,351.6	11.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8.5 COSMETICS

8.5.1 ANTIMICROBIAL AND ANTIOXIDANT PROPERTIES OF PLANT EXTRACTS LEADING TO ADOPTION OF HERBAL COSMETICS

Plant extracts or herbal extracts are primarily added to cosmetic formulations due to several associated properties, such as antioxidants, anti-inflammatory, antiseptic, and antimicrobial properties. Even today, people in rural and urban areas depend on herbs for traditional cosmetics. Plant extracts, as a whole or part, have been used for various skin ailments, hair, and dental care for overall appearance. Herbal cosmetics claim to have efficacy and intrinsic acceptability due to routine use and avoid the side-effects, which are commonly seen in synthetic products. Owing to the awareness of the environmental damage caused by industrialization, a trend has developed to use products with natural ingredients. Various adverse effects may occur in the form of acute toxicity, percutaneous absorption, skin irritation, eye irritation, skin sensitization and photosensitization, sub-chronic toxicity, mutagenicity, and phototoxicity by the usage of synthetic products owing to which consumers today prefer herbal cosmetics for hair, skin, and dental care. Plant extracts also show beneficial features, i.e., mildness, efficacy, biodegradability, low toxicity, cleansing ability, emulsification, moisturization, skin appearance, feel, fragrance, and lubrication when used in cosmetics.

Melanin shields the skin from UV light damage by absorbing UV rays and removing reactive oxygen species. Flavonoids have a pigment-reducing action due to their ROS-scavenging activity and ability to chelate metals at the active site of metalloenzymes. Aloesin, hydroxystilbene derivatives, and licorice extracts are among the flavonoids commonly used in skin-lightening preparations. Consumers are increasingly interested in natural products, particularly cosmetic products. Several works, on the other hand, discuss the benefits of plant extracts, such as antioxidant capacity, tyrosinase inhibition, and antimicrobial activity, which can be beneficial for the treatment and prevention of various skin conditions.

Various cosmetic products are launched by market players with plant extracts due to their beneficial properties. In April 2022, Symrise introduced Hydrolite 8 green, which is ideal for a variety of traditional and green cosmetic formulations. This new caprylyl glycol has a higher level of sustainability due to its 100% bio-based supply. This new launch aims to broaden the company's green ingredient offering. In May 2019, Vidya Herbs launched Skincera, a Japanese Konjac extract standardized with 5% ceramides and clinically proven for its beauty benefits. The product is clinically proven to treat dryness, hyperpigmentation, redness, itching, and oiliness. This expands the company's cosmetics extract portfolio.

Asia Pacific is one of the largest markets for cosmetics after Europe. It is one of the most diverse and dynamic markets in the cosmetics industry. The growing use of products, such as body moisturizing lotions, sunscreen lotions & creams, facial wipes, face washes, hair styling gels, and make-up products for men and women, fuel the growth of the Asia Pacific cosmetics industry. The increase in consciousness related

to personal appearance among individuals drives the industry growth. The demand for natural and organic cosmetic products supplements the market growth of various plant extracts in the cosmetics industry.

TABLE 72 COSMETICS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	404.8	468.4	534.6	634.5	722.4	824.3	15.3%
Europe	702.2	794.3	886.5	1,028.8	1,145.4	1,278.1	12.7%
Asia Pacific	1,672.6	1,904.9	2,140.1	2,500.2	2,802.0	3,147.5	13.5%
RoW	265.8	313.4	364.0	439.1	507.4	586.9	17.2%
Total	3,045.5	3,481.1	3,925.2	4,602.5	5,177.2	5,836.8	13.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 73 COSMETICS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	941.9	1,078.9	1,238.8	1,425.9	1,640.6	1,889.6	14.9%
Europe	1,428.0	1,599.4	1,795.7	2,021.1	2,273.9	2,539.4	12.2%
Asia Pacific	3,540.2	3,991.5	4,511.3	5,111.6	5,789.0	6,465.2	12.8%
RoW	679.1	786.8	912.9	1,060.8	1,231.0	1,438.9	16.2%
Total	6,589.3	7,456.6	8,458.7	9,619.5	10,934.5	12,333.0	13.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 74 COSMETICS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	29.1	33.3	37.5	43.9	49.4	55.7	13.8%
Europe	49.8	55.5	61.1	70.0	76.8	84.5	11.2%
Asia Pacific	125.6	141.2	156.7	180.7	199.9	221.7	12.0%
RoW	21.7	25.3	29.0	34.6	39.5	45.1	15.7%
Total	226.2	255.3	284.3	329.2	365.6	407.0	12.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 75 COSMETICS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	62.9	71.2	80.7	91.8	104.7	121.1	14.0%
Europe	93.2	103.1	114.2	126.8	141.2	156.0	10.8%
Asia Pacific	246.3	274.4	306.4	342.9	384.7	427.9	11.7%
RoW	51.5	59.0	67.7	77.7	89.4	100.0	14.2%
Total	454.0	507.6	569.0	639.2	719.9	804.9	12.1%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

8.6 OTHER APPLICATIONS

This segment includes personal care products, home care, feed, and toiletries. Plant extracts are projected to witness increased demand in the personal care & toiletry industry. Various home care and dental care formulations are built around the utilization of various plant extracts, such as essential oils and aromatic fragrances. A wide variety of extracts can be used in cold process soap. Aloe leaf, chamomile, calendula, green tea, grapefruit seed, mallow, and evening primrose are some of the more popular extracts. Extracts can be made from a variety of plants and have several properties. Advantages include antioxidant properties, skin conditioning, anti-inflammatory properties, and added moisture to soaps and body products.

With the growing health concerns about synthetic fragrances, there has been a shift toward natural perfumes in recent years, as many people seek safe and natural ingredients in their favorite perfumes. Botanical extracts are natural and safe, making them ideal for improving the quality of perfumery products. The most widely used natural ingredients in the natural perfumery industry are essential oils.

The use of plant extracts as feed additives is a significant application of animal nutrition science. The most common feed additive is herb-based. It uses the herbs blend with feed materials directly without extraction processing. The products are simple to produce and have the benefit of being low-cost for livestock management. Plant essential oils (EOs) are another type of feed additives. EO is a monomer or natural essential oil obtained through a chemical extraction process. EOs were extracted from herbs, such as oregano leaves, thyme, parsley, eucalyptus oil, cumin, pine needles, houttuynia, artemisia, and bay leaves. Comparing the biotechnological routes of enzyme preparations, prebiotics, functional proteins, and peptides, EOs are applied to the feed through natural active substances, and the animals absorb the active ingredients after eating to exert the medicinal effects of plants. The plant extracts for feed additives are cost-effective, thereby driving the growth of the market.

Plant extracts are also used for home care as detergents, conditioners, and air care. Iberchem is a leading fragrance manufacturer that specializes in home care fragrances. The use of plant extracts in home care formulations allows for the creation of one-of-a-kind products. Furthermore, the use of fragrances encourages consumers to explore sensorial experiences.

The changing habits of baby boomers and consumers who prefer personal care products from renewable and natural materials will aid healthy gains in the personal care & toiletry industry. The expansion of retail channels, due to urbanization in Asian countries, is expected to boost the sales of these personal care products and toiletries.

TABLE 76 OTHER APPLICATIONS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	219.9	247.4	274.3	316.2	349.5	386.9	12.0%
Europe	420.8	463.6	503.6	568.7	615.7	667.8	9.7%
Asia Pacific	815.9	901.4	982.0	1,111.6	1,206.4	1,311.3	10.0%
RoW	189.5	218.2	247.4	291.1	328.1	370.1	14.3%
Total	1,646.1	1,830.6	2,007.3	2,287.6	2,499.7	2,736.1	10.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 77 OTHER APPLICATIONS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	428.6	475.6	528.7	588.8	655.0	751.3	11.9%
Europe	725.0	788.5	859.2	937.9	1,022.9	1,137.6	9.4%
Asia Pacific	1,426.1	1,553.4	1,694.9	1,852.1	2,021.0	2,248.3	9.5%
RoW	417.4	471.2	532.5	602.4	680.3	791.7	13.7%
Total	2,997.1	3,288.8	3,615.2	3,981.3	4,379.1	4,928.9	10.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 78 OTHER APPLICATIONS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	15.3	17.5	19.7	23.1	25.9	29.1	13.7%
Europe	26.8	29.7	32.6	37.1	40.5	44.4	10.6%
Asia Pacific	65.3	73.6	81.8	94.5	104.7	116.2	12.2%
RoW	11.4	13.3	15.3	18.2	20.7	23.6	15.6%
Total	118.8	134.1	149.3	172.8	191.8	213.3	12.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 79 OTHER APPLICATIONS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	32.8	37.0	41.9	47.5	53.9	60.5	13.0%
Europe	48.7	53.5	59.0	65.2	72.2	79.8	10.4%
Asia Pacific	129.2	143.9	160.7	179.7	201.5	222.7	11.5%
RoW	26.9	30.8	35.2	40.3	46.2	51.3	13.8%
Total	237.6	265.3	296.7	332.7	373.8	414.4	11.8%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

9 PLANT EXTRACTS MARKET, BY SOURCE

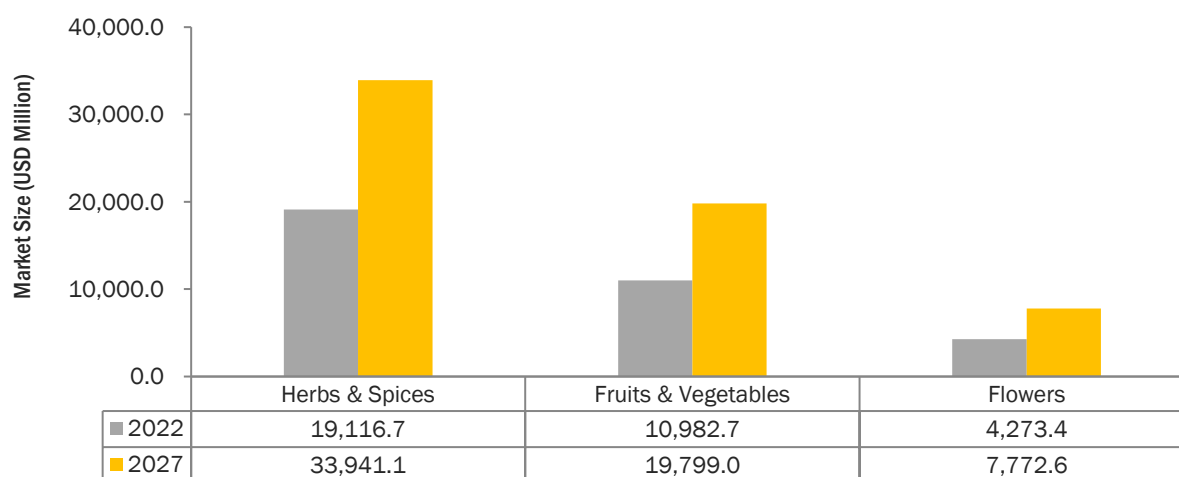
KEY FINDINGS

- The herbs & spices segment dominated the plant extracts market in 2021 and is projected to record a growth rate of 12.2% in terms of value during the forecast period. This growth is because of different herbs & spices extracts that find applications in the food and cosmetics industries, respectively.
- The fruits & vegetables segment was the second-largest in 2021, in terms of value sales, as the extracts from fruits & vegetables make excellent and powerful ingredients in skin care and hair care.
- The flowers segment accounted for a share of 12.4% in 2021, in terms of value, with the highest CAGR of 13.1%. Flower extracts containing phytochemical compounds such as carotenoids and flavonoids are primarily used in food & beverages, cosmetics, homecare, and others, fueling the demand for plant extracts.

9.1 INTRODUCTION

The active ingredients present in extracts vary according to the plant part it is derived from, which includes fruits & vegetables, herbs & spices, and flowers. These extracts find wide applications in different industries, with their major functions in skin care and cosmetics, including anti-irritant, anti-inflammatory, wound healing, anti-infective, sterilization, and protecting the skin. No matter whether it is a single plant extract or compound extract, it often has multiple effects. The clinical efficacy of the compound extract is reflected in the combined and the overall effect of compound compatibility. Its efficacy is sometimes better than mixed materials of isolated purified ingredients. For instance, plant parts, such as flowers, have always been popular in traditional medicine. Examples include clove and chamomile flowers. Different parts of flowers are also used, such as saffron stamens, the stigmas of maize, or pollen. The leaves of plants, shrubs, and trees can be used for medicinal properties. Some of the commonly used extraction methods include maceration; solvent extraction; and microwave-assisted extraction, ultrasound-assisted extraction, ASE, and supercritical fluid extraction.

FIGURE 47 PLANT EXTRACTS MARKET, BY SOURCE, 2022 VS. 2027 (USD MILLION)



Source: Secondary Research, Primary Interviews, Related Research Publications, Industry Journals, Press Releases, and MarketsandMarkets Analysis

TABLE 80 PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	5,346.8	6,058.5	6,772.2	7,872.3	8,779.0	9,812.5	12.9%
Herbs & Spices	9,478.5	10,707.4	11,932.4	13,828.4	15,374.2	17,131.8	12.6%
Flowers	2,057.5	2,335.9	2,616.1	3,046.7	3,403.8	3,811.3	13.1%
Total	16,882.9	19,101.8	21,320.7	24,747.3	27,557.0	30,755.7	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 81 PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (USD MILLION)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	10,982.7	12,322.4	13,859.6	15,628.0	17,614.4	19,799.0	12.5%
Herbs & Spices	19,116.7	21,383.2	23,977.6	26,954.6	30,288.3	33,941.1	12.2%
Flowers	4,273.4	4,802.8	5,411.0	6,111.4	6,899.3	7,772.6	12.7%
Total	34,372.8	38,508.4	43,248.2	48,694.1	54,802.0	61,512.6	12.3%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 82 PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (KT)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	397.7	445.3	491.7	564.6	621.8	686.4	11.5%
Herbs & Spices	685.4	765.0	842.2	964.1	1,058.8	1,165.2	11.2%
Flowers	164.5	184.3	203.8	234.3	258.4	285.5	11.7%
Total	1,247.6	1,394.6	1,537.7	1,763.0	1,938.9	2,137.2	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 83 PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (KT)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	759.4	842.1	936.0	1,043.0	1,165.0	1,303.2	11.4%
Herbs & Spices	1,285.3	1,421.1	1,574.9	1,749.5	1,948.4	2,172.1	11.1%
Flowers	316.3	351.1	390.7	435.8	487.3	544.5	11.5%
Total	2,361.0	2,614.3	2,901.6	3,228.3	3,600.7	4,019.7	11.2%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

9.2 FRUITS & VEGETABLES

9.2.1 INTRODUCTION OF NEWER TECHNOLOGIES

Fruits & vegetables are important foods for human health because they provide a variety of flavors and are linked to a higher quality of life. Their consumption has been linked to health benefits due to the presence of phenolic compounds with antioxidant activity, which may help to reduce the risk of cardiovascular diseases and cancer. The fruit extracts such as amla prevent hyperlipidemia, osteoporosis, and several other disorders. It shows strong antioxidant properties through ascorbic acid due to the presence of various tannins, such as emblicanin A, emblicanin B, punigluconin, and gallic acid. These extracts, when used in chemoprevention therapy, may prevent the development or recurrence of cancer. The extracts from fruits & vegetables also make excellent and powerful ingredients in skin care and hair care. The number of cosmetics containing fruit extracts is rising, particularly in the skin care category. Orange and red vegetable extracts are incorporated into new cosmetic formulations because they are high in beta-

carotene and are rich in essential antioxidant properties. These applications of fruits & vegetables are driving the growth of the market.

The extraction process for plant extracts from fruits & vegetables is determined by several factors, including the technique used, the raw material, and the organic solvent. Conventional techniques generally require large amounts of organic solvents, a lot of energy, and a lot of time, which has sparked interest in new technologies known as clean or green technologies. These can help to reduce or eliminate the use of toxic solvents, preserving the natural environment and its resources. Due to the advancement of technologies and the emergence of new extraction processes, the extraction of plant extracts from fruits & vegetables has become more convenient.

TABLE 84 FRUITS & VEGETABLES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	710.8	815.2	922.3	1,085.2	1,224.9	1,385.8	14.3%
Europe	1,232.8	1,382.5	1,529.5	1,759.7	1,942.2	2,148.6	11.8%
Asia Pacific	2,936.6	3,315.3	3,692.3	4,276.3	4,751.4	5,291.3	12.5%
RoW	466.7	545.5	628.1	751.0	860.4	986.7	16.2%
Total	5,346.8	6,058.5	6,772.2	7,872.3	8,779.0	9,812.5	12.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 85 FRUITS & VEGETABLES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,570.0	1,782.9	2,029.8	2,316.6	2,642.8	3,033.5	14.1%
Europe	2,380.2	2,643.1	2,942.3	3,283.6	3,663.0	4,076.7	11.4%
Asia Pacific	5,900.7	6,596.1	7,391.8	8,304.4	9,325.5	10,378.9	12.0%
RoW	1,131.9	1,300.2	1,495.8	1,723.4	1,983.1	2,309.9	15.3%
Total	10,982.7	12,322.4	13,859.6	15,628.0	17,614.4	19,799.0	12.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 86 FRUITS & VEGETABLES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	51.4	58.2	65.0	75.5	84.2	94.1	12.9%
Europe	86.0	95.2	104.0	118.1	128.7	140.5	10.3%
Asia Pacific	222.8	248.5	273.4	312.8	343.2	377.5	11.1%
RoW	37.5	43.3	49.2	58.1	65.7	74.3	14.6%
Total	397.7	445.3	491.7	564.6	621.8	686.4	11.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 87 FRUITS & VEGETABLES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	105.3	118.2	133.0	149.9	169.5	194.4	13.0%
Europe	153.8	168.7	185.5	204.5	225.9	250.0	10.2%
Asia Pacific	416.1	459.7	509.0	565.1	628.9	700.5	11.0%
RoW	84.2	95.5	108.5	123.5	140.7	158.2	13.4%
Total	759.4	842.1	936.0	1,043.0	1,165.0	1,303.2	11.4%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

9.3 FLOWERS

9.3.1 RISING USAGE OF FLOWER EXTRACTS FOR FRAGRANCES IN THE COSMETICS INDUSTRY

Plant extracts hold a lot of usage for active cosmetics, food & beverages, pharmaceuticals, and dietary supplements, but the colorful flowers of some plants also have natural dye potential in addition to providing bioactive compounds. Extracts from flowers are used in several applications, ranging from food & beverages to pharmaceuticals & dietary supplements. In the food & beverage application, flowers and fruits are generally used to provide flavors and aroma. Essential oils from flowers, when incorporated into finished products, impart various benefits, such as a pleasant aroma in perfumery, shine, or conditioning effects in hair care products, and improving the elasticity of the skin.

Flower extracts contain active compounds that ensure the stability of color due to the presence of antioxidants that prevent oxidation. Extracts from some other flowers such as Hibiscus, Chamomile, Magnolia, and Echinacea are highly used in pharmaceuticals and dietary supplements due to their various therapeutic effects. End Fatigue Revitalizing Sleep Formula, a multi-ingredient product from Integrative Therapeutics, is beneficial anecdotally. This product contains L-theanine, valerian root extract, passionflower leaf and flower extract, hops flower extract, wild lettuce leaf and leaf extract, Jamaica dogwood root extract, and passionflower leaf and flower extract. The rise of vegan perfumes in line with a strong desire for cosmetics and fragrances to be cruelty-free will contribute to the overall sales of floral extracts. Manufacturers are further exploring the capabilities of floral extracts as colorants and fragrance enhancers in skin care products and body care & toiletries. In July 2022, Symrise introduced Maison Lautier 1795 (fragrance), a natural brand, and its first three product lines. The emphasis is on artisan, sustainably grown Mediterranean flowers and plants. This adds to the company's scent product portfolio.

TABLE 88 FLOWERS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	274.6	315.4	357.3	420.9	475.6	538.7	14.4%
Europe	515.6	578.9	641.1	738.3	815.7	903.3	11.9%
Asia Pacific	1,041.8	1,178.0	1,313.9	1,524.0	1,695.7	1,891.1	12.7%
RoW	225.4	263.7	303.8	363.5	416.8	478.3	16.2%
Total	2,057.5	2,335.9	2,616.1	3,046.7	3,403.8	3,811.3	13.1%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 89 FLOWERS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	611.0	694.7	791.7	904.6	1,033.1	1,187.0	14.2%
Europe	1,001.6	1,113.2	1,240.4	1,385.5	1,546.9	1,723.1	11.5%
Asia Pacific	2,111.8	2,363.9	2,652.6	2,984.0	3,355.3	3,739.0	12.1%
RoW	549.0	631.0	726.3	837.3	964.0	1,123.4	15.4%
Total	4,273.4	4,802.8	5,411.0	6,111.4	6,899.3	7,772.6	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 90 FLOWERS: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	19.9	22.6	25.3	29.4	32.8	36.7	13.0%
Europe	35.5	39.4	43.0	48.9	53.4	58.3	10.4%
Asia Pacific	91.1	101.7	112.0	128.2	140.8	154.9	11.2%
RoW	17.9	20.7	23.5	27.8	31.4	35.6	14.7%
Total	164.5	184.3	203.8	234.3	258.4	285.5	11.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 91 FLOWERS: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	41.2	46.3	52.1	58.8	66.5	76.4	13.2%
Europe	63.9	70.1	77.2	85.1	94.1	104.3	10.3%
Asia Pacific	170.9	189.0	209.4	232.6	259.1	288.8	11.1%
RoW	40.3	45.8	52.0	59.2	67.5	75.0	13.2%
Total	316.3	351.1	390.7	435.8	487.3	544.5	11.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

9.4 HERBS & SPICES

9.4.1 BIOACTIVE FUNCTION OF HERBS & SPICES

Herbs & spices have been used to fortify foods as preservatives, flavors, and therapeutic agents. They are used as food additives, not only to improve the organoleptic properties of food, but also to extend the shelf-life by reducing or eliminating foodborne pathogens. Herbs & spices can be sourced from different parts of plants as mentioned below:

TABLE 92 SOURCES OF HERBS & SPICES

PART OF THE PLANT	HERBS & SPICES
Leaves	Basil, oregano, bay leaf, thyme, tarragon, mint, marjoram, sage, curry leaf
Bark	Cinnamon, cassia
Flower/Bud, pistil	Clove, saffron
Fruits/Berries	Clove, chili, black pepper, allspice
Bulbs	Onion, garlic, leek
Root	Ginger, turmeric
Aril	Mace
Seed	Ajowan, aniseed, caraway, celery, coriander, dill, fennel, fenugreek, mustard

Source: ncbi.nlm.nih.gov

Spices & herbs have been used for flavoring, coloring, aroma, enhancing agents, and food preservation. They contain bioactive compounds that have the potential to reduce or prevent the risk of degenerative diseases such as diabetes, obesity, cancer, and cardiovascular diseases. Herbal and spice phenolic compounds are effective substitutes for artificial antimicrobial agents used in food production. Tea catechins, oleuropein, ferulic acid, ellagic acid, and coumaric acid have been found to inhibit the growth of some pathogenic bacteria. Some of the bioactive functions of herbs and spices are mentioned below.

TABLE 93 BIOACTIVE FUNCTIONS OF SPICES & HERBS

SPICES & HERBS	EFFECT & FUNCTION
Cayenne pepper	Capsaicin in cayenne pepper has been shown to reduce appetite and increase fat burning (weight management), help combat lung, liver, and prostate cancer
Cinnamon	Potent antioxidant activity, help fight inflammation, lower cholesterol and triglycerides in the blood
Fenugreek	Improve function of insulin, lower blood sugar levels
Garlic	Help combat sickness and cold, reduce cholesterol and LDL, improve heart health, reduce blood pressure in hypertensive people
Ginger	Treat nausea caused by morning and seasickness, and chemotherapy, strong anti-inflammatory, pain management
Rosemary	Prevent allergies and nasal congestion, strong antioxidant
Sage	Improve brain function and memory, inhibit the breakdown of acetylcholine, a chemical messenger in the brain
Turmeric	Powerful antioxidant and help fight antioxidant damage in the body, strong anti-inflammatory, fight Alzheimer's

Source: ncbi.nlm.nih.gov

Processing technology advancements have been used to ensure that raw herbs & spices are free of microbial and other contamination and to extend their shelf-life. Radiation treatment is currently used for the extraction of plant extracts from herbs & spices for a wide range of food and ingredient categories. Irradiated herbs and spices help to extend the shelf-life of herbs and reduce the food-related health risks caused by pathogenic microorganisms. Also, for the separation of organic compounds from herbs & spices, Supercritical Fluid Extraction (SFE) has gained acceptance as an alternative to conservative solvent extraction. It can be used professionally to fight insect, pesticide, and microbial contamination.

Essential oils from various herbs & spices have been shown to inhibit microbial growth while also having significant antioxidant activity, making them ideal candidates for dairy products, particularly cheese preservation. In contrast to various pathogenic and spoilage microorganisms, essential oils have a broad spectrum of usage in cosmetics, home care, and feed.

TABLE 94 HERBS & SPICES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	1,260.0	1,440.7	1,625.1	1,906.3	2,145.2	2,419.5	13.9%
Europe	2,185.3	2,443.3	2,694.9	3,091.1	3,401.4	3,751.3	11.4%
Asia Pacific	5,205.8	5,859.2	6,505.7	7,511.8	8,320.9	9,238.2	12.2%
RoW	827.3	964.1	1,106.7	1,319.2	1,506.7	1,722.8	15.8%
Total	9,478.5	10,707.4	11,932.4	13,828.4	15,374.2	17,131.8	12.6%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 95 HERBS & SPICES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	2,732.7	3,093.9	3,511.5	3,995.6	4,544.4	5,200.3	13.7%
Europe	4,142.9	4,586.6	5,090.2	5,663.5	6,298.5	6,988.6	11.0%
Asia Pacific	10,270.8	11,446.4	12,788.1	14,323.1	16,035.4	17,792.4	11.6%
RoW	1,970.2	2,256.3	2,587.7	2,972.5	3,410.0	3,959.8	15.0%
Total	19,116.7	21,383.2	23,977.6	26,954.6	30,288.3	33,941.1	12.2%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 96 HERBS & SPICES: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	90.1	101.8	113.4	131.4	146.0	162.6	12.5%
Europe	150.2	165.8	180.5	204.4	222.0	241.7	10.0%
Asia Pacific	379.5	422.0	462.8	527.8	577.4	633.1	10.8%
RoW	65.5	75.4	85.4	100.5	113.3	127.8	14.3%
Total	685.4	765.0	842.2	964.1	1,058.8	1,165.2	11.2%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 97 HERBS & SPICES: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	181.5	203.1	227.7	256.0	288.5	330.0	12.7%
Europe	263.7	288.4	316.2	347.4	382.7	422.2	9.9%
Asia Pacific	695.7	766.3	846.0	936.3	1,038.9	1,153.7	10.6%
RoW	144.4	163.3	185.0	209.8	238.3	266.2	13.0%
Total	1,285.3	1,421.1	1,574.9	1,749.5	1,948.4	2,172.1	11.1%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

10 PLANT EXTRACTS MARKET, BY FORM

KEY FINDINGS

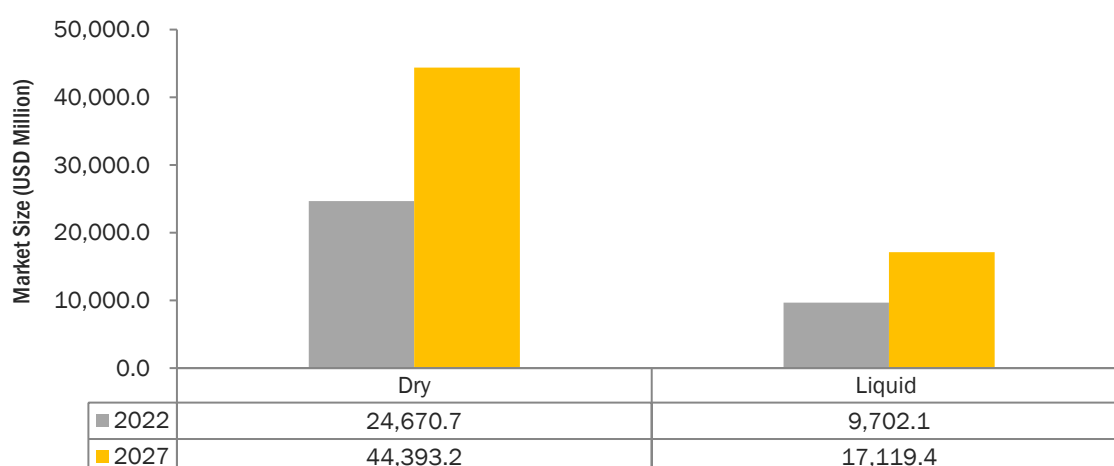
- The dry segment dominated the plant extracts market with a value of USD 22,052.2 million in 2021. It is projected to grow at a CAGR of 12.5% during the forecast period.
- Dry extracts have numerous advantages, including less storage space, stability, and ease of standardization for plant active ingredients, leading to increased demand.
- The liquid segment is expected to grow at a CAGR of 12.0%, owing to the increased demand for cosmetics. This is because liquid extracts enable cosmetics companies to use natural plant ingredients to capitalize on the growing demand for natural and plant-based products.

10.1 INTRODUCTION

Plant extracts, whether dry or liquid, are gaining traction as a versatile ingredient as millennials and Gen Z consumers become more conscious of their eating habits and food choices. Customers want to live a healthy lifestyle and use nutrition to improve their gut health or immune system. At the same time, they want products that make healthy eating more enjoyable, pleasurable, and convenient. Furthermore, veganism is becoming more popular as consumers become more conscious of their health and daily lifestyles. All these factors have made food manufacturers increase the use of plant extracts in their products to substitute and minimize the use of chemical materials. SAPNOV (offered by Nature (under Givaudan)), for example, is a purified extract of quillaia, which acts as an emulsifier in beverages as well as in flavor and color emulsions.

Premium plant extract products such as food & beverages and dietary supplements are gaining popularity. The growing demand for herbal medicines and natural cosmetics has resulted in the increased use of herbal extracts in the preparation of medicines and beauty products over the already existing chemical-based products. These factors will help the market propel in the coming years.

FIGURE 48 PLANT EXTRACTS MARKET (VALUE), BY FORM, 2022 VS. 2027



Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 98 PLANT EXTRACTS MARKET, BY FORM, 2016–2021 (USD MILLION)

Form	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Dry	12,048.1	13,643.8	15,242.8	17,709.3	19,739.1	22,052.2	12.9%
Liquid	4,834.8	5,457.9	6,077.9	7,038.0	7,817.9	8,703.5	12.5%
Total	16,882.9	19,101.8	21,320.7	24,747.3	27,557.0	30,755.7	12.7%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 99 PLANT EXTRACTS MARKET, BY FORM, 2022–2027 (USD MILLION)

Form	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Dry	24,670.7	27,667.7	31,106.3	35,061.2	39,502.9	44,393.2	12.5%
Liquid	9,702.1	10,840.6	12,141.9	13,632.8	15,299.1	17,119.4	12.0%
Total	34,372.8	38,508.4	43,248.2	48,694.1	54,802.0	61,512.6	12.3%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

10.2 DRY

10.2.1 INCREASING USE IN PHARMACEUTICALS AND DIETARY SUPPLEMENTS

Powder extracts are standardized and tested to ensure a certain percentage of 'active' ingredient content. The standardized extract is spray-dried to form a consistent powder after being extracted with ethanol and water. Spray-dried powders are stable and do not require any special storage conditions. A cool, dry location away from heat, sunlight, and moisture is adequate. Apart from the storage condition, dry extracts have numerous advantages, including less storage space, stability, and ease of standardization for plant active ingredients. These dry extracts are used in various applications; for instance, the food & beverage industry uses these extracts to improve the functionality of their products and as additives, flavoring, and coloring agents. Because of the growing popularity of clean-label, chemical-free products, their usage has increased dramatically in recent years, particularly in developed countries.

Dry extract is also used in the manufacturing of pharmaceuticals and dietary supplements. For thousands of years, medicinal plants have provided therapeutic agents and continue to be a great source of novel pharmaceutical products. Furthermore, as a result of an aging population, increasing interest of consumers in natural ingredients, and increased consumer awareness about general health, the demand for medicinal plant products is increasing in developing and developed countries. Yerba mate, catuaba, and muirapuama are some of the popular choices in the medicinal plant category. Despite this, there is widespread concern about the safety, efficacy, and quality of herbal medicines. Dietary supplements with additional health benefits remain popular in developed countries. Increasing active lifestyles and globalization are expected to drive the market in developing countries.

Since the outbreak of the pandemic, there has been a greater emphasis on health and nutrition. With improved health and immunity, it is easier to avoid and recover from a variety of illnesses. While eating healthy vegetables and fruits is important, many people have turned to nutritional supplements to meet their body's nutritional needs. A survey by the Council of Responsible Nutrition (CRN) in 2021 found that 80% of Americans consume dietary supplements, up 7% from the previous year. The increased emphasis on health and well-being, followed by endurance, filling the nutrient gap, and for hair, skin, and nails, can be attributed to the popularity of dietary supplements. This increased demand, due to the rising awareness, projects a bright future for the use of dry extracts in the above-mentioned applications.

TABLE 100 DRY: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	1,605.5	1,843.8	2,088.8	2,461.0	2,781.5	3,151.0	14.4%
Europe	2,895.2	3,243.5	3,584.6	4,119.7	4,542.2	5,019.5	11.6%
Asia Pacific	6,493.3	7,322.8	8,146.9	9,425.6	10,461.6	11,638.0	12.4%
RoW	1,054.1	1,233.8	1,422.5	1,703.1	1,953.7	2,243.6	16.3%
Total	12,048.1	13,643.8	15,242.8	17,709.3	19,739.1	22,052.2	12.9%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 101 DRY: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	3,574.5	4,064.7	4,633.6	5,295.4	6,049.1	6,952.5	14.2%
Europe	5,554.6	6,161.6	6,851.8	7,638.6	8,512.0	9,463.5	11.2%
Asia Pacific	12,964.5	14,477.2	16,206.3	18,187.7	20,402.6	22,683.1	11.8%
RoW	2,577.2	2,964.3	3,414.6	3,939.5	4,539.1	5,294.1	15.5%
Total	24,670.7	27,667.7	31,106.3	35,061.2	39,502.9	44,393.2	12.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

10.3 LIQUID

10.3.1 SHIFTING CONSUMER PREFERENCES

Historically, liquid plant extracts were used for both medicinal and cosmetics purposes. These liquid extracts have an advantage over powdered extracts because they can be added directly to formulations without prior dissolving. Liquid extracts enable cosmetics and personal care companies to use non-synthetic ingredients while still incorporating naturally occurring colors in their products. They also add unique plant-derived benefits to formulations for the skin, hair, and body. In recent years, the use of these extracts has increased due to factors such as changing consumer preferences, cost-effectiveness, increasing cosmetics demand, and additional benefits associated with plant extracts, which have driven this renewed interest in the use of natural, plant-based extracts in cosmetics formulations. This changes consumer demand and preferences as they become more concerned with the ingredients in beauty products. Also, synthetic additives are perceived to be less safe and unappealing for long-term use. When used for commercial cosmetics purposes, only small amounts are generally required, making them cost-effective. Integrity Ingredients Corporation (IIC), a supplier of sustainable, clean beauty ingredients, announced the addition of Chaga Mushroom liquid extract and Chaga Mushroom liquid extract BW to its botanical portfolio in July 2022, capitalizing on the mushroom's antioxidant and anti-aging properties. Similarly, Givaudan introduced Nootropics in February 2021, a collection of six natural and traceable extracts that include guarana, ginger, green tea, gotu kola, ginkgo, and green coffee. Each extract has powerful benefits when formulated into cosmetics, providing consumers with natural and sustainable alternatives.

Apart from the previously mentioned applications, extracts are increasingly being used in a beverage category known as relaxation drinks, which is a relatively new but thriving category. This category arose as a result of companies seeking to differentiate their products as the energy drink market in developed countries became saturated and further gained traction due to the pandemic. The pandemic has increased mental health awareness, with more consumers turning to food and beverages that can improve their mood. Givaudan, for example, launched an ashwagandha infusion in 2021, capitalizing on the COVID-19-induced megatrend for mood-enhancing food & beverage products. The liquid extract allows for easier formulation into functional food & beverages, with improved solubility and taste, expanding the ingredient's application beyond supplements.

Tinctures and essential oils are also included in the liquid plant extract category. Tinctures are alcohol-based liquid extracts that may contain other food-grade solvents. Alcohol extracts a wide range of phytochemicals and is an excellent preservative. Essential oils, on the other hand, are the volatile components extracted from an aromatic herb. Essential oils are plant-based compounds and potent preparations.

Tincture is extremely popular in Ayurveda and homeopathy. Apart from its use for medical purposes, it is also used for recreational applications like cannabis tinctures. Cannabis tincture has historically been used recreationally and as an herbal medication. There is a growing need for marijuana delivery methods that do not rely on the inhalation of smoke. Tincture has been one of the most popular methods. It is a simple, unobtrusive method for a user to consume marijuana without drawing attention or inhaling possibly harmful ingredients.

Furthermore, the recent growth in the number of medical marijuana dispensaries has aided in the supply of tinctures, making them more widely available. They are also used in food and beverage applications where drops of cannabis tinctures are added to drinks for liquid edibles. Recent developments in the space of tincture are related to an increase in its delivery mechanism with the help of nanotechnology. In October 2021, Heavy Hitters, California's cannabis brand, debuted HVY Grand, the first large format distilled cannabis tincture. It employs the nano-emulsion technology. This product was created in response to cannabis consumers' desire for a pleasing flavor profile and full-body elevation that was previously only available through inhaled experiences. Apart from cannabis tincture, other variants like oil are also gaining popularity due to their cost-effective nature. CBD oils, like tinctures, contain the CBD mixture in a liquid. However, in this case, a carrier oil is used. Apart from its medical application, it is increasingly used in cosmetics. Ilesol Pharmaceuticals (Croatia) launched two new CBD cosmetics ingredients in March 2022. CBD Skin Complex is a multifunctional cosmetic ingredient for skin health and natural nutrition. CBD is dissolved in almond oil with the addition of Serenoa Serrulata fruit extract in this new ingredient. Macadamia oil is enriched with Cupuacu Butter and CBD in CBD HSL Complex 3in1. This anti-inflammatory cosmetic ingredient is ideal for dry and damaged skin and repairs the skin barrier.

On the other hand, the essential oil market is steadily expanding, driven by the industrialization and urbanization of emerging economies, as well as new demand avenues such as aromatherapy in the markets. Furthermore, consumer interest in organic products and increased awareness regarding sustainability have propelled the market. Essential oils are popular home remedies for various physical and mental ailments. They are also used in food & beverage, cosmetics, perfumes, and room fragrances, among other things. Citrus essential oils, tea tree, rosemary, and lavender essential oils are some popular choices. Market advancements like the increased use of essential oil in functional beverages are expected to boost the growth of this segment during the forecast period.

TABLE 102 LIQUID: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
North America	639.9	727.5	815.8	951.4	1,064.2	1,193.0	13.3%
Europe	1,038.5	1,161.2	1,281.0	1,469.4	1,617.1	1,783.7	11.4%
Asia Pacific	2,691.0	3,029.7	3,365.0	3,886.5	4,306.5	4,782.6	12.2%
RoW	465.3	539.5	616.1	730.7	830.2	944.2	15.2%
Total	4,834.8	5,457.9	6,077.9	7,038.0	7,817.9	8,703.5	12.5%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

TABLE 103 LIQUID: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,339.2	1,506.8	1,699.4	1,921.4	2,171.2	2,468.2	13.0%
Europe	1,970.1	2,181.3	2,421.1	2,694.0	2,996.4	3,325.0	11.0%
Asia Pacific	5,318.7	5,929.3	6,626.2	7,423.7	8,313.6	9,227.1	11.6%
RoW	1,074.0	1,223.3	1,395.2	1,593.7	1,818.0	2,099.0	14.3%
Total	9,702.1	10,840.6	12,141.9	13,632.8	15,299.1	17,119.4	12.0%

Source: Secondary Research, Primary Interviews, Related Research Publications, Press Releases, Industry Journals, and MarketsandMarkets Analysis

11 PLANT EXTRACTS MARKET, BY REGION

KEY FINDINGS

- The plant extracts market was valued at USD 30,755.7 million in 2021 and projected to reach USD 61,512.6 million by 2027, at a CAGR of 12.3% during the forecast period.
- Asia Pacific is estimated to account for the largest market share, in terms of value and volume, of 53.4% and 54.5%, respectively, in the global plant extracts market in 2021. The surge for vegan food products is one of the key factors driving the demand for plant extracts in Asia Pacific.
- In Europe, the plant extracts market was valued at USD 6,803.2 million in 2021 and projected to reach USD 12,788.5 million by 2027, at a CAGR of 11.2% during the forecast period.
- The plant extracts market in RoW is projected to grow at the highest CAGR of 15.2% during the forecast period. Factors such as the rising number of vegans and health-conscious consumers have helped fuel the demand for food processed with plant extracts as a key ingredient.

11.1 INTRODUCTION

Plant extracts are used as ingredients in food & beverages for various functions such as enhancing aromas, flavors, and nutritive quality due to the presence of bioactive compounds which act as antimicrobials and antioxidants. Due to their natural origin, they can be used to replace synthetic compounds, which are generally considered to have toxicological and carcinogenic effects. A high cost associated with health care has led consumers to maintain a healthy diet with increased consumption of fortified food, beverages, and dietary supplements. The growth of the plant extracts market is also influenced by their varying applications adopted in different regions; some of these applications include pharmaceuticals, dietary supplements, food & beverages, and cosmetics. Prior to the growing health food trend, busy consumers opted for high-calorie, high-cholesterol on-the-go foods. As health concerns grew, the food industry was impacted by the growing demand for healthier food options. Hence, busy lifestyles and the rapid economic development of countries have encouraged the consumption of fortified food & beverage products as a healthier alternative. The application of plant extracts in the cosmetics sector is also expected to drive the market. The introduction of new trends in food & beverages such as veganism, plant-forward focus, clean label, organic, and natural additive is also driving the market growth. North American consumers are becoming increasingly health conscious and pay close attention to the labels of food products. This has led to a heightened awareness of the food ingredients that they consume; this factor fuels the market growth for plant extracts.

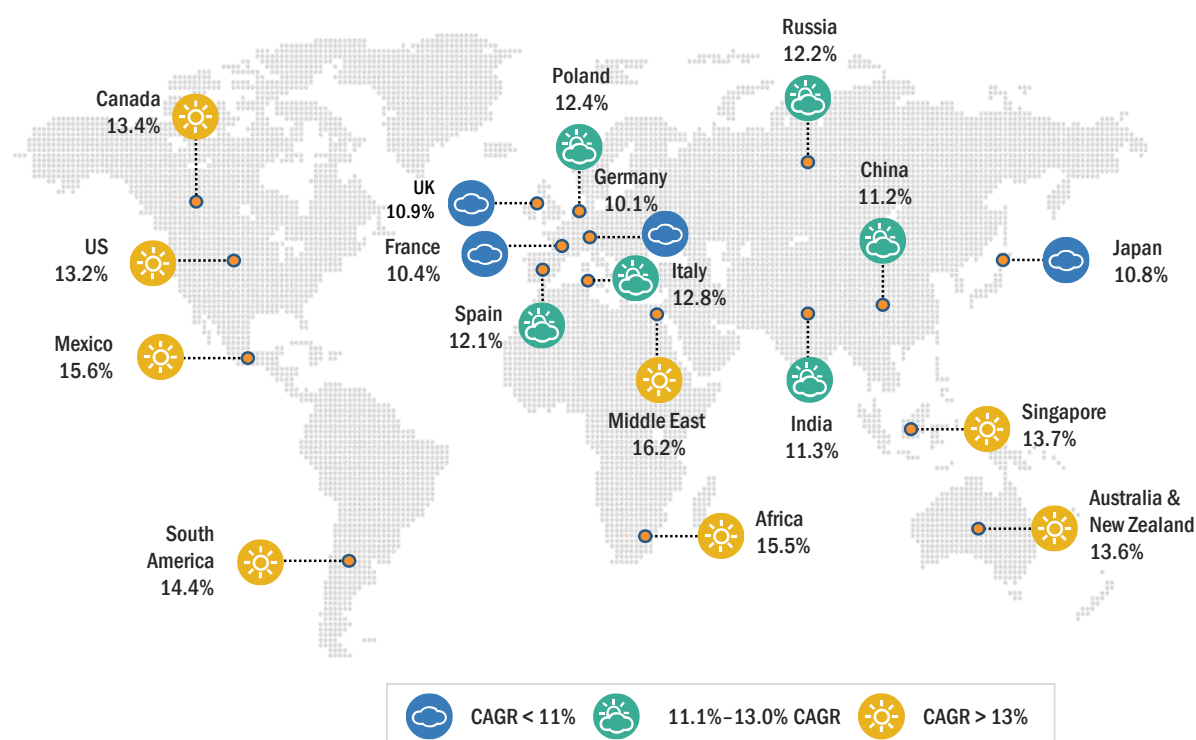
Oleoresins find wide usage to provide a natural fragrance and flavor in food & beverages. The high inclination of consumers toward natural ingredients and the use of plant bioactive in the production of food products have resulted in an increase in the consumption of oleoresins. These are usually used in the preparation of soups, meats, sauces, and dairy products. Emerging countries like India and China prefer a variety of spices in the preparation of meat and vegetables. Hence, the high demand for spice oils in the region aids the plant extracts market.

The plant extracts market is experiencing strong growth trends across the globe, especially in Asia Pacific. In 2021, Asia Pacific dominated the plant extracts market in terms of value and volume. Based on type, phytomedicines & herbal extracts dominated the plant extracts market across all regions due to the increasing inclination of consumers toward organic products and the adoption of green and healthy lifestyles. However, factors such as adherence to international quality standards & regulations and lack of awareness about plant extracts in developing countries restrain the growth of the market. Europe is the second-fastest growing market; this growth is attributed to the increasing innovations by local plant extract manufacturers to produce new products. In September 2020, Symrise, a Germany-based manufacturer and supplier of flavors, fragrances, cosmetic bases, and ingredients, launched SymControl Scalp, which normalizes the sebum production on the scalp and promotes healthy hair. To solve the issue of sebum overproduction at its source, Tetraselmis Suecica extract was identified as being highly effective in reducing sebum levels. Combined with the naturally occurring Mannitol, the algae can also protect the scalp and hair from environmental stress. This helps the company to cater to customers with a new product portfolio. Further, the market is driven by the growing consumer awareness related to the benefits of plant extracts. Also, government initiatives support the growth of the market. A government body in Coimbatore, India, is encouraging medicinal plant cultivation by incentivizing farmers, entrepreneurs, and traders. In February 2022, a two-day workshop was organized on 'Entrepreneurial Scope in Medicinal Plants Cultivation and Extraction' by the government to motivate the farmers for cultivation.

The geographical segmentation of the plant extracts market is provided below:

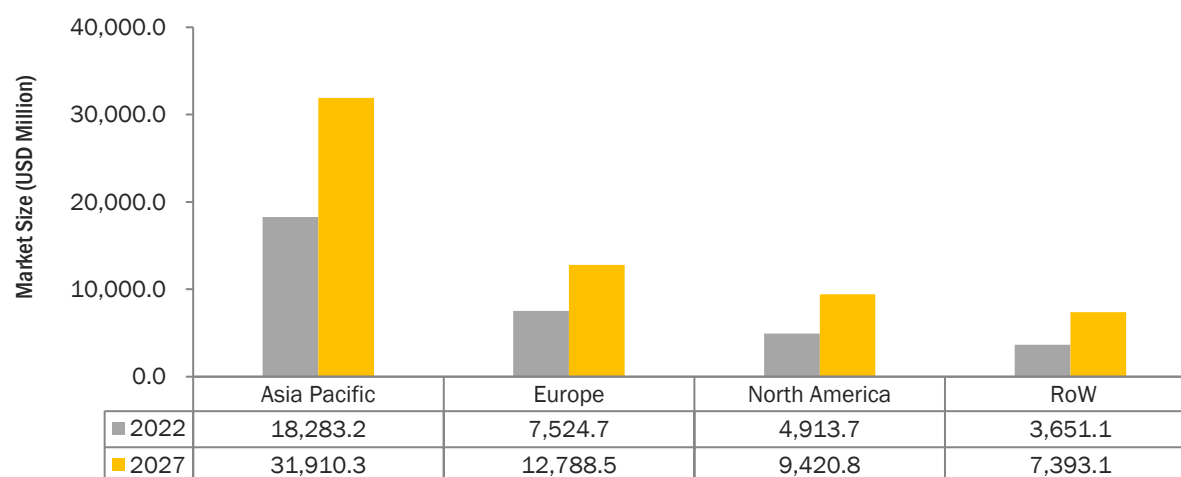
- North America (the US, Canada, and Mexico)
- Europe (Spain, France, Germany, the UK, Italy, Russia, Poland, and the Rest of Europe)
- Asia Pacific (China, India, Japan, Australia & New Zealand, Singapore, and the Rest of Asia Pacific)
- RoW (Brazil, Rest of South America, South Africa, Rest of Africa, and the Middle East)

FIGURE 49 GEOGRAPHIC SNAPSHOT (2022–2027):
RAPIDLY GROWING MARKETS EMERGING AS NEW HOTSPOTS



Source: Company Press Releases, Investor Presentations, Food Journals, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 50 PLANT EXTRACTS MARKET, BY REGION, 2022 VS. 2027 (USD MILLION)



Source: Secondary Research, Primary Interviews, Related Research Publications, Industry Journals, Press Releases, and MarketsandMarkets Analysis

TABLE 104 PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016-2021)
North America	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%
Europe	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%
Asia Pacific	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%
RoW	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%
Total	16,882.9	19,101.8	21,320.7	24,747.3	27,557.0	30,755.7	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 105 PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022-2027)
North America	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%
Europe	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%
Asia Pacific	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%
RoW	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%
Total	34,372.8	38,508.4	43,248.2	48,694.1	54,802.0	61,512.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 106 PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016-2021)
North America	161.4	182.6	203.7	236.4	263.0	293.4	12.7%
Europe	271.7	300.4	327.6	371.5	404.1	440.6	10.1%
Asia Pacific	693.5	772.2	848.2	968.8	1,061.4	1,165.5	10.9%
RoW	121.0	139.4	158.1	186.3	210.4	237.7	14.5%
Total	1,247.6	1,394.6	1,537.7	1,763.0	1,938.9	2,137.2	11.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 107 PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

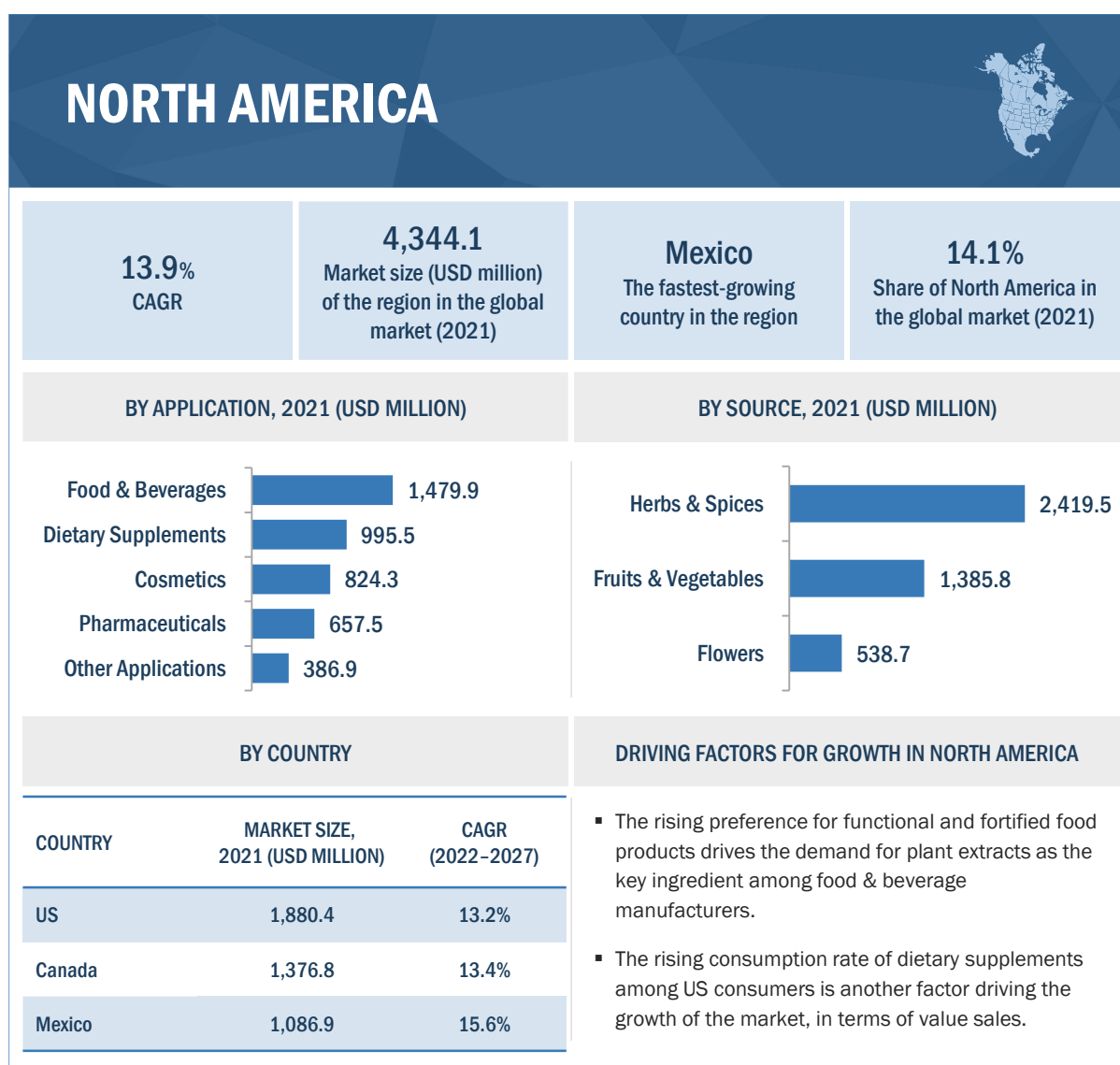
Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	328.0	367.5	412.8	464.7	524.5	600.8	12.9%
Europe	481.4	527.2	578.8	637.0	702.8	776.6	10.0%
Asia Pacific	1,282.7	1,414.9	1,564.5	1,734.0	1,926.8	2,143.0	10.8%
RoW	268.9	304.6	345.5	392.5	446.6	499.4	13.2%
Total	2,361.0	2,614.3	2,901.6	3,228.3	3,600.7	4,019.7	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.2 NORTH AMERICA

The North American region holds potential for the consumption of plant extracts and is expected to play an important role in the global market. Plant extracts have various health benefits against chronic diseases such as cancer, diabetes, hypertension, and obesity. In recent years, due to increasing awareness among consumers regarding the health benefits of plant extracts, they are shifting toward herbal medicines to treat these diseases. Consumer concerns related to these diseases have created a huge market for nutraceuticals that incorporate natural plant extract ingredients. The region is one of the major importers of various types of plant extracts due to the continued acceptance of essential oils, oleoresins, and flavonoids, since oleoresins and essential oils are used in almost every food application in the US to add natural spice attributes such as flavor, color, or as a natural antioxidant. Local and organic consumer goods are expected to gain market share as consumers seek these green and ethical attributes in their dietary supplements, cosmetics, and food & beverage products to fit the new lifestyle trend. These trends are further projected to attract investments from plant extract-based product manufacturers across the globe. North America dominates the flavonoids segment due to high incidences of high cholesterol high blood pressure, coronary heart diseases, atherosclerosis, and congestive heart failures. Flavonoids help protect heart and blood vessels from getting damaged, driving the market growth.

The North American plant extracts market focuses on investing more in research programs, to introduce new ingredients to address the diversified demand for varied applications and potential customers in the region. The key players in the North American plant extracts market include International Flavors & Fragrances (US) and Sensient Technologies (US). Technological advancements by these companies have made plant extracts available for several applications, which is projected to drive growth in the region.

FIGURE 51 NORTH AMERICA: MARKET SNAPSHOT

Source: Company Press Releases, Investor Presentations, Food Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 NORTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016-2021 (USD MILLION)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016-2021)
US	997.0	1,135.9	1,276.6	1,492.2	1,673.1	1,880.4	13.5%
Canada	725.3	827.4	931.1	1,089.7	1,223.5	1,376.8	13.7%
Mexico	523.2	608.0	696.9	830.4	949.1	1,086.9	15.7%
Total	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 109 NORTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	2,116.2	2,387.3	2,699.8	3,061.0	3,469.0	3,941.9	13.2%
Canada	1,551.5	1,752.5	1,984.5	2,252.9	2,556.5	2,911.0	13.4%
Mexico	1,246.1	1,431.7	1,648.6	1,902.8	2,194.8	2,567.9	15.6%
Total	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 110 NORTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (KT)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
US	71.5	80.5	89.4	103.2	114.3	126.9	12.2%
Canada	52.2	58.8	65.3	75.4	83.5	92.7	12.2%
Mexico	37.8	43.4	49.1	57.8	65.2	73.8	14.3%
Total	161.4	182.6	203.7	236.4	263.0	293.4	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 111 NORTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (KT)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	141.1	157.4	175.9	197.1	221.4	252.6	12.3%
Canada	103.2	115.1	128.7	144.3	162.1	185.0	12.4%
Mexico	83.6	95.0	108.1	123.3	141.0	163.2	14.3%
Total	328.0	367.5	412.8	464.7	524.5	600.8	12.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 112 NORTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	373.3	428.1	484.4	570.0	643.4	727.9	14.3%
Essential Oils	595.2	678.6	763.2	892.6	1,001.6	1,126.3	13.6%
Flavonoids	435.4	499.7	565.7	666.0	752.2	851.6	14.4%
Alkaloids	283.7	326.7	371.0	438.2	496.5	563.9	14.7%
Carotenoids	244.7	280.4	317.0	372.7	420.3	475.2	14.2%
Other Types	313.0	357.7	403.3	472.8	531.7	599.3	13.9%
Total	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 113 NORTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	824.6	936.5	1,066.1	1,216.8	1,388.2	1,593.3	14.1%
Essential Oils	1,268.4	1,431.8	1,620.3	1,838.2	2,084.5	2,372.0	13.3%
Flavonoids	965.3	1,096.9	1,249.5	1,426.9	1,628.8	1,880.4	14.3%
Alkaloids	641.3	731.1	835.5	957.2	1,096.1	1,263.1	14.5%
Carotenoids	537.9	610.3	694.3	791.7	902.5	1,044.7	14.2%
Other Types	676.3	765.0	867.4	986.0	1,120.2	1,267.3	13.4%
Total	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 114 NORTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	28.5	32.3	36.1	41.9	46.7	52.2	12.9%
Essential Oils	42.8	48.2	53.5	61.8	68.5	76.1	12.2%
Flavonoids	30.7	34.8	38.9	45.2	50.4	56.4	12.9%
Alkaloids	20.0	22.7	25.5	29.7	33.3	37.3	13.3%
Carotenoids	17.4	19.7	22.0	25.6	28.5	31.8	12.8%
Other Types	22.0	24.9	27.7	32.1	35.6	39.6	12.5%
Total	161.4	182.6	203.7	236.4	263.0	293.4	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 115 NORTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	58.5	65.6	73.8	83.2	94.1	107.9	12.9%
Essential Oils	84.7	94.4	105.6	118.4	133.0	152.0	12.2%
Flavonoids	63.1	70.9	79.8	90.1	101.9	117.5	12.9%
Alkaloids	42.0	47.3	53.4	60.4	68.6	78.9	13.3%
Carotenoids	35.5	39.9	44.8	50.5	57.0	65.9	12.8%
Other Types	44.2	49.4	55.4	62.2	70.0	78.5	12.5%
Total	328.0	367.5	412.8	464.7	524.5	600.8	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 116 NORTH AMERICA: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (USD MILLION)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	776.4	886.4	998.4	1,169.4	1,314.0	1,479.9	13.8%
Pharmaceuticals	336.8	386.4	437.3	514.6	581.0	657.5	14.3%
Dietary Supplements	507.5	582.7	660.1	777.6	878.8	995.5	14.4%
Cosmetics	404.8	468.4	534.6	634.5	722.4	824.3	15.3%
Other Applications	219.9	247.4	274.3	316.2	349.5	386.9	12.0%
Total	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 117 NORTH AMERICA: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	1,669.0	1,886.9	2,138.4	2,429.6	2,759.3	3,143.4	13.5%
Pharmaceuticals	745.1	846.3	963.7	1,100.2	1,255.4	1,441.4	14.1%
Dietary Supplements	1,129.1	1,283.8	1,463.4	1,672.2	1,910.0	2,195.0	14.2%
Cosmetics	941.9	1,078.9	1,238.8	1,425.9	1,640.6	1,889.6	14.9%
Other Applications	428.6	475.6	528.7	588.8	655.0	751.3	11.9%
Total	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 118 NORTH AMERICA: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (KT)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	55.8	62.9	70.0	81.0	89.9	99.9	12.4%
Pharmaceuticals	24.7	27.5	30.2	34.5	37.8	41.4	10.9%
Dietary Supplements	36.5	41.4	46.3	53.9	60.1	67.2	13.0%
Cosmetics	29.1	33.3	37.5	43.9	49.4	55.7	13.8%
Other Applications	15.3	17.5	19.7	23.1	25.9	29.1	13.7%
Total	161.4	182.6	203.7	236.4	263.0	293.4	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 119 NORTH AMERICA: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (KT)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	111.4	124.5	139.4	156.5	176.1	201.3	12.6%
Pharmaceuticals	45.6	50.2	55.4	61.3	68.0	77.9	11.3%
Dietary Supplements	75.4	84.7	95.4	107.7	121.9	140.0	13.2%
Cosmetics	62.9	71.2	80.7	91.8	104.7	121.1	14.0%
Other Applications	32.8	37.0	41.9	47.5	53.9	60.5	13.0%
Total	328.0	367.5	412.8	464.7	524.5	600.8	12.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 120 NORTH AMERICA: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	710.8	815.2	922.3	1,085.2	1,224.9	1,385.8	14.3%
Herbs & Spices	1,260.0	1,440.7	1,625.1	1,906.3	2,145.2	2,419.5	13.9%
Flowers	274.6	315.4	357.3	420.9	475.6	538.7	14.4%
Total	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 121 NORTH AMERICA: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (USD MILLION)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	1,570.0	1,782.9	2,029.8	2,316.6	2,642.8	3,033.5	14.1%
Herbs & Spices	2,732.7	3,093.9	3,511.5	3,995.6	4,544.4	5,200.3	13.7%
Flowers	611.0	694.7	791.7	904.6	1,033.1	1,187.0	14.2%
Total	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 122 NORTH AMERICA: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (KT)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	51.4	58.2	65.0	75.5	84.2	94.1	12.9%
Herbs & Spices	90.1	101.8	113.4	131.4	146.0	162.6	12.5%
Flowers	19.9	22.6	25.3	29.4	32.8	36.7	13.0%
Total	161.4	182.6	203.7	236.4	263.0	293.4	12.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 123 NORTH AMERICA: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (KT)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	105.3	118.2	133.0	149.9	169.5	194.4	13.0%
Herbs & Spices	181.5	203.1	227.7	256.0	288.5	330.0	12.7%
Flowers	41.2	46.3	52.1	58.8	66.5	76.4	13.2%
Total	328.0	367.5	412.8	464.7	524.5	600.8	12.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 124 NORTH AMERICA: PLANT EXTRACTS MARKET, BY FORM, 2016–2021 (USD MILLION)

Form	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Dry	1,605.5	1,843.8	2,088.8	2,461.0	2,781.5	3,151.0	14.4%
Liquid	639.9	727.5	815.8	951.4	1,064.2	1,193.0	13.3%
Total	2,245.4	2,571.3	2,904.7	3,412.3	3,845.7	4,344.1	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 125 NORTH AMERICA: PLANT EXTRACTS MARKET, BY FORM, 2022–2027 (USD MILLION)

Form	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Dry	3,574.5	4,064.7	4,633.6	5,295.4	6,049.1	6,952.5	14.2%
Liquid	1,339.2	1,506.8	1,699.4	1,921.4	2,171.2	2,468.2	13.0%
Total	4,913.7	5,571.5	6,333.0	7,216.8	8,220.3	9,420.8	13.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.2.1 US

11.2.1.1 Concept of healthy aging and rising trend of health-conscious diets

The US is one of the leading importers of plant extracts from Asian countries such as China and India due to the consumption of healthy foods by most of its population. It is projected to grow well in terms of plant extract production. During COVID-19, when supply chain complexities exacerbated, especially the reduced global availability of tocopherol (vitamin E), many food producers searched for an alternative in naturally sourced antioxidants to meet the high consumer demand. IFF (US) introduced a new GUARDIAN family of products, a line of plant extract-based formulations and synergistic oxidation control solutions that can be used as alternatives of tocopherol. In food & beverage products, it controls oxidation that helps protect the product's quality, freshness, and stability. This is further expected to be driven by the development of new plant-based pharmaceuticals and the rise of the beverage industry, where consumer preferences continue to shift away from carbonated soft drinks toward alternative beverages with natural flavors.

In the US, the FDA regulates functional foods from plant extracts and label claims that appear on foods and beverages. As stated by the CDC, in the US, one in six women is iron deficient during pregnancy;

deficiency is higher among non-Hispanic blacks and Hispanics. According to the CDC, in 2020, the obesity prevalence was 39.8% among adults aged 20 to 39 years, 44.3% among adults aged 40 to 59 years, and 41.5% among adults aged 60 and older, which can put them at risk for heart diseases, type 2 diabetes, and some cancers. In addition, obesity costs the US health care system USD 147 billion a year. The CDC released the 2022 National Diabetes Statistics Report. This report estimates that more than 130 million adults are living with diabetes or prediabetes in the US. To combat this, the population and government are taking collaborative efforts to enhance eating habits by reducing unhealthy saturated fats and carbohydrates in food and encouraging the consumption of plant extracts that meet the daily nutrition intake needs.

The markets for pharmaceuticals and dietary supplements accounted for 38% of the market share. The country has been witnessing a higher level of demand for plant dietary supplements, specifically among the geriatric population. There has been a rise in preference for fortified and functional food products among general consumers. These food products are processed with different types of plant extracts that drive the overall growth of the market in terms of value sales.

Organic and natural food products have become priorities for food processors and consumers, owing to the food quality, safety, sustainability and social acceptance. Consumers in the US are now aware about their purchasing habits as food labeling is often checked for ingredient origin, impact of supply chain over the environment, and manufacturing practices. Such factors drive the market for plant extracts in the US.

The growing industrial consumption of plant extracts is largely driven by an increase in consumer demand for fortified food & beverage products and dietary supplements, which impart health benefits and aid in dealing with the rising health care costs; the concept of healthy aging and the rising trend of health-conscious diets are also driving this market.

TABLE 126 US: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	164.5	187.7	211.3	247.4	277.9	312.8	13.7%
Essential Oils	266.2	301.9	337.7	392.9	438.6	490.6	13.0%
Flavonoids	192.1	219.3	247.0	289.3	325.0	366.0	13.8%
Alkaloids	125.2	143.4	162.0	190.3	214.5	242.4	14.1%
Carotenoids	108.0	123.1	138.4	161.9	181.6	204.2	13.6%
Other Types	141.0	160.5	180.2	210.3	235.5	264.3	13.4%
Total	997.0	1,135.9	1,276.6	1,492.2	1,673.1	1,880.4	13.5%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 127 US: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	352.6	398.4	451.3	512.6	581.8	662.2	13.4%
Essential Oils	549.6	617.2	694.8	784.2	884.6	997.3	12.7%
Flavonoids	412.7	466.5	528.7	600.6	682.0	780.5	13.6%
Alkaloids	274.2	310.9	353.5	402.9	459.0	524.3	13.8%
Carotenoids	230.0	259.6	293.8	333.2	377.9	433.6	13.5%
Other Types	297.0	334.6	377.7	427.5	483.7	544.0	12.9%
Total	2,116.2	2,387.3	2,699.8	3,061.0	3,469.0	3,941.9	13.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 128 US: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	11.8	13.3	14.8	17.1	19.0	21.1	12.3%
Essential Oils	19.1	21.4	23.6	27.2	30.0	33.1	11.6%
Flavonoids	13.8	15.5	17.3	20.0	22.2	24.7	12.4%
Alkaloids	9.0	10.2	11.3	13.2	14.7	16.4	12.7%
Carotenoids	7.7	8.7	9.7	11.2	12.4	13.8	12.2%
Other Types	10.1	11.4	12.6	14.5	16.1	17.8	12.0%
Total	71.5	80.5	89.4	103.2	114.3	126.9	12.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 129 US: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	23.5	26.3	29.4	33.0	37.1	42.2	12.4%
Essential Oils	36.7	40.7	45.3	50.5	56.5	63.9	11.8%
Flavonoids	27.5	30.8	34.5	38.7	43.5	50.0	12.7%
Alkaloids	18.3	20.5	23.0	25.9	29.3	33.6	12.9%
Carotenoids	15.3	17.1	19.1	21.5	24.1	27.8	12.6%
Other Types	19.8	22.1	24.6	27.5	30.9	35.1	12.1%
Total	141.1	157.4	175.9	197.1	221.4	252.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.2.2 CANADA

11.2.2.1 Rising consumption of plant-based products

Canadian plant extracts come from a wide range of sources, including fruits, flowers, and stems. The plant extracts market is projected to grow during the forecast period due to the increasing interest of consumers in herbal/traditional products. This is due to the rising concerns over the potential side effects of Over-The-Counter (OTC) products. Consequently, an increasing number of consumers have started to look for herbal/traditional products, as they are perceived to be natural and safe alternatives. Plant extracts can be used as dietary supplements. According to Health Canada, a dietary supplement is “A food sold or represented as a supplement to a diet that may be inadequate in energy and essential nutrients.” The market growth of the food industry and the increased usage of healthy food in the country have driven the plant extracts market. There are several factors responsible for consumers changing their attitude toward foods and reshaping food supply trends. These include the consumer’s desire to enhance their personal health, aging population, and the growing awareness that diets can alter disease prevalence and progression. Dietary supplements are usually used by Canadian Armed Forces (CAF) personnel to enhance their performance. These products promise weight loss, muscle gain, increased muscle power, increased endurance, improved appearance, reduced fatigue, accelerated workout recovery, and slowed aging.

The rising awareness related to the health benefits associated with the consumption of plant-based products among Canadians drives their demand. Most products are processed with plant extracts as an ingredient. As a result, the rising demand for plant-based food products eventually triggers the demand for plant extracts among food & beverage manufacturers. Canada is driven by the strong support for R&D by the government, which has further led to the growth of the industry. In August 2021, investment of USD 20 million was made by Natural Products Canada (Charlottetown-based not-for-profit organization established in 2015) to support current and future innovations in the sector, create jobs, and help protect the environment. Natural health products need to be approved by Health Canada to be sold in the country but receive less scrutiny over product claims compared with OTC drugs. Hence, this encourages new plant extract manufacturers to enter the market.

In response to the growing demand for plant extracts, Health Canada has developed regulations and policies governing health claims for food. After extensive consultations with Canadian consumers, academics, health care practitioners, and industry stakeholders, the Natural Health Products Regulations were developed. To be legally sold in Canada, all NHPs require a product license, as do the Canadian facilities that manufacture, package, label, and import these products. The safety and efficacy of NHPs, as well as their health claims, must be supported by appropriate evidence so that consumers and Health Canada can be confident that the products are safe and effective. This has led to the use of naturally sourced ingredients such as plant extracts in applications such as food, beverage, and dietary supplements.

A new technology to extract valuable nutrients from plants, which has potential benefits for both producers and consumers, is getting a boost by the Canadian government. This project is funded under the Agricultural Innovation Program (AIP) - a USD 50 million initiative announced as part of Canada's Economic Action Plan. This will help manufacturers to optimize and increase the efficiency of their plant extract extraction for product development.

TABLE 130 CANADA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	123.3	140.9	158.9	186.2	209.4	236.1	13.9%
Essential Oils	195.1	221.7	248.6	289.9	324.2	363.5	13.3%
Flavonoids	138.4	158.2	178.4	209.2	235.3	265.3	13.9%
Alkaloids	90.2	103.4	117.0	137.6	155.3	175.7	14.3%
Carotenoids	77.8	88.8	99.9	117.0	131.5	148.0	13.7%
Other Types	100.6	114.4	128.4	149.8	167.8	188.2	13.4%
Total	725.3	827.4	931.1	1,089.7	1,223.5	1,376.8	13.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 131 CANADA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	266.4	301.5	341.9	388.8	441.9	504.0	13.6%
Essential Oils	408.0	459.2	518.0	585.8	662.1	748.1	12.9%
Flavonoids	299.6	339.1	384.7	437.6	497.6	570.6	13.8%
Alkaloids	199.0	226.0	257.2	293.6	334.9	383.3	14.0%
Carotenoids	166.9	188.7	213.8	242.8	275.7	317.0	13.7%
Other Types	211.5	238.2	268.9	304.4	344.3	387.9	12.9%
Total	1,551.5	1,752.5	1,984.5	2,252.9	2,556.5	2,911.0	13.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 132 CANADA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	8.9	10.0	11.1	12.9	14.3	15.9	12.4%
Essential Oils	14.0	15.7	17.4	20.1	22.1	24.5	11.8%
Flavonoids	10.0	11.2	12.5	14.5	16.1	17.9	12.4%
Alkaloids	6.5	7.3	8.2	9.5	10.6	11.8	12.8%
Carotenoids	5.6	6.3	7.0	8.1	9.0	10.0	12.3%
Other Types	7.2	8.1	9.0	10.4	11.5	12.7	11.9%
Total	52.2	58.8	65.3	75.4	83.5	92.7	12.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 133 CANADA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	17.7	19.8	22.2	24.9	28.0	32.0	12.6%
Essential Oils	27.1	30.2	33.6	37.5	42.0	47.6	11.9%
Flavonoids	19.9	22.3	25.0	28.0	31.6	35.9	12.5%
Alkaloids	13.2	14.8	16.7	18.8	21.2	24.4	13.0%
Carotenoids	11.1	12.4	13.9	15.5	17.5	20.2	12.7%
Other Types	14.1	15.6	17.4	19.5	21.8	25.0	12.2%
Total	103.2	115.1	128.7	144.3	162.1	185.0	12.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.2.3 MEXICO

11.2.3.1 Growing health and wellness trend to drive market for phytomedicines and herbal extracts

The Mexican economy is growing and experiencing various socio-economic transitions, which have led to the increased spending power on food, driving the demand for plant extracts. There was degradation in health of Mexican people due to more sedentary lifestyles and sociocultural traditions or beliefs leading to overfeeding. Mexicans used to eat less traditional dishes made from fresh plant ingredients and far more commercial products and fast food, high in saturated fat. However, the health market in Mexico is rapidly developing due to the increased Foreign Direct Investments (FDI), increased per capita income, and consumer awareness of the benefits of a healthy diet which have encouraged people to adopt a plant-based diet. These changing eating habits and rising awareness drive the demand for plant extracts.

Mexico has a wide variety of medicinal and aromatic plants; their use is rooted in its culture, and they are used to solve health problems in areas where medicines are scarce. Mexico has a large number of medicinal endemic plants. The consumption of flavonoids has increased due to their medicinal properties. The growing health and wellness trend in this country is driving the market for nutraceutical ingredients. A growing number of chronic lifestyle diseases have also urged several consumers to adopt healthy food products. In 2017, according to the Organisation for Economic Co-operation and Development (OECD), Mexico was the second-most obese country in the world, after the US. The obesity issue in Mexico has prompted various players to raise health awareness. Hence, Mexican consumers tend to look for products to reduce the effects of obesity and help them maintain the required fitness levels. These trends are leading to an increased demand for organic products for food, beverage, and dietary supplement industries in the country.

According to ITC Trade Map, Mexico is the leading avocado producer and exporter in 2022; avocado is a rich source of vitamins such as C, E, B6, and minerals, including magnesium and potassium which is used in cosmetics and personal care products. The demand for such fruits provides a growth opportunity for cosmetics and food & beverage product manufacturers to innovate health products from fruit extracts, increasing the demand for plant extracts during the forecast period.

TABLE 134 MEXICO: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	85.5	99.5	114.2	136.3	156.1	179.0	15.9%
Essential Oils	133.9	155.0	176.8	209.8	238.7	272.2	15.2%
Flavonoids	105.0	122.2	140.4	167.6	191.9	220.2	16.0%
Alkaloids	68.4	79.9	92.1	110.3	126.7	145.8	16.4%
Carotenoids	59.0	68.6	78.7	93.8	107.2	122.9	15.8%
Other Types	71.5	82.9	94.8	112.7	128.4	146.7	15.5%
Total	523.2	608.0	696.9	830.4	949.1	1,086.9	15.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 135 MEXICO: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	205.6	236.6	272.9	315.4	364.4	427.1	15.7%
Essential Oils	310.7	355.4	407.5	468.3	537.7	626.6	15.1%
Flavonoids	253.0	291.3	336.1	388.7	449.2	529.3	15.9%
Alkaloids	168.1	194.1	224.7	260.7	302.3	355.5	16.2%
Carotenoids	141.0	162.1	186.7	215.7	248.9	294.1	15.8%
Other Types	167.8	192.2	220.8	254.1	292.2	335.3	14.9%
Total	1,246.1	1,431.7	1,648.6	1,902.8	2,194.8	2,567.9	15.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 136 MEXICO: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	6.2	7.1	8.0	9.5	10.7	12.2	14.5%
Essential Oils	9.7	11.1	12.5	14.6	16.4	18.5	13.8%
Flavonoids	7.6	8.7	9.9	11.7	13.2	15.0	14.6%
Alkaloids	4.9	5.7	6.5	7.7	8.7	9.9	14.9%
Carotenoids	4.3	4.9	5.5	6.5	7.4	8.3	14.4%
Other Types	5.2	5.9	6.7	7.8	8.8	10.0	14.1%
Total	37.8	43.4	49.1	57.8	65.2	73.8	14.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 137 MEXICO: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	13.8	15.7	17.9	20.4	23.4	27.3	14.7%
Essential Oils	20.9	23.6	26.7	30.3	34.5	39.6	13.7%
Flavonoids	17.0	19.3	22.0	25.2	28.9	33.7	14.7%
Alkaloids	11.3	12.9	14.7	16.9	19.4	22.8	15.1%
Carotenoids	9.5	10.8	12.2	14.0	16.0	18.7	14.6%
Other Types	11.3	12.8	14.5	16.5	18.8	21.0	13.3%
Total	83.6	95.0	108.1	123.3	141.0	163.2	14.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3 EUROPE

The European plant extracts market was valued at USD 6,803.2 million in 2021 and is projected to reach USD 12,788.5 million by 2027, at a CAGR of 11.2% during the forecast period. The key country-level markets identified in the European plant extracts market include Germany, France, the UK, Italy, Spain, Russia, Poland, and the Rest of Europe.

Europe is the second-largest region in the global plant extracts market; it is majorly influenced by the large production of medicinal and aromatic plants. Aromatherapy and natural remedies are becoming increasingly popular in Europe as consumers become more aware of the importance of living a healthy lifestyle. Cosmetics manufacturers respond to this trend by using essential oils, including high-value and low-volume oils, to create aromatherapy-infused cosmetics. Europe is the largest essential oils market and one of the two most important markets for aromatherapy products. Aromatherapy has thriving markets in France, Germany, Austria, and Switzerland. It is also widely used as a medicine in these countries.

Organizations such as EPIC (European Prospective Investigation into Cancer and Nutrition) are conducting R&D on flavonoids related to isolation, identification, characterization, functions, and application on health benefits. Flavonoid-containing drugs have been extensively used to treat disorders of the peripheral circulation, radiation poisoning, intoxications, and liver diseases. It is now considered an indispensable component in various nutraceutical, pharmaceutical, medicinal, and cosmetics applications.

Paprika oleoresin is an oil-soluble extract from the fruits of *Capsicum annum* Linn or *Capsicum frutescens*, which is primarily used as a coloring and flavoring in food products. An increased demand for natural food extracts with coloring properties in Europe is one of the drivers behind the increase in the demand for paprika oleoresin. Salt reduction is another significant driver of the increase in the demand for paprika oleoresin. According to the European Food Information Council, average salt intake per day should be 5 g. However, Europeans consume 8–12 g salt per day. Therefore, the demand for products containing less salt is strong, contributing to the growth of paprika oleoresin as it is used instead of salt in meats, fish, poultry, and vegetables, driving the demand for plant extracts. According to CBI-The Ministry of Foreign Affairs, Europe is one of the major markets for oleoresins worldwide. According to Trade Map ITC data, Germany is among the top ten oleoresin importers, where it majorly imports from the Asia Pacific region. The major spices consumed in the region are pepper, paprika, and allspice (pimento), while the leading herbs include thyme and oregano. The growing flavoring and confectionery industries in Europe, coupled with established players such as Givaudan (Switzerland), Symrise (Germany), and MANE (France), which are the major users of oleoresins for flavoring purposes, have made the region a market of opportunities for plant extracts.

Many Europeans, particularly the elderly, are incorporating nutritional supplements into their diets to reduce the negative effects of aging on their overall health. A growing number of plant-based nutritional supplements are developed with ingredients such as vitamin D3 and vitamin K2, with the goal of addressing bone and muscle problems in seniors or encouraging a more balanced diet. This trend opens opportunities for the plant extracts market.

The culmination of factors such as the rising vegan population and the increase in the demand for natural cosmetics drive the growth of the plant extracts market in terms of value sales. The rising awareness among health-conscious consumers related to different ingredients used by companies in food products and their preference for food products derived from natural sources have been propelling the demand for plant-based food ingredients in the region. This consumer adoption of various functional food & beverage products drives the plant extracts market in Europe.

The major companies operating in Europe include Döhler (Germany), Blue Sky Botanicals (UK), Symrise AG (Germany), Givaudan (Switzerland), Robertet SA (France), Mane (France), and NHR Organic Oils (UK).

TABLE 138 EUROPE: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (USD MILLION)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Germany	932.5	1,032.5	1,127.9	1,281.2	1,396.3	1,525.1	10.3%
UK	864.6	965.5	1,063.7	1,218.6	1,339.3	1,475.4	11.3%
France	494.0	550.7	605.5	692.4	759.6	835.2	11.1%
Italy	487.9	554.5	621.7	724.9	810.9	909.2	13.3%
Spain	436.6	492.7	548.6	635.1	705.3	785.2	12.5%
Russia	271.4	304.8	337.6	389.0	429.9	476.2	11.9%
Poland	236.0	266.0	295.7	341.9	379.2	421.6	12.3%
Rest of Europe	210.7	238.0	264.8	306.0	338.8	375.4	12.2%
Total	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 139 EUROPE: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	1,668.1	1,828.9	2,010.3	2,215.1	2,439.8	2,703.0	10.1%
UK	1,627.5	1,799.7	1,995.0	2,217.1	2,462.8	2,734.0	10.9%
France	919.6	1,014.9	1,122.9	1,245.6	1,381.0	1,505.7	10.4%
Italy	1,020.7	1,148.7	1,296.0	1,465.8	1,657.2	1,861.9	12.8%
Spain	875.2	978.0	1,095.5	1,230.3	1,381.0	1,550.0	12.1%
Russia	528.2	587.3	654.6	731.5	817.1	938.5	12.2%
Poland	469.3	523.7	585.9	657.0	736.5	842.6	12.4%
Rest of Europe	416.0	461.6	512.7	570.2	633.0	652.7	9.4%
Total	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 140 EUROPE: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (KT)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Germany	63.6	69.5	74.9	84.0	90.3	97.3	8.9%
UK	58.5	64.4	70.0	79.1	85.7	93.0	9.7%
France	33.2	36.5	39.6	44.7	48.4	52.5	9.6%
Italy	34.1	38.3	42.4	48.8	53.9	59.7	11.9%
Spain	30.9	34.5	38.0	43.5	47.8	52.6	11.2%
Russia	19.5	21.6	23.6	26.8	29.2	31.9	10.3%
Poland	16.9	18.8	20.6	23.6	25.8	28.3	10.9%
Rest of Europe	15.1	16.8	18.4	21.0	23.0	25.1	10.8%
Total	271.7	300.4	327.6	371.5	404.1	440.6	10.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 141 EUROPE: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (KT)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	105.1	113.8	123.5	134.4	146.6	160.4	8.8%
UK	101.3	110.5	120.8	132.4	145.5	159.2	9.5%
France	57.1	62.2	68.0	74.5	81.8	88.8	9.2%
Italy	66.3	73.7	82.2	91.9	103.0	114.6	11.6%
Spain	58.0	64.2	71.1	79.0	88.0	98.0	11.0%
Russia	34.9	38.3	42.1	46.4	51.3	57.5	10.5%
Poland	31.2	34.4	38.1	42.2	46.9	52.6	11.0%
Rest of Europe	27.5	30.1	33.0	36.3	39.9	45.6	10.7%
Total	481.4	527.2	578.8	637.0	702.8	776.6	10.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 142 EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	638.2	715.7	791.8	910.9	1,005.4	1,112.3	11.8%
Essential Oils	1,024.4	1,142.3	1,256.6	1,437.4	1,577.5	1,735.1	11.1%
Flavonoids	745.3	836.2	925.4	1,065.1	1,176.0	1,301.4	11.8%
Alkaloids	485.7	546.6	606.9	700.8	776.2	861.8	12.2%
Carotenoids	418.8	469.2	518.5	596.0	657.1	726.2	11.6%
Other Types	621.3	694.7	766.3	878.9	967.1	1,066.5	11.4%
Total	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 143 EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	1,232.1	1,368.2	1,523.1	1,699.7	1,896.1	2,110.0	11.4%
Essential Oils	1,911.2	2,110.2	2,335.6	2,591.7	2,874.6	3,155.5	10.5%
Flavonoids	1,442.2	1,602.0	1,784.0	1,991.7	2,222.5	2,486.8	11.5%
Alkaloids	958.1	1,067.7	1,192.8	1,336.0	1,495.7	1,670.4	11.8%
Carotenoids	803.6	891.4	991.3	1,105.1	1,231.5	1,381.6	11.4%
Other Types	1,177.6	1,303.4	1,446.1	1,608.4	1,788.1	1,984.1	11.0%
Total	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 144 EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	45.4	50.3	54.9	62.4	68.0	74.2	10.3%
Essential Oils	70.7	77.9	84.6	95.5	103.5	112.3	9.7%
Flavonoids	52.5	58.2	63.6	72.2	78.7	86.0	10.4%
Alkaloids	34.2	38.0	41.7	47.5	52.0	56.9	10.7%
Carotenoids	29.5	32.6	35.6	40.4	44.0	48.0	10.2%
Other Types	39.3	43.4	47.2	53.4	58.0	63.1	9.9%
Total	271.7	300.4	327.6	371.5	404.1	440.6	10.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 145 EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	81.2	89.1	98.0	108.0	119.3	132.0	10.2%
Essential Oils	122.2	133.3	145.7	159.7	175.4	192.8	9.5%
Flavonoids	94.1	103.3	113.6	125.3	138.5	154.1	10.4%
Alkaloids	62.5	68.8	76.0	84.0	93.2	103.5	10.6%
Carotenoids	52.5	57.5	63.1	69.5	76.7	85.6	10.3%
Other Types	68.8	75.2	82.4	90.5	99.6	108.5	9.5%
Total	481.4	527.2	578.8	637.0	702.8	776.6	10.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 146 EUROPE: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (USD MILLION)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	1,346.5	1,503.3	1,655.6	1,896.2	2,083.5	2,294.5	11.2%
Pharmaceuticals	584.2	655.3	725.1	834.5	921.3	1,019.4	11.8%
Dietary Supplements	880.1	988.2	1,094.7	1,261.0	1,393.5	1,543.4	11.9%
Cosmetics	702.2	794.3	886.5	1,028.8	1,145.4	1,278.1	12.7%
Other Applications	420.8	463.6	503.6	568.7	615.7	667.8	9.7%
Total	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 147 EUROPE: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	2,530.3	2,797.1	3,099.8	3,443.8	3,824.4	4,224.5	10.8%
Pharmaceuticals	1,129.6	1,254.6	1,397.0	1,559.5	1,740.0	1,937.1	11.4%
Dietary Supplements	1,711.8	1,903.2	2,121.3	2,370.2	2,647.3	2,949.9	11.5%
Cosmetics	1,428.0	1,599.4	1,795.7	2,021.1	2,273.9	2,539.4	12.2%
Other Applications	725.0	788.5	859.2	937.9	1,022.9	1,137.6	9.4%
Total	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 148 EUROPE: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (KT)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	93.5	103.1	112.1	126.8	137.6	149.7	9.9%
Pharmaceuticals	40.9	44.7	48.1	53.8	57.7	62.0	8.7%
Dietary Supplements	60.8	67.4	73.7	83.8	91.4	99.9	10.5%
Cosmetics	49.8	55.5	61.1	70.0	76.8	84.5	11.2%
Other Applications	26.8	29.7	32.6	37.1	40.5	44.4	10.6%
Total	271.7	300.4	327.6	371.5	404.1	440.6	10.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 149 EUROPE: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (KT)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	163.2	178.3	195.2	214.3	235.9	260.6	9.8%
Pharmaceuticals	66.8	72.1	78.0	84.6	91.9	101.0	8.6%
Dietary Supplements	109.5	120.3	132.4	146.1	161.7	179.1	10.3%
Cosmetics	93.2	103.1	114.2	126.8	141.2	156.0	10.8%
Other Applications	48.7	53.5	59.0	65.2	72.2	79.8	10.4%
Total	481.4	527.2	578.8	637.0	702.8	776.6	10.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 150 EUROPE: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	1,232.8	1,382.5	1,529.5	1,759.7	1,942.2	2,148.6	11.8%
Herbs & Spices	2,185.3	2,443.3	2,694.9	3,091.1	3,401.4	3,751.3	11.4%
Flowers	515.6	578.9	641.1	738.3	815.7	903.3	11.9%
Total	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 151 EUROPE: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (USD MILLION)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	2,380.2	2,643.1	2,942.3	3,283.6	3,663.0	4,076.7	11.4%
Herbs & Spices	4,142.9	4,586.6	5,090.2	5,663.5	6,298.5	6,988.6	11.0%
Flowers	1,001.6	1,113.2	1,240.4	1,385.5	1,546.9	1,723.1	11.5%
Total	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 152 EUROPE: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (KT)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	86.0	95.2	104.0	118.1	128.7	140.5	10.3%
Herbs & Spices	150.2	165.8	180.5	204.4	222.0	241.7	10.0%
Flowers	35.5	39.4	43.0	48.9	53.4	58.3	10.4%
Total	271.7	300.4	327.6	371.5	404.1	440.6	10.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 153 EUROPE: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (KT)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	153.8	168.7	185.5	204.5	225.9	250.0	10.2%
Herbs & Spices	263.7	288.4	316.2	347.4	382.7	422.2	9.9%
Flowers	63.9	70.1	77.2	85.1	94.1	104.3	10.3%
Total	481.4	527.2	578.8	637.0	702.8	776.6	10.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 154 EUROPE: PLANT EXTRACTS MARKET, BY FORM, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Dry	2,895.2	3,243.5	3,584.6	4,119.7	4,542.2	5,019.5	11.6%
Liquid	1,038.5	1,161.2	1,281.0	1,469.4	1,617.1	1,783.7	11.4%
Total	3,933.7	4,404.7	4,865.5	5,589.1	6,159.3	6,803.2	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 155 EUROPE: PLANT EXTRACTS MARKET, BY FORM, 2022–2027 (USD MILLION)

Form	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Dry	5,554.6	6,161.6	6,851.8	7,638.6	8,512.0	9,463.5	11.2%
Liquid	1,970.1	2,181.3	2,421.1	2,694.0	2,996.4	3,325.0	11.0%
Total	7,524.7	8,342.9	9,272.9	10,332.6	11,508.4	12,788.5	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.1 GERMANY

11.3.1.1 Thriving food & beverage industry and growing demand for natural ingredients

The plant extract market in Germany occupied a share of 22.4% in 2021, in terms of value. Germany has a vast number of consumers, making it one of the most lucrative markets for food & beverage manufacturers. The country is a hub for bakery products. Manufacturers in the bakery industry face challenges to maintain consistent natural flavoring in their products. While using natural flavors such as raw spices and herbs, it becomes difficult for manufacturers to maintain the consistency and standardization of the tastes in the products. Oleoresin is an easy and natural solution, thereby encouraging manufacturers to make use of spice or herb oleoresins instead of its raw form. Germany also has some of the world's largest food processing companies, which drive the demand for oleoresins for producing food products with consistent taste and flavor. In Germany, veganism has become increasingly popular, and it is home to the most vegan product launches worldwide. Because of this, there is an emphasis in the vegan market on unprocessed, natural, and wholesome products. Much of the vegan population sees veganism as a lifestyle choice. A wide range of spices is used in meat alternative products. The most popular mixes include dried onions, dried garlic, black pepper, garlic, chili, paprika, ginger, and curry powder. Many also contain a wide range of herbs such as sage, rosemary, and thyme. Thus, an increase in preference for vegan food products provides immense opportunity for the plant extracts market to grow significantly in terms of value sales.

Besides the above-stated factors, the rising aging population fuels the demand for healthy and natural products to prevent diseases such as diabetes, high blood pressure, and cholesterol. There is also a growing innovation in the beverage industry together with consumer inclination toward functional beverages, leading to product offerings such as smoothies, enhanced waters, and organic beverages with plant-based ingredients, which have led to rising consumer interests. Dietary supplements are also being adopted by end consumers as a preventive health measure and avoid exposure to several diseases associated with the bones and heart. As a result, the country is expected to continue being one of the important contributors to the European plant extracts market. In May 2018, BMBF (Bundesministerium für Forschung und Bildung) (Germany) started a joint international initiative, which recognizes that new active substances are urgently needed against infectious diseases. Hence, the plant extracts market shows enormous potential within the country to provide dietary supplements.

The country is also one of the major markets for essential oil products in Europe. It is the third-largest country globally, which imports and exports essential oils. Among different types, orange oil is used extensively in various end-use applications, such as food & beverages, health care, home care, and aromatherapy. However, the usage of essential oils is more in aromatherapy due to its various health benefits, such as improvement in immunity and speedy recovery from illnesses. Companies such as Symrise (Germany) are expected to grow further, with the increasing sales of its products and new product launches. In July 2021, Symrise introduced Lilybelle, a lily of the valley fragrance ingredient made from renewable sources. It can be used for personal care, cleaning, and laundry care goods. It is made from by-

products of the orange juice industry. With this, Symrise is expanding its line of special fragrance compounds with new products, which include a sustainable, easily biodegradable fragrance element.

TABLE 156 GERMANY: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	153.8	170.6	186.6	212.3	231.8	253.6	10.5%
Essential Oils	247.1	272.6	296.6	335.6	364.4	396.5	9.9%
Flavonoids	179.6	199.3	218.1	248.3	271.1	296.7	10.6%
Alkaloids	117.0	130.3	143.0	163.3	178.9	196.5	10.9%
Carotenoids	100.9	111.8	122.2	138.9	151.5	165.6	10.4%
Other Types	134.0	148.0	161.3	182.7	198.6	216.3	10.0%
Total	932.5	1,032.5	1,127.9	1,281.2	1,396.3	1,525.1	10.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 157 GERMANY: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	81.2	89.1	98.0	108.0	119.3	132.0	10.3%
Essential Oils	122.2	133.3	145.7	159.7	175.4	192.8	9.7%
Flavonoids	94.1	103.3	113.6	125.3	138.5	154.1	10.4%
Alkaloids	62.5	68.8	76.0	84.0	93.2	103.5	10.7%
Carotenoids	52.5	57.5	63.1	69.5	76.7	85.6	10.2%
Other Types	68.8	75.2	82.4	90.5	99.6	108.5	9.9%
Total	481.4	527.2	578.8	637.0	702.8	776.6	10.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 158 GERMANY: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	10.5	11.5	12.4	13.9	15.0	16.2	9.1%
Essential Oils	16.8	18.3	19.7	22.0	23.6	25.3	8.5%
Flavonoids	12.2	13.4	14.5	16.3	17.5	18.9	9.1%
Alkaloids	8.0	8.8	9.5	10.7	11.6	12.5	9.5%
Carotenoids	6.9	7.5	8.1	9.1	9.8	10.6	9.0%
Other Types	9.1	10.0	10.7	12.0	12.8	13.8	8.6%
Total	63.6	69.5	74.9	84.0	90.3	97.3	8.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 159 GERMANY: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	17.5	19.0	20.6	22.5	24.6	26.9	9.0%
Essential Oils	27.2	29.4	31.7	34.4	37.4	40.6	8.3%
Flavonoids	20.5	22.2	24.2	26.4	28.8	31.7	9.1%
Alkaloids	13.6	14.8	16.2	17.7	19.4	21.3	9.4%
Carotenoids	11.4	12.4	13.4	14.6	16.0	17.6	9.1%
Other Types	14.9	16.0	17.4	18.8	20.5	22.2	8.4%
Total	105.1	113.8	123.5	134.4	146.6	160.4	8.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

The market for phytomedicines & herbal extracts accounts for a 37% value share. With the rise in the vegan population in the country, there has been a significant rise in the demand for plant-based food products processed with plant extracts that drive growth in terms of value sales.

11.3.2 UNITED KINGDOM

11.3.2.1 Herbal tea gaining popularity

The market for plant extracts in the country occupied the second-largest share in the region, at 21.7% in 2021, in terms of value. According to the Food and Drink Federation, the food & beverage industry was one of the largest in the UK, with a value of about USD 14.82 billion in 2020. The growing food & beverage industry and the changing consumer trends toward natural ingredients demand many plant-based ingredients, including plant extracts, to be used in food products to cater to the growing inclination toward functional food & beverage products. Further, the UK government has been focusing on awareness programs to promote a healthy diet and healthy living, which drives the demand for plant extracts in dietary supplements and nutraceutical products in the country. According to the British Nutrition Foundation, approximately 48% of consumers in the UK consume plant-based dietary supplements. The increase in the consumption rate of dietary supplements in the country eventually triggers the demand for plant extracts as an ingredient by dietary supplement manufacturers.

The country's huge consumer base provides manufacturers with rampant growth opportunities, thereby enabling them to develop newer plant extract-based functional food & beverage products. Some of the spices consumed in the country include pepper and cardamom, while herbs include thyme and oregano. Due to the South Asian population, the UK is Europe's leading turmeric market. Turmeric is a key ingredient in many Indian dishes, particularly curry powder, given the UK's large South Asian population and its influence on British life and food. This increases the consumption of oleoresins in the UK, encouraging the growth of the plant extracts market.

The British Herbal Medicine Association was established in 1964 with the goal of furthering the science and practice of herbal medicine in the UK. It encourages the use of herbal medicinal products that are manufactured to pharmaceutical standards to ensure consistent high quality and effectiveness for the consumer. Herbal drinks, including herbal tea, are gaining popularity in the country. The growth in the health-conscious population also drives the preference for herbal tea and green tea in the country. Alkaloids are used in the UK as diet ingredients, supplements, and pharmaceuticals, in medicine and other applications. Alkaloids tend to have a pronounced effect when ingested by humans. Some may cause stimulation (caffeine, for instance), others intoxication (cocaine), and some can even be toxic (solanine).

North Macedonia's drug maker Alkaloid established its subsidiary in the UK, Alkaloid UK Limited. It was established to strengthen the existing positions in the market for alkaloids in the country.

The UK is one of the most prominent countries in Europe and among the key consumers of plant extracts. Essential oils such as jasmine and lavender are used by manufacturers to cater to the demand from consumers. Aromatherapy is one of the major application areas where manufacturers are focusing on delivering essential oil products due to their growing demand in the UK.

TABLE 160 UK: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	140.8	157.5	173.8	199.4	219.6	242.3	11.5%
Essential Oils	231.7	257.8	282.9	322.9	353.5	388.0	10.9%
Flavonoids	164.4	184.0	203.1	233.2	256.8	283.5	11.5%
Alkaloids	107.2	120.3	133.2	153.4	169.5	187.7	11.9%
Carotenoids	92.4	103.3	113.8	130.5	143.5	158.2	11.3%
Other Types	128.0	142.7	156.8	179.2	196.5	215.9	11.0%
Total	864.6	965.5	1,063.7	1,218.6	1,339.3	1,475.4	11.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 161 UK: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	267.7	296.5	329.2	366.4	407.7	453.4	11.1%
Essential Oils	426.3	469.7	518.6	574.2	635.4	675.8	9.7%
Flavonoids	313.3	347.2	385.6	429.4	477.9	534.3	11.3%
Alkaloids	208.1	231.4	257.8	288.0	321.6	358.9	11.5%
Carotenoids	174.6	193.2	214.3	238.2	264.8	296.8	11.2%
Other Types	237.5	261.9	289.5	320.8	355.3	414.8	11.8%
Total	1,627.5	1,799.7	1,995.0	2,217.1	2,462.8	2,734.0	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 162 UK: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	9.5	10.5	11.4	12.9	14.0	15.3	9.9%
Essential Oils	15.7	17.2	18.6	21.0	22.6	24.5	9.3%
Flavonoids	11.1	12.3	13.4	15.1	16.4	17.9	9.9%
Alkaloids	7.3	8.0	8.8	10.0	10.8	11.8	10.3%
Carotenoids	6.3	6.9	7.5	8.5	9.2	10.0	9.8%
Other Types	8.7	9.5	10.3	11.6	12.6	13.6	9.4%
Total	58.5	64.4	70.0	79.1	85.7	93.0	9.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 163 UK: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	16.7	18.2	19.9	21.9	24.1	26.4	9.7%
Essential Oils	26.5	28.8	31.4	34.3	37.5	40.9	9.1%
Flavonoids	19.5	21.3	23.3	25.6	28.2	31.1	9.8%
Alkaloids	12.9	14.2	15.6	17.2	19.0	20.9	10.0%
Carotenoids	10.9	11.9	13.0	14.2	15.6	17.3	9.7%
Other Types	14.8	16.1	17.5	19.2	21.0	22.6	8.8%
Total	101.3	110.5	120.8	132.4	145.5	159.2	9.5%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.3 FRANCE

11.3.3.1 Consumption of plant extracts for production of dietary supplements and functional foods

Plant extracts in France occupied the second-largest share in the European dietary supplements market, at 19.9% in 2021, in terms of value. Consumers here also prefer dietary supplements that contain natural ingredients and offer several health benefits. Consumers and manufacturers are inclining toward natural products or supplement antibiotics as an alternative that contain natural health ingredients. Hence, the demand for plant extracts in the production of dietary supplements and functional foods is projected to remain high in the coming years. Apart from the country's consumer interest in dietary supplements, it has also been successful at generating interest in manufacturing dietary supplements by inviting investments. Natural antioxidants such as phenolic compounds found in plant extracts are reported to have beneficial bioactivity due to their capacity to act as antioxidants. In France, herbal extracts are increasingly utilized in pharmaceuticals compared to any other applications. In France, any dietary supplement, before being bought to the retail sale, needs to be notified to the DGCCRF (Directorate General for Competition, Consumer Affairs and Fraud Prevention). In the pharmaceutical industry, nutritional supplements are recognized as the prescribed treatment. For instance, B9 vitamin in pregnancy, vitamin D in childhood, vitamin B12 in certain vegetarian diets, and micro- and macronutrient complexes in malnutrition are

consumed. Due to the least side-effects of herbal dietary supplements, they are consumed by the population.

The country is one of the major producers of dried herbs such as basil, coriander, marjoram, oregano, parsley, rosemary, and sage, driving the demand for oleoresins. Herbs, such as thyme, are widely used as flavoring agents in domestic food preparations. Fytosan (France), a local player in the botanical extracts market, serves dried herbs through its dried herb workshops.

France also contributes significantly to the essential oils market in Europe. It is the second-largest country in terms of imports & exports of essential oils. Essential oils such as orange, corn mint, citronella, and eucalyptus are used in various end-use applications. The demand for essential oils in France is driven by food & beverage, aromatherapy, and health care industries. Companies, such as Robertet SA (France) and Mane (France), extensively use essential oils to cater to consumer demand.

In France, medicinal and aromatic plants (MAP) or herbal, medicinal, and aromatic plants (HMAP) industries traditionally refer to culinary herbs, medicinal plants, and plants cultivated for the fragrance and perfume industries. According to the International Horticulture Congress 2022, the organic production for these crops is larger than traditional crops in French agriculture. This production is very diverse and dynamic, depending on strong traditional skills and benefiting from new techniques and high value processing. With a growing demand for natural products and unfolding opportunities for new markets, HMAP is facing exciting prospects and hence aiding the growth of the plant extracts market.

The sectors that drive the demand for plant extracts include food, animal nutrition, and cosmetics. Apart from the country's consumer interest in food & beverage products, it has also been successful at generating manufacturer's interest to expand their business due to investor-friendly policies and free trade agreements.

TABLE 164 FRANCE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	79.7	88.9	97.9	112.2	123.3	135.8	11.3%
Essential Oils	126.5	140.4	153.7	174.9	191.1	209.2	10.6%
Flavonoids	93.0	103.9	114.5	131.2	144.2	158.9	11.3%
Alkaloids	60.6	67.9	75.1	86.3	95.2	105.2	11.7%
Carotenoids	52.3	58.3	64.1	73.4	80.6	88.6	11.1%
Other Types	82.0	91.3	100.2	114.4	125.3	137.6	10.9%
Total	494.0	550.7	605.5	692.4	759.6	835.2	11.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 165 FRANCE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	149.7	165.5	183.4	203.8	226.3	247.2	10.5%
Essential Oils	229.3	252.0	277.5	306.5	338.3	366.2	9.8%
Flavonoids	175.3	193.8	214.9	238.8	265.3	291.3	10.7%
Alkaloids	116.4	129.2	143.7	160.2	178.5	195.7	10.9%
Carotenoids	97.6	107.8	119.4	132.5	147.0	161.8	10.6%
Other Types	151.2	166.6	184.0	203.7	225.5	243.5	10.0%
Total	919.6	1,014.9	1,122.9	1,245.6	1,381.0	1,505.7	10.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 166 FRANCE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	5.4	5.9	6.4	7.2	7.9	8.5	9.8%
Essential Oils	8.5	9.3	10.1	11.3	12.2	13.2	9.1%
Flavonoids	6.2	6.9	7.5	8.5	9.2	10.0	9.8%
Alkaloids	4.1	4.5	4.9	5.6	6.1	6.6	10.2%
Carotenoids	3.5	3.9	4.2	4.7	5.1	5.6	9.7%
Other Types	5.5	6.1	6.6	7.4	8.0	8.7	9.5%
Total	33.2	36.5	39.6	44.7	48.4	52.5	9.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 167 FRANCE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	9.3	10.2	11.1	12.2	13.4	14.6	9.4%
Essential Oils	14.2	15.5	16.8	18.3	20.0	21.6	8.7%
Flavonoids	10.9	11.9	13.0	14.3	15.7	17.2	9.6%
Alkaloids	7.2	7.9	8.7	9.6	10.6	11.5	9.8%
Carotenoids	6.1	6.6	7.2	7.9	8.7	9.5	9.5%
Other Types	9.4	10.2	11.1	12.2	13.3	14.4	8.9%
Total	57.1	62.2	68.0	74.5	81.8	88.8	9.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.4 ITALY

11.3.4.1 Growth of food supplements

In Italy, the demand for plants is primarily driven by the growth in food supplements and functional food industries. According to the FAO, the country has been witnessing a growing production of certain plants with therapeutic benefits, including cinnamon, citrus fruits, poppy, bay leaf, saffron, thyme, and tea. The dietary supplements industry in Italy is established and witnessing significant growth. This growth is due to the increasing adoption of healthy lifestyles and the increasing awareness of consumers, which have encouraged the demand for health benefiting and natural ingredients used in products they consume. In response to this trend, food supplements and functional food manufacturers in this country are opting for products with plant-based ingredients. The rising aging population and the high prevalence of vitamin D deficiency among Italian consumers are the two major reasons driving the demand for dietary supplements. According to the Italian Society for Osteoporosis, Mineral Metabolism and Bone Diseases (SIOMMMS), approximately 50% of Italian adults have vitamin D insufficiency during the winter months. The prevalence of deficiency increases with age, affecting almost all elderly people who do not consume vitamin D supplements.

Market globalization and the migration phenomenon have both contributed to changes in consumer food preferences, as they gradually incorporate foods and recipes from other countries into their diets. In this scenario, in Italy, the use of "foreign products" will increase, resulting in the fusion of traditional cuisine techniques and recipes with ingredients from other countries and cultures. Foods and ingredients originally consumed in Asian or South American countries are increasingly consumed by Italians, driving the demand for oleoresins in the country.

Italy also has one of the major essential oils markets in Europe. Different types of essential oils, such as rosemary and lavender, are used widely by manufacturers to produce cosmetics and personal care products. The country's essential oil consumption is primarily driven by the increasing demand from various home care, health care, and cosmetics & toiletries applications. The increasing need for natural ingredients in these applications is projected to drive the demand for plant extracts in the cosmetics industry, as it offers several skin benefits such as anti-inflammation, anti-aging, antioxidants, and prevents the skin from any damage. The country is witnessing a high demand for essential oils in the aromatherapy industry as well. All these factors tend to drive the market for plant extracts in the country.

TABLE 168 ITALY: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	77.1	87.8	98.6	115.1	129.0	144.8	13.4%
Essential Oils	125.9	142.5	159.1	184.8	205.9	230.0	12.8%
Flavonoids	90.0	102.5	115.2	134.6	150.9	169.5	13.5%
Alkaloids	58.7	67.0	75.6	88.6	99.6	112.2	13.9%
Carotenoids	50.6	57.5	64.6	75.3	84.3	94.6	13.3%
Other Types	85.6	97.1	108.7	126.5	141.2	158.1	13.1%
Total	487.9	554.5	621.7	724.9	810.9	909.2	13.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 169 ITALY: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	162.9	183.6	207.5	235.1	266.2	299.6	13.0%
Essential Oils	257.2	288.3	324.0	365.0	411.0	459.0	12.3%
Flavonoids	190.7	215.0	243.1	275.5	312.0	353.1	13.1%
Alkaloids	126.7	143.3	162.5	184.8	210.0	237.2	13.4%
Carotenoids	106.2	119.6	135.0	152.8	172.9	196.2	13.0%
Other Types	177.1	198.9	223.9	252.7	285.1	316.9	12.3%
Total	1,020.7	1,148.7	1,296.0	1,465.8	1,657.2	1,861.9	12.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 170 ITALY: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	5.4	6.1	6.7	7.8	8.6	9.5	12.1%
Essential Oils	8.8	9.8	10.8	12.4	13.7	15.1	11.4%
Flavonoids	6.3	7.1	7.9	9.1	10.0	11.1	12.1%
Alkaloids	4.1	4.6	5.1	6.0	6.6	7.4	12.5%
Carotenoids	3.5	4.0	4.4	5.1	5.6	6.2	12.0%
Other Types	6.0	6.7	7.4	8.5	9.4	10.4	11.7%
Total	34.1	38.3	42.4	48.8	53.9	59.7	11.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 171 ITALY: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	10.6	11.8	13.2	14.7	16.5	18.4	11.8%
Essential Oils	16.7	18.5	20.6	22.9	25.5	28.2	11.1%
Flavonoids	12.4	13.8	15.4	17.3	19.4	21.7	11.9%
Alkaloids	8.2	9.2	10.3	11.6	13.1	14.6	12.2%
Carotenoids	6.9	7.7	8.6	9.6	10.7	12.1	11.8%
Other Types	11.5	12.8	14.2	15.8	17.7	19.5	11.1%
Total	66.3	73.7	82.2	91.9	103.0	114.6	11.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.5 SPAIN

11.3.5.1 Increasing focus of food manufacturers on producing functional food & beverages

Spanish cuisine is dependent on dozens of spices and herbs. Unlike Mexican food, Spanish food does not consume a lot of chilies and paprika; however, it is dependent mainly on pepper, garlic, and herbs. Food manufacturers in Spain are focusing on the reduction of salt, sugar, and fat. The Ministry of Health, Social Services and Equality in Spain has launched the Collaboration Plan for Improving the Composition of Food and Beverages and Other Measures (2017–2020) that would be controlled by the Spanish Agency for Consumer Affairs, Food Safety and Nutrition (AECOSAN). It aims to reduce the level of salt, sugar, and saturated fat by 10% by 2020. Such initiatives provide a great opportunity for players to launch innovative functional food & beverage products using spice-extracted oleoresins, fueling the consumption of plant extracts in the country.

According to the World Bank figures, Spain's GDP reached USD 1,281 billion in 2020. It ranks 14th in the world and 5th in Europe. According to the Maddison Project Database of the University of Groningen, an authoritative source of data on global long-term growth, its GDP per capita in volume increased more than sixfold between 1958 and 2018, while the population increased by 58%. The economy's structure has been modernized, and new or strengthened industries and services have been created or strengthened. Today, Spanish agriculture is especially competitive in fruits & vegetables enabling the manufacturing of plant extracts from fruits & vegetables, flowers, herbs & spices. The demand for orange and lime oils is rapidly growing in the country due to their demand in the food & beverage industry. The plant extracts market here is therefore projected to have a positive outlook during the forecast period.

The dietary supplements market in Spain is expected to grow rapidly in the coming years because of the growing popularity of sports and personalized nutrition and the growing aging population. The increased incidence of dietary intolerances, vitamin-related conditions, and new eating habits (vegetarian, vegan, flexitarian, etc.) have increased the sales of dietary supplements in Spain. Moreover, there is an ongoing trend of consumers preferring clean-label ingredients. This trend represents consumers looking to avoid refined and processed foods and artificial preservatives and additives.

Plant extracts are also used among various applications of cosmetics; the skin care segment is a major application, accounting for 28% in the cosmetics market, and witnessing strong growth, as per STANPA: Asociacion Nacional de Perfumería y Cosmética. Hence, with the growing cosmetics industry, particularly the skin care segment, the demand for plant extracts in Spain is also projected to grow.

TABLE 172 SPAIN: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	71.8	81.2	90.5	105.0	116.8	130.2	12.6%
Essential Oils	111.3	125.0	138.4	159.4	176.1	195.0	11.9%
Flavonoids	83.9	94.9	105.8	122.8	136.6	152.4	12.7%
Alkaloids	54.7	62.0	69.4	80.8	90.2	100.9	13.0%
Carotenoids	47.1	53.2	59.3	68.7	76.3	85.0	12.5%
Other Types	67.8	76.5	85.1	98.5	109.3	121.6	12.4%
Total	436.6	492.7	548.6	635.1	705.3	785.2	12.5%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 173 SPAIN: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	145.4	162.7	182.6	205.4	230.9	259.6	12.3%
Essential Oils	216.3	240.4	267.9	299.3	334.2	373.9	11.6%
Flavonoids	170.2	190.6	213.9	240.7	270.7	306.0	12.4%
Alkaloids	113.1	127.0	143.0	161.4	182.2	205.5	12.7%
Carotenoids	94.8	106.0	118.8	133.5	150.0	170.0	12.4%
Other Types	135.5	151.3	169.3	189.9	213.0	235.0	11.6%
Total	875.2	978.0	1,095.5	1,230.3	1,381.0	1,550.0	12.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 174 SPAIN: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	5.1	5.7	6.3	7.2	7.9	8.7	11.4%
Essential Oils	7.9	8.8	9.6	10.9	11.9	13.1	10.6%
Flavonoids	5.9	6.7	7.3	8.4	9.3	10.2	11.4%
Alkaloids	3.9	4.3	4.8	5.5	6.1	6.8	11.8%
Carotenoids	3.3	3.7	4.1	4.7	5.2	5.7	11.3%
Other Types	4.8	5.4	5.9	6.7	7.4	8.1	11.1%
Total	30.9	34.5	38.0	43.5	47.8	52.6	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 175 SPAIN: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	9.6	10.7	11.9	13.2	14.7	16.4	11.2%
Essential Oils	14.3	15.8	17.4	19.2	21.3	23.6	10.5%
Flavonoids	11.3	12.5	13.9	15.5	17.2	19.1	11.1%
Alkaloids	7.5	8.3	9.3	10.4	11.6	13.0	11.6%
Carotenoids	6.3	7.0	7.7	8.6	9.6	10.5	10.7%
Other Types	9.0	9.9	11.0	12.2	13.6	15.4	11.3%
Total	58.0	64.2	71.1	79.0	88.0	98.0	11.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.6 RUSSIA

11.3.6.1 Consumption of herbal dietary supplements to create dynamic opportunities

Russia is a highly developed country with a high standard of living and a growing middle-class population. This is expected to create dynamic growth opportunities in the country's fast-moving consumer goods, pharmaceuticals, cosmetics, and food & beverage sectors, which, in turn, would fuel the demand for plant extracts in the country. In the Russian pharmaceuticals market, herbal medicinal products (HMP) and dietary supplements based on plant raw materials have similar compositions, dosage forms, and application scope. The country has gradually been moving toward the adoption of healthier food & beverage products, like its other European counterparts. Consumers take supplements in the form of tablets, capsules, syrups, and powders that are rich in minerals, amino acids, fibers, and vitamins. According to the Food and Food Supplements Regulatory Services in Russia, all nutritional and dietary supplements are known as Biologically Active Supplements (BAS) in Russia, and these are subject to mandatory federal registration by the Russian Federal Service for Surveillance on Consumer Rights Protection and Human Wellbeing (ROSPOTREBNADZOR). The increase in the consumption rate of dietary supplements in the country paves opportunity for the plant extracts market to grow exponentially in terms of value sales during the forecast period.

The EFCO Group, a leading Russian producer of yogurt, mayonnaise, and oils, launched new products, including plant-based patties and cutlets (which can be used as part of a traditional Russian dish made with minced meat and onion, garlic, and herbs). This increased the consumption of oleoresins for plant-based alternative products, driving the demand for plant extracts.

TABLE 176 RUSSIA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	43.3	48.7	54.1	62.4	69.1	76.6	12.1%
Essential Oils	68.9	77.1	85.1	97.7	107.6	118.8	11.5%
Flavonoids	50.6	56.9	63.2	72.9	80.8	89.7	12.1%
Alkaloids	33.0	37.2	41.4	48.0	53.3	59.4	12.5%
Carotenoids	28.4	31.9	35.4	40.8	45.1	50.0	12.0%
Other Types	47.2	52.8	58.4	67.1	73.9	81.7	11.6%
Total	271.4	304.8	337.6	389.0	429.9	476.2	11.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 177 RUSSIA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	85.1	94.8	105.9	118.5	132.6	152.5	12.4%
Essential Oils	131.3	145.5	161.5	179.9	200.2	229.7	11.8%
Flavonoids	99.7	111.0	124.0	138.9	155.4	179.8	12.5%
Alkaloids	66.2	74.0	82.9	93.1	104.6	120.7	12.8%
Carotenoids	55.5	61.8	68.9	77.0	86.1	99.9	12.5%
Other Types	90.4	100.2	111.4	124.1	138.2	155.9	11.5%
Total	528.2	587.3	654.6	731.5	817.1	938.5	12.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 178 RUSSIA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	3.1	3.5	3.8	4.3	4.7	5.1	10.5%
Essential Oils	5.0	5.5	6.0	6.7	7.3	8.0	9.9%
Flavonoids	3.6	4.0	4.4	5.0	5.5	6.0	10.5%
Alkaloids	2.4	2.6	2.9	3.3	3.6	4.0	10.9%
Carotenoids	2.0	2.3	2.5	2.8	3.1	3.4	10.4%
Other Types	3.4	3.7	4.1	4.6	5.0	5.5	10.0%
Total	19.5	21.6	23.6	26.8	29.2	31.9	10.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 179 RUSSIA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	5.6	6.2	6.8	7.5	8.3	9.3	10.7%
Essential Oils	8.7	9.5	10.4	11.4	12.6	14.1	10.1%
Flavonoids	6.6	7.2	8.0	8.8	9.8	11.0	10.8%
Alkaloids	4.4	4.8	5.3	5.9	6.6	7.4	11.0%
Carotenoids	3.7	4.0	4.4	4.9	5.4	6.1	10.7%
Other Types	6.0	6.5	7.2	7.9	8.7	9.5	9.8%
Total	34.9	38.3	42.1	46.4	51.3	57.5	10.5%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.7 POLAND

11.3.7.1 Reduced consumption of meat leading to plant-based alternatives

Poland has a long history of meat consumption. However, in the recent past, the country has witnessed a gradual rise in vegan food products. According to the European Commission, in 2019, the per capita meat consumption dropped to 60 kg from 62 kg compared to the previous year. According to New Zealand Foreign Affairs and Trade, in 2020, there was a trend in Poland of declining meat consumption and rapidly rising sales of vegetarian replacement products. During 2020 alone, sales of plant-based meat substitutes in Poland doubled. The rising demand for different types of plant-based food products has led food & beverage manufacturers in the country to strategize incorporating plant ingredients into their products. As a result, different types of plant extracts from various sources such as fruits & vegetables, flowers, and herbs & spices are expected to have a higher level of traction in the country. As per the recent scenario, the country accounts for a lower value share in the plant extracts market; however, during the forecast period, the country is expected to witness significant value sales growth.

The Polish Herbal Committee founded in 1930 coordinates R&D, cultivation, processing and marketing activities of plants with medicinal and other benefits. It brings together pharmaceutical industry representatives and entities involved in the distribution of herbal medicines and promotes cooperative work between the science and industry sectors with the aim of effective sharing of knowledge and technologies from universities to companies. The future development in the country has to focus on the implementation of technologies for upgrading the herbal extracts industry, the organization of controlled large-scale cultivation, and the production of homogeneous batches to meet the regulations on quality assurance binding for the European Union.

TABLE 180 POLAND: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	37.0	41.8	46.5	53.9	59.9	66.7	12.5%
Essential Oils	59.5	66.8	73.9	85.1	94.0	104.1	11.8%
Flavonoids	43.2	48.8	54.4	63.0	70.0	78.0	12.5%
Alkaloids	28.2	31.9	35.7	41.5	46.2	51.7	12.9%
Carotenoids	24.3	27.4	30.5	35.3	39.1	43.5	12.4%
Other Types	43.8	49.3	54.7	63.2	70.0	77.6	12.1%
Total	236.0	266.0	295.7	341.9	379.2	421.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 181 POLAND: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	74.3	83.1	93.1	104.6	117.4	134.6	12.6%
Essential Oils	115.4	128.3	142.9	159.6	178.2	202.6	11.9%
Flavonoids	87.0	97.3	109.0	122.5	137.6	158.6	12.8%
Alkaloids	57.8	64.8	72.9	82.2	92.6	106.5	13.0%
Carotenoids	48.5	54.1	60.6	68.0	76.3	88.1	12.7%
Other Types	86.3	96.1	107.3	120.1	134.3	152.2	12.0%
Total	469.3	523.7	585.9	657.0	736.5	842.6	12.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 182 POLAND: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	2.6	3.0	3.2	3.7	4.1	4.5	11.1%
Essential Oils	4.3	4.7	5.2	5.9	6.4	7.0	10.5%
Flavonoids	3.1	3.4	3.8	4.3	4.8	5.2	11.2%
Alkaloids	2.0	2.3	2.5	2.9	3.1	3.5	11.5%
Carotenoids	1.7	1.9	2.1	2.4	2.7	2.9	11.0%
Other Types	3.1	3.5	3.8	4.4	4.8	5.2	10.8%
Total	16.9	18.8	20.6	23.6	25.8	28.3	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 183 POLAND: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	4.9	5.5	6.0	6.7	7.5	8.4	11.2%
Essential Oils	7.7	8.4	9.3	10.2	11.3	12.6	10.6%
Flavonoids	5.8	6.4	7.1	7.9	8.8	9.9	11.3%
Alkaloids	3.8	4.3	4.7	5.3	5.9	6.6	11.6%
Carotenoids	3.2	3.6	3.9	4.4	4.9	5.5	11.3%
Other Types	5.7	6.3	7.0	7.7	8.5	9.5	10.4%
Total	31.2	34.4	38.1	42.2	46.9	52.6	11.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.3.8 REST OF EUROPE

The Rest of European market for plant extracts comprises Ireland, the Netherlands, Sweden, Turkey, Belgium, and other EU & non-EU countries. The essential oils market in these countries is rapidly expanding in line with major European countries. The major drivers and opportunities for the essential oils market in this region are rapid urbanization, the rise in the standard of living of the middle class, the increase in the purchasing power & GDP, and the growth in demand from aromatherapy, cosmetics & toiletries, health care, and food & beverage industries.

The Swedish populace is adopting healthier food and living options to have a healthy lifestyle. A vegan or vegetarian diet is another factor for the growth of the plant extracts market. However, vegans must seek alternatives to compensate for the nutrient deficiency. Plant-based supplements contain replacements for vegans. Botanical Healthcare, a manufacturer of herbal extracts in Sweden, provides a product line for the pharmaceutical, food, cosmetics, personal care, and dietary supplements industries. Plant extracts are also used for feed nutrition in Sweden. According to the Swedish University of Agricultural Sciences, Sesamin, a sesame oil component added to fish feed, may allow salmonid fish to produce long-chain omega-3 fatty acid DHA from linolenic acid in plant oils, potentially doubling fish feed production.

Turkey, with its geographical location, rich plant diversity, ecological structure, and large area, has enormous potential in the global medicinal and aromatic plant trade. According to the National Library of Medicine survey in 2021, as a result of the pandemic, Turkey has seen an increase in demand for herbal products that support the immune system and have a wide range of applications. According to the International Alliance of Dietary/Food Supplement Association, lowering restrictions and increasing the working relationship between governments in 2008 have helped markets in countries such as Turkey to boom. This will provide new opportunities to plant extract manufacturers in the country.

In the Netherlands, according to MAP-EXPO, 45% of physicians consider homeopathic medicines highly effective; homeopathic medicines are generally made using plants and mineral extracts. The growing awareness of the importance of consuming healthy products and the rising aging population in these countries are expected to boost the demand for plant extracts. The improvement of gastronomic functioning, immune system, and cardiovascular health and support in weight control is a major factor promoting the usage of plant extracts in dietary supplements. The growing consumer preference for healthy lifestyles reflects on the demand for plant extracts.

According to the Ministry of Foreign Affairs, Dutch imports of oleoresins increased at a triple-digit rate between 2015 and 2019 because of increasing demand in Europe. According to the Dutch Food Economic Report 2018, the output of the Dutch food and drink sector was USD 58.73 billion in 2017; this presents an opportunity for oleoresins exporters in developing countries. Important oleoresins importers in the country include Holland Ingredients B.V., De Lange, and Will & Co. The rising popularity of spices like dried thyme in the Netherlands is driven by the Mediterranean cuisine and interest in thyme's health benefits. Dried thyme is a traditional culinary aromatic herb. It is also used in herbal teas and a raw material to produce essential oils and other health care and cosmetics products. This rise in the consumption of oleoresin in the country drives the growth of the plant extracts market.

TABLE 184 REST OF EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	34.7	39.2	43.7	50.6	56.1	62.3	12.4%
Essential Oils	53.5	60.2	66.8	76.9	84.8	93.7	11.8%
Flavonoids	40.5	45.8	51.1	59.1	65.6	72.9	12.5%
Alkaloids	26.4	30.0	33.5	38.9	43.3	48.2	12.8%
Carotenoids	22.7	25.7	28.6	33.1	36.7	40.7	12.3%
Other Types	32.9	37.1	41.1	47.4	52.3	57.7	11.9%
Total	210.7	238.0	264.8	306.0	338.8	375.4	12.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 185 REST OF EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	69.1	76.8	85.5	95.2	105.8	109.3	9.6%
Essential Oils	103.4	114.3	126.5	140.2	155.1	164.6	9.7%
Flavonoids	80.9	89.9	100.1	111.5	124.1	128.9	9.8%
Alkaloids	53.7	59.9	66.9	74.8	83.5	86.6	10.0%
Carotenoids	45.1	50.0	55.6	61.9	68.7	71.6	9.7%
Other Types	63.8	70.5	78.1	86.5	95.7	91.8	7.5%
Total	416.0	461.6	512.7	570.2	633.0	652.7	9.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 186 REST OF EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2021	CAGR (2016–2021)
Oleoresins	2.5	2.8	3.0	3.5	3.8	11.0%
Essential Oils	3.8	4.2	4.6	5.3	5.7	10.4%
Flavonoids	2.9	3.2	3.6	4.1	4.4	11.0%
Alkaloids	1.9	2.1	2.3	2.7	2.9	11.4%
Carotenoids	1.6	1.8	2.0	2.3	2.5	10.8%
Other Types	2.4	2.6	2.9	3.3	3.5	10.4%
Total	15.1	16.8	18.4	21.0	23.0	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 187 REST OF EUROPE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	4.6	5.0	5.5	6.1	6.7	7.6	10.8%
Essential Oils	6.8	7.5	8.2	8.9	9.8	11.5	11.1%
Flavonoids	5.3	5.9	6.4	7.1	7.8	8.9	10.8%
Alkaloids	3.6	3.9	4.3	4.8	5.3	6.0	11.2%
Carotenoids	3.0	3.3	3.6	3.9	4.3	5.0	10.9%
Other Types	4.2	4.6	5.0	5.5	6.0	6.4	8.8%
Total	27.5	30.1	33.0	36.3	39.9	45.6	10.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

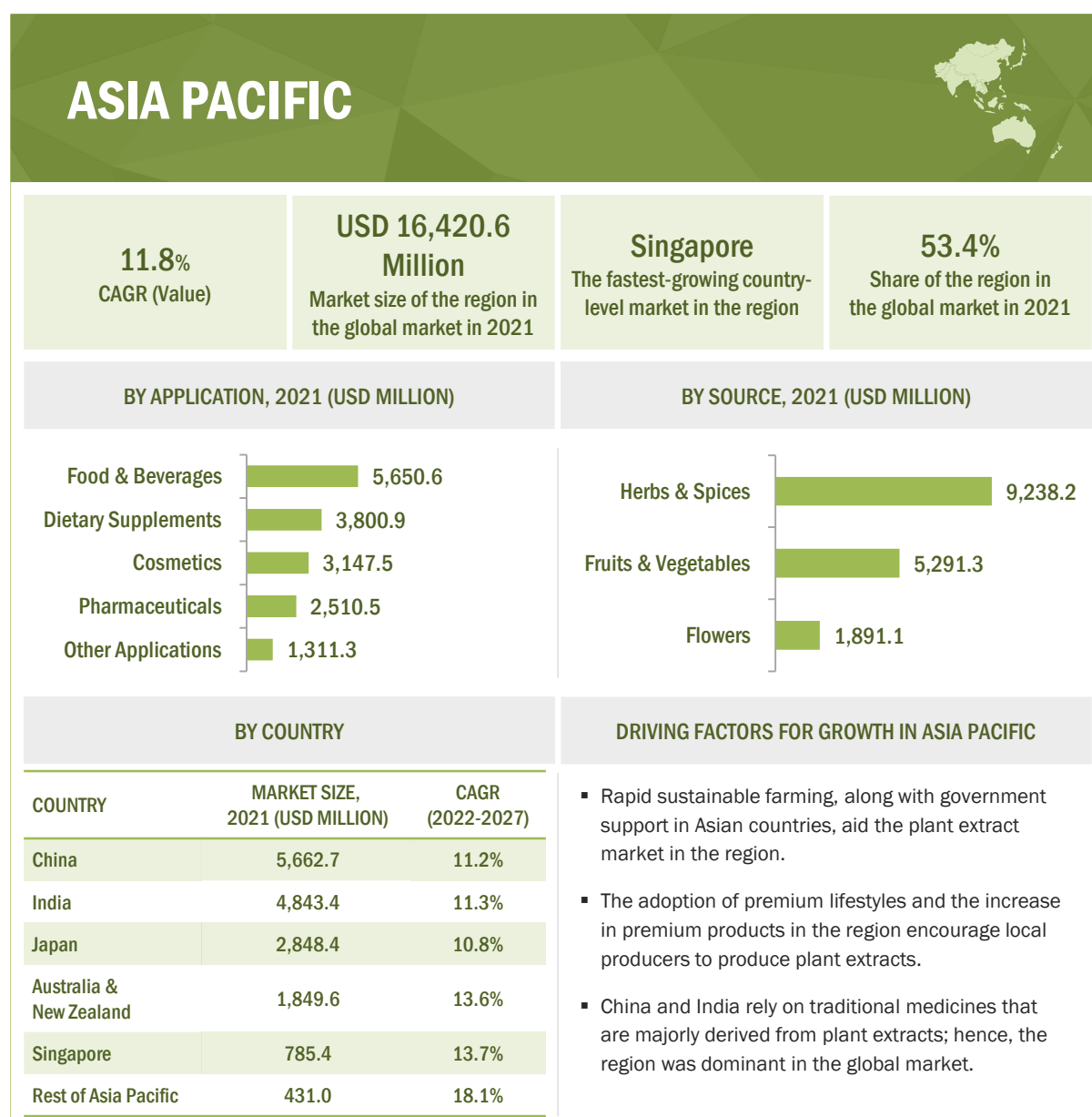
11.4 ASIA PACIFIC

Asia Pacific dominated the plant extracts market in 2021 and is projected to continue its dominance during the forecast period. The market for plant extracts in the region is thriving due to strong local and international demand. Plant extract as an ingredient has been gaining a higher level of traction in the region, thus driving the overall value sales of the market.

The availability of international cuisines in the major economies, shifting consumer focus of customers toward natural products, and the increasing demand for convenience foods and other end-user products are the key drivers of this market. The region has several small and medium-sized manufacturers of plant extracts that are well-versed with the processing of plant extracts and aware of the benefits offered by them. Some of these manufacturers include Alchemy Chemicals, Vital Herbs, Sydler, and Plantnat. Manufacturers in Indonesia, India, and Thailand are continuously investing in the market and focusing on innovation to expand their product range. This is due to the rise in the trade of plant extracts between Asia and international manufacturers of spices and essential oils.

Asia Pacific has always been known as the land of spices. Some of the major spice-producing countries in the Asia Pacific region are India, China, Sri Lanka, Indonesia, and Vietnam. With the increasing R&D in the spice industry, the benefits of extracts have been realized. Oleoresins are an example of such extracts, which were first introduced in countries such as the UK, the US, and Canada. However, now the US has become one of the major importers of oleoresins from Asia Pacific countries, especially India. Oleoresins manufacturers are expanding their capabilities and conducting regional diversification in Southeast Asia. Synthite Industries (manufacturer and exporter of spice oleoresin and essential oils) and Plant Lipids (producer of spice oils and extracts) started their units in China and Sri Lanka, respectively. Thus, Asian countries have a high production of oleoresins, but a majority of the production is exported to European and North American countries.

Synthite Industries (India), PT. Indesso Aroma (Indonesia), Vidya Herbs Private Limited (India), Arjuna Natural Extracts Ltd. (India), Kangcare Bioindustry Co. Ltd. (China), and Jiaherb, Inc. (China) are a few major Asian companies focusing on natural/organic extracts as they are becoming popular in every segment of the industry and offer plenty of scope for the market.

FIGURE 52 ASIA PACIFIC: MARKET SNAPSHOT

Source: Company Press Releases, Investor Presentations, Food Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 188 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (USD MILLION)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
China	3,260.4	3,653.9	4,039.6	4,644.3	5,122.5	5,662.7	11.7%
India	2,755.3	3,095.2	3,430.2	3,953.2	4,370.7	4,843.4	11.9%
Japan	1,625.6	1,825.0	2,021.2	2,327.9	2,572.1	2,848.4	11.9%
Australia & New Zealand	955.2	1,094.0	1,236.1	1,452.4	1,637.1	1,849.6	14.1%
Singapore	404.1	463.2	523.7	615.8	694.7	785.4	14.2%
Rest of Asia Pacific	183.7	221.2	261.0	318.5	370.9	431.0	18.6%
Total	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 189 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
China	6,268.6	6,956.1	7,738.1	8,629.7	9,619.9	10,658.0	11.2%
India	5,374.5	5,978.4	6,666.5	7,452.5	8,327.7	9,190.2	11.3%
Japan	3,158.7	3,511.3	3,913.0	4,371.5	4,824.3	5,265.2	10.8%
Australia & New Zealand	2,092.6	2,373.2	2,698.0	3,075.1	3,503.4	3,956.9	13.6%
Singapore	889.3	1,009.3	1,148.3	1,309.7	1,493.2	1,691.2	13.7%
Rest of Asia Pacific	499.5	578.2	668.6	772.8	947.6	1,148.8	18.1%
Total	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 190 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (KT)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
China	246.6	273.3	298.8	339.7	370.5	404.9	10.4%
India	208.4	231.4	253.3	288.4	315.0	344.7	10.6%
Japan	120.8	133.8	146.1	165.8	180.6	197.2	10.3%
Australia & New Zealand	72.8	82.3	91.8	106.4	118.4	132.0	12.6%
Singapore	30.8	34.8	38.8	45.0	50.1	55.8	12.6%
Rest of Asia Pacific	14.0	16.7	19.4	23.4	26.9	30.9	17.2%
Total	693.5	772.2	848.2	968.8	1,061.4	1,165.5	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 191 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (KT)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
China	443.6	487.0	536.0	591.4	654.1	720.2	10.2%
India	378.2	415.8	458.3	506.3	560.8	617.5	10.3%
Japan	215.8	236.6	260.1	286.6	312.9	344.6	9.8%
Australia & New Zealand	147.4	165.1	185.4	208.6	235.3	261.7	12.2%
Singapore	62.3	69.8	78.3	88.1	99.4	111.7	12.4%
Rest of Asia Pacific	35.4	40.5	46.3	52.9	64.3	87.4	19.8%
Total	1,282.7	1,414.9	1,564.5	1,734.0	1,926.8	2,143.0	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 192 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	1,533.9	1,731.8	1,928.9	2,234.3	2,482.7	2,765.0	12.5%
Essential Oils	2,409.5	2,704.2	2,994.0	3,447.1	3,807.5	4,215.2	11.8%
Flavonoids	1,791.5	2,023.4	2,254.5	2,612.3	2,903.8	3,235.2	12.5%
Alkaloids	1,167.4	1,322.8	1,478.6	1,718.8	1,916.7	2,142.4	12.9%
Carotenoids	1,006.7	1,135.4	1,263.3	1,461.7	1,622.6	1,805.2	12.4%
Other Types	1,275.4	1,435.0	1,592.6	1,838.0	2,034.8	2,257.6	12.1%
Total	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 193 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	3,083.7	3,447.5	3,863.7	4,341.1	4,875.2	5,426.1	12.0%
Essential Oils	4,673.0	5,193.1	5,785.3	6,461.4	7,213.6	7,988.5	11.3%
Flavonoids	3,609.4	4,036.7	4,525.7	5,086.7	5,714.7	6,395.1	12.1%
Alkaloids	2,397.9	2,690.4	3,026.0	3,412.1	3,845.7	4,295.7	12.4%
Carotenoids	2,011.2	2,246.1	2,514.6	2,822.4	3,166.4	3,552.8	12.1%
Other Types	2,508.0	2,792.7	3,117.2	3,487.8	3,900.7	4,252.1	11.1%
Total	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 194 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	121.8	135.8	149.5	171.0	187.6	206.4	11.1%
Essential Oils	181.9	201.7	220.6	250.9	273.7	299.2	10.5%
Flavonoids	136.6	152.5	167.8	192.0	210.8	232.0	11.2%
Alkaloids	89.0	99.7	110.0	126.4	139.2	153.6	11.5%
Carotenoids	76.8	85.5	94.0	107.5	117.8	129.4	11.0%
Other Types	87.3	97.0	106.3	121.1	132.4	144.9	10.7%
Total	693.5	772.2	848.2	968.8	1,061.4	1,165.5	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 195 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	227.5	251.4	278.4	309.1	344.0	383.2	11.0%
Essential Oils	327.9	360.1	396.5	437.5	484.1	532.1	10.2%
Flavonoids	255.8	282.7	313.2	347.9	387.3	433.8	11.1%
Alkaloids	169.9	188.4	209.4	233.3	260.7	291.4	11.4%
Carotenoids	142.5	157.3	174.0	193.0	214.6	241.0	11.1%
Other Types	159.1	175.0	192.9	213.2	236.2	261.5	10.4%
Total	1,282.7	1,414.9	1,564.5	1,734.0	1,926.8	2,143.0	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 196 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (USD MILLION)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	3,207.6	3,604.9	3,996.8	4,608.1	5,097.0	5,650.6	12.0%
Pharmaceuticals	1,391.6	1,571.4	1,750.5	2,027.9	2,253.8	2,510.5	12.5%
Dietary Supplements	2,096.6	2,369.9	2,642.6	3,064.3	3,408.9	3,800.9	12.6%
Cosmetics	1,672.6	1,904.9	2,140.1	2,500.2	2,802.0	3,147.5	13.5%
Other Applications	815.9	901.4	982.0	1,111.6	1,206.4	1,311.3	10.0%
Total	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 197 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	6,272.9	6,980.7	7,787.5	8,709.5	9,736.5	10,755.1	11.4%
Pharmaceuticals	2,800.3	3,131.1	3,509.6	3,943.9	4,430.0	4,931.6	12.0%
Dietary Supplements	4,243.8	4,749.7	5,329.2	5,994.4	6,739.8	7,510.2	12.1%
Cosmetics	3,540.2	3,991.5	4,511.3	5,111.6	5,789.0	6,465.2	12.8%
Other Applications	1,426.1	1,553.4	1,694.9	1,852.1	2,021.0	2,248.3	9.5%
Total	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 198 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (KT)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	239.8	266.3	291.8	332.4	363.2	397.8	10.7%
Pharmaceuticals	104.5	114.3	123.3	138.2	148.6	160.1	8.9%
Dietary Supplements	158.3	176.8	194.7	223.0	245.0	269.8	11.3%
Cosmetics	125.6	141.2	156.7	180.7	199.9	221.7	12.0%
Other Applications	65.3	73.6	81.8	94.5	104.7	116.2	12.2%
Total	693.5	772.2	848.2	968.8	1,061.4	1,165.5	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 199 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (KT)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	436.7	480.4	529.8	585.7	649.1	721.9	10.6%
Pharmaceuticals	172.8	186.8	202.5	219.9	239.3	266.1	9.0%
Dietary Supplements	297.7	329.3	365.2	405.8	452.2	504.4	11.1%
Cosmetics	246.3	274.4	306.4	342.9	384.7	427.9	11.7%
Other Applications	129.2	143.9	160.7	179.7	201.5	222.7	11.5%
Total	1,282.7	1,414.9	1,564.5	1,734.0	1,926.8	2,143.0	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 200 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	2,936.6	3,315.3	3,692.3	4,276.3	4,751.4	5,291.3	12.5%
Herbs & Spices	5,205.8	5,859.2	6,505.7	7,511.8	8,320.9	9,238.2	12.2%
Flowers	1,041.8	1,178.0	1,313.9	1,524.0	1,695.7	1,891.1	12.7%
Total	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 201 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (USD MILLION)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	5,900.7	6,596.1	7,391.8	8,304.4	9,325.5	10,378.9	12.0%
Herbs & Spices	10,270.8	11,446.4	12,788.1	14,323.1	16,035.4	17,792.4	11.6%
Flowers s	2,111.8	2,363.9	2,652.6	2,984.0	3,355.3	3,739.0	12.1%
Total	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 202 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (KT)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	222.8	248.5	273.4	312.8	343.2	377.5	11.1%
Herbs & Spices	379.5	422.0	462.8	527.8	577.4	633.1	10.8%
Flowers	91.1	101.7	112.0	128.2	140.8	154.9	11.2%
Total	693.5	772.2	848.2	968.8	1,061.4	1,165.5	10.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 203 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (KT)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	416.1	459.7	509.0	565.1	628.9	700.5	11.0%
Herbs & Spices	695.7	766.3	846.0	936.3	1,038.9	1,153.7	10.6%
Flowers	170.9	189.0	209.4	232.6	259.1	288.8	11.1%
Total	1,282.7	1,414.9	1,564.5	1,734.0	1,926.8	2,143.0	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 204 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY FORM, 2016–2021 (USD MILLION)

Form	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Dry	6,493.3	7,322.8	8,146.9	9,425.6	10,461.6	11,638.0	12.4%
Liquid	2,691.0	3,029.7	3,365.0	3,886.5	4,306.5	4,782.6	12.2%
Total	9,184.3	10,352.5	11,511.9	13,312.1	14,768.1	16,420.6	12.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 205 ASIA PACIFIC: PLANT EXTRACTS MARKET, BY FORM, 2022–2027 (USD MILLION)

Form	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Dry	12,964.5	14,477.2	16,206.3	18,187.7	20,402.6	22,683.1	11.8%
Liquid	5,318.7	5,929.3	6,626.2	7,423.7	8,313.6	9,227.1	11.6%
Total	18,283.2	20,406.4	22,832.5	25,611.4	28,716.3	31,910.3	11.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.1 CHINA

11.4.1.1 Increasing rejection of chemical products

China is one of the leading producers of plant extracts and an exporter of diverse plant extracts to various Asian, European, and North American countries. With the improvement of living standards and the enhancement of health awareness, people prefer safe green medicinal and food products. Functional health care is becoming more important in the country, and natural and healthy plant extract products are attracting more consumers, making China one of the leading plant extracts markets not just in Asia Pacific but globally. The industry prospects for plant extracts are promising in the country due to factors such as national industrial regulatory policy support, the increase in the degree of rejection toward chemical products, and adopting plant-based products, majorly in the US and Europe.

Chinese traditional medicines have a long history; however, since the 21st century, the Chinese plant extract industry entered a golden age. The improved living standards and the awareness of healthy lifestyles have spurred strong demand for plant extract products in the country. The adoption of advanced plant extraction techniques and the application of new technologies, such as separation technology, have improved production efficiency. With rapid economic development, the quantity and variety of foods have expanded, nutritional deficiencies have declined, and an epidemic of chronic disease has emerged. In China, dietary supplements are regulated similar to health foods. According to the China Food Additives & Ingredients Association (CFAA) in 2021, the Chinese health foods market is one of the fastest-growing food segments in Asia Pacific. The middle-aged and elderly are focusing on retaining good health and boosting immunity. Therefore, they are shifting toward plant-based diet to maintain their health. Also, with the advent of the aging population, hypertension has become a global health burden to be addressed. Although there are numerous drugs and procedures available to treat hypertension symptoms, people nowadays are shifting to herbal medicines. Flavonoids, which are common plant extracts found in fruits and vegetables, are active components in Chinese herbal medicine. Flavonoids have cardiovascular benefits and are endowed with anti-hypertension activities, helping in treating hypertension symptoms.

China is also known for its cuisines, which involve various spices and flavors. Paprika and black pepper are two of the major spices used in China. The country has rising paprika production coupled with government concessions to sell paprika oleoresin at a lower price since China is the largest paprika producer in the world. Growers use sunlight to dry out paprika instead of any gas or fuel, making it less expensive, beating India in paprika oleoresins production. This also encourages key market players to set up their facilities in China. Under bilateral trade between India and China, Synthite inaugurated its 46,000 sqm facility at Wucheng County in China to cater to the demand for oleoresins.

Owing to the gradual popularization of the concept 'returning to nature' and 'green consumption,' plant extract manufacturers in China have started to increase their production. The export of plant extracts in China has spurred, majorly to the US and Europe, due to the consumption of plant ingredient-based products in diverse applications, such as dietary supplements, pharmaceuticals, food & beverages, and cosmetics, in these regions. China has a vast territory, having a wide variety of plant resources, which can be used for industrial extraction, which is projected to drive the market further during the forecast period.

TABLE 206 CHINA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	538.0	603.9	668.7	770.1	850.8	942.0	11.9%
Essential Oils	857.5	957.3	1,054.2	1,207.3	1,326.5	1,460.7	11.2%
Flavonoids	628.3	705.5	781.6	900.4	995.1	1,102.2	11.9%
Alkaloids	409.4	461.2	512.6	592.4	656.8	729.9	12.3%
Carotenoids	353.1	395.9	438.0	503.8	556.0	615.0	11.7%
Other Types	474.2	530.1	584.5	670.3	737.3	812.8	11.4%
Total	3,260.4	3,653.9	4,039.6	4,644.3	5,122.5	5,662.7	11.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 207 CHINA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	1,044.5	1,161.0	1,293.6	1,445.1	1,613.5	1,790.6	11.4%
Essential Oils	1,610.7	1,780.5	1,973.0	2,191.8	2,433.8	2,696.5	10.9%
Flavonoids	1,222.6	1,359.4	1,515.3	1,693.3	1,891.3	2,110.3	11.5%
Alkaloids	812.2	906.0	1,013.2	1,135.8	1,272.8	1,417.5	11.8%
Carotenoids	681.2	756.4	841.9	939.5	1,047.9	1,172.4	11.5%
Other Types	897.3	992.8	1,101.1	1,224.3	1,360.5	1,470.8	10.4%
Total	6,268.6	6,956.1	7,738.1	8,629.7	9,619.9	10,658.0	11.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 208 CHINA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	40.7	45.2	49.5	56.3	61.5	67.4	10.6%
Essential Oils	64.9	71.6	78.0	88.3	95.9	104.4	10.0%
Flavonoids	47.5	52.8	57.8	65.9	72.0	78.8	10.6%
Alkaloids	31.0	34.5	37.9	43.3	47.5	52.2	11.0%
Carotenoids	26.7	29.6	32.4	36.9	40.2	44.0	10.5%
Other Types	35.9	39.7	43.2	49.0	53.3	58.1	10.1%
Total	246.6	273.3	298.8	339.7	370.5	404.9	10.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 209 CHINA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	73.9	81.3	89.6	99.0	109.7	121.0	10.4%
Essential Oils	114.0	124.7	136.7	150.2	165.5	182.2	9.8%
Flavonoids	86.5	95.2	105.0	116.0	128.6	142.6	10.5%
Alkaloids	57.5	63.4	70.2	77.8	86.5	95.8	10.8%
Carotenoids	48.2	53.0	58.3	64.4	71.3	79.2	10.4%
Other Types	63.5	69.5	76.3	83.9	92.5	99.4	9.4%
Total	443.6	487.0	536.0	591.4	654.1	720.2	10.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.2 INDIA

11.4.2.1 Shifting trend from synthetic to herbal medicines

According to India Brand Equity Foundation (IBEF), in 2020, India had 15 agroclimatic zones that comprise ~18,000 types of plants, of which 6,000–7,000 have therapeutic properties. These medicinal plants are used in numerous applications and are used to make medicines in traditional medical practices. 80% of medicinal plants are extracted from the wild, while 69% plants are collected using destructive farming practices. The Indian plant extracts market is experiencing significant growth due to the shift in universal trends from synthetic to herbal medicines. As synthetic products have adverse side-effects, herbal products are gaining popularity in the Indian market due to well-practiced herbal medicine knowledge and the presence of many medicinal plants. India is bestowed with a variety of plant species and has one of the world's richest medicinal plant heritages. Plants with therapeutic and medicinal properties are highly valued globally, as they help in the prevention of diseases.

Changing lifestyles, health consciousness, changing trends in population demographics, increasing consumer affluence, and increased life expectancy fuel the use of dietary supplements in India. Dietary supplements are used for various diseases like cancer, diabetes, obesity, and arthritis and by the vegan population to fulfill their basic dietary needs. Nutrazee is a vegan dietary supplement brand focused on developing innovative plant-based vegan supplements. India is also one of the world's fastest-growing

economies. Rising incomes have prompted households to increase their budgets and spend more on health and nutrition. According to a survey conducted by Kerry Group, in 2021, India had a potential market for plant-based food. It was found that 63% of Indian consumers are willing to buy plant-based foods regularly, 60% are even ready to pay a premium for it.

India is the hub of spices, spice oils, and oleoresins, due to the large spice production in the country. Chili, cumin, turmeric, and ginger hold the major share in the overall spice production in India. With the rising demand for spices, the demand for value-added spice extracts has also increased in domestic as well as international markets. Factors such as shift in consumer demand toward an herbal and natural product, government support toward economic farming and international trade, and the increase in investments and expansions by Indian plant extract manufacturers propel the growth of the market in India.

The substantial demand for plant extracts in international markets, such as North America and Europe, and the consequent increase in investments have created an opportunity for existing and new players such as Vidya Herbs (India) and Arjuna Natural Extract Limited (India) to tap the growing market.

TABLE 210 INDIA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	467.1	525.6	583.4	673.4	745.8	827.8	12.1%
Essential Oils	732.9	819.9	904.8	1,038.3	1,143.1	1,261.4	11.5%
Flavonoids	545.5	614.0	681.8	787.4	872.3	968.6	12.2%
Alkaloids	355.5	401.4	447.2	518.1	575.8	641.4	12.5%
Carotenoids	306.5	344.5	382.1	440.6	487.4	540.4	12.0%
Other Types	347.8	389.8	431.0	495.4	546.3	603.8	11.7%
Total	2,755.3	3,095.2	3,430.2	3,953.2	4,370.7	4,843.4	11.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 211 INDIA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	920.1	1,025.1	1,145.0	1,282.1	1,435.0	1,586.2	11.5%
Essential Oils	1,393.9	1,543.9	1,714.4	1,908.4	2,123.6	2,333.6	10.9%
Flavonoids	1,076.9	1,200.3	1,341.2	1,502.3	1,682.1	1,869.5	11.7%
Alkaloids	715.4	800.0	896.7	1,007.7	1,132.0	1,255.8	11.9%
Carotenoids	600.1	667.9	745.2	833.6	932.0	1,038.6	11.6%
Other Types	668.2	741.1	824.0	918.4	1,023.1	1,106.5	10.6%
Total	5,374.5	5,978.4	6,666.5	7,452.5	8,327.7	9,190.2	11.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 212 INDIA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	35.3	39.3	43.1	49.1	53.7	58.9	10.8%
Essential Oils	55.4	61.3	66.8	75.7	82.4	89.8	10.1%
Flavonoids	41.3	45.9	50.4	57.4	62.9	68.9	10.8%
Alkaloids	26.9	30.0	33.0	37.8	41.5	45.7	11.2%
Carotenoids	23.2	25.8	28.2	32.1	35.1	38.5	10.7%
Other Types	26.3	29.1	31.8	36.1	39.4	43.0	10.3%
Total	208.4	231.4	253.3	288.4	315.0	344.7	10.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 213 INDIA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	64.7	71.3	78.7	87.1	96.6	106.6	10.5%
Essential Oils	98.1	107.4	117.9	129.7	143.0	156.8	9.8%
Flavonoids	75.8	83.5	92.2	102.1	113.3	125.6	10.6%
Alkaloids	50.3	55.6	61.6	68.5	76.2	84.4	10.9%
Carotenoids	42.2	46.5	51.2	56.6	62.8	69.8	10.6%
Other Types	47.0	51.5	56.6	62.4	68.9	74.3	9.6%
Total	378.2	415.8	458.3	506.3	560.8	617.5	10.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.3 JAPAN

11.4.3.1 Increasing need for natural ingredients in health care, cosmetics, and toiletries

In Japan, plants that are most aromatic and medicinal are used in the crude drug system, Kampo, the traditional Japanese medicine system, and herbal tea, as alternatives or complementary medicines. Various manufacturers produce Kampo medicines in Japan. However, every manufacturer follows the same formula composed of the same ingredients under the Ministry of Health, Labour and Welfare standardization methodology. One of the factors attributed to this formulation of guidelines is the increase in consumer inclination for healthy foods, such as dietary supplements and herbal extracts in the country. According to a report by the EU-Japan Centre for Industrial Cooperation (a joint venture between European Commission and the Japanese Government), in Japan, the dietary supplements market is strong, trying to incorporate traditional herbal ingredients (most often ayurvedic and Kampo) into the nutraceuticals portfolio. Japan is one of the foremost countries to implement legislation for functional foods, namely, the Foods for Specified Health Use (FOSHU). This early recognition and awareness about the consumption of functional food products have aided the market for these products. Moreover, the rising aging population has triggered the demand for dietary supplements in the country.

The increase in the aging population in developed countries and the rising prevalence of chronic diseases such as diabetes, cardiovascular diseases, and neurological disorders are the key drivers. Unlike China and India, Japan imports large amounts of plant extracts from other Asian countries and is largely focusing on its traditional medicine, Kampo. This trend is limiting the plant extracts market growth in the country.

TABLE 214 JAPAN: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	271.1	304.8	338.1	390.1	431.7	478.9	12.1%
Essential Oils	421.0	470.1	517.7	593.0	651.5	717.5	11.3%
Flavonoids	316.6	356.1	395.2	456.1	504.9	560.3	12.1%
Alkaloids	206.3	232.8	259.2	300.1	333.3	371.0	12.5%
Carotenoids	177.9	199.8	221.4	255.2	282.1	312.6	11.9%
Other Types	232.8	261.4	289.6	333.5	368.5	408.1	11.9%
Total	1,625.6	1,825.0	2,021.2	2,327.9	2,572.1	2,848.4	11.9%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 215 JAPAN: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	531.9	592.2	661.1	739.7	817.7	893.9	10.9%
Essential Oils	791.3	874.7	969.4	1,077.0	1,182.0	1,277.7	10.1%
Flavonoids	622.6	693.5	774.3	866.8	958.5	1,053.5	11.1%
Alkaloids	413.6	462.2	517.7	581.4	645.0	707.7	11.3%
Carotenoids	346.9	385.9	430.2	481.0	531.1	585.3	11.0%
Other Types	452.5	502.9	560.2	625.6	690.0	747.1	10.6%
Total	3,158.7	3,511.3	3,913.0	4,371.5	4,824.3	5,265.2	10.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 216 JAPAN: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	20.1	22.3	24.4	27.8	30.3	33.2	10.5%
Essential Oils	31.3	34.5	37.4	42.2	45.8	49.7	9.7%
Flavonoids	23.5	26.1	28.6	32.5	35.5	38.8	10.5%
Alkaloids	15.3	17.1	18.7	21.4	23.4	25.7	10.9%
Carotenoids	13.2	14.6	16.0	18.2	19.8	21.6	10.4%
Other Types	17.3	19.2	20.9	23.8	25.9	28.3	10.3%
Total	120.8	133.8	146.1	165.8	180.6	197.2	10.3%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 217 JAPAN: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	36.3	39.9	43.9	48.5	53.0	58.5	10.0%
Essential Oils	54.1	58.9	64.4	70.6	76.7	83.6	9.1%
Flavonoids	42.5	46.7	51.5	56.8	62.2	68.9	10.1%
Alkaloids	28.3	31.1	34.4	38.1	41.8	46.3	10.4%
Carotenoids	23.7	26.0	28.6	31.5	34.4	38.3	10.1%
Other Types	30.9	33.9	37.2	41.0	44.8	48.9	9.6%
Total	215.8	236.6	260.1	286.6	312.9	344.6	9.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.4 AUSTRALIA & NEW ZEALAND

11.4.4.1 Rising investments by governments

In Australia & New Zealand, plant extract manufacturers are finding it increasingly difficult to compete against cheaper imports. They are opting to source these imported raw materials due to better supply consistency, quality, and price. The Australian plant extracts market is too small to support an active industry, due to which growers are concentrating on overseas opportunities. The unpredictability of supply and demand, coupled with significant price fluctuation, has led to diminishing local grower interest in the cultivation of medicinal herbs.

According to the Australian Institute of Health and Welfare, in 2022, more men were overweight than women (75% of men and 60% of women), and higher rates of obesity (33% of men and 30% of women). The presence of top dietary supplements manufacturers in Australia such as LAVIDA Pharmaceutical (Wetherill Park, New South Wales), Vianature Nutrition (Granville, New South Wales), and Tripak Pharmaceuticals (Brisbane) drives the growth of the plant extracts market as they are developing new products to cater to the needs of the population. According to the National Nutrition Survey in 2021, about 2% of Australians are vegan. This is attributed to the rising number of health-conscious consumers in the country. As a result, the rise in the vegan population corresponds to the increase in the demand for plant extracts as an ingredient.

New Zealand has been a prime food market for companies seeking good margins with no tariff barriers. Meat alternatives and clean-eating vegan diets are the key focus areas for consumers in New Zealand who are already seeking to embrace a significant shift in dietary preferences toward sustainable plant extract-based products, thereby increasing the consumption of oleoresins.

The infancy and fragmentation of the industry mean that limited available resources should be invested in industry development initiatives. Appropriate support infrastructure is, therefore, needed to help growers spread risks and investments associated with growing medicinal herbs in Australia. Under RIRDC's Essential Oils and Plant Extracts R&D program, significant investment has been made to assist growers in developing the necessary expertise to produce high-quality medicinal herbs in Australia. Such initiatives by the government will boost manufacturers' confidence and stimulate market growth.

TABLE 218 AUSTRALIA & NEW ZEALAND: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	160.5	184.1	208.3	245.2	276.8	313.3	14.3%
Essential Oils	246.4	281.1	316.4	370.3	415.8	467.9	13.7%
Flavonoids	187.4	215.1	243.5	286.7	323.8	366.6	14.4%
Alkaloids	122.1	140.6	159.7	188.6	213.7	242.7	14.7%
Carotenoids	105.3	120.7	136.4	160.4	180.9	204.5	14.2%
Other Types	133.4	152.4	171.7	201.2	226.1	254.6	13.8%
Total	955.2	1,094.0	1,236.1	1,452.4	1,637.1	1,849.6	14.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 219 AUSTRALIA & NEW ZEALAND: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	355.0	403.3	459.2	524.3	598.3	676.8	13.8%
Essential Oils	527.2	595.6	674.4	765.6	868.8	979.7	13.2%
Flavonoids	415.5	472.2	537.9	614.3	701.3	797.7	13.9%
Alkaloids	276.1	314.7	359.7	412.1	471.9	535.8	14.2%
Carotenoids	231.5	262.7	298.9	340.9	388.6	443.2	13.9%
Other Types	287.2	324.6	367.9	417.9	474.4	523.7	12.8%
Total	2,092.6	2,373.2	2,698.0	3,075.1	3,503.4	3,956.9	13.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 220 AUSTRALIA & NEW ZEALAND: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	12.2	13.8	15.5	18.0	20.0	22.4	12.8%
Essential Oils	18.8	21.1	23.5	27.1	30.1	33.4	12.2%
Flavonoids	14.3	16.2	18.1	21.0	23.4	26.2	12.9%
Alkaloids	9.3	10.6	11.9	13.8	15.5	17.3	13.2%
Carotenoids	8.0	9.1	10.1	11.8	13.1	14.6	12.7%
Other Types	10.2	11.5	12.7	14.7	16.3	18.2	12.3%
Total	72.8	82.3	91.8	106.4	118.4	132.0	12.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 221 AUSTRALIA & NEW ZEALAND: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	25.0	28.1	31.6	35.6	40.2	44.8	12.3%
Essential Oils	37.1	41.4	46.3	51.9	58.4	64.8	11.8%
Flavonoids	29.3	32.9	37.0	41.7	47.1	52.8	12.5%
Alkaloids	19.5	21.9	24.7	28.0	31.7	35.4	12.7%
Carotenoids	16.3	18.3	20.5	23.1	26.1	29.0	12.2%
Other Types	20.2	22.6	25.3	28.3	31.9	34.9	11.5%
Total	147.4	165.1	185.4	208.6	235.3	261.7	12.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.5 SINGAPORE

11.4.5.1 Rising concerns over health risks

The vegan movement has been growing significantly in the Asia Pacific region. The same trend has been noticed in Singapore since consumers have been gradually shifting to a vegan diet. According to YouGov survey in February 2020, two in five (39%) Singaporeans are into flexitarian diets and one in twenty (7%) follow vegetarian or vegan diet. Factors such as the rising concerns over health risks over the consumption of processed meat products and the increase in the number of health-conscious consumers have been driving the demand for vegan food products among Singaporeans. The majority of these plant-based food products are processed with different types of plant extracts. Though in the recent scenario, Singapore accounts for a lower share compared to other Asian countries; however, during the forecast period, the country is expected to witness significant growth in terms of value sales.

According to a survey published by the National Library of Medicine in 2021, Singaporeans suffer from a high prevalence of chronic diseases such as diabetes, heart diseases, and obesity. After COVID-19, many Singaporeans had begun to participate in sports to maintain health and prevent from diseases. As a result, the health and wellness industry has seen a long-term shift toward healthier products and categories, such as increased demand for dietary supplements. Singapore-based Tabernacle Health Group (THG) developed an herbal supplement to support the immune system.

TABLE 222 SINGAPORE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	67.0	76.9	87.1	102.6	116.0	131.3	14.4%
Essential Oils	104.3	119.0	134.1	157.0	176.4	198.7	13.8%
Flavonoids	78.3	89.9	101.8	120.0	135.6	153.7	14.4%
Alkaloids	51.0	58.8	66.8	78.9	89.5	101.7	14.8%
Carotenoids	44.0	50.4	57.1	67.1	75.8	85.7	14.3%
Other Types	59.6	68.1	76.8	90.1	101.4	114.3	13.9%
Total	404.1	463.2	523.7	615.8	694.7	785.4	14.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 223 SINGAPORE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	148.9	169.3	192.9	220.4	251.7	285.6	13.9%
Essential Oils	224.1	253.3	287.0	326.1	370.3	418.2	13.3%
Flavonoids	174.3	198.2	226.0	258.3	295.1	336.5	14.1%
Alkaloids	115.8	132.1	151.1	173.2	198.6	226.1	14.3%
Carotenoids	97.1	110.3	125.6	143.3	163.5	187.0	14.0%
Other Types	129.1	146.0	165.7	188.4	214.1	237.9	13.0%
Total	889.3	1,009.3	1,148.3	1,309.7	1,493.2	1,691.2	13.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 224 SINGAPORE: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	5.1	5.8	6.5	7.5	8.4	9.3	12.8%
Essential Oils	8.0	9.0	9.9	11.5	12.7	14.1	12.2%
Flavonoids	6.0	6.8	7.6	8.8	9.8	10.9	12.8%
Alkaloids	3.9	4.4	5.0	5.8	6.5	7.2	13.2%
Carotenoids	3.4	3.8	4.2	4.9	5.5	6.1	12.7%
Other Types	4.5	5.1	5.7	6.6	7.3	8.1	12.3%
Total	30.8	34.8	38.8	45.0	50.1	55.8	12.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 225 SINGAPORE: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	10.4	11.7	13.2	14.8	16.8	18.9	12.5%
Essential Oils	15.7	17.5	19.6	21.9	24.6	27.6	11.9%
Flavonoids	12.2	13.7	15.4	17.4	19.6	22.2	12.7%
Alkaloids	8.1	9.1	10.3	11.7	13.2	14.9	13.0%
Carotenoids	6.8	7.6	8.6	9.6	10.9	12.3	12.6%
Other Types	9.0	10.1	11.3	12.7	14.3	15.7	11.7%
Total	62.3	69.8	78.3	88.1	99.4	111.7	12.4%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.4.6 REST OF ASIA PACIFIC

The Rest of Asia Pacific includes Vietnam, Malaysia, Thailand, Indonesia, the Philippines, and South Korea. These countries are well known for their abundant natural resources. Indonesia represents huge potential for spices extracts. It is considered one of the major producers and exporters of oleoresins in the Asia Pacific region. This is due to the presence of many key players in the region. The value chains in Indonesia for extracts destined for use in the pharmaceutical or herbal medicine sectors are transparent. There is coordination among value chain stakeholders, which include suppliers of cosmetics, fragrance and flavors, pharmaceuticals, and supplements, each with its own industry association, namely, the Indonesian Cosmetics Association (PERKOSMI), the Indonesian Essential Oils Association (DAI), and the Association of Herbs, and Traditional Pharmaceutical Producers (GPJ), which enable them to keep up with the demand. Government organizations do not consider the extracts sector as a whole in their policies or support activities, and the service industry to the sector is underdeveloped in key services such as efficacy research. It is a developing country, with a growing middle-class segment. Due to its huge population, the food & beverage market in the country is dominant. According to the Food Export Association of the Midwest USA and Food Export USA–Northeast, Thais spend 25% of total expenditures on food and beverages. It encourages manufacturers to make use of natural and organic products, largely in processed food, which drives the market for oleoresins in Indonesia.

The sales of Thai herbal medicinal products have also shown promising growth. Thailand currently has 6,300 companies registered with the Ministry of Commerce as producers of food supplements for health care and beauty. Around 90% are small and medium-sized businesses, leaving room for big corporations to compete because they have more investment and technology. Plant extract manufacturers in South Korea, Vietnam, Malaysia, and other Asian countries are influenced by the growing plant extracts market in China and India and are looking forward to increasing the production of plant extracts. To add to that, the governments in these countries are supporting economic farming and taking initiatives to attract international demand. Such positive trends in this market would provide promising growth opportunities for the Rest of Asia Pacific.

TABLE 226 REST OF ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	30.3	36.6	43.2	52.8	61.6	71.7	18.8%
Essential Oils	47.4	56.8	66.8	81.2	94.2	109.0	18.1%
Flavonoids	35.4	42.7	50.5	61.8	72.1	83.9	18.8%
Alkaloids	23.1	27.9	33.1	40.6	47.6	55.6	19.2%
Carotenoids	19.9	24.0	28.3	34.6	40.3	46.8	18.7%
Other Types	27.6	33.1	39.0	47.5	55.1	63.9	18.3%
Total	183.7	221.2	261.0	318.5	370.9	431.0	18.6%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 227 REST OF ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	83.3	96.5	111.8	129.5	159.0	193.1	18.3%
Essential Oils	125.9	145.1	167.1	192.4	235.0	282.7	17.6%
Flavonoids	97.5	113.0	131.0	151.7	186.4	227.6	18.5%
Alkaloids	64.8	75.3	87.6	101.8	125.4	152.9	18.7%
Carotenoids	54.3	62.9	72.8	84.2	103.3	126.4	18.4%
Other Types	73.8	85.2	98.3	113.3	138.5	166.1	17.6%
Total	499.5	578.2	668.6	772.8	947.6	1,148.8	18.1%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 228 REST OF ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	2.3	2.8	3.2	3.9	4.5	5.1	17.4%
Essential Oils	3.6	4.3	5.0	6.0	6.8	7.8	16.7%
Flavonoids	2.7	3.2	3.8	4.5	5.2	6.0	17.4%
Alkaloids	1.8	2.1	2.5	3.0	3.5	4.0	17.8%
Carotenoids	1.5	1.8	2.1	2.5	2.9	3.4	17.2%
Other Types	2.1	2.5	2.9	3.5	4.0	4.6	16.9%
Total	14.0	16.7	19.4	23.4	26.9	30.9	17.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 229 REST OF ASIA PACIFIC: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	5.9	6.8	7.7	8.9	10.8	14.7	20.0%
Essential Oils	8.9	10.2	11.6	13.2	15.9	21.5	19.2%
Flavonoids	6.9	7.9	9.1	10.4	12.6	17.3	20.2%
Alkaloids	4.6	5.3	6.1	7.0	8.5	11.6	20.4%
Carotenoids	3.9	4.4	5.0	5.8	7.0	9.6	20.1%
Other Types	5.2	6.0	6.8	7.8	9.4	12.6	19.3%
Total	35.4	40.5	46.3	52.9	64.3	87.4	19.8%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.5 REST OF THE WORLD

The Rest of the World covers the potential countries in South America, Africa, and the Middle East. The market in the region is still in its early stage of growth and is projected to be a potential market for plant extracts in the coming years. The pharmaceuticals and dietary supplements segments have contributed to the growth of the plant extracts market in RoW. The rising health consciousness among consumers, locally and internationally, is one of the major factors driving the demand for plant extracts. South America holds the largest share in the plant extracts market in RoW. South American plant extract manufacturers are more technologically equipped and well versed with the regulations of major importing countries. These manufacturers have easy availability of various herbs and plants that grow indigenously. South America, Africa, and the Middle East have different environmental conditions, and each region is bestowed with certain specific tropical plants. However, all regions produce plant extracts, majorly sourced from fruits, flowers, and spices & herbs due to diverse applications, which provide growth opportunities for plant extract export. Product packaging also influences consumers' buying behavior, driving them to the purchase of healthier food and other consumer products.

TABLE 230 ROW: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (USD MILLION)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
South America	668.6	775.5	886.1	1,051.5	1,195.4	1,360.5	15.3%
Africa	648.8	759.3	875.3	1,048.0	1,202.3	1,380.8	16.3%
Middle East	202.1	238.5	277.1	334.3	386.2	446.4	17.2%
Total	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 231 ROW: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
South America	1,548.8	1,765.6	2,015.7	2,304.8	2,631.8	3,038.6	14.4%
Africa	1,586.3	1,825.0	2,102.8	2,426.8	2,797.1	3,260.4	15.5%
Middle East	516.0	596.9	691.4	801.7	928.1	1,094.2	16.2%
Total	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 232 ROW: PLANT EXTRACTS MARKET, BY REGION, 2016–2021 (KT)

Region	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
South America	52.0	59.6	67.2	78.7	88.3	99.1	13.8%
Africa	52.8	61.0	69.4	81.9	92.7	105.0	14.7%
Middle East	16.1	18.8	21.6	25.7	29.4	33.5	15.8%
Total	121.0	139.4	158.1	186.3	210.4	237.7	14.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 233 ROW: PLANT EXTRACTS MARKET, BY REGION, 2022–2027 (KT)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
South America	111.5	125.5	141.5	159.7	180.6	202.1	12.6%
Africa	119.1	135.3	153.9	175.3	199.9	223.7	13.4%
Middle East	38.3	43.8	50.2	57.5	66.0	73.7	14.0%
Total	268.9	304.6	345.5	392.5	446.6	499.4	13.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 234 ROW: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	270.9	316.7	364.7	436.1	499.7	573.1	16.2%
Essential Oils	395.2	459.2	525.6	624.8	711.6	811.3	15.5%
Flavonoids	313.4	366.5	422.1	505.0	578.8	664.2	16.2%
Alkaloids	204.2	239.6	276.9	332.3	382.1	439.8	16.6%
Carotenoids	176.1	205.6	236.5	282.6	323.4	370.6	16.0%
Other Types	159.7	185.7	212.7	253.0	288.3	328.8	15.5%
Total	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 235 ROW: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	657.5	755.3	869.0	1,001.4	1,152.4	1,342.4	15.3%
Essential Oils	925.2	1,056.6	1,208.4	1,384.2	1,583.5	1,830.5	14.6%
Flavonoids	762.3	876.0	1,008.2	1,162.3	1,338.0	1,567.2	15.5%
Alkaloids	506.4	583.9	674.2	779.6	900.4	1,052.7	15.8%
Carotenoids	424.7	487.4	560.2	644.9	741.4	870.7	15.4%
Other Types	375.1	428.3	489.8	560.9	641.4	729.7	14.2%
Total	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 236 ROW: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	22.2	25.6	29.2	34.4	38.9	44.0	14.7%
Essential Oils	31.5	36.1	40.8	47.8	53.8	60.5	14.0%
Flavonoids	24.7	28.5	32.4	38.3	43.3	49.0	14.7%
Alkaloids	16.1	18.6	21.3	25.2	28.6	32.5	15.1%
Carotenoids	13.9	16.0	18.2	21.4	24.2	27.4	14.5%
Other Types	12.6	14.5	16.4	19.2	21.6	24.3	14.0%
Total	121.0	139.4	158.1	186.3	210.4	237.7	14.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 237 ROW: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	49.9	56.6	64.3	73.2	83.4	93.3	13.3%
Essential Oils	68.1	76.8	86.8	98.2	111.2	124.4	12.8%
Flavonoids	55.6	63.1	71.7	81.6	93.0	104.1	13.4%
Alkaloids	36.9	42.0	47.9	54.7	62.6	70.0	13.7%
Carotenoids	31.0	35.1	39.8	45.3	51.6	57.7	13.2%
Other Types	27.4	30.9	34.9	39.5	44.7	50.0	12.8%
Total	268.9	304.6	345.5	392.5	446.6	499.4	13.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 238 ROW: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (USD MILLION)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	509.8	593.2	679.9	809.3	922.9	1,053.7	15.6%
Pharmaceuticals	221.1	258.6	297.8	356.1	408.1	468.2	16.2%
Dietary Supplements	333.2	389.9	449.5	538.2	617.3	708.8	16.3%
Cosmetics	265.8	313.4	364.0	439.1	507.4	586.9	17.2%
Other Applications	189.5	218.2	247.4	291.1	328.1	370.1	14.3%
Total	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 239 ROW: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	1,203.3	1,376.0	1,575.8	1,807.5	2,070.5	2,393.6	14.7%
Pharmaceuticals	537.2	617.2	710.2	818.5	942.0	1,097.5	15.4%
Dietary Supplements	814.1	936.3	1,078.4	1,244.0	1,433.2	1,671.4	15.5%
Cosmetics	679.1	786.8	912.9	1,060.8	1,231.0	1,438.9	16.2%
Other Applications	417.4	471.2	532.5	602.4	680.3	791.7	13.7%
Total	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 240 ROW: PLANT EXTRACTS MARKET, BY APPLICATION, 2016–2021 (KT)

Application	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Food & Beverages	41.6	47.8	54.2	63.7	71.7	80.8	14.2%
Pharmaceuticals	19.7	22.3	24.8	28.7	31.9	35.4	12.4%
Dietary Supplements	26.5	30.6	34.9	41.2	46.6	52.9	14.8%
Cosmetics	21.7	25.3	29.0	34.6	39.5	45.1	15.7%
Other Applications	11.4	13.3	15.3	18.2	20.7	23.6	15.6%
Total	121.0	139.4	158.1	186.3	210.4	237.7	14.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 241 ROW: PLANT EXTRACTS MARKET, BY APPLICATION, 2022–2027 (KT)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Food & Beverages	91.2	103.1	116.6	132.1	150.0	167.7	13.0%
Pharmaceuticals	39.3	43.6	48.6	54.1	60.3	67.5	11.4%
Dietary Supplements	60.0	68.1	77.5	88.2	100.7	112.9	13.5%
Cosmetics	51.5	59.0	67.7	77.7	89.4	100.0	14.2%
Other Applications	26.9	30.8	35.2	40.3	46.2	51.3	13.8%
Total	268.9	304.6	345.5	392.5	446.6	499.4	13.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 242 ROW: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (USD MILLION)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	466.7	545.5	628.1	751.0	860.4	986.7	16.2%
Herbs & Spices	827.3	964.1	1,106.7	1,319.2	1,506.7	1,722.8	15.8%
Flowers	225.4	263.7	303.8	363.5	416.8	478.3	16.2%
Total	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 243 ROW: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (USD MILLION)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	1,131.9	1,300.2	1,495.8	1,723.4	1,983.1	2,309.9	15.3%
Herbs & Spices	1,970.2	2,256.3	2,587.7	2,972.5	3,410.0	3,959.8	15.0%
Flowers	549.0	631.0	726.3	837.3	964.0	1,123.4	15.4%
Total	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 244 ROW: PLANT EXTRACTS MARKET, BY SOURCE, 2016–2021 (KT)

Source	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Fruits & Vegetables	37.5	43.3	49.2	58.1	65.7	74.3	14.6%
Herbs & Spices	65.5	75.4	85.4	100.5	113.3	127.8	14.3%
Flowers	17.9	20.7	23.5	27.8	31.4	35.6	14.7%
Total	121.0	139.4	158.1	186.3	210.4	237.7	14.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 245 ROW: PLANT EXTRACTS MARKET, BY SOURCE, 2022–2027 (KT)

Source	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Fruits & Vegetables	84.2	95.5	108.5	123.5	140.7	158.2	13.4%
Herbs & Spices	144.4	163.3	185.0	209.8	238.3	266.2	13.0%
Flowers	40.3	45.8	52.0	59.2	67.5	75.0	13.2%
Total	268.9	304.6	345.5	392.5	446.6	499.4	13.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 246 ROW: PLANT EXTRACTS MARKET, BY FORM, 2016–2021 (USD MILLION)

Form	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Dry	1,054.1	1,233.8	1,422.5	1,703.1	1,953.7	2,243.6	16.3%
Liquid	465.3	539.5	616.1	730.7	830.2	944.2	15.2%
Total	1,519.5	1,773.3	2,038.6	2,433.8	2,783.9	3,187.8	16.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

TABLE 247 ROW: PLANT EXTRACTS MARKET, BY FORM, 2022–2027 (USD MILLION)

Form	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Dry	2,577.2	2,964.3	3,414.6	3,939.5	4,539.1	5,294.1	15.5%
Liquid	1,074.0	1,223.3	1,395.2	1,593.7	1,818.0	2,099.0	14.3%
Total	3,651.1	4,187.6	4,809.8	5,533.3	6,357.0	7,393.1	15.2%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, and MarketsandMarkets Analysis

11.5.1 SOUTH AMERICA

The South American plant extracts market has huge growth potential as they find applications as ingredients for foods and other consumer products. The region has a widely distributed tropical forest, having several plant species with health benefits. Plant extract manufacturers in countries such as Brazil and Peru are focusing on plant extract production due to various factors such as the rise in expenditure on premium consumer products in local and international markets. Brazil is one of the leading producers of spice and herbal extracts in South America. The stevia extract that is used as a natural sugar-free sweetener is commercially produced in South America and has a high global demand, mainly from Japan.

TABLE 248 SOUTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (USD MILLION)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Brazil	364.7	423.2	483.8	574.4	653.4	744.1	15.3%
Rest of South America	303.9	352.3	402.3	477.0	542.0	616.5	15.2%
Total	668.6	775.5	886.1	1,051.5	1,195.4	1,360.5	15.3%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 249 SOUTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Brazil	847.5	966.6	1,104.1	1,263.1	1,443.0	1,667.9	14.5%
Rest of South America	701.4	799.0	911.6	1,041.7	1,188.8	1,370.6	14.3%
Total	1,548.8	1,765.6	2,015.7	2,304.8	2,631.8	3,038.6	14.4%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 250 SOUTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (KT)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Brazil	28.3	32.4	36.6	42.9	48.2	54.3	13.9%
Rest of South America	23.7	27.1	30.6	35.7	40.0	44.9	13.6%
Total	52.0	59.6	67.2	78.7	88.3	99.1	13.8%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 251 SOUTH AMERICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (KT)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Brazil	61.1	68.9	77.8	87.9	99.6	111.0	12.7%
Rest of South America	50.4	56.6	63.7	71.8	81.1	91.1	12.6%
Total	111.5	125.5	141.5	159.7	180.6	202.1	12.6%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.1.1 Brazil

11.5.1.1.1 Biodiversity to enable development of variety of plant extract products

Brazil is one of the hot spots in the world in terms of biodiversity, but little is known about the chemical and pharmacological properties of most of the plants found in the Amazon rainforest and the Atlantic Forest. Brazilian plant extracts belonging to different species have been used in the preparation of various medicines across the globe. For instance, the use of natural antioxidants in cancer therapy has increased due to the potential of natural antioxidants to kill tumor cells and also because of their capacity to protect healthy cells from the damage caused by chemotherapy. The antioxidant properties of extracts obtained from medicinal plants from the Brazilian Cerrado help induce the cell death profile by each of these extracts in malignant cells.

Brazil offers a wide range of medicinal plants, vegetable oils, and essential oils. There are over 50 species of the tropical *Copaifera* genus used for wounds and ulcer healing, antimicrobial, and anti-inflammatory properties for skin diseases. It can be present in both essential oil and oleoresins. According to the OECD, in 2020, Brazil exported USD 274 million of essential oils, making it the 5th largest exporter in the world. The main destination of essential oils exports from Brazil are: US, Netherlands, Germany, China, and India. There is a continuous rise in imports of herbal and medicinal plant extracts by European and North American countries due to a consumer shift toward a healthy lifestyle, thereby driving the plant extracts market in Brazil.

Established players are acquiring Brazilian companies to enhance their product offerings. In January 2020, ADM acquired Yerbalatina, a manufacturer of plant-based extracts and ingredients based in Brazil. Yerbalatina extracts acerola, acai, guarana, green coffee, and other plants using a technology known as cool drying. Its product line includes organic ingredients such as food colorings and vegetable milks. This acquisition will broaden ADM's offerings in the health and wellness segment, which targets USD 750 billion in annual food, beverage, and supplement markets.

Increasing consumer interest in nutritionally enriched products and all-natural foods promotes the demand for high-value premium and natural additives, which is expected to propel the plant extracts (spices and natural flavors) market in Brazil. The naturally flavored products and alternative sweeteners remain the largest product types in demand. The changing lifestyles of people in Brazil also impact their food habits. Dietary supplement consumption patterns in Brazil put it as one of the quickest developing in the country, mostly because of the expanded interest of the population in quality of life, balanced nutrition, complementing everyday diet, weight management, and well-being. Natural herbs and spices were earlier used in domestic cooking to add flavors to the food, which is turning out to be an industrial need to cater to the big food processors in the local market. Organizations in the country are constantly making efforts to ensure the circulation of value-based and clean-label products, acting as a driver for the plant extracts market, compelling it to maintain the transparency between consumers and final tailored products.

TABLE 252 BRAZIL: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	63.9	74.3	85.1	101.2	115.3	131.5	15.5%
Essential Oils	95.5	110.4	125.6	148.4	168.0	190.4	14.8%
Flavonoids	76.2	88.6	101.5	120.7	137.6	157.0	15.6%
Alkaloids	49.6	57.9	66.5	79.4	90.8	103.9	15.9%
Carotenoids	42.8	49.7	56.8	67.5	76.9	87.6	15.4%
Other Types	36.6	42.4	48.3	57.2	64.9	73.7	15.0%
Total	364.7	423.2	483.8	574.4	653.4	744.1	15.3%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 253 BRAZIL: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	150.0	171.4	196.0	224.6	257.1	297.6	14.7%
Essential Oils	215.9	245.1	278.6	317.2	360.8	414.9	14.0%
Flavonoids	179.1	204.7	234.3	268.6	307.5	357.9	14.8%
Alkaloids	119.0	136.4	156.7	180.2	206.9	240.4	15.1%
Carotenoids	99.8	113.9	130.2	149.0	170.4	198.8	14.8%
Other Types	83.7	95.1	108.3	123.4	140.5	158.3	13.6%
Total	847.5	966.6	1,104.1	1,263.1	1,443.0	1,667.9	14.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 254 BRAZIL: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	5.0	5.7	6.4	7.6	8.5	9.6	14.1%
Essential Oils	7.4	8.5	9.5	11.1	12.4	13.9	13.4%
Flavonoids	5.9	6.8	7.7	9.0	10.2	11.4	14.2%
Alkaloids	3.8	4.4	5.0	5.9	6.7	7.6	14.5%
Carotenoids	3.3	3.8	4.3	5.0	5.7	6.4	14.0%
Other Types	2.8	3.2	3.7	4.3	4.8	5.4	13.6%
Total	28.3	32.4	36.6	42.9	48.2	54.3	13.9%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 255 BRAZIL: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	10.8	12.2	13.8	15.6	17.7	19.8	12.9%
Essential Oils	15.6	17.5	19.6	22.1	24.9	27.6	12.1%
Flavonoids	12.9	14.6	16.5	18.7	21.2	23.8	13.0%
Alkaloids	8.6	9.7	11.0	12.5	14.3	16.0	13.3%
Carotenoids	7.2	8.1	9.2	10.4	11.8	13.2	12.9%
Other Types	6.0	6.8	7.6	8.6	9.7	10.5	11.8%
Total	61.1	68.9	77.8	87.9	99.6	111.0	12.7%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.1.2 Rest of South America

11.5.1.2.1 Rising demand for organic products in food, cosmetics, and pharma industries

The Rest of South America includes Argentina, Chile, Peru, and Colombia, among others. The demand for health products is high and growing in South America due to high awareness of healthy diets and well-being. The growing purchasing power of the working class has also supported market development. Moreover, as per the National Center for Biotechnology Information, in 2020, South America showed a high prevalence of vitamin A deficiency (VAD), providing a growth avenue for the dietary supplements market based on plant extracts. Certain South American plants listed in the European Commission's cosmetic ingredient database are high in antioxidants or have protective properties, suggesting that they could be used in cosmetics. Studies are conducted to identify the potential cosmetic use of South American indigenous plants.

The South American oleoresins market has high growth potential, as it has native production of ingredients used for food & beverages and other consumer products. Oleoresins of paprika, chili, and black pepper are the prime products that are exported globally, mainly to European and American countries. Although the region has extensive plant production, its usage at a commercial level is minimal and has started to pick up recently; however, with the increasing technological innovation by the local medicinal and herbal plant extract producers, it is expected to grow during the forecast period.

TABLE 256 REST OF SOUTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	54.3	63.1	72.2	85.7	97.6	111.1	15.4%
Essential Oils	80.5	92.9	105.6	124.7	141.0	159.6	14.7%
Flavonoids	62.2	72.3	82.7	98.2	111.8	127.5	15.4%
Alkaloids	40.5	47.2	54.2	64.6	73.8	84.4	15.8%
Carotenoids	35.0	40.5	46.3	55.0	62.5	71.1	15.3%
Other Types	31.3	36.2	41.3	48.8	55.3	62.7	14.9%
Total	303.9	352.3	402.3	477.0	542.0	616.5	15.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 257 REST OF SOUTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	126.7	144.5	165.2	189.0	216.1	249.5	14.5%
Essential Oils	180.8	205.0	232.8	264.8	300.8	346.6	13.9%
Flavonoids	145.3	165.9	189.6	217.1	248.2	288.3	14.7%
Alkaloids	96.5	110.5	126.8	145.6	167.1	193.6	14.9%
Carotenoids	81.0	92.3	105.4	120.5	137.5	160.1	14.6%
Other Types	71.2	80.8	91.9	104.7	119.1	132.5	13.2%
Total	701.4	799.0	911.6	1,041.7	1,188.8	1,370.6	14.3%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 258 REST OF SOUTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	4.2	4.9	5.5	6.4	7.2	8.1	13.8%
Essential Oils	6.3	7.2	8.0	9.3	10.4	11.6	13.0%
Flavonoids	4.9	5.6	6.3	7.4	8.3	9.3	13.8%
Alkaloids	3.2	3.6	4.1	4.8	5.4	6.1	14.2%
Carotenoids	2.7	3.1	3.5	4.1	4.6	5.2	13.6%
Other Types	2.4	2.8	3.1	3.7	4.1	4.6	13.3%
Total	23.7	27.1	30.6	35.7	40.0	44.9	13.6%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 259 REST OF SOUTH AMERICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	9.1	10.2	11.5	13.0	14.7	16.6	12.8%
Essential Oils	13.0	14.5	16.3	18.2	20.5	23.0	12.2%
Flavonoids	10.4	11.7	13.2	15.0	16.9	19.2	12.9%
Alkaloids	6.9	7.8	8.9	10.0	11.4	12.9	13.2%
Carotenoids	5.8	6.5	7.4	8.3	9.4	10.6	12.9%
Other Types	5.1	5.7	6.4	7.2	8.1	8.8	11.5%
Total	50.4	56.6	63.7	71.8	81.1	91.1	12.6%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.2 AFRICA

Plant extracts are known to play a crucial role in African traditional medicines for the treatment of infectious diseases. The use of herbal medicines in Africa has greatly elevated and enhanced the primary health care system, but the standardization and quantification/dosing of herbal medicines in the treatment of diseases and infections have left much to be desired.

In Africa, undernutrition has been a persistent factor, with major implications for health, particularly among poor and vulnerable populations. Poor infrastructure and limited resources compounded with economic uncertainties and poor access to health services contribute to malnutrition and food insecurity on the continent. According to the WHO, vitamin A deficiency (VAD) is the leading cause of preventable blindness in children and increases the risk of several other fatal diseases. In pregnant women, VAD causes night blindness and may increase the risk of maternal mortality. Vitamin A deficiency is widely prevalent in Africa and Southeast Asia. As per the WHO, supplying adequate vitamin A and other essential food minerals for maternal and child survival would significantly reduce mortality in such a high-risk area. This has pushed non-profit organizations operating parallel to the WHO to supply these regions with dietary supplements. The WHO estimates that approximately 80% of the population in Africa uses traditional medicines. Approximately 85% of traditional medicines involve the use of plant extracts. This would imply that the reliance on herbal medicine is immense. To appreciate the extent of this dependence, it is estimated that in Sub-Saharan Africa, there is one traditional healer for every 500 people, whereas there is only one medical doctor for every 40,000 people. Hence, the importance of herbal medicines in the life of Africans cannot be overemphasized. The re-insurgence of interest in herbal medicines in Africa is backed by several reasons, namely, the increasingly expensive and unavailability of orthodox drugs to average income earners. Such support is expected to offer immense opportunities for the plant extracts market.

TABLE 260 AFRICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (USD MILLION)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
South Africa	420.9	489.9	561.8	669.0	763.3	871.8	15.7%
Rest of Africa	227.9	269.3	313.5	379.0	439.0	509.0	17.4%
Total	648.8	759.3	875.3	1,048.0	1,202.3	1,380.8	16.3%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 261 AFRICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
South Africa	996.0	1,139.4	1,305.3	1,497.8	1,716.4	1,998.9	15.0%
Rest of Africa	590.4	685.6	797.5	929.0	1,080.7	1,261.4	16.4%
Total	1,586.3	1,825.0	2,102.8	2,426.8	2,797.1	3,260.4	15.5%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 262 AFRICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2016–2021 (KT)

Country	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
South Africa	34.2	39.2	44.4	52.2	58.8	66.2	14.2%
Rest of Africa	18.6	21.7	24.9	29.7	33.9	38.8	15.8%
Total	52.8	61.0	69.4	81.9	92.7	105.0	14.7%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 263 AFRICA: PLANT EXTRACTS MARKET, BY COUNTRY, 2022–2027 (KT)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
South Africa	74.7	84.4	95.5	108.2	122.8	137.0	12.9%
Rest of Africa	44.4	50.9	58.4	67.1	77.2	86.7	14.3%
Total	119.1	135.3	153.9	175.3	199.9	223.7	13.4%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.2.1 South Africa

11.5.2.1.1 Unique qualities of plant extracts leading to their increased use in cosmetics

In South Africa, a variety of herbs and spices are used for culinary and medicinal purposes. South African cuisine combines food traditions from various parts of the world, including indigenous dishes from Africa, Dutch, Indian, and Portuguese. While it is not as rich or as distinctive as the culinary cultures of older countries, South African cookery often produces complex flavors from its liberal use of spices. Popular South African spices include peri, cape malay curry powder, baari salt, and bay leaves.

South African medicinal plants, such as *Tabernaemontana ventricosa*, *Cussonia spicata*, *Rapanea melanophloeos*, *Pittosporum viridiflorum*, and *Clerodendrum glabrum*, are used traditionally for the treatment of several diseases including inflammatory and respiratory diseases. Influenza infection remains a major health threat for animals and humans in the region, which requires effective antiviral remedies. The usage of herbal medications as readily available alternatives for their compatibility with the body and fewer side effects compared to synthetic chemical treatments has become popular globally. Nowadays, various medicinal plants from the East Cape of South Africa are being incorporated into soaps, sanitizers, creams, and sunscreens due to their antimicrobial activity, antioxidant potential, photo-protective effect,

and anti-tyrosinase properties. All these factors are expected to drive the market for plant extracts in South Africa. South Africa's health supplement market has tapped into a vein of vitality that has seen unprecedented growth in recent years, owing to consumers' increased interest in healthy living and a desire to reduce their risk of illness. Many experts attribute the boom in supplement sales to increased consumer awareness of health, an older population, and greater access to health information through digital technology.

Essential oil is produced in South Africa as a plant extract; most of it is exported, with only a small percentage sold locally. Europe and the UK are the largest export markets for South African essential oils, followed by the US and Japan. Zuplex Botanicals is a South African company that produces active extracts from African botanicals. It is involved in botanical cultivation, product R&D, and extract manufacturing. Its primary market is cosmetic and nutraceutical extracts, but also conducts research and product development in other industries such as flavor and food. Due to the vast application of South African plant extracts, there is growth in the market.

TABLE 264 SOUTH AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	76.0	88.6	101.8	121.4	138.8	158.8	15.9%
Essential Oils	108.6	125.8	143.5	169.9	192.9	219.2	15.1%
Flavonoids	85.3	99.5	114.3	136.4	155.9	178.4	15.9%
Alkaloids	55.6	65.0	75.0	89.7	102.9	118.2	16.3%
Carotenoids	47.9	55.8	64.0	76.3	87.1	99.6	15.7%
Other Types	47.5	55.2	63.2	75.2	85.7	97.7	15.5%
Total	420.9	489.9	561.8	669.0	763.3	871.8	15.7%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 265 SOUTH AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	181.7	208.2	238.9	274.6	315.1	367.6	15.1%
Essential Oils	249.1	283.5	323.1	368.9	420.5	485.7	14.3%
Flavonoids	204.3	234.1	268.8	309.0	354.8	416.2	15.3%
Alkaloids	135.7	156.0	179.7	207.3	238.8	279.6	15.6%
Carotenoids	113.8	130.3	149.3	171.5	196.6	231.2	15.2%
Other Types	111.4	127.2	145.5	166.6	190.5	218.7	14.4%
Total	996.0	1,139.4	1,305.3	1,497.8	1,716.4	1,998.9	15.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 266 SOUTH AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	6.2	7.1	8.0	9.5	10.7	12.1	14.3%
Essential Oils	8.8	10.1	11.3	13.3	14.9	16.7	13.6%
Flavonoids	6.9	8.0	9.0	10.6	12.0	13.6	14.4%
Alkaloids	4.5	5.2	5.9	7.0	7.9	9.0	14.8%
Carotenoids	3.9	4.5	5.1	6.0	6.7	7.6	14.2%
Other Types	3.9	4.4	5.0	5.9	6.6	7.4	14.0%
Total	34.2	39.2	44.4	52.2	58.8	66.2	14.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 267 SOUTH AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	13.6	15.4	17.5	19.8	22.5	25.2	13.1%
Essential Oils	18.7	21.0	23.6	26.6	30.1	33.3	12.2%
Flavonoids	15.3	17.3	19.7	22.3	25.4	28.5	13.2%
Alkaloids	10.2	11.6	13.1	15.0	17.1	19.2	13.5%
Carotenoids	8.5	9.7	10.9	12.4	14.1	15.8	13.2%
Other Types	8.4	9.4	10.6	12.0	13.6	15.0	12.4%
Total	74.7	84.4	95.5	108.2	122.8	137.0	12.9%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.2.2 Rest of Africa

11.5.2.2.1 Inability of western medicine to provide cures for some diseases and infections

Numerous African medicinal plant species are important aromatic and ornamental plants in addition to being medicinal. Many of these species produce essential oils, which are used in traditional and modern medicine, cosmetics, and pharmaceuticals. Essential oils have a broad spectrum of bioactivity due to the presence of several active ingredients that act in different ways. These oils contain a variety of volatile molecules due to their extraction method, which is mostly distillation, such as terpenes and phenol-derived aromatic and aliphatic components. Aromatic and medicinal plants of Africa belong to important families that are currently receiving phytochemical attention due to their biological and chemical diversity. The genera are found worldwide and are important plants in African traditional medicine, where they are used to treat diseases such as malaria, hepatitis, cancer, inflammation, and infections caused by fungi, bacteria, and viruses. Extensive research on aromatic and medicinal plants from Africa has resulted in the discovery of many new compounds as well as essential oils, driving the growth of the plant extracts market.

The use of phyto-cosmetics in the African culture is perhaps the oldest and most diverse of all other herbal pharmacopeias. The region is a preferred exporter of various plant extract types from across the world. There are several African botanical ingredients sold in North America that are considered Generally Recognized as Safe (GRAS) or have received novel food approval in Canada. One example of GRAS approval from the FDA and Health Canada is baobab ingredients. The other African natural products currently available in the North American market include rooibos, honey bush, and other herbal teas, and devil's claw, which is marketed as a dietary supplement product in the US, but as a natural health product (medicine) in Canada.

TABLE 268 REST OF AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	41.6	49.2	57.4	69.5	80.6	93.6	17.6%
Essential Oils	57.2	67.4	78.3	94.4	109.1	126.2	17.2%
Flavonoids	46.7	55.2	64.4	78.0	90.6	105.2	17.7%
Alkaloids	30.4	36.1	42.3	51.4	59.8	69.7	18.0%
Carotenoids	26.2	31.0	36.1	43.7	50.6	58.7	17.5%
Other Types	25.9	30.3	35.0	42.0	48.3	55.5	16.5%
Total	227.9	269.3	313.5	379.0	439.0	509.0	17.4%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 269 REST OF AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	108.8	126.5	147.4	172.0	200.4	234.3	16.6%
Essential Oils	146.1	169.2	196.3	228.1	264.8	305.2	15.9%
Flavonoids	122.3	142.3	165.9	193.6	225.7	265.3	16.7%
Alkaloids	81.3	94.9	110.9	129.9	151.9	178.2	17.0%
Carotenoids	68.1	79.2	92.2	107.4	125.0	147.4	16.7%
Other Types	63.9	73.5	84.8	97.9	112.9	131.0	15.5%
Total	590.4	685.6	797.5	929.0	1,080.7	1,261.4	16.4%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 270 REST OF AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	3.4	4.0	4.6	5.5	6.2	7.1	16.0%
Essential Oils	4.7	5.4	6.2	7.4	8.4	9.6	15.5%
Flavonoids	3.8	4.5	5.1	6.1	7.0	8.0	16.0%
Alkaloids	2.5	2.9	3.4	4.0	4.6	5.3	16.4%
Carotenoids	2.2	2.6	2.9	3.5	4.0	4.6	15.8%
Other Types	2.1	2.4	2.7	3.2	3.7	4.1	14.8%
Total	18.6	21.7	24.9	29.7	33.9	38.8	15.8%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 271 REST OF AFRICA: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	8.2	9.4	10.8	12.4	14.3	16.1	14.5%
Essential Oils	11.0	12.6	14.4	16.5	18.9	21.0	13.8%
Flavonoids	9.2	10.6	12.1	14.0	16.1	18.2	14.7%
Alkaloids	6.1	7.0	8.1	9.4	10.8	12.2	14.9%
Carotenoids	5.2	6.0	6.9	7.9	9.1	10.3	14.6%
Other Types	4.7	5.3	6.1	6.9	7.9	8.8	13.4%
Total	44.4	50.9	58.4	67.1	77.2	86.7	14.3%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

11.5.3 MIDDLE EAST

11.5.3.1 Growing import demand for plant extracts from North America

The major factors driving the market growth in the Middle East include the growing industrial base due to investments in food, beverage, pharmaceutical, and personal care sectors, along with the increasing domestic demand for nutraceutical and functional products. The region serves as the origin of the majority of plant-based medical compounds; however, only a minority of the materials was imported due to the lack of commercialization and the exposure of local plant compound manufacturers. Being a largely unorganized and fragmented market, the region offers profitable opportunities for growth, as various small and large food and other consumer product processing players are entering this marketplace. Nutraceuticals, functional foods, and wellness products are going through a significant growth phase in the Middle East. Plant extract manufacturers in the region are actively looking to cater to the evolving customer needs, technologically and in terms of raw materials supply, leading to market opportunities for plant extracts in functional food and health & wellness products, boosting the plant extracts market in the region.

Saudis have increased their focus on maintaining health and protection from diseases. Much of the young population in the Middle East consume dietary supplements and view them as a solution to deficiencies in different vitamins and minerals. Most sports personnel in Saudi Arabia rely on dietary supplements for endurance and enhanced physical performance. As a result of the increased focus on immunity, Aster Group, which operates 201 pharmacies across the UAE, stated that the demand for supplements and immunity-boosting pills has increased. Meanwhile, VPS Healthcare, one of the UAE's largest healthcare providers, has seen a significant increase in demand for multivitamins and wellness supplements.

TABLE 272 MIDDLE EAST: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (USD MILLION)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	35.1	41.4	48.2	58.3	67.4	78.1	17.4%
Essential Oils	53.4	62.7	72.7	87.3	100.5	115.8	16.8%
Flavonoids	43.1	50.9	59.3	71.7	82.9	96.1	17.4%
Alkaloids	28.1	33.3	38.9	47.1	54.8	63.6	17.8%
Carotenoids	24.2	28.6	33.2	40.1	46.3	53.6	17.2%
Other Types	18.4	21.5	24.8	29.8	34.1	39.2	16.4%
Total	202.1	238.5	277.1	334.3	386.2	446.4	17.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 273 MIDDLE EAST: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (USD MILLION)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	90.4	104.8	121.5	141.2	163.7	193.3	16.4%
Essential Oils	133.4	153.8	177.5	205.2	236.7	278.1	15.8%
Flavonoids	111.3	129.0	149.7	173.9	201.8	239.6	16.6%
Alkaloids	73.9	86.0	100.1	116.7	135.8	160.9	16.8%
Carotenoids	62.0	71.8	83.2	96.5	111.8	133.1	16.5%
Other Types	45.0	51.6	59.3	68.3	78.4	89.2	14.7%
Total	516.0	596.9	691.4	801.7	928.1	1,094.2	16.2%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 274 MIDDLE EAST: PLANT EXTRACTS MARKET, BY TYPE, 2016–2021 (KT)

Type	2016	2017	2018	2019	2020	2021	CAGR (2016–2021)
Oleoresins	2.8	3.3	3.8	4.5	5.1	5.9	15.9%
Essential Oils	4.3	5.0	5.7	6.7	7.6	8.7	15.4%
Flavonoids	3.4	4.0	4.6	5.5	6.3	7.2	16.0%
Alkaloids	2.2	2.6	3.0	3.6	4.2	4.8	16.4%
Carotenoids	1.9	2.3	2.6	3.1	3.5	4.0	15.8%
Other Types	1.5	1.7	1.9	2.3	2.6	2.9	15.0%
Total	16.1	18.8	21.6	25.7	29.4	33.5	15.8%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

TABLE 275 MIDDLE EAST: PLANT EXTRACTS MARKET, BY TYPE, 2022–2027 (KT)

Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Oleoresins	6.7	7.7	8.8	10.1	11.6	13.0	14.2%
Essential Oils	9.9	11.3	12.9	14.7	16.8	18.7	13.6%
Flavonoids	8.3	9.5	10.9	12.5	14.3	16.1	14.3%
Alkaloids	5.5	6.3	7.3	8.4	9.7	10.8	14.6%
Carotenoids	4.6	5.3	6.0	6.9	7.9	9.0	14.2%
Other Types	3.3	3.8	4.3	4.9	5.6	6.0	12.5%
Total	38.3	43.8	50.2	57.5	66.0	73.7	14.0%

Source: Secondary Research, Primary Interviews, Industry Journals, Related Research Publications, Press Releases, and MarketsandMarkets Analysis

12 COMPETITIVE LANDSCAPE

12.1 OVERVIEW

The competitive landscape provides an overview of the relative market position of major players operating in the plant extracts market based on the strength of their product offerings and business strategies analyzed based on a proprietary model.

The relative robustness of product offerings has been analyzed objectively using the information available in the public domain pertaining to portfolio strength and geographical reach. An equal weightage is assigned to the factors mentioned above to ascertain an overall score for the selected players in the plant extracts market.

Similarly, the relative robustness of business strategies has been analyzed objectively using the information available in the public domain pertaining to industry coverage, operating and financial strength, and growth attempt endeavors. An equal weightage is assigned to the above-mentioned factors to ascertain an overall score for the selected players in the plant extracts market.

A competitive leadership mapping is prepared based on the ascertained scores on these parameters, which depicts the differentiation of selected players as visionary leaders, innovators, dynamic differentiated, and emerging companies.

The key players in this market include International Flavors & Fragrances Inc. (US), Givaudan (Switzerland), Symrise (Germany), Kerry Group Plc (Ireland), ADM (US), Synthite Industries Ltd (India), Kalsec Inc. (US), Kangcare Bioindustry Co. Ltd. (China), Carbery Group (Ireland), DSM (Netherlands), Döhler (Germany), Indesso (Indonesia), and Vidya Herbs (India).

MAJOR STRATEGIES IDENTIFIED INCLUDE THE FOLLOWING:

- New product launches
- Deals (agreements, collaborations, partnerships, and acquisitions)
- Others (expansions)

12.2 MARKET SHARE ANALYSIS, 2021

While studying any market, it is important to identify the key players. A company's market share helps analyze its share relative to the total market. The analysis related to the key players and their shares in the market helps identify the extent to which they provide a premium-quality product, premium service, or price advantage. A company with a large market share is most likely to have higher profit margins. The market share explains the consumers' preference for a product over similar products. A larger market share also signifies that the company has achieved economies of scale in procurement, manufacturing, marketing, and other cost components.

Companies are focusing on expanding their production facilities by entering into partnerships and agreements as well as by launching new products to grow their businesses and their market shares. New product launches as a result of extensive research & development (R&D) initiatives, geographical expansion to tap the potential of emerging economies, and strategic acquisitions to gain a foothold over the large extent of the supply chain are the key strategies adopted by companies in the plant extracts market. For instance, in July 2022, Symrise introduced Maison Lautier 1795 (fragrance), a natural brand, and its first three product lines. The emphasis is on artisan, sustainably grown Mediterranean flowers and plants. This adds to the company's scent product portfolio. Moreover, in December 2021, Symrise established an innovation lab in the United Kingdom. The on-site resources enable local application

specialists to decipher the market, build flavor solutions that align with current consumer trends, and produce application products that exceed client expectations. The facility includes bakery, beverage, and dairy application labs, as well as a taste creation lab. It also helps UK manufacturers and producers develop locally. This new facility will allow the organization to provide greater service to its consumers. Companies are also investing in new technology. For instance, Givaudan, in February 2012, introduced Advanced Tools for Modelling (ATOM), which uses artificial intelligence to optimize food and flavor formulation to generate new options and insights based on consumer preferences. The tools investigate ingredient synergies and identify flavor drivers that are positive and negative. This will support the company's strategy and intent to lead the way in digitalization; these pioneering tools open up new avenues for creative development and have the potential to shorten the time to market for new products.

TABLE 276 PLANT EXTRACTS MARKET: DEGREE OF COMPETITION (COMPETITIVE)

DEGREE OF COMPETITION	COMPETITIVE
Givaudan	11%-12%
International Flavors & Fragrances	10%-12%
Sensient Technologies	2%-4%
Symrise	7%-8%
Synthite Industries Ltd.	1%-3%
Others	61%-69%

Note: The companies in the above table are ranked based on their revenue growth, segmental revenue growth, product portfolio, the number of developments (January 2018 to August 2022), geographical footprint, and product innovations.

DEGREE OF COMPETITION IS DEFINED AS BELOW:

Fragmented: When the top 5 players have a total market share < 25%.

Competitive: When the top 5 players have a total of 25%–50% of the market share.

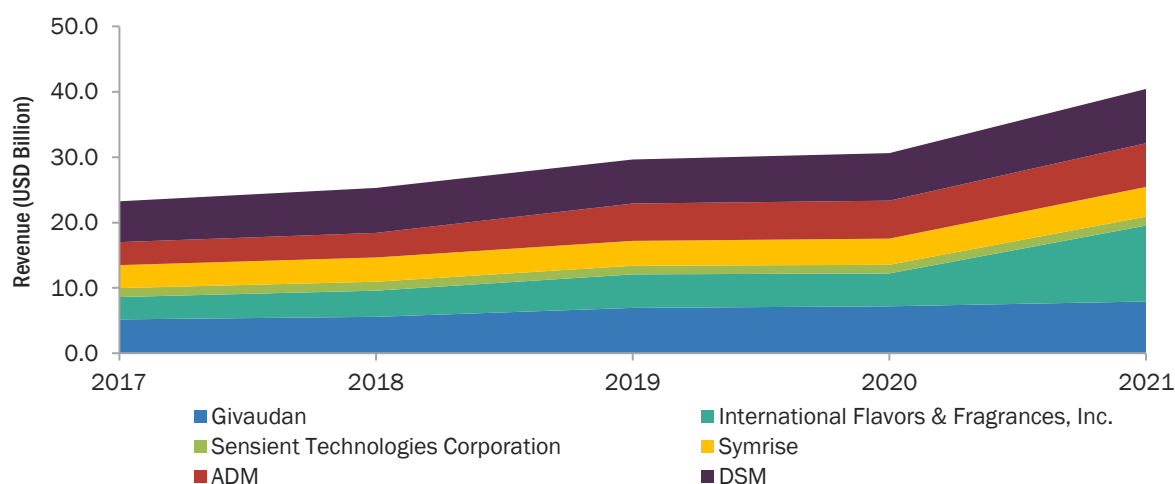
Consolidated: When the top 5 players have a total market share > 50%.

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

12.3 SEGMENTAL REVENUE ANALYSIS OF KEY PLAYERS

This section provides insights into key players' revenues in the plant extracts market from 2017 to 2021.

FIGURE 53 SEGMENTAL REVENUE ANALYSIS OF KEY PLAYERS, 2017–2021 (USD MILLION)



Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

The segmental revenue analysis of key players in the plant extracts market indicates that Givaudan and International Flavors & Fragrances Inc. (US) accounted for the largest share of the segmental revenue. ADM has adopted the development of innovative products as a strategy to acquire the major market share. Their products meet the heightened demand for plant-based solutions and can deliver on evolving consumer expectations. Symrise also ranks among the top five and has adopted new product launch and partnership strategies and strategic plans, such as they have partnered with CRIEPPAM on a bio-regeneration initiative for white lavender. Its work with CRIEPPAM will result in the next generation of Symrise's White Lavender plants. Within three years, they must be verified by "Plants Sains." This ensures a more sustainable supply chain with full traceability. This collaboration aims to fulfill Symrise's commitment to sustainability.

12.4 KEY PLAYER STRATEGIES

International Flavors & Fragrances Inc. (US), Givaudan (Switzerland), Symrise (Germany), Kerry Group Plc (Ireland), and ADM (US) are among the key players in the global plant extracts market. To strengthen their market position, they are now focusing on strategies such as product developments, mergers and acquisitions, product innovations, joint ventures, collaborations, and partnerships.

TABLE 277 STRATEGIES ADOPTED BY KEY PLANT EXTRACT MANUFACTURERS

KEY PLAYER	PRODUCT SOURCE	STRATEGIC DEVELOPMENT/PARTNERSHIP	MERGER AND ACQUISITION/NEW PRODUCT LAUNCH	REGION
Givaudan (Switzerland)	- Herbs & Spices - Fruits & Vegetables - Flowers	Givaudan opened a new digital space in China, one of the most digitalized markets, as part of its long-term strategy to expand its digital capabilities. It will enable the company to capitalize on transformation opportunities while also supporting growth with local and regional customers.	Givaudan Active Beauty innovated Synchronight, an active cosmetic ingredient made from gardenia fruit extract designed to protect the skin from digital stress. This will help the company strengthen its skin care ingredients product portfolio.	Givaudan is present globally in 148 locations, manufacturing its products through 64 production sites. Its presence traverse from regions including Asia Pacific, Europe, Africa, the Middle East, North America, and South America.
Symrise (Germany)	- Herbs & Spices - Fruits & Vegetables - Flowers	Symrise entered into a strategic partnership with Evoxx Technologies. The partners plan to create biotechnological procedures for ingredients used in cosmetics. Evoxx provides its expertise in research and development, while Symrise extends its expertise and capabilities in developing novel and long-lasting cosmetic components. This collaboration assists Symrise in further expanding its biotech platform. It contributes to the company's strategic goal of continuously expanding its line of sustainable cosmetic components.	Symrise introduced its Diana Food™ blueberry components and active compounds. This strengthens the taste, nutrition, and health segment offerings.	Symrise has customers across 150 countries and manufacturing facilities in approximately 40 countries. The company operates in Europe, Africa, the Middle East, North America, Asia Pacific, and Latin America.

Kerry Group Plc (Ireland)	▪ Herbs & Spices ▪ Fruits & Vegetables ▪ Flowers	Kerry launched a new e-commerce platform, KerrySelect.com, in North America, featuring a targeted portfolio of Kerry products. KerrySelect.com's portfolio includes dairy powders and sauces, edible decorations, seasoning blends, coatings, beverage mixes, flavor ingredients, and emulsifiers.	Kerry completed the acquisition of Natreon, a leading supplier of branded Ayurvedic botanical ingredients, based in the US. Kerry's leadership position and the ProActive Health portfolio of science-backed branded ingredients are significantly expanded as a result of the acquisition, which accelerates the company's technological growth.	Kerry Group has its manufacturing units in 32 countries. The company sells its products across 200 countries, such as Singapore, Brazil, Mexico, Ireland, and the UK.
ADM (US)	▪ Herbs & Spices ▪ Fruits & Vegetables	-	Flavor Infusion International, S.A. (FISA), a provider of flavor and specialty ingredient solutions in Latin America and the Caribbean, was acquired by ADM. The addition of FISA will provide ADM with new growth opportunities in Latin America and the Caribbean.	The company operates through 46 innovation centers, 271 processing plants, 450 crop procurement facilities, and 200 bulk storage facilities in nearly 200 countries.
International Flavors & Fragrances Inc. (US)	▪ Herbs & Spices ▪ Fruits & Vegetables ▪ Flowers	-	IFF Health, a part of International Flavors & Fragrances Inc., and DoCas Biotech, partnered on the sales and marketing of Curcugen, a next-gen, all-natural turmeric extract for both dietary supplements and food & beverage industries. It aims to increase the interest in curcumin and expand the activities in the market for nutraceuticals.	The company is in more than 40 countries and serves in 160 countries. IFF has 210 manufacturing facilities, creative centers, and application laboratories in 45 different countries. It has its major manufacturing units in the US, the Netherlands, Spain, Great Britain, Germany, Indonesia, Turkey, Brazil,

			Mexico, Slovenia, China, India, Ireland, Finland, Denmark, Belgium, and Singapore.
Sensient Technologies Corporation (US)	- Herbs & Spices - Fruits & Vegetables - Flowers	The company is committed to sustainability and is taking various steps in its operations to achieve it, such as increasing the share of electrical power consumed from renewable sources whenever possible. All three of its UK manufacturing facilities were entirely powered by renewable energy in 2021. As a result, carbon dioxide emissions reduced by approximately 4,481 metric tons in 2021.	The company acquired Flavor Solutions, Inc., a New Jersey-based flavors company. This business is now part of the flavors and extracts segment, and to assist the company in expanding its flavor product portfolio and adding technological expertise in product formulation.
			The company has a presence in 35 countries and operates from 75 locations serving customers in more than 150 countries.

Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

12.5 COMPANY EVALUATION QUADRANT (KEY PLAYERS)

The company evaluation matrix provides an overview of the relative market position of major players operating in the plant extracts market based on the strength of their product offerings and business strategies, which are analyzed based on a proprietary model. The relative robustness of product offerings has been analyzed objectively using the information available in the public domain about portfolio strength, innovative strength, and geographical reach. Equal weightage is assigned to the factors mentioned above to ascertain an overall score for the selected players in the plant extracts market.

Similarly, the relative robustness of business strategies has been analyzed objectively using the information available in the public domain about industry coverage, operating and financial strength, and growth attempt endeavors. Equal weightage has been assigned to the factors mentioned above to ascertain an overall score for the selected players in the plant extracts market.

The company evaluation matrix is prepared based on scores ascertained on these parameters that depict the differentiation of selected players as stars, pervasive players, emerging leaders, and participants.

12.5.1 STARS

Stars include the leading market players in terms of the strength of their product portfolios and business strategies. These players focus on new product & technology launches as well as acquiring a leading market position through the provision of broad portfolios, catering to the varied requirements of the market, along with a focus on the diverse end-user segments. They are also focused on innovation and are geographically diversified. They have broad industry coverage and strong operational & financial strength.

International Flavors & Fragrances Inc. (US), Givaudan (Switzerland), Symrise (Germany), Kerry Group Plc (Ireland), ADM (US), and Sensient Technologies Corporation (US) are some of the star companies identified in the plant extracts market.

12.5.2 EMERGING LEADERS

Emerging leaders are players with a strong focus on innovation and invest strongly in their R&D endeavors. These players hold market-specific technology developments and have launched new products in the recent past.

Kalsec Inc. (US), Kangcare Bioindustry Co. Ltd. (China), MartinBauer (Germany), and Carbery Group (Ireland) are emerging leaders in the plant extracts market.

12.5.3 PERVASIVE PLAYERS

Pervasive players have a relatively high score on the business strategy robustness criteria compared with their peers and their score on product offering robustness.

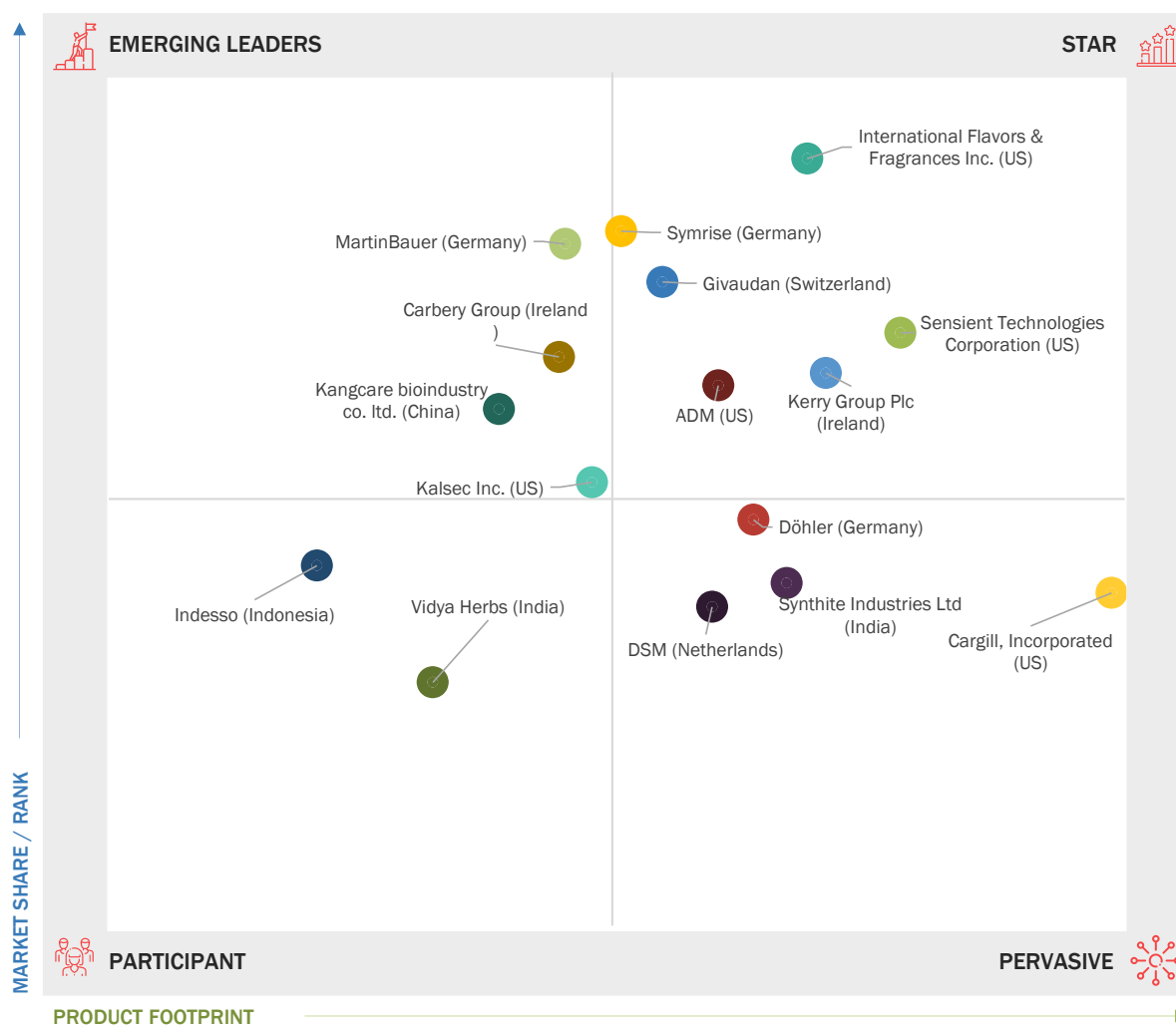
DSM (Netherlands), Döhler (Germany), Cargill, Incorporated (US), and Synthite Industries Ltd (India) are pervasive players in the plant extracts market.

12.5.4 PARTICIPANTS

The plant extracts market includes some participants who have recently included the market-related services in their portfolios, have a limited product portfolio catering to the market, or are comparatively small in operations. In terms of geographical reach as well, they have a limited footprint. At present, their operational & financial strength is limited.

Indesso (Indonesia) and Vidya Herbs (India) are the major participants in the plant extracts market.

FIGURE 54 PLANT EXTRACTS MARKET: COMPANY EVALUATION QUADRANT, 2021 (KEY PLAYERS)



Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

12.5.5 PRODUCT FOOTPRINT

TABLE 278 COMPANY FOOTPRINT, BY SOURCE

COMPANY	FRUITS & VEGETABLES	HERBS & SPICES	FLOWERS	OVERALL SOURCE FOOTPRINT
Givaudan (Switzerland)	Y	Y	Y	4.00
International Flavors & Fragrances Inc. (US)	Y	Y	Y	4.00
Sensient Technologies Corporation (US)	Y	Y	Y	4.00
Symrise (Germany)	Y	Y	Y	4.00
Döhler (Germany)	Y	Y	Y	4.00
Synthite Industries Ltd (India)	Y	Y	Y	4.00
Indesso (Indonesia)	N	Y	Y	2.67
Kangcare Bioindustry Co. Ltd. (China)	Y	Y	Y	4.00
Vidya Herbs (India)	Y	Y	Y	4.00
Carbery Group (Ireland)	Y	Y	Y	4.00
ADM (US)	Y	Y	Y	4.00
DSM (Netherlands)	Y	Y	Y	2.67
Kerry Group Plc (Ireland)	Y	Y	Y	4.00
Kalsec Inc. (US)	Y	Y	N	2.67
MartinBauer (Germany)	Y	Y	Y	4.00
Cargill, Incorporated (US)	Y	Y	Y	4.00

Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

TABLE 279 COMPANY FOOTPRINT, BY APPLICATION

COMPANY	PHARMACEUTICALS	DIETARY SUPPLEMENTS	FOOD & BEVERAGES	COSMETICS	OTHER APPLICATIONS	OVERALL APPLICATION FOOTPRINT
Givaudan (Switzerland)	Y	Y	Y	Y	Y	4.00
International Flavors & Fragrances Inc. (US)	N	Y	Y	Y	Y	3.20
Sensient Technologies Corporation (US)	N	N	Y	Y	N	1.60
Symrise (Germany)	Y	Y	Y	Y	Y	4.00
Döhler (Germany)	N	N	Y	Y	Y	2.40
Synthite Industries Ltd (India)	Y	Y	Y	Y	N	3.20
Indesso (Indonesia)	N	N	Y	N	N	0.80
Kangcare Bioindustry Co. Ltd. (China)	Y	Y	Y	Y	N	3.20
Vidya Herbs (India)	Y	Y	Y	Y	N	3.20
Carbery Group (Ireland)	N	N	Y	N	N	0.80
ADM (US)	N	Y	Y	N	N	1.60
DSM (Netherlands)	N	Y	Y	N	Y	2.40
Kerry Group Plc (Ireland)	N	N	Y	N	Y	1.60
Kalsec Inc. (US)	N	N	Y	N	N	0.80
MartinBauer (Germany)	Y	Y	Y	N	N	2.40
Cargill, Incorporated (US)	N	Y	Y	N	N	1.60

Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

TABLE 280 COMPANY FOOTPRINT, BY REGION

COMPANY	NORTH AMERICA	EUROPE	ASIA PACIFIC	SOUTH AMERICA	ROW	OVERALL REGIONAL FOOTPRINT
Givaudan (Switzerland)	Y	Y	Y	Y	Y	4.00
International Flavors & Fragrances Inc. (US)	Y	Y	Y	Y	Y	4.00
Sensient Technologies Corporation (US)	Y	Y	Y	Y	Y	4.00
Symrise (Germany)	Y	Y	Y	Y	Y	4.00
Döhler (Germany)	Y	Y	Y	Y	Y	4.00
Synthite Industries Ltd (India)	Y	Y	Y	N	N	2.40
Indesso (Indonesia)	Y	Y	Y	Y	Y	4.00
Kangcare Bioindustry Co. Ltd. (China)	Y	N	N	N	N	0.80
Vidya Herbs (India)	Y	Y	Y	N	N	2.40
Carbery Group (Ireland)	Y	Y	Y	Y	Y	4.00
ADM (US)	Y	Y	Y	Y	Y	4.00
DSM (Netherlands)	Y	N	N	N	N	0.80
Kerry Group Plc (Ireland)	Y	Y	Y	Y	Y	4.00
Kalsec Inc. (US)	Y	Y	Y	N	N	2.40
MartinBauer (Germany)	Y	Y	Y	Y	Y	4.00
Cargill, Incorporated (US)	Y	Y	Y	Y	Y	4.00

Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

TABLE 281 OVERALL COMPANY FOOTPRINT

COMPANY	SOURCE FOOTPRINT	APPLICATION FOOTPRINT	REGION FOOTPRINT	COMPANY FOOTPRINT
	33%	33%	33%	100%
Givaudan (Switzerland)	4.00	2.40	4.00	3.47
International Flavors & Fragrances Inc. (US)	4.00	3.20	4.00	3.73
Sensient Technologies Corporation (US)	4.00	1.60	4.00	3.20
Symrise (Germany)	4.00	4.00	4.00	4.00
Döhler (Germany)	4.00	2.40	4.00	3.47
Synthite Industries Ltd (India)	4.00	3.20	2.40	3.20
Indesso (Indonesia)	2.67	0.80	4.00	2.49
Kangcare Bioindustry Co. Ltd. (China)	4.00	3.20	0.80	2.67
Vidya Herbs (India)	4.00	3.20	2.40	3.20
Carbery Group (Ireland)	4.00	0.80	4.00	2.93
ADM (US)	4.00	1.60	4.00	3.20
DSM (Netherlands)	2.67	2.40	0.80	1.96
Kerry Group Plc (Ireland)	4.00	1.60	4.00	3.20
Kalsec Inc. (US)	2.67	0.80	2.40	1.96
MartinBauer (Germany)	4.00	2.40	4.00	3.47
Cargill, Incorporated (US)	4.00	1.60	4.00	3.20

Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

12.6 PLANT EXTRACTS MARKET: EVALUATION QUADRANT FOR START-UPS/SMES, 2021

12.6.1 PROGRESSIVE COMPANIES

The plant extracts market includes certain progressive companies, which, though they are in the initial stages of development, have done exceedingly well in product portfolio and business strategy. These companies have well-developed marketing channels and extensive funding to build their product portfolios. These companies are instrumental in achieving high-growth prospects and gaining higher partner attractiveness.

Arjuna Natural (India), Blue Sky Botanics (UK), and Prinova Group LLC (US) are progressive companies operating in the plant extracts market.

12.6.2 STARTING BLOCKS

These companies are start-ups with niche offerings. They are still focusing on gaining a leading position in the market. Unlike other start-ups, they do not have strong business strategies or product portfolios. They are constantly seeking to tap growth opportunities in the market.

Sydler India Pvt. Ltd. (India) and Alchemy Chemicals (India) are the starting blocks in the plant extracts market.

12.6.3 RESPONSIVE COMPANIES

These companies are technology disruptors offering innovative products. They constantly work on developing their product portfolios. However, they do not have a strong business strategy and can be responsive if backed by appropriate marketing techniques, suitable partnerships, and VC funding.

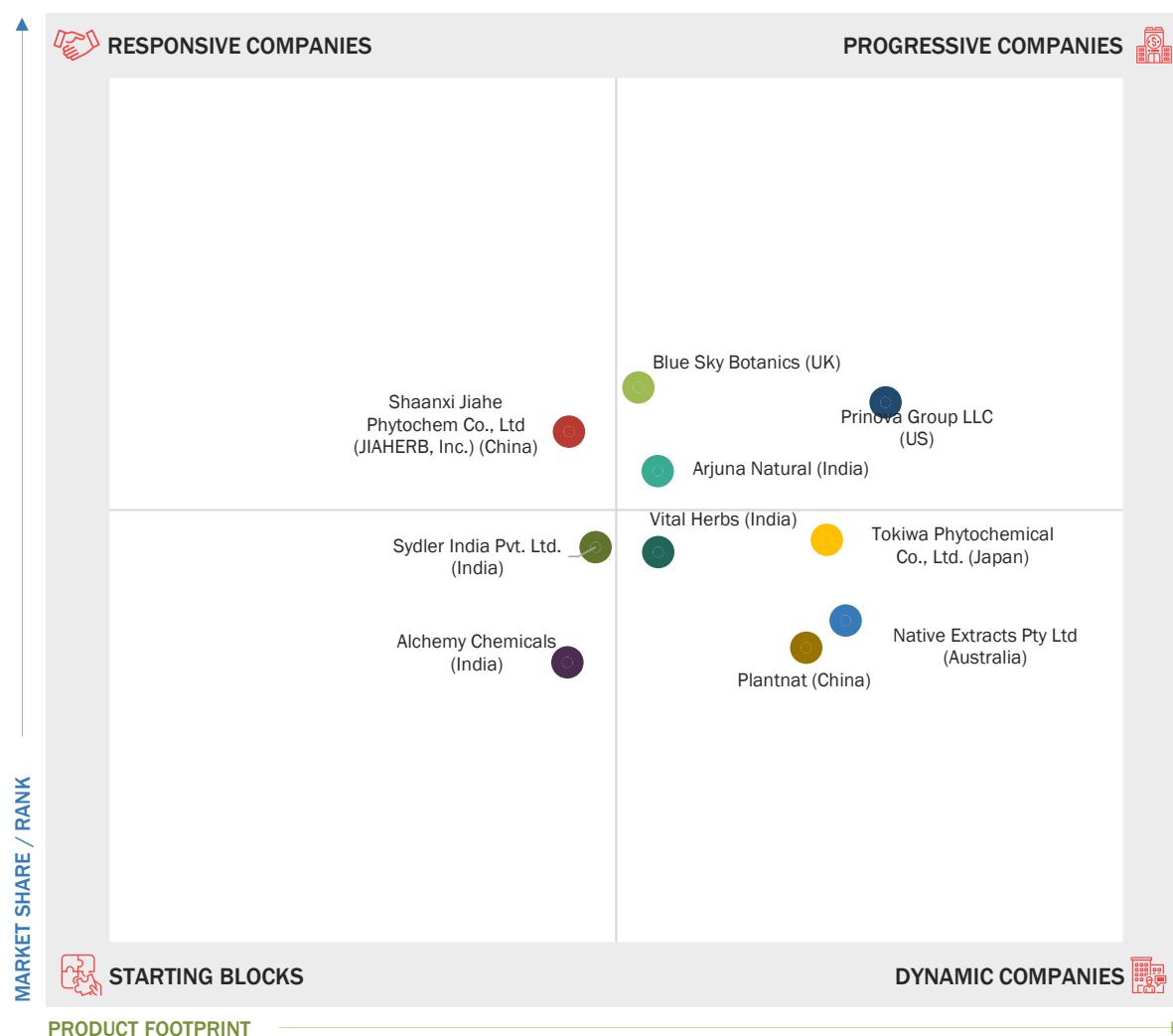
Shaanxi Jiahe Phytochem Co., Ltd (JIAHERB, Inc.) (China) is a key responsive company in the market.

12.6.4 DYNAMIC COMPANIES

These start-ups perform substantially in the business excellence parameter. They generally focus on a specific type of technology related to product offerings.

Vital Herbs (India), Tokiwa Phytochemical Co., Ltd. (Japan), Native Extracts Pty Ltd (Australia), and Plantnat (China) are the dynamic companies in the plant extracts market.

FIGURE 55 PLANT EXTRACTS MARKET: COMPANY EVALUATION QUADRANT, 2021 (START-UPS/SMES)



Source: Company Websites, Annual Reports, Presentations, and MarketsandMarkets Analysis

12.6.5 COMPETITIVE BENCHMARKING OF KEY START-UPS/SMES

TABLE 282 PLANT EXTRACTS MARKET: DETAILED LIST OF KEY START-UPS/SMES

COMPANY	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	FOUNDED	EMPLOYEES	FINANCIAL STATUS	LATEST FUNDING ROUND	TOTAL FUNDING (USD MN)
Native Extracts Pty Ltd (Australia)	Manufacturer	Private	Alstonville, New South Wales, Australia	2012	-	-	-	-
Arjuna Natural (India)	Manufacturer	Private	Aluva, Kerala, India	1989	-	-	-	-
Blue Sky Botanics (UK)	Manufacturer	Private	Herefordshire, England, UK	2005	-	-	-	-
Tokiwa Phytochemical Co., Ltd. (Japan)	Manufacturer	Private	Sakura-shi, Chiba, Japan	1949	110	-	-	-
Shaanxi Jiahe Phytochem Co., Ltd (JIAHERB, Inc.) (China)	Manufacturer	Private	Xi'an, China	2000	-	-	-	-
Alchemy Chemicals (India)	Manufacturer	Private	Ujjain, MP, India	1993	-	-	-	-
Prinova Group LLC (US)	Manufacturer	Private	Carol Stream, IL, US	2002	-	-	-	-
Vital Herbs (India)	Manufacturer	Private	Delhi, India	2018	-	-	-	-
Sydler India Pvt. Ltd. (India)	Manufacturer	Private	Mumbai, Maharashtra, India	2004	-	-	-	-
Plantnat (China)	Manufacturer	Private	Chang Sha City, Hunan Province, China	2004	-	-	-	-

Source: Company Websites and MarketsandMarkets Analysis

TABLE 283 PLANT EXTRACTS MARKET: COMPETITIVE BENCHMARKING OF KEY START-UPS/SMES

COMPANY	SOURCE	APPLICATION							REGION				
	FRUITS & VEGETABLES	HERBS & SPICES	FLOWERS	PHARMACEUTICALS	DIETARY SUPPLEMENTS	FOOD & BEVERAGES	COSMETICS	OTHER APPLICATIONS	NORTH AMERICA	EUROPE	ASIA PACIFIC	MIDDLE EAST & AFRICA	SOUTH AMERICA
Native Extracts Pty Ltd (Australia)	Y	Y			Y	Y	Y				Y		
Arjuna Natural (India)	Y	Y		Y	Y	Y	Y		Y	Y	Y	Y	
Blue Sky Botanicals (UK)	Y	Y				Y	Y			Y			
Tokiwa Phytochemical Co., Ltd. (Japan)	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y
Shaanxi Jiahe Phytochem Co., Ltd (JIAHERB, Inc.) (China)	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y		
Alchemy Chemicals (India)	Y	Y	Y	Y	Y						Y		
Prinova Group LLC (US)	Y	Y	Y		Y	Y	Y	Y	Y	Y	Y	Y	Y
Vital Herbs (India)	Y	Y	Y		Y	Y					Y		
Sydler India Pvt. Ltd. (India)	Y	Y	Y	Y	Y	Y				Y	Y	Y	
Plantnat (China)	Y	Y	Y	Y	Y	Y							

Source: Company Websites and MarketsandMarkets Analysis

12.7 COMPETITIVE SCENARIO

12.7.1 NEW PRODUCT LAUNCHES

Adopting this strategy ensured the development of new products and helped strengthen companies' product portfolios.

TABLE 284 PLANT EXTRACTS: NEW PRODUCT LAUNCHES, 2021–2022

MONTH & YEAR	COMPANY NAME	PRODUCT NAME	PRODUCT TYPE	PRODUCT DESCRIPTION
July 2022	Symrise	Fragrance	Maison Lautier 1795	Symrise introduced Maison Lautier 1795 (fragrance), a natural brand, and its first three product lines. The emphasis is on artisan, sustainably grown Mediterranean flowers and plants. This adds to the company's scent product portfolio.
June 2022	Symrise	Fruit Ingredients	Diana food™ blueberry components and active compounds	Symrise introduced its Diana food™ blueberry components and active compounds. This strengthens the taste, nutrition, and health segment offerings.
November 2021	Sensient Technologies Corporation	Oleoresin	Boundless (line of chili extracts)	Sensient debuted Boundless, a line of chili-based extracts, in its flavors & extracts segment. Boundless offers a variety of chili profiles, including Ancho Chili, Chipotle Chili, Green Jalapeno, Habanero Chili, and Red Jalapeno. This will help the company meet the growing demand for new flavors.
October 2021	Cargill, Incorporated	Essential oils from oregano and other herbs	Nutrena NatureWise	Cargill introduced improved Nutrena NatureWise formulas that improve the overall wellness and productivity of chickens. This poultry feed contains essential oils (oregano and other herbs) that promote flock health and egg production. Furthermore, the feed contains Yucca schidigera extract to help with manure odor control. This launch aided the company expands its product portfolio and also helped meet its customers' demand for wholesome feed.
February 2021	Givaudan	Software	Advanced Tools for Modelling (ATOM)	Givaudan Taste & Wellbeing introduced Advanced Tools for Modelling (ATOM), which uses artificial intelligence to optimize food and flavor formulation to generate new options and insights based on consumer preferences. The tools investigate ingredient synergies and identify flavor drivers that are both positive and negative. This will support the company's strategy and intent to lead the way in digitalization; these pioneering tools open new avenues for creative development and have the potential to shorten the time to market for new products.

Source: Company Websites, Company Publications, and Press Releases

12.7.2 DEALS

Companies adopted this strategy to expand their product range and geographical presence in developing markets.

TABLE 285 PLANT EXTRACTS: DEALS, 2019–2022

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
June 2022	Partnership	Symrise	Evoxx Technologies (Germany)	Symrise entered into a strategic partnership with Evoxx Technologies. The partners plan to create biotechnological procedures for ingredients used in cosmetics. Evoxx provides its expertise in R&D and Symrise extends their expertise and capabilities in developing novel and long-lasting cosmetic components. This collaboration assists Symrise in further expanding its biotech platform. It contributes to the company's strategic goal of continuously expanding its line of sustainable cosmetic components.	
March 2022	Partnership	Symrise	CRIEPPAM (France)	Symrise partnered with CRIEPPAM on a bio-regeneration initiative for white lavender. Its work with CRIEPPAM will result in the next generation of Symrise's White Lavender plants. Within three years, they must be verified by "Plants Sains." This ensures a more sustainable supply chain with full traceability. This collaboration aims to fulfill Symrise's commitment to sustainability.	
December 2021	Acquisition	Givaudan	Custom Essence (US)	Custom Essence was purchased by the company. It is a family-owned company that specializes in the creation of natural fragrances. The acquisition talks began in November 2021 and was completed in December 2021. This will strengthen Givaudan's access to local and regional customers while also expanding its natural perfumery capabilities in North America.	NA
December 2021	Acquisition	ADM	Flavor Infusion International, S.A. (FISA) (Panama)	Flavor Infusion International, S.A. (FISA), a provider of flavor and specialty ingredient solutions in Latin America and the Caribbean, was acquired by ADM. The addition of FISA will provide ADM with new growth opportunities in Latin America and the Caribbean.	NA
July 2021	Acquisition	Sensient Technologies Corporation	Flavor Solutions, Inc. (US)	The company acquired Flavor Solutions, Inc., a New Jersey-based flavors company. This business is now part of the flavors and extracts segment, and to assist the company in expanding its flavor product portfolio and adding technological expertise to its product formulation.	USD 14.9 Million

February 2021	Merger	International Flavors & Fragrances Inc.	DuPont's Nutrition & Biosciences (Denmark)	DuPont's N&B merged with IFF, which supplies binders, plant-based proteins, and texturants which has increased the revenue exponentially and broadened the customer base for sustainable growth.	USD 7.3 billion
March 2020	Acquisition	Givaudan	Alderys (France)	Givaudan acquired Alderys to expand its bioengineering capabilities. Alderys is a French biotechnology firm that pioneers novel approaches to the biological engineering of valuable compounds from renewable feedstock. Furthermore, the acquisition will allow Givaudan to diversify its product portfolio with natural and bio-sourced ingredients.	NA
September 2019	Acquisition	Givaudan	Drom Fragrances (Germany)	Givaudan bought Drom Fragrances. The acquisition was completed in September 2019, after negotiations began in July 2019. This global perfume house develops fragrances for consumer goods and fine fragrance customers worldwide. Drom has four production sites in Germany, China, the US, and Brazil. This will further strengthen Givaudan's position in the global fragrance market.	NA

Source: Company Websites, Company Publications, and Press Releases

12.7.3 OTHERS

Companies have undertaken expansions by establishing R&D, innovation centers, and production facilities in various regions to meet the increasing customer demands.

TABLE 286 PLANT EXTRACTS: OTHERS, 2021-2022

MONTH & YEAR	COMPANY NAME	TYPE	LOCATION	DESCRIPTION
January 2022	Symrise	Expansion	New manufacturing site	Symrise purchased a new manufacturing facility in Granada, Spain. There, the company will manufacture cosmetic product protection ingredients. This expansion is in response to the increased demand for 1,2-alkanediols. The plant can now generate cosmetic chemicals and a wide range of fragrance ingredients from renewable natural materials. The increased output at the Granada manufacturing site will provide Symrise's clients with considerable expansion potential in Europe and around the world, as well as ensuring customers' continuing access to high-quality Symrise goods.
December 2021	Symrise	Expansion	R&D facility	Symrise established an innovation lab in the UK. The on-site resources enable local application specialists to decipher the market, build flavor solutions that align with current consumer trends, and produce application products that exceed client expectations. The facility includes bakery, beverage, and dairy application labs, as well as a taste creation lab. It also helps UK manufacturers and producers develop locally. This new facility will allow the organization to provide greater service to its consumers.
November 2021	Givaudan	Expansion	Digital space	Givaudan opened a new digital space in China, one of the most digitalized markets, as part of its long-term strategy to expand its digital capabilities. It will enable the company to capitalize on transformation opportunities while also supporting growth with local and regional customers.

Source: Company Website, Company Publications, and Press Releases

13 COMPANY PROFILES

13.1 INTRODUCTION

Company profiles provide details about major companies in the plant extracts market. Initially, companies present in the plant extracts ecosystem were identified and analyzed. Then, major players, based on their product portfolio, revenue, recent developments, R&D, geographic presence, and export potential, among other factors, were shortlisted. Along with this, countries with significant demand were considered as a key selection criterion. The above approach resulted in the selection of the final set of companies. Each company profile enlists its business overview, product offerings, and recent developments.

The business overview section provides basic information about the companies and their key business segments in this market. It also contains snapshots of companies, which offer a glimpse into their financial performance. The financials block depicts the company revenue and the percentage of the total revenue invested in R&D activities. It also displays the geographic revenue mix and business revenue mix percentage in terms of the overall income of the company. The products offered by various companies operating in the plant extracts market are also listed in this chapter. The recent developments section contains the latest and most important developments undertaken by companies operating in the market.

13.2 KEY PLAYERS

13.2.1 SYMRISE

13.2.1.1 Business overview

Symrise is a global manufacturer and supplier of flavors, fragrances, cosmetic bases, and ingredients. The company was founded after the merger of Dragoco Gerberding & Co. AG (Germany) and Haarmann & Reimer GmbH (Germany). It serves clients ranging from food and beverage, pharmaceutical, and manufacturers of nutritional supplements and pet food. It operates through two business segments: taste, nutrition & health, and scent & care. The company, through its subsidiary Diana Group, offers various plant extracts and other ingredients for the food & beverage, cosmetics, and personal care industries.

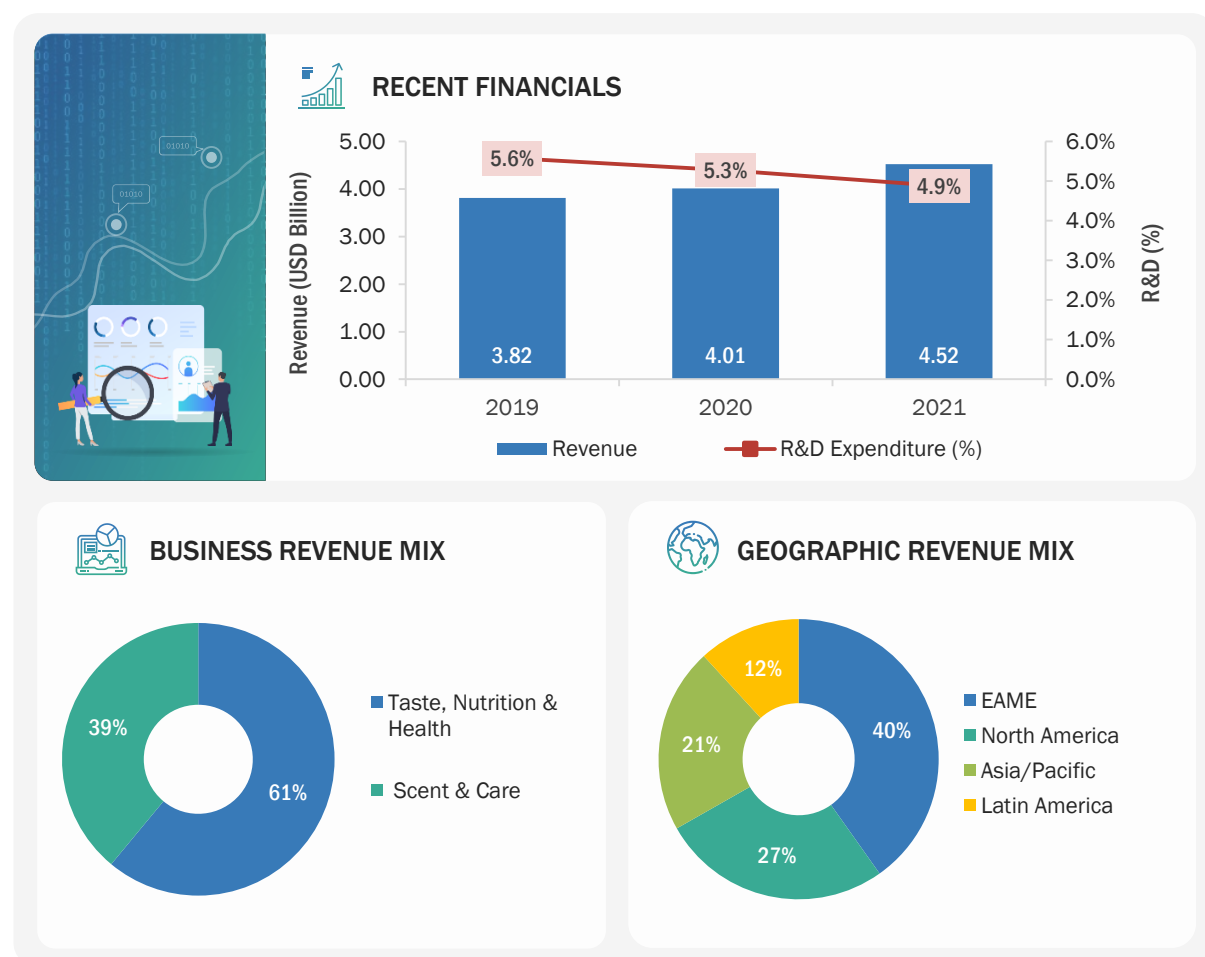
Diana Food is a leading global ingredient solutions provider of organoleptic and nutritional solutions, based on natural ingredients for numerous industrial sectors such as food, pet food, health, nutraceutical, aquaculture, aquafeed, and cosmetic markets. The company offers functional solutions that are based on natural ingredients such as fruits, vegetables, meat, and seafood, and provides other products such as ready-to-use sauces, coating and decoration products, culinary solutions, and coloring solutions for food and pet food industry. The company's flavor and fragrance businesses are operated by Diana and its subsidiaries. Diana Naturals has 30 industrial sites and 29 sales offices in around 23 countries, working closely to integrate the company's flavor and fragrance business, to establish a significant position in the food & beverage market.

Symrise has customers across 150 countries and manufacturing facilities in approximately 40 countries. The company operates in Europe, Africa, the Middle East, North America, Asia Pacific, and Latin America. The subsidiaries of this company include Symrise S.R.L. (Argentina), Symrise Ltd. (UK), Symrise, Inc. (US), Symrise Pty. Ltd. (Australia), Symrise Pvt. Ltd. (India), Symrise Vertriebs GmbH (Austria), Symrise (France), Aromatics S.A.S. (France), and Symrise K.K. (Japan).

TABLE 287 SYMRISE: BUSINESS OVERVIEW

SYMRISE-2021	
Founded	2003
Country	Germany
City	Holzminden
Total No. Of Employees	11,000
R&D Expenditure	USD 261.2 Million
Overall Revenue	USD 4.52 Billion
Ownership	Public (ETR: SY1)

Source: Company Website and Annual Reports

FIGURE 56 SYMRISE: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.1.2 Products/Solutions/Services offered

TABLE 288 SYMRISE: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Actipone	It is a premium collection of high-quality, primarily standardized botanicals. The Symrise Actipone product line includes a wide range of natural, plant extracts from Asia, Europe, and Africa. Herbal extracts (such as Actipone Rosemary), fruit concentrates (such as Actipone Alpha Pulp), tea and coffee extracts (such as Actipone Green Coffee GW), marine plants (Actipone Laminaria Saccharina), and traditional Chinese medicine root and flower extracts (such as Actipone Honeysuckle Flower (Jin Yin Hua) are examples.	- Fruits & Vegetables - Flowers - Herbs	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Cosmetics
Extrapone	Extrapone is a range of aqueous glycolic preparations of natural plant extracts. Some of the extracts from this range include Extrapone Acacia GW, Extrapone Acacia GW, and Extrapone Acai.	Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Cosmetics
Flowerpone	Designed for aqueous applications, soluble in water and ethanol. Some of the products from this range include Salvia sclarea extract, Plumeria acutifolia flower extract, Jasminum officinale flower extract, Rosa damascena flower extract, and Lavandula angustifolia flower extract.	Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Cosmetics
Flowerconcentrole	Designed for emulsions and oil formulations; soluble in oil, glycols, emulsions, and dispersible in water	Flowers	Essential Oils	Cosmetics
Simply Vanilla	They are certified extracts and are cost-effective natural and nature-identical vanilla flavorings.	Spices	- Oleoresins - Essential Oils	Cosmetics Food & Beverages
Naturally Citrus	These extracts are certified citrus oils/extracts with natural citrus flavors and are cost-effective.	Fruits & Vegetables	- Flavonoids - Alkaloids - Essential Oils	Food & Beverages

SymColor	Natural colorings from SymColor give authentic colorants for delectable items. Natural raw materials such as grapes, hibiscus, tomatoes, beets, and other plants are used to make SymColor colorant foods.	▪ Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids ▪ Carotenoids ▪ Essential Oils	Cosmetics Food & Beverages
Hydrolite 5 green	Hydrolite 5 green is a multifunctional substance made from renewable sugar cane raw materials that has the same properties as the original pentylene glycol. It moisturizes the skin and increases the effectiveness of active chemicals in cosmetic compositions.	▪ Herbs	▪ Oleoresins ▪ Essential Oils	Cosmetics
Hydrolite 8	It normalizes the sebum production on the scalp and promotes healthy hair.	▪	▪ Oleoresins	Cosmetics
Cranberry Extracts	Cranberry extracts include soluble saccharides, which are thought to work as prebiotics for beneficial endogenous bacteria in the human gastrointestinal tract.	▪ Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids	Pharmaceutical Dietary Supplement
Aronia Extract	Contains at least 50% total polyphenols and at least 10% anthocyanins. It has an intense and typical Aronia fruity flavor and is made in Canada.	▪ Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids	Pharmaceutical Dietary Supplement
Blueberry Extract	V. angustifolium is a species that contains up to 40% polyphenols. It is simple to incorporate into any delivery system (capsules, pills, gummies, sticks, etc.).	▪ Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins	Pharmaceutical Dietary Supplement

Source: Company Website

13.2.1.3 Recent developments

13.2.1.3.1 New product launches

TABLE 289 SYMRISE: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
July 2022	Symrise	Fragrance	Maison Lautier 1795	Symrise introduced Maison Lautier 1795 (fragrance), a natural brand, and its first three product lines. The emphasis is on artisan, sustainably grown Mediterranean flowers and plants. This adds to the company's scent product portfolio.

June 2022	Symrise	Fruit Ingredients	Diana food™ blueberry components and active compounds	Symrise introduced its Diana food™ blueberry components and active compounds. This strengthens the taste, nutrition, and health segment offerings.
May 2022	Symrise	Fruit Ingredients	Aronia health actives	Symrise released a new line of Aronia health actives. The collection includes an Aronia extract and an Aronia juice powder, both high in polyphenols and anthocyanins. Aronia, often known as black chokeberry, is rich in antioxidants like polyphenols, particularly anthocyanins. This product line adds to the Diana food™ portfolio of health actives, which is part of the Taste, Nutrition, and Health division.
April 2022	Symrise	Cosmetic Ingredient	Hydrolite 8	Symrise introduced Hydrolite 8 green, which is ideal for a variety of traditional and green cosmetic formulations. This new caprylyl glycol has a higher level of sustainability due to its 100% biobased supply. This new launch aims to broaden the company's green ingredient offering.
July 2021	Symrise	Fragrance ingredient	Lilybelle	Symrise introduced Lilybelle, a lily of the valley fragrance ingredient made from renewable sources. It can be used for personal care, cleaning, and laundry care goods. It is made from by-products of the orange juice industry. With this, Symrise is expanding its line of special fragrance compounds with new products, which include a sustainable, easily biodegradable fragrance element.
June 2021	Symrise	Cosmetic Ingredient	Hydrolite 7 Green	Hydrolite 7 green is a one-of-a-kind seven-carbon alkanediol made entirely from bio-based sources, specifically sustainably obtained castor bean oil. Symrise provides an exclusive high-purity bio-based liquid component with Hydrolite 7 green.
September 2020	Symrise	Cosmetic ingredient	Hydrolite 8	Symrise launched SymControl Scalp, which normalizes the sebum production on the scalp and promotes healthy hair. To solve the issue of sebum overproduction at its source, Tetraselmis Suecica extract was selected as it is highly effective in reducing sebum levels. Combined with the naturally occurring Mannitol, the algae can also protect the scalp and hair from environmental stress. This expands the company's hair ingredients portfolio.

Source: Company Website

13.2.1.3.2 Deals

TABLE 290 SYMRISE: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
June 2022	Partnership	Symrise	Evoxx Technologies (Germany)	Symrise entered into a strategic partnership with Evoxx Technologies. The partners plan to create biotechnological procedures for ingredients used in cosmetics. Evoxx provides its expertise in R&D and Symrise extends their expertise and capabilities in developing novel and long-lasting cosmetic components. This collaboration assists Symrise in further expanding its biotech platform. It contributes to the company's strategic goal of continuously expanding its line of sustainable cosmetic components.	NA
March 2022	Partnership	Symrise	CRIEPPAM (France)	Symrise partnered with CRIEPPAM on a bio-regeneration initiative for white lavender. Its work with CRIEPPAM will result in the next generation of Symrise's White Lavender plants. Within three years, they must be verified by "Plants Sains." This ensures a more sustainable supply chain with full traceability. This collaboration aims to fulfill Symrise's commitment to sustainability.	NA
December 2021	Acquisition	Symrise	Giraffe Foods Inc. (Canada)	Giraffe Foods was acquired by Symrise. It manufactures customized sauces, dips, dressings, syrups, and beverages for business-to-business customers in the home meal replacement, food service, and retail markets. This acquisition is viewed by Symrise as a strategic opportunity to extend its portfolio of tailored flavor and taste improvement products. Symrise aspires to be the North American leader in comprehensive flavor solutions. The purchase of Giraffe Foods complements Symrise and expands the value proposition of the taste, nutrition & health division.	NA
November 2021	Partnership	Symrise	Antofénol (France)	Symrise entered into a three-year R&D collaboration with the French company, Antofénol. The company's primary focus is on natural alternatives to traditional agro-extracts obtained by innovative, sustainable eco-extraction using microwave, ultrasound, and vacuum technologies. The parties intend to collaborate on the development of environmentally friendly products. They will concentrate on cosmetic components and scents, as well as the extraction of useful compounds from the Flavor & Nutrition division's by-products. This will assist Symrise in expanding its product offering of sustainable ingredients.	NA

November 2020	Acquisition	Symrise	Sensient Technologies Corporation (US)	Symrise bought the fragrance and aroma chemicals operations of Sensient Technologies Corporation. These include a variety of scent molecules and aromas derived from natural and renewable sources. Symrise's acquisition expands its influence, particularly in EAME and Latin America.	NA
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Source: Company Website

13.2.1.3.3 Others

TABLE 291 SYMRISE: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
January 2022	Symrise	Expansion	New manufacturing site	Symrise purchased a new manufacturing facility in Granada, Spain. There, the company will manufacture cosmetic product protection ingredients. This expansion is in response to the increased demand for 1,2-alkanediols. The plant can now generate cosmetic chemicals and a wide range of fragrance ingredients from renewable natural materials. The increased output at the Granada manufacturing site will provide Symrise's clients with considerable expansion potential in Europe and around the world, as well as ensuring customers' continuing access to high-quality Symrise goods.
December 2021	Symrise	Expansion	R&D facility	Symrise established an innovation lab in the UK. The on-site resources enable local application specialists to decipher the market, build flavor solutions that align with current consumer trends, and produce application products that exceed client expectations. The facility includes bakery, beverage, and dairy application labs, as well as a taste creation lab. It also helps UK manufacturers and producers develop locally. This new facility will allow the organization to provide greater service to its consumers.
June 2021	Symrise	Expansion	Innovation center	In Dubai, Symrise established cutting-edge development, application, and sensory laboratories. The corporation has spent approximately USD 1.2 million. To accommodate its increasing personnel and suit the needs of its partners, the company relocated to the renowned gold tower building in the Dubai multi commodities center (DMCC). Its location will allow it to service the Middle East and Pakistan, as well as support activities in Africa. Furthermore, its expansion aligns with the four sustainability pillars toward carbon neutrality.

Source: Company Website

13.2.1.4 MnM view

13.2.1.4.1 Right to win

The company has a vast range of plant extract products and also has a strong focus on its growth strategies. It is focusing on developing new products using plant extracts to meet the growing consumer demand; for instance, in 2020, it developed SymControl, made from algae extract to reduce the sebum level on the scalp. The company is also making investments in upgrading its systems; for instance, its software investments, to implement ERP software in the old Nutrition business. Moreover, Symrise seeks to expand its sustainability in terms of footprint, innovation, sourcing, and care thus, decreasing risks and encouraging economic success. For example, it adopted Water and Effluents (GRI 303) and Occupational Health and Safety (GRI 403) indicators to improve its performance in the areas of environment, health, and safety. In 2022, Symrise became the first firm in the industry to source Hibiscus that has been verified at the gold level of the SAI Platform's Farm Sustainability Assessment (FSA). Symrise has pledged to promote sustainable agriculture techniques across a variety of crops and areas, including its hibiscus supply chain. This moves the company closer to its objective of acquiring all strategic biological raw resources responsibly by 2025.

13.2.1.4.2 Strategic choices

The company has a strong focus on its growth strategies. In 2020, it acquired Sensient Technologies Corporation (US). The company is a fragrance and aroma chemical manufacturing company. This was aimed to strengthen its backward integration and acquire more clients and enhance its footprint, notably in EAME and Latin America. It also focuses on expansion; it increased its fragrance production capacity in Colonels Island and its cosmetic ingredients and aroma molecules production capacity in Charleston. Its acquisitions with well-known companies and expansion activities have helped strengthen its position in the plant extracts market.

13.2.1.4.3 Weaknesses and competitive threats

Other market leaders, such as Givaudan, pose a threat to the corporation. Since the company procures raw materials from many countries, it is constantly exposed to harvest risk, political risk, and currency risk. Moreover, the political climate within the country affects its business. The amount spent on R&D activities by the company is decreasing each year; it reached 4.6% of its total revenue in 2021 from 5.6% in 2019. This might affect the R&D activities of the company and new product launches.

13.2.2 SENSIENT TECHNOLOGIES CORPORATION

13.2.2.1 Business overview

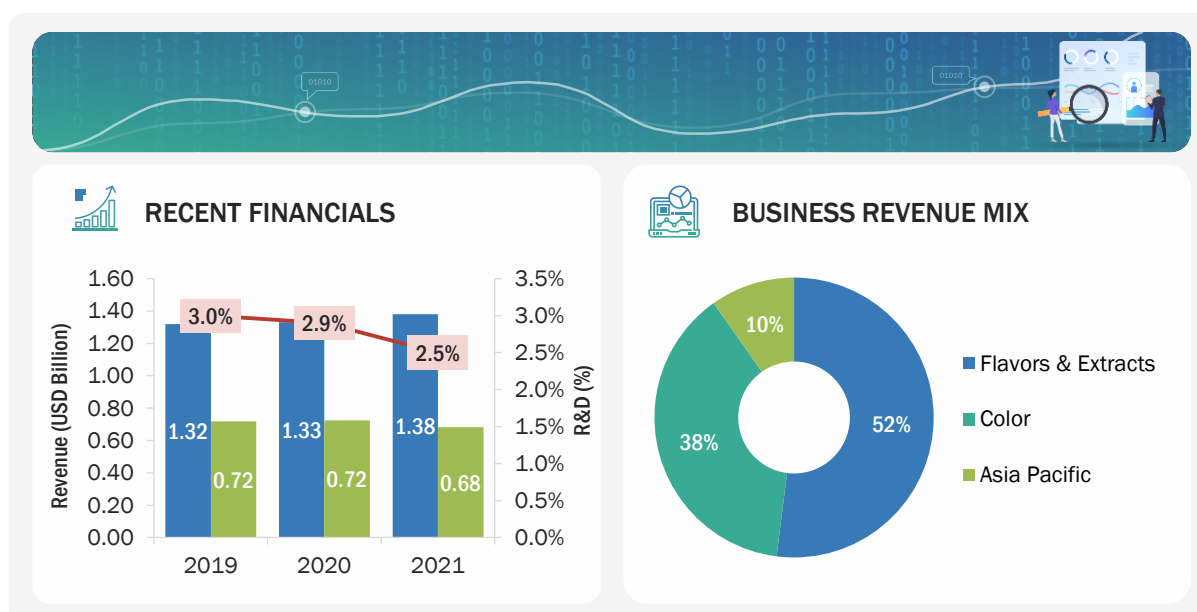
The company is a global leader in the manufacture and marketing of colors, flavors, and other specialty ingredients. The flavors & extracts group, and the color group were the company's reportable segments based on its product line in 2021, and on the geographic basis, it was Asia Pacific. The company's flavor & extract group develops, manufactures, and sells flavor systems to the food, beverage, personal care, and household-products industries. The Group supplies multinational and regional companies under the unified brand names of Sensient Flavors and Sensient Natural Ingredients. Its manufacturing sites for the flavor & extract segment are in the US, Belgium, Costa Rica, Mexico, Germany, and the UK. The company offers natural and synthetic color systems for application in foods, drinks, pharmaceuticals, and nutraceuticals through its color group division. Its manufacturing sites for color group segment are in the US (Missouri), Brazil, Canada, China, France, Germany, Italy, Mexico, Peru, and the UK.

In July 2018, the company acquired Mazza Innovation Limited (Canada), which increased its presence in the plant extracts and natural ingredients space. The company has its presence in 35 countries and operates from 75 locations serving customers in more than 150 countries. Its subsidiaries include Sensient India Private Limited (India), Sensient Flavors International, Inc. (US), Sensient Imaging Technologies, Inc. (US), Sensient Colors U.K. Ltd (UK), Sensient Colors South Africa Ltd. (South Africa), Sensient Flavors South Africa (Pty) Ltd. (South Africa), Sensient Flavors Austria GmbH (Austria), and Sensient Flavors Belgium N.V. (Belgium).

TABLE 292 SENSIENT TECHNOLOGIES CORPORATION: BUSINESS OVERVIEW

SENSIENT TECHNOLOGIES CORPORATION-2021	
Founded	1882
Country	US
City	Milwaukee, Wisconsin
Total No. of Employees	3,844
R&D Expenditure	USD 34.3 Million
Overall Annual Revenue	USD 1.38 Billion
Ownership	Public (NYSE: SXT)

Source: Company Website and Annual Reports

FIGURE 57 SENSIENT TECHNOLOGIES CORPORATION: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.2.2 Products/Solutions/Services offered

TABLE 293 SENSIENT TECHNOLOGIES CORPORATION: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
PhytoClean Extracts	PhytoClean Extracts are a clean label alternative to chemical extraction solvents. They are phytochemicals from plant sources.	- Fruits - Herbs & Spices - Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils - Carotenoids	Food & Beverages
Sensient's Natural Origins	Natural Origins by Sensient is a highly concentrated extract. Citrus, herbs, spices, vanilla, tea, and floral compounds are among the natural botanical extracts offered by Sensient Natural Origins.	- Fruits - Herbs & Spices - Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Food & Beverages
Flavor Extracts	The company offers a variety of flavor extracts, each with its set of characteristics. Citrus fruits have a sweet and clean flavor that is refreshing and crisp. Teas, coffees, and cocoa provide the earthy essence of the leaves, while coffees and cocoa provide the bold character of the beans, while herbs provide freshness.	- Fruits & Vegetables - Herbs & Spices - Flowers	- Flavonoids - Oleoresins - Essential Oils	Food & Beverages

Essential Oil	The company provides a wide range of essential oils like Ambrette Seed Oil, Amyris Oil, Angelica Root Oil, Aniseed Oil Pimpinella, Basil Oil (Linalool type), Basil Oil (Methylchavicol type), Bay Oil, Bergamot Oil Reggio, Bergamot Oil Reggio Bergapten, Blood Orange Oil, Cade Oil, Cajeput Oil, Camphor Oil Light, Cananga Oil, Caraway Oil Rectified, Cardamom Oil, Carrot Seed Oil, Cassia Oil, Cedar Leaf Oil, Cedarwood Oil Atlas, Cedarwood Oil Texas, Cedarwood Oil Virginia, Chamomile Oil Blue, Chamomile Oil Roman, Cardamom Oil, Bay Oil, and Lemon Oil.	▪ Fruits & Vegetables ▪ Herbs & Spices ▪ Flowers	▪ Essential Oil	Food & Beverages Cosmetics
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Source: Company Website

13.2.2.3 Recent developments

13.2.2.3.1 New product launches

TABLE 294 SENSIENT TECHNOLOGIES CORPORATION: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
November 2021	Sensient Technologies Corporation	Oleoresin	Boundless (line of chili extracts)	Sensient debuted Boundless, a line of chili-based extracts, in its flavors & extracts segment. Boundless offers a variety of chili profiles, including Ancho Chili, Chipotle Chili, Green Jalapeno, Habanero Chili, and Red Jalapeno. This will help the company meet the growing demand for new flavors.

Source: Company Website

13.2.2.3.2 Deals

TABLE 295 SENSIENT TECHNOLOGIES CORPORATION: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
July 2021	Acquisition	Sensient Technologies Corporation	Flavor Solutions, Inc. (US)	The company acquired Flavor Solutions, Inc., a New Jersey-based flavors company. This business is now part of the flavors and extracts segment, and to assist the company in expanding its flavor product portfolio and adding technological expertise to its product formulation.	USD 14.9 Million

Source: Company Website

13.2.2.4 MnM view

13.2.2.4.1 Right to win

The company is committed to sustainability and is taking various steps in its operations to achieve it, such as increasing the share of electrical power consumed from renewable sources whenever possible. All three of its UK manufacturing facilities were entirely powered by renewable energy in 2021. As a result, carbon dioxide emissions reduced by approximately 4,481 metric tons in 2021. It has also installed lighting systems with timers, daylight sensors, and infrared sensors to reduce the amount of electricity used for lighting. Furthermore, it has enhanced the sludge dewatering process at a wastewater treatment plant, resulting in a 36% decline in sludge sent to landfill.

13.2.2.4.2 Strategic choices

The company invests in R&D which further provides it with a competitive advantage in creating exceptional products. Development activity in the flavors & extracts group is focused on ingredients, flavors, natural extracts, and essential oils, as well as flavor systems that respond to consumer trends and the processing needs of food & beverage customers. Among these activities are the creation of functional ingredient systems for foods and beverages, as well as savory flavors.

13.2.2.4.3 Weaknesses and competitive threats

Due to intense market competition, the company's sales and profitability may suffer. These competitors range from large multinational flavor companies with broad and sophisticated product portfolios such as Symrise, to smaller and specialized regional companies that focus on a single product line or offering. To remain competitive, the company must provide product development assistance, on-time delivery, regulatory assistance, and after-sales product support to its customers.

13.2.3 INTERNATIONAL FLAVORS & FRAGRANCES INC.

13.2.3.1 Business overview

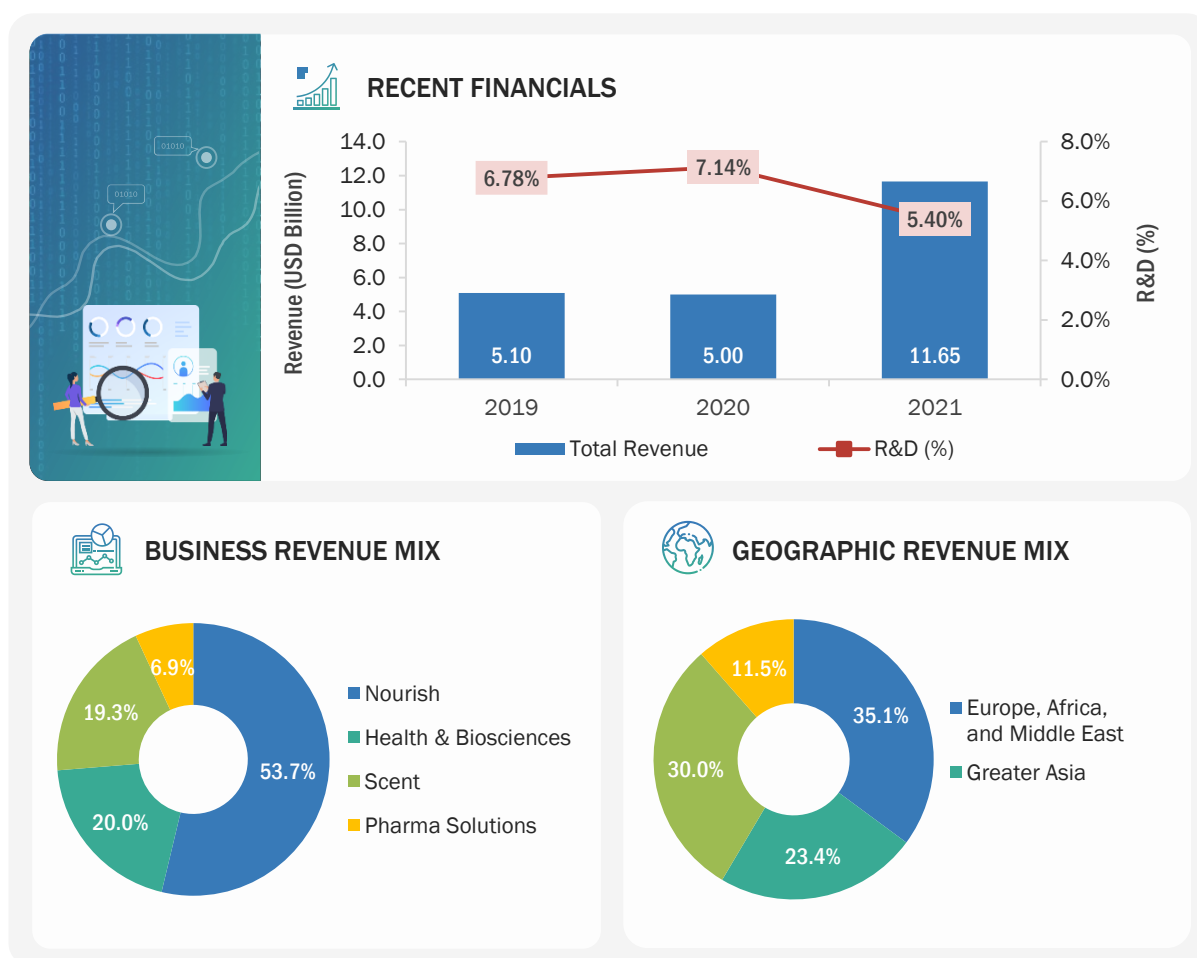
International Flavors & Fragrances Inc. was the result of a merger between Polak & Schwarz (Netherlands) and van Ameringen-Haebleris (US) in 1985, which made it a global supplier of flavors and fragrances. The company is one of the leading producers and manufacturers of food, beverage, health & biosciences, scent and pharma solutions, cosmetic active, and natural health ingredients. There are majorly four domains in which the business operates: Nourish, Health & Biosciences, Scent, and Pharma Solutions. Dietary supplements and food are the major domains where plant extract products are marketed along with perfumery. In February 2021, DuPont's Nutrition & Biosciences ("N&B") business combined with IFF leading to various legal entities associated with the N&B business being transferred to IFF.

The company is in more than 40 countries and serves 160 countries. IFF has 210 manufacturing facilities, creative centers, and application laboratories in 45 countries. It has its major manufacturing units in the US, the Netherlands, Spain, Great Britain, Germany, Indonesia, Turkey, Brazil, Mexico, Slovenia, China, India, Ireland, Finland, Denmark, Belgium, and Singapore. The company operates through its subsidiaries, such as Frutarom (Israel), International Flavors & Fragrances Ltd. (China), International Flavors & Fragrances IFF Ltd. (UK), International Flavors & Fragrances IFF A.G. (Switzerland), International Flavors & Fragrances Ltd. (Hong Kong), International Flavors & Fragrances Ltd. (Japan), and International Flavors & Fragrances Pty. Ltd. (Australia).

TABLE 296 INTERNATIONAL FLAVORS & FRAGRANCES INC.: BUSINESS OVERVIEW

INTERNATIONAL FLAVORS & FRAGRANCES INC.-2021	
Founded	1958
Country	US
City	New York, New York State
Total No. of Employees	24,000
R&D Expenditure	USD 629.0 Million
Overall Annual Revenue	USD 11.65 Billion
Ownership	Public (NYSE: IFF)

Source: Company Website and Annual Reports

FIGURE 58 INTERNATIONAL FLAVORS & FRAGRANCES INC.: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.3.2 Products/Solutions/Services offered

TABLE 297 INTERNATIONAL FLAVORS & FRAGRANCES INC.: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Pepper Sichuan Absolute CO2 Extract	Made from the seasonal crop of Pepper Sichuan berries, it gives a fresh and boosting effect	Herbs & Spices	- Oleoresins - Essential Oils	Cosmetics
Lavandin Enfleurage 2.0 Organic	The product is made by using only the floral head and extracted with 100% organic, vegetal origin solvents, which has a long-lasting fresh smell.	Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Cosmetics
GRINDSTED Pectin CF 120 PRM	GRINDSTED Pectin is a range of formulate plant-based supplements, sourced from citrus fruit	Fruits & Vegetables	- Oleoresins - Essential Oils	Dietary supplements
GRINDSTED				

Pectin CF 130 B PRM	peels and are used as supplements for providing consistent quality, short setting time. and uniform gel strength.			
GRINDSTED Pectin CF 140 B PRM				
GUARDIAN Rosemary Extract		Herbs & Spices	▪ Oleoresins ▪ Essential Oils	
GUARDIAN Green Tea Extract		Herbs & Spices	▪ Flavonoids ▪ Oleoresins ▪ Essential Oils	
GUARDIAN Toco (Tocopherol)		Natural plant extracts	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	
GUARDIAN 74 Blend of Rosemary & Green Tea Extracts Ginkgo	GUARDIAN Natural Extracts & Flavorants are plant extracts with flavoring and antioxidant properties to prevent rancid off-flavor development, decrease color change and contribute to the fresh flavor of foods throughout the shelf-life.	Herbs & Spices	▪ Oleoresins ▪ Essential Oils	Food & Beverages
GUARDIAN 81 Blend of Acerola & Green Tea Extracts		Herbs & Spices Fruits & Vegetables	▪ Flavonoids ▪ Oleoresins ▪ Essential Oils	
GUARDIAN 83 Blend of Acerola & Rosemary Extracts		Herbs & Spices Fruits & Vegetables	▪ Oleoresins ▪ Essential Oils	
GUARDIAN CHELOX L Blend of Chamomile & Rosemary Extracts		Herbs & Spices	▪ Oleoresins ▪ Essential Oils	
Curcugen	It is derived from turmeric oleoresin as the major ingredient with 50% curcuminoid concentration which has enhanced bioavailability and dispersion properties.	Herbs & Spices	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Dietary supplements

Source: Company Website

13.2.3.3 Recent developments

13.2.3.3.1 Deals

TABLE 298 INTERNATIONAL FLAVORS & FRAGRANCES INC.: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
February 2021	Merger	International Flavors & Fragrances Inc.	DuPont's Nutrition & Biosciences (Denmark)	DuPont's N&B merged with IFF, which supplies binders, plant-based proteins, and texturants which has increased the revenue exponentially and broadened the customer base for sustainable growth.	USD 7.3 billion
August 2020	Partnership	International Flavors & Fragrances, Inc.	DolCas Biotech (US)	IFF Health, a part of International Flavors & Fragrances Inc. and DolCas Biotech, partnered together on the sales and marketing of Curcugen, a next-gen, all-natural turmeric extract for both dietary supplements and food & beverage industries. It aims to increase the interest in curcumin and expand the activities in the market for nutraceuticals.	NA

Source: Company Website

13.2.3.4 MnM view

13.2.3.4.1 Right to win

The company has an array of product portfolios in the domains of taste, texture, scent, nutrition, enzymes, and cultures. It has its presence in major countries with manufacturing units set up to increase its customer base. The company has made strategic acquisitions and mergers with key players which have increased its global market share. It is into sustainability and responsible sourcing of ingredients for its products. IFF has been a member of the Upcycled Food Association (UFA) since 2019, with 10 products verified for the association body to create a positive impact on the environment.

13.2.3.4.2 Strategic choices

IFF is one of the largest companies in the flavors & fragrances industry. It holds a wide product portfolio and caters to clients across the food, beverages, and cosmetic industries. The major growth strategy adopted by the company in the natural extracts segment is merger. Its revenue has increased by 129% due to the company's strategic merger with DuPont's Nutrition & Biosciences. In addition, there was an increase in sales across the Nourish, Health & Biosciences and Scent operating units and increased prices in the Nourish segment. Owing to a strong product portfolio and strategic mergers, it has the potential to be one of the largest players in the flavor and fragrance industry.

13.2.3.4.3 Weaknesses and competitive threats

Cultivation of some seasonal plants can hamper the natural ingredients product sales, which might affect the company's profit margin. The company is expanding at a greater scale for which it needs to address the capital integrated with machinery and processing to meet the demand of the consumers. There is a drop in the percent of investment in R&D domain from 7.14% in 2020 to 5.40% in 2021, which may affect new product launches in the coming years.

13.2.4 GIVAUDAN

13.2.4.1 Business overview

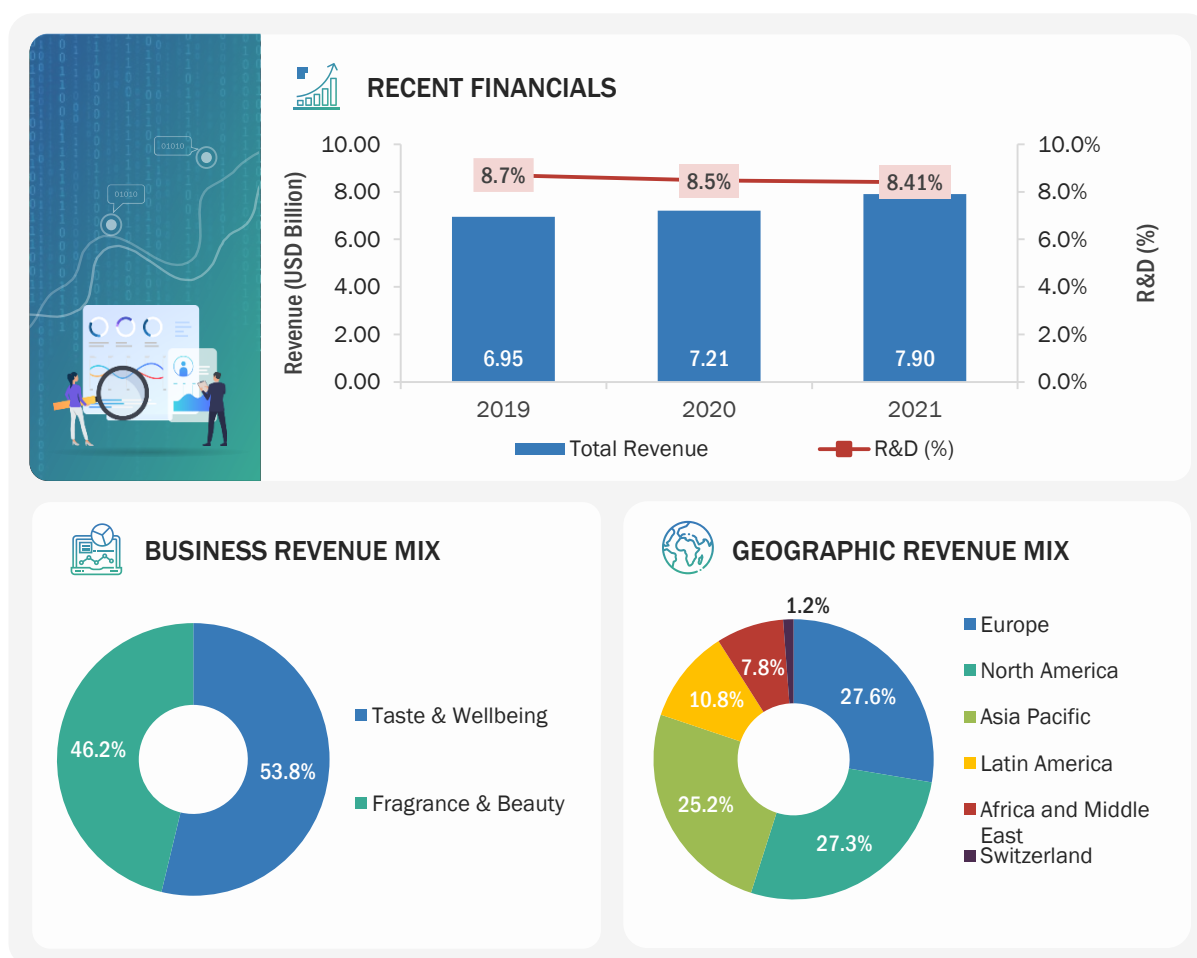
Givaudan is one of the largest players in the manufacturing and marketing of flavors & fragrances. It is involved in the manufacturing of botanical fragrance and flavors. The company operates through 2 segments: taste & wellbeing and fragrance & beauty. Its fragrance & beauty division offers fine fragrances, consumer products, active beauty products, and fragrance ingredients, whereas taste & wellbeing offers a wide range of natural and synthetic flavors used in beverages, dairy products, savory, snacks, and sweet goods. Flavors offered by the company include citrus, seafood, vanilla, and natural products. The company has 2,200 different strawberry flavors in its portfolio, which makes it a global leader in the strawberry flavoring market. With the acquisition of Naturex (France) in 2018, the company strengthened its product portfolio of plant extracts used in various applications, including food & beverages, cosmetics, personal care, and dietary supplements. It uses more than 60 varieties of plants to manufacture extracts for various applications.

Givaudan is present globally in 148 locations, manufacturing its products through 64 production sites. It has its presence in Asia Pacific, Europe, Africa, the Middle East, North America, and South America. Some of Givaudan's subsidiaries include Naturex (France), Givaudan France Fragrances SAS (France), Ungerer & Company Inc. (US), and Golden Frog Flavor and Fragrance (Vietnam).

TABLE 299 GIVAUDAN: BUSINESS OVERVIEW

GIVAUDAN-2021	
Founded	1895
Country	Switzerland
City	Zurich
Total No. Of Employees	16,842
Overall Annual Revenue	USD 7.90 Billion
Ownership	Public (SWX: GIVN)

Source: Company Website and Annual Reports

FIGURE 59 GIVAUDAN: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.4.2 Products/Solutions/Services offered

TABLE 300 GIVAUDAN: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Apricot oil	Apricot oil derived from kernels that were previously discarded during fruit preservation. Apricot oil hydrates and soothes the skin, as well as repairs the lips. Used in face care, lip care, hair care, anti-aging products, and baby massage oil	Fruits & Vegetables	- Flavonoids - Oleoresins - Essential Oils	Cosmetics
Commiphoroline	Commiphoroline is a resin extract derived from the Commiphora mukul tree, which grows in arid areas of Northwest India. It aids in the restoration of skin volume and the softening of wrinkles.	Tree	- Oleoresins - Essential Oils	Cosmetics
Depollutine	Depollutine is a natural product that combines a peptidic extract	Algae	Flavonoids	Cosmetics

	of a microalgae, <i>Phaeodactylum tricornutum</i> , and arginine with pyrrolidone carboxylic acid. It reduces chronic microinflammation caused by pollutant exposure.		▪ Oleoresins ▪ Essential Oils ▪ Others	
Glossyliance	Glossyliance is a blend of natural extracts (sugar cane and lemon zest) that are by-products of the food industry processed in accordance with Givaudan active beauty's philosophy. The product significantly improves hair shine and softness.	Herbs Fruits	▪ Flavonoids ▪ Alkaloids ▪ Essential Oils	Cosmetics
Green Tea Extract	Provides the best antioxidant activity; an excellent cosmetic active for well-aging	Herbs	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Dietary Supplements Cosmetics
Ginger Extract	It is extracted from <i>Zingiber Officinale</i> (Ginger). Perfect for regenerating and calming skin care products	Spices	▪ Oleoresins ▪ Essential Oils	Dietary Supplements Cosmetics
Ellagi-C	Ellagi-C is a natural ingredient derived from the bark of <i>Anogeissus leiocarpus</i> , an African tree. It assists the skin in regaining its youthful resilience, thereby delaying the onset of aging signs. Increases collagen production by stimulating vitamin C incorporation in fibroblasts	Tree (bark)	▪ Flavonoids ▪ Alkaloids	Cosmetics
BlurHD	BlurHD is derived from the fruit of the gardenia flower. It is a natural source of Crocin, a sugar derivative used as a natural food colorant in Japanese cuisine. Crocin has anti-aging properties that include wrinkle reduction and skin smoothing.	Fruits	▪ Flavonoids ▪ Carotenoids	▪ Food & Beverage ▪ Cosmetics
Mariliance	Mariliance, a powerful neuro-soothing active, was created by Givaudan marine biotechnology by extracting the essence of the red microalgae, <i>Rhodorus marinus</i> . It is suitable for use in neuro-cosmetic creams, soothing products, day care, and night care.	Herb	▪ Oleoresins ▪ Essential Oils	Cosmetics
Megassane	Megassane is a lipidic extract of microalgae of the <i>Diatomophyceae</i> order. It is found	Herb	▪ Oleoresins ▪ Essential Oils	COSMETICS

	in sunblock and skin lightening creams.			
Pongamia Extract	Pongamia oil is a strong natural antioxidant that is used in sun care and anti-stress treatments.	Plant (Seed)	▪ Oleoresins ▪ Essential Oils	▪ Pharmaceutical ▪ Dietary Supplements ▪ Cosmetics
RitualEssenz Brazil	The collection is made of three pure extracts Guarana, Mate, and Jambu. It is used in eye contour creams, slimming jellies cream, day and night creams, and shampoos.	Fruits Herbs	▪ Flavonoids ▪ Oleoresins ▪ Essential Oils	▪ Pharmaceutical ▪ Dietary Supplements ▪ Food & Beverages ▪ Cosmetics
RitualEssenz Brazilian Guarana Extract	Rich in caffeine (9%–11%), a much higher percentage than coffee beans	Plant (Seed)	Alkaloids	▪ Dietary supplements ▪ Cosmetics
Green Coffee Extract	Rich in phenolic and chlorogenic acids, green coffee is known for its topical anti-aging and firming applications, as well as a marked slimming action.	Fruits Seeds	Alkaloids	Dietary Supplements Cosmetics
Synchronight	Synchronight is a cosmetic active ingredient derived from gardenia fruit extract that is intended to protect the skin from digital stress. Anti-aging serum or cream, night cream, and well-being products contain this ingredient.	Fruits	▪ Flavonoids ▪ Essential Oils	Dietary supplements Cosmetics
Vetivine	Vetivine is an extract derived from the exhausted roots of vetiver, a fragrant plant. To increase fragrance longevity, it is used in serums for dry skin, anti-aging perilabial serum, skin tonifying mask, anti-acne cream, and body lotion.	Plant (Root)	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Dietary supplements Cosmetics
Ginkgo Biloba Extract	Cultivated in China's Jiangsu province and used in traditional Chinese medicine. It boosts vital energy and is an antioxidant.	Plant (leaf)	ESSENTIAL OILS	Dietary Supplements Cosmetics
Wakamine	Wakamine extract is of a brown macroalgae that inhibits tyrosinase activity, limiting melanin production. Its associated claims include whitening, brightening, and lightening.	Herb	▪ Oleoresins ▪ Essential Oils	Cosmetics
Uninontan	Uninontan is a plant-based skin	Fruits	▪ Oleoresins	Cosmetics

	lightener derived from lemon and cucumber. It reduces melanin synthesis.		▪ Essential Oil	
Patchouli Oil	Patchouli is used in fine fragrance and consumer products. The company has established a collection network in Indonesia, where it purchases patchouli directly from well-known local producers.	Plant (Flower)	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Cosmetics
Vallina Extract	The company offers a diverse range of vanilla profiles as well as cost-effective vanilla and vanilla-based natural flavor solutions. From extracts made from certified Madagascar vanilla beans to natural alternatives to ethyl-vanillin	Spice	▪ Oleoresins ▪ Essential Oils	▪ Food & Beverages ▪ Cosmetics
Albert Vieille Essential oil	Albert Vieille, which was acquired by Givaudan in 2019, offers a diverse range of essential oils, including basil essential oil, black pepper essential oil, carrot seed essential oil, clove bud essential oil, coriander seed essential oil, lemon essential oil, and rose essential oil.	Fruits & Vegetables Herbs & Spices Flowers	Essential Oils	▪ Food & Beverage ▪ Cosmetics

Source: Company Website

13.2.4.3 Recent developments

13.2.4.3.1 New product launches

TABLE 301 GIVAUDAN: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
February 2021	Givaudan	Software	Advanced Tools for Modelling (ATOM)	Givaudan Taste & Wellbeing introduced Advanced Tools for Modelling (ATOM), which uses artificial intelligence to optimize food and flavor formulation to generate new options and insights based on consumer preferences. The tools investigate ingredient synergies and identify flavor drivers that are both positive and negative. This will support the company's strategy and intent to lead the way in digitalization; these pioneering tools open new avenues for creative development and have the potential to shorten the time to market for new products.
October 2021	Givaudan	Software	DigiPulse	Givaudan introduced DigiPulse, the first olfactive social listening tool designed to collect and decode online spontaneous comments about fine fragrances from consumers. DigiPulse is a digitally strategic method of gathering feedback from online consumer posts. It

				collects consumer voices using a special algorithm that captures and analyzes their views with a unique focus on olfactory driven perceptions, which is powered by artificial intelligence (AI). Givaudan perfumers and creative teams use the decoded olfactive data to tailor their fragrance to the preferences of their customers. This would allow the company to better understand the needs of its customers.
March 2020	Givaudan	Fruit extract	Synchronight	Givaudan Active Beauty innovated Synchronight, an active cosmetic ingredient made from gardenia fruit extract designed to protect the skin from digital stress. This will help the company strengthen its skincare ingredients product portfolio.
November 2019	Givaudan	Botanical extract	RitualEssenz	RitualEssenz™ Brazil, a new collection inspired by Brazil's natural treasures, was introduced by Givaudan Active Beauty. The collection includes three new botanical extracts from Brazilian biodiversity: RitualEssenz™ Brazilian Jambu, RitualEssenz™ Brazilian Maté, and RitualEssenz™ Brazilian Leaf extract. The goal of this launch was to promote clean and natural beauty products.

Source: Company Website

13.2.4.3.2 Deals

TABLE 302 GIVAUDAN: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
December 2021	Acquisition	Givaudan	Custom Essence (Us)	Custom Essence was purchased by the company. It is a family-owned company that specializes in the creation of natural fragrances. The acquisition talks began in November 2021 and was completed in December 2021. This will strengthen Givaudan's access to local and regional customers while also expanding its natural perfumery capabilities in North America.	NA
October 2021	Acquisition	Givaudan	DDW The Color House (Us)	Givaudan acquired DDW, The color House, a natural color company based in the United States. Givaudan will become a global leader in natural colors because of the acquisition, which is also part of the company's strategy to expand its global taste and wellbeing business portfolio.	NA
July 2021	Acquisition	Givaudan	B.Kolar Makeup & Skincare S.P.A. (Italy)	Givaudan purchased a 25% stake in B.kolar makeup & skincare S.P.A. B.Kolar makeup & skincare S.P.A. is an Italian company that develops and manufactures end-to-end makeup and skincare products. This acquisition aims to strengthen Givaudan's position in skin care and cosmetics.	NA

February 2021	Acquisition	Givaudan	Myrissi (France)	Givaudan acquired Myrissi as part of its long-term strategy to expand its capabilities in artificial intelligence (AI). Myrissi has created a patented AI technology that can translate fragrances into color patterns and images that are relevant to the consumer, as well as predict their emotional response. Myrissi's AI expertise will help the company propose new visual and verbal storytelling approaches to customers.	NA
May 2020	Acquisition	Givaudan	Indena (Italy)	As part of its long-term strategy to expand the capabilities of its active beauty business, Givaudan completed the acquisition of the cosmetics business of Indena (Italy).	NA
February 2020	Acquisition	Givaudan	Ungerer & Company (Us)	Givaudan completed the acquisition of Ungerer & Company, a US-based flavor, fragrance, and specialty ingredients company. This will help expand its product portfolio.	NA
March 2020	Acquisition	Givaudan	Alderys (France)	Givaudan acquired Alderys to expand its bioengineering capabilities. Alderys is a French biotechnology firm that pioneers novel approaches to the biological engineering of valuable compounds from renewable feedstock. furthermore, the acquisition will allow Givaudan to diversify its product portfolio with natural and bio-sourced ingredients.	NA
September 2019	Acquisition	Givaudan	Drom Fragrances (Germany)	Givaudan bought Drom Fragrances. The acquisition was completed in September 2019, after negotiations began in July 2019. this global perfume house develops fragrances for consumer goods and fine fragrance customers worldwide. Drom has four production sites in Germany, China, the US, and Brazil. This strengthens Givaudan's position in the global fragrance market.	NA
May 2019	Acquisition	Givaudan	Golden Frog (Vietnam)	Givaudan signed an agreement to acquire golden frog (Vietnam) to strengthen its product portfolio in natural flavors, extracts, and essential oils for the food & beverage industry. The company caters to the ASEAN market with its Vietnam-originated natural ingredients, including herbs, spices, fruit & vegetable extracts, and essential oils.	NA
May 2019	Acquisition	Givaudan	Albert Vieille (France)	Albert Vieille, a French company specializing in natural ingredients used in the fragrance and aromatherapy markets, was acquired by Givaudan. This acquisition will help the company to expand its leadership in the naturals space while also strengthening its presence in Grasse, France.	NA

Source: Company Website

13.2.4.4 Others

TABLE 303 GIVAUDAN: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
November 2021	Givaudan	Expansion	Digital space	Givaudan opened a new digital space in China, one of the most digitalized markets, as part of its long-term strategy to expand its digital capabilities. It will enable the company to capitalize on transformation opportunities while also supporting growth with local and regional customers.
July 2021	Givaudan	Expansion	Online b-2-b store	Givaudan expanded its e-commerce presence in China by launching its flagship business-to-business (B2B) store on China's leading wholesale website, 1688.com. The business-to-business collaboration is a first for Givaudan in Asia and for Givaudan fragrance & beauty globally. Givaudan's partnership with 1688.com will broaden its reach among China's growing number of local, regional, and niche brands.
October 2020	Givaudan	Investment	Production facility	Givaudan opened its newest state-of-the-art fragrance production facility in Changzhou, China. In line with the recently announced 2025 strategy – committed to growth, with purpose – China remains a key growth market for innovative and consumer-preferred fragranced products.
June 2019	Givaudan	Investment	Innovation center	Givaudan opened a new flagship innovation center in Switzerland. The workspace is 12,000 square meters and is aimed to create differentiated and sustainable flavor, taste, and fragrance solutions for the food & beverage, beauty, personal care, and home care industries.
January 2019	Givaudan	Expansion	Flavors technical and commercial center	Givaudan opened a new flavors technical and commercial center in Casablanca, Morocco. The new center was designated to meet the evolving needs of its African customers with innovative taste and flavor solutions.
November 2021	Givaudan	Expansion	Digital space	Givaudan opened a new digital space in China, one of the most digitalized markets, as part of its long-term strategy to expand digital capabilities. It will enable the company to capitalize on transformation opportunities while also supporting growth with local and regional customers.

Source: Company Website

13.2.4.5 MnM view

13.2.4.5.1 Right to win

To maintain its market position, the company focuses on sustainable and profitable expansion. As a part of its expansion strategy to enter highly desirable markets like China, it strengthened its e-commerce offering in the country by opening a flagship business-to-business (B2B) store on 1688.com, China's leading wholesale website. It also opened a new digital space to drive innovation and growth with its Chinese customers.

13.2.4.5.2 Strategic choices

The company has a wide geographic network. It has strong R&D investments. It offers a wide range of flavorings for various culinary applications. However, the company has limited new product launches for its flavor division compared to its fragrance division. Acquisitions and agreements are the key strategies adopted by the company to grow in the market. For instance, it acquired Custom Essence. It is a family-owned company that specializes in the creation of natural fragrances. This acquisition will strengthen Givaudan's access to local and regional customers while also expanding its natural perfumery capabilities in North America.

13.2.4.5.3 Weaknesses and competitive threats

Givaudan faces a threat from competitors, both existing and new market entrants. Climate change may have several consequences like inconsistent supply of raw material, affecting its ability to operate or incurring additional costs. The threat of cyber-attacks has also grown significantly, making information security a critical component of business resilience.

13.2.5 ADM

13.2.5.1 Business overview

ADM is a food manufacturing company that is primarily involved in the production of food & beverage ingredients, feed & feed ingredients, industrial ingredients, biofuels, and naturally derived alternatives to industrial chemicals. The company operates through four business segments, namely, Ag services & oilseeds, carbohydrate solutions, nutrition, and others. It offers plant extracts under its nutrition segment.

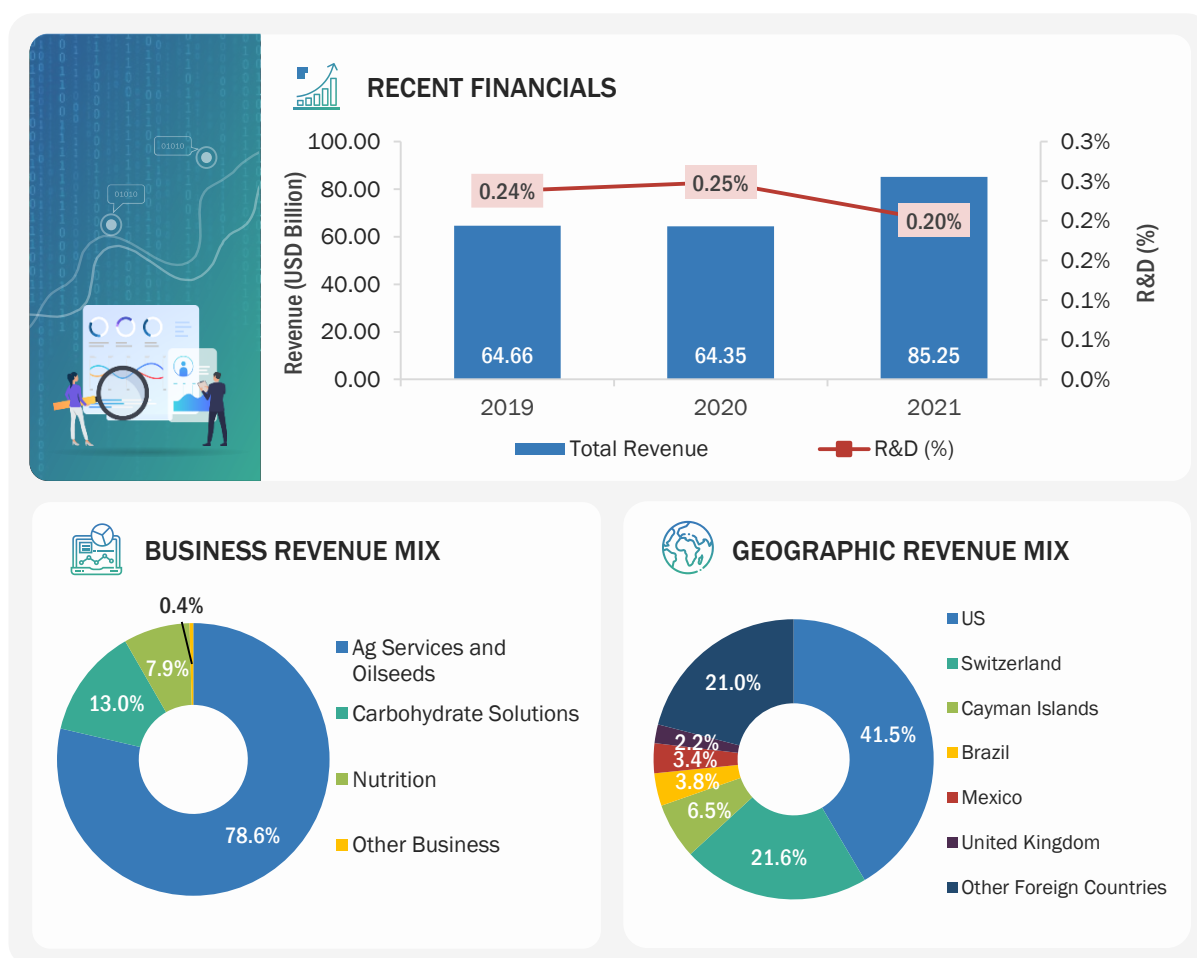
The company's nutrition segment is involved in the manufacturing, sales, and distribution of a wide array of products from nature, including plant-based proteins, natural flavor ingredients, flavor systems, natural colors, emulsifiers, soluble fiber, polyols, hydrocolloids, natural health, and nutrition products, including probiotics, prebiotics, enzymes, and botanical extracts, and other specialty food and feed ingredients. It offers a range of natural food flavors and colors, emulsifiers and stabilizers, specialty starches, and sweeteners.

The company operates through 46 innovation centers, 271 processing plants, 450 crop procurement facilities, and 200 bulk storage facilities in nearly 200 countries across regions. ADM operates through numerous subsidiaries, such as Ab Mauri (UK), ABF Ingredients (UK), ABITEC Corporation (US), Ohly (Germany), Golden Peanut Company LLC (US), ADM Milling (US), Wild Flavors, Inc. (US), and ADM Hamburg Ag (Germany).

TABLE 304 ADM: BUSINESS OVERVIEW

ADM-2021	
Founded	1865
Country	US
City	Chicago, Illinois
Total No. Of Employees	41,000
Overall Annual Revenue	USD 85.25 Billion
Ownership	Public (NYSE: ADM)

Source: Company Website and Annual Reports

FIGURE 60 ADM: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.5.2 Products/Solutions/Services offered

TABLE 305 ADM: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
NaturePlus Range	The NaturePlus range consists of acerola, anthocyanin-focused extract blend, beta-carotene-focused extract blend, black, white, green tea with standardized polyphenol concentrations, guarana, and yerba mate.	Fruits Herbs	- Carotenoids - Alkaloids	Food & Beverages Dietary Supplements
NatureEdge Range	NatureEdge portfolio contains selected botanical extracts.	-	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Food & Beverages Dietary Supplements
NatureSource Range	NatureSource products are designed to meet consumers' desires for fruit and vegetable-forward products.	Fruits & Vegetables	- Flavonoids - Alkaloids	Food & Beverages Dietary Supplements

Tea Extract	The company provides certified tea extract options, which serve as a delicious base for a variety of tea beverages.	Herbs	Alkaloids	Food & Beverages
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Source: Company Website

13.2.5.3 Recent developments

13.2.5.3.1 Deals

TABLE 306 ADM: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
December 2021	Acquisition	ADM	Flavor Infusion International, S.A. (FISA) (Panama)	Flavor Infusion International, S.A. (FISA), a provider of flavor and specialty ingredient solutions in Latin America and the Caribbean, was acquired by ADM the addition of FISA will provide ADM with new growth opportunities in Latin America and the Caribbean.	NA
September 2021	Acquisition	ADM	Petdine, Pedigree Ovens, The Pound Bakery and Nutradine (P4 Companies) (US)	ADM Purchased A 75% Stake in Petdine, Pedigree ovens, the pound bakery, and Nutradine. This will allow both companies to better meet the growing global demand for high-quality, nutritious pet treats, and supplements.	USD 600 Billion
January 2020	Acquisition	ADM	Yerbalatina Phytoactives (Brazil)	ADM acquired Yerbalatina Phytoactives, a Brazilian manufacturer of plant-based extracts and ingredients. To stay ahead of health and wellness trends, ADM is expanding its portfolio of ingredients derived from natural sources that promote health. This acquisition expands ADM's already substantial presence in Brazil.	NA

Source: Company Website

13.2.5.4 MnM view

13.2.5.4.1 Right to win

The company invests on R&D as seen by the USD 171 million in R&D expenses for the fiscal year ended December 31, 2021. It also has a large, consolidated network of subsidiaries spread across 70 countries. The majority of the company's subsidiaries are based outside the US. The company has built a complete transformation competence that allows it to efficiently carry raw materials and processed goods. Thousands of producers, grain elevators, and wholesale merchants across North America, South America, EMEA, Asia, and Australia provide the company with raw materials.

13.2.5.4.2 Strategic choices

The company has been focusing on various collaborations and acquisitions to strengthen its market position. It has collaborated with various firms to strengthen its product portfolio. To be competitive, ADM works on controlling unit costs and increasing efficiency through technological advancements, productivity increases, and regular asset portfolio evaluations. ADM aims to increase its footprint in most countries to develop a significant global network by adopting “acquisition” as its key strategy. The company wants to help the environment by developing renewable products and processes. ADM's strategy plan for long-term growth takes advantage of sustainability trends and technologies to help it thrive and create value for its stakeholders.

13.2.5.4.3 Weaknesses and competitive threats

ADM has a strong presence in North America and Latin America. Thus, it faces strong competition from other players in the regions. However, it has a limited presence in Asia Pacific, with a strong potential for plant extracts. In the markets where it competes, the company faces strong competition driven by price, foreign currency rates, quality, worldwide supply, and alternative goods; some of which are created from different raw materials than those used by the company.

13.2.6 CARGILL, INCORPORATED

13.2.6.1 Business overview

Cargill, Incorporated is one of the largest globally diversified services companies, catering to the demand in various industries. The company has its global operations in major key segments: Food Ingredients & Bio-Industrial, Animal Nutrition, Protein & Salt, Agricultural Supply Chain, and Metals & Shipping. It offers essential oil products under its animal nutrition segment to support animal health and performance.

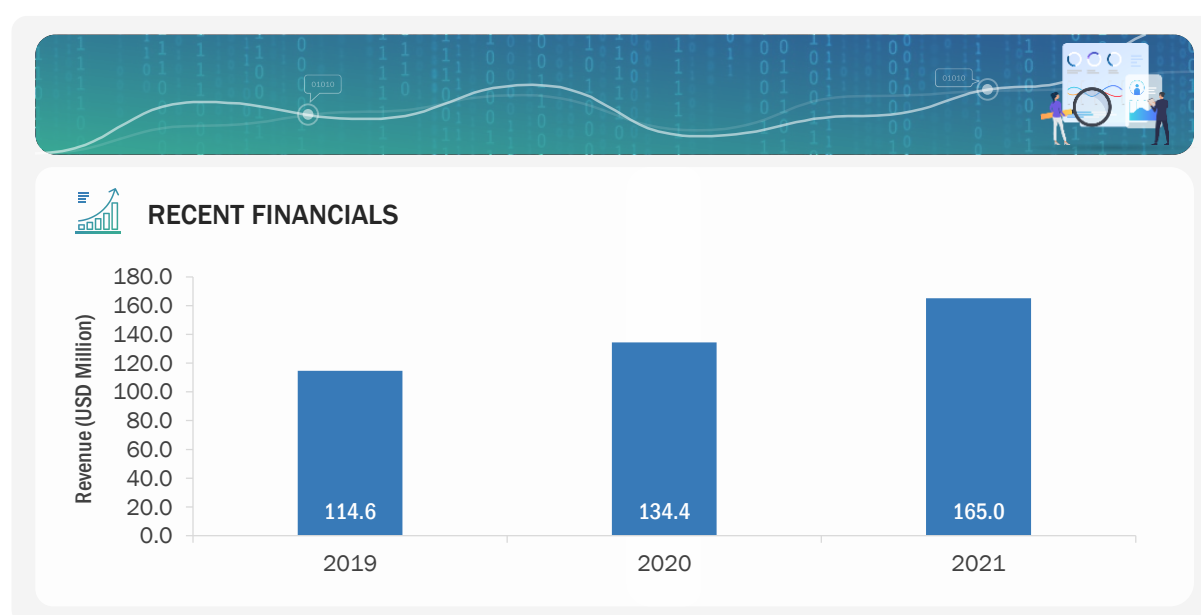
The company operates in 70 countries and has a sales presence in nearly 125 countries across North America, Latin America, Asia Pacific, Europe, the Middle East, and Africa. Some of the subsidiaries of the company include Cargill Limited (Canada), Cargill Enterprises Inc. (Russia), Cargill Asia Pacific Holdings Pte Limited (Singapore), Cargill Investments (China) Ltd, Cargill España SA (Spain), and Cargill Nordic A/S (Sweden). With R&D presence over 200 locations, it provides a spectrum of services encompassing technical service, applications, development, research, intellectual asset management, and scientific and regulatory affairs.

TABLE 307 CARGILL, INCORPORATED: BUSINESS OVERVIEW

CARGILL, INCORPORATED-2022	
Founded	1865
Country	US
City	Wayzata, Minnesota
Total No. Of Employees	155,000
Overall Revenue	USD 165 Billion
Ownership	Private

Source: Company Website

FIGURE 61 CARGILL, INCORPORATED: COMPANY SNAPSHOT



Notes: The business revenue and geographic segmentation were unavailable for the company.

Source: Company Website and Company Publications

13.2.6.2 Products/Solutions/Services offered

TABLE 308 CARGILL, INCORPORATED: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	TYPE	APPLICATION
Cinergy FIT 3S	It is beneficial for the gut health of swine, improves the animal performance along with growth and feed intake enhancement.	Blend of stabilized essential oil compounds	Swine Feed Additive

Source: Company Website

13.2.6.3 Recent developments

13.2.6.3.1 New product launches

TABLE 309 CARGILL, INCORPORATED: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
January 2022	Cargill, Incorporated	Essential oils from oregano, thyme, rosemary, and star anise	Nutrena NatureWise	Essential oils from oregano, thyme, rosemary, and star anise, maximize birds' laying performance and reduce ammonia emissions. This increases the customer base with a preference for natural ingredients in the poultry feed.
October 2021	Cargill, Incorporated	Essential oils from oregano and other herbs	Nutrena NatureWise	Cargill introduced improved Nutrena NatureWise formulas that improve the overall wellness and productivity of chickens. This poultry feed contains essential oils (oregano and other herbs) that promote flock health and egg production. Furthermore, the feed contains Yucca schidigera extract to help with manure odor control. This launch aided the company expands its product portfolio and also helped meet its customers' demand for wholesome feed.

Source: Company Website

13.2.6.4 MnM view

13.2.6.4.1 Right to win

Cargill, Incorporated has a dynamic product portfolio, which has been strengthening its market position with wide applicability. With the company's regional and local presence, it provides its customers with applications and technical service support to offer products tailored to the local needs. The company is keen on improving its product quality by understanding and conducting research on animal health in recent years by which it has introduced new formulations for poultry under its brand, Nutrena NatureWise. The company has also adopted advanced technologies to help improve the product portfolio of essential oils. It mainly focuses on developing innovative and new products and solutions to meet the requirements of the food industry.

13.2.6.4.2 Strategic choices

The company has a strong brand image, which enables it to strengthen its position in the market. There is a 23% increase in its revenue since the last year which is the result of its strategic investments along with continuously upgrading business-enabling capabilities. It focuses on tapping the rural and local markets of emerging economies to increase its global presence. The company has a sustainability strategy focused on climate, land & water, and people, many of which provide multiple environmental and social benefits. Cargill focuses on becoming a key supplier in the global oils market, which will also help the company penetrate further into the market.

13.2.6.4.3 Weaknesses and competitive threats

Cargill, Incorporated has minimal product offering in the essential oil segment within the plant extracts market in comparison to other leaders in the market due to which the company is facing tough competition. It has integrated technology in the backend processes but is still not able to harness the power of technology in the front-end processes, hence lacking to launch more products in the market. There are no significant acquisitions, partnerships, collaborations, and alliances to discern its growth in the products offered.

13.2.7 DSM

13.2.7.1 Business overview

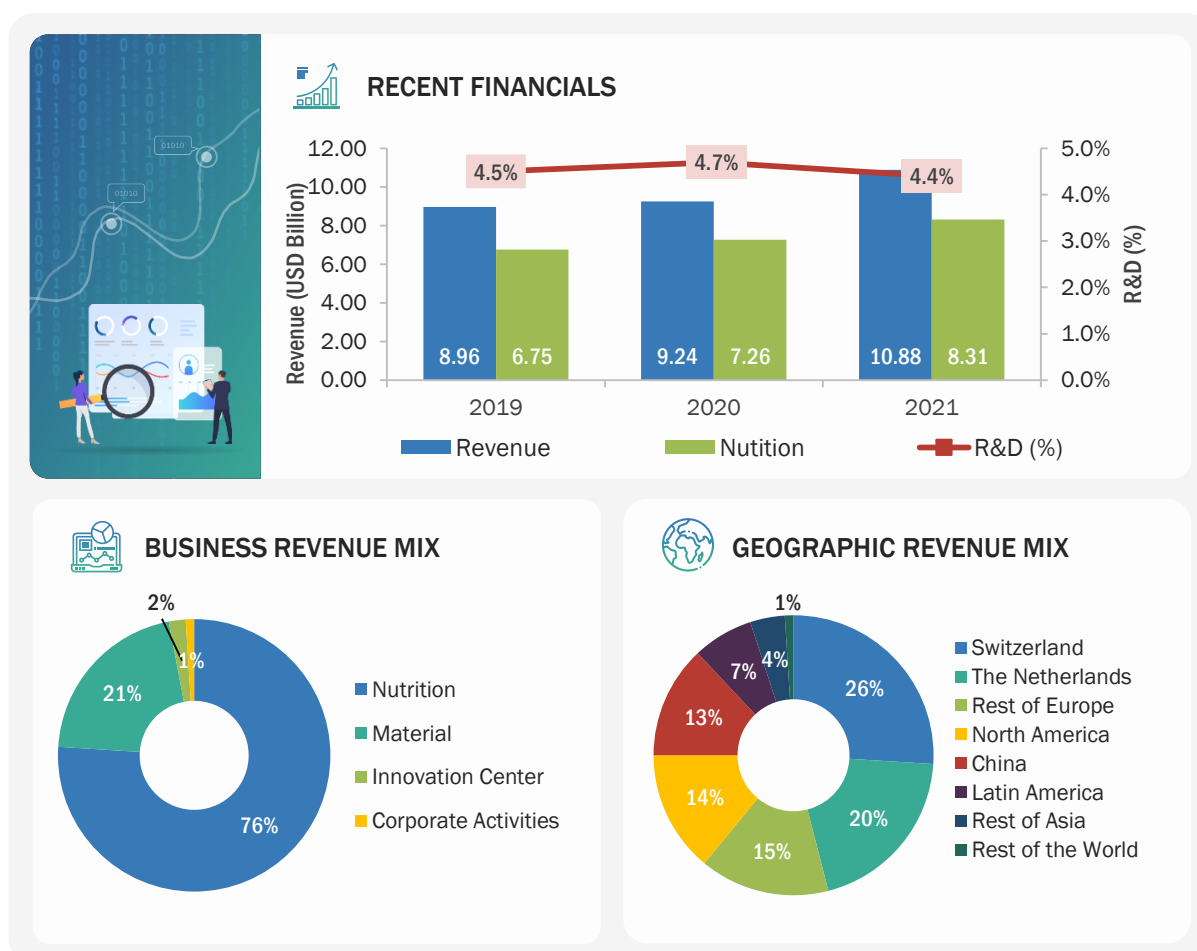
DSM is a global science-based company engaged in the health, nutrition, and materials businesses. The company operates through four segments: nutrition, materials, innovation center, and corporate activities. It offers functional food ingredients through its nutrition segment. The nutrition segment of the company includes the following business units: animal nutrition & health, human nutrition & health, and personal care. Plant extracts are provided under human nutrition & health. DSM and Firmenich announced that they have entered into a business combination agreement for a merger to establish a creation and innovation partner in nutrition, beauty, and well-being.

The company has its presence in Austria, China, Japan, the Netherlands and South Africa. Its production sites are located in Grenzach, Jiangshan, Maroussi, Sisseln, and Village-Neuf. The subsidiaries of the company include DSM Services Inc. (US), DSM Holding Company Inc. (US), DSM Limited (China), and DSM Food Specialties (US). The key competitors of the company are Cargill (US) and DuPont (US).

TABLE 310 DSM: BUSINESS OVERVIEW

DSM-2021	
Founded	1902
Country	Netherlands
City	Heerlen
Total No. Of Employees	21,358
Overall Annual Revenue	USD 10.88 Billion
Ownership	Public (ASM: DSM)

Source: Company Website, Company Publications, and Company Annual Reports

FIGURE 62 DSM: COMPANY SNAPSHOT

Source: Company Website, Company Publications, and Company Annual Reports

13.2.7.2 Products/Solutions/Services offered

TABLE 311 DSM: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
CRINA Poultry Plus	Improves gut performance in poultry	Herbs & Spices	- Oleoresins - Essential Oils	Poultry Feed
CRINA Poultry	Stimulates the production of digestive enzymes and improves the balance of gut microflora	Herbs & Spices	- Oleoresins - Essential Oils	Poultry Feed
CRINA Digest	CRINA Digest ensures gradual diffusion of the essential oil actives along the gastrointestinal tract for better digestion.	Herbs & Spices	- Oleoresins - Essential Oils	Poultry Feed
ROVIMIX® β-Carotene	It is used as a supplement in feed.	Fruits & Vegetables	Carotenoids	Feed

CAROPHYLL	This variety of carotenoid additives enables manufacturers to reliably and consistently deliver precisely colored and pigmented food.	Fruits & Vegetables	Carotenoids	Food & Beverages Feed
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Source: Company Website

13.2.7.3 Recent developments

13.2.7.3.1 Deals

TABLE 312 DSM: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
May 2022	Merger	DSM	Firmenich SA	In May 2022, DSM and Firmenich announced that they entered into a business combination agreement for a merger to establish a creation and innovation partner in nutrition, beauty, and well-being. The DSM nutrition, taste, texture, and fragrance segments will be leveraged in this merger and will continue to foster innovation in sectors with strong growth potential.	USD 21 Billion

Source: Company Website

13.2.7.4 MnM view

The company has a strong product portfolio on a global scale; DSM has one of the broadest portfolios of highly competitive formulations and active ingredients. It should focus on adopting strategies such as mergers & acquisitions, partnerships, and collaborations, which will help strengthen its foothold in the market. Moreover, the company must expand its offerings in other applications.

13.2.8 KERRY GROUP PLC

13.2.8.1 Business overview

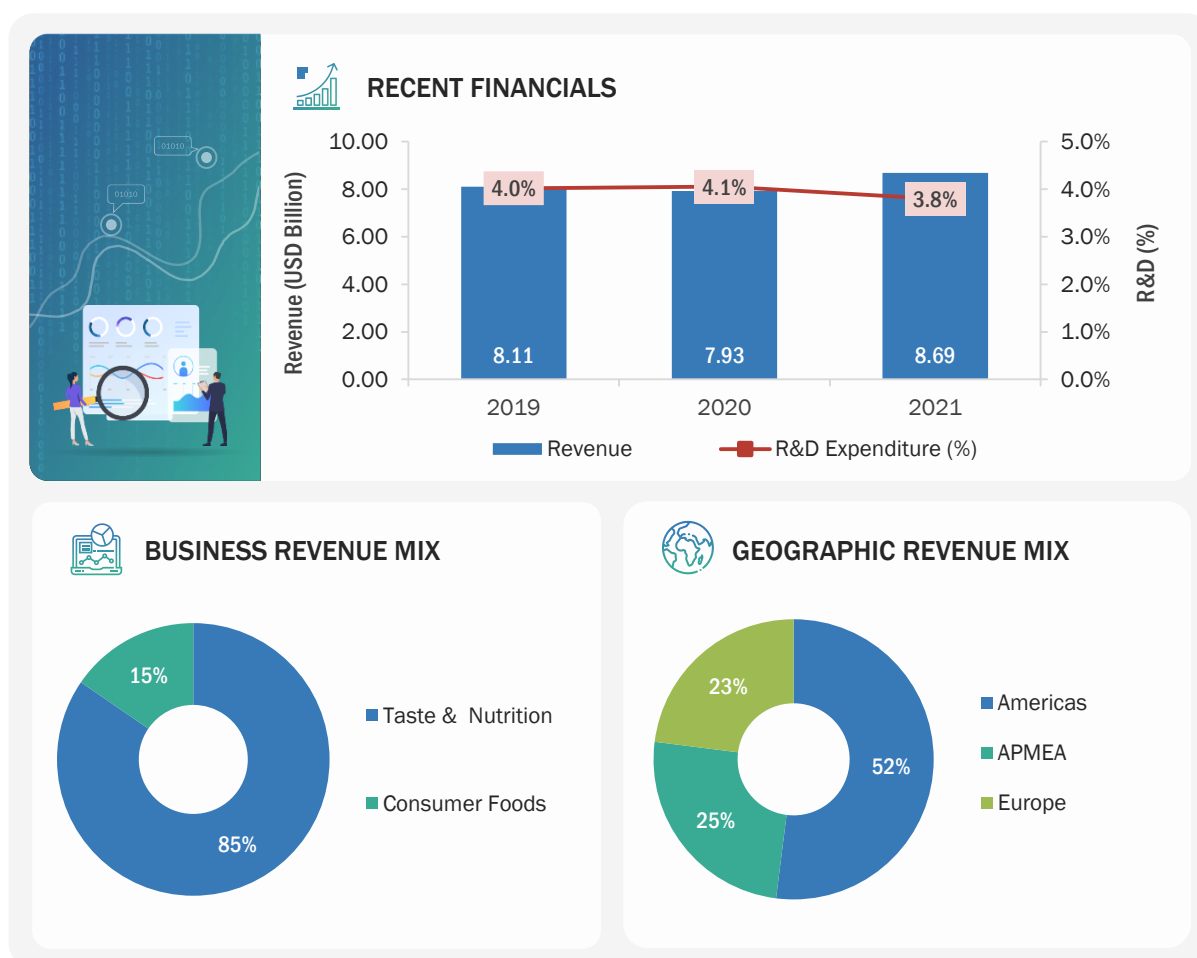
Kerry Group is the world's leading taste and nutrition company, serving the food, beverage, and pharmaceutical industries. Initially a small dairy company, it now operates through two business segments, namely, taste & nutrition and consumer foods. Through its taste & nutrition segment, Kerry focuses on meals and beverages made from trustworthy ingredients, while its consumer foods segment is mainly focused on supplying chilled and frozen food products. Its vision is to be the customers' most valued partner, creating a world of sustainable nutrition.

Kerry Group has its manufacturing units in 32 countries. The company sells its products across 200 countries, such as Singapore, Brazil, Mexico, Ireland, and the UK.

TABLE 313 KERRY GROUP PLC: BUSINESS OVERVIEW

KERRY GROUP PLC-2021	
Founded	1972
Country	Ireland
City	Tralee
Total No. Of Employees	25,195
Overall Annual Revenue	USD 8.69 Billion
Ownership	Public (LON: KYGA)

Source: Company Website and Annual Reports

FIGURE 63 KERRY GROUP PLC: COMPANY SNAPSHOT

Source: Company Website, Company Publications, Annual Reports, and Press Releases

13.2.8.2 Products/Solutions/Services offered

TABLE 314 KERRY GROUP PLC: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Simply Nature Cocoa Extracts and Chocolate Flavors	A comprehensive range of cocoa and chocolate extracts for beverage, dairy, bakery, and savory applications	Fruits	Alkaloids	Food & Beverages
Simply Nature Vanilla	Kerry obtains premium vanilla beans from key growing regions such as Madagascar, Mexico, Papua New Guinea, and Tahiti.	Herbs & Spices	- Oleoresins - Essential Oils	Food & Beverages
Simply Nature Coffee Extracts and Flavors	The range includes cold brew and hot brew coffee extracts, aromatic coffee extracts, natural coffee flavorings, and coffee flavorings.	Fruits & Vegetables	Alkaloids	Food & Beverages

Simply Nature Botanicals	Florals, herbs, roots and barks, spices, fruits and vegetables, teas, and blends are among the options available in the botanical range.	▪ Fruits & Vegetables ▪ Herbs & Spices ▪ Flowers	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Food & Beverages
Simply Nature Citrus	Kerry natural citrus ingredients include oils, fractions, aromas, specialty ingredients, emulsions, natural flavors, and extracts.	Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids	Food & Beverages

Source: Company Website

13.2.8.3 Recent developments

13.2.8.3.1 Deals

TABLE 315 KERRY GROUP PLC: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
April 2022	Acquisition	Kerry Group Plc	Natreon (US)	Kerry completed the acquisition of Natreon, a leading supplier of branded Ayurvedic botanical ingredients based in the US. Kerry's leadership position and the ProActive Health portfolio of science-backed branded ingredients of Kerry are significantly expanded as a result of the acquisition, which accelerates the company's technological growth.	NA
February 2022	Acquisition	Kerry Group Plc	Enmex (Mexico)	Enmex is a well-known enzyme manufacturer based in Mexico that offers a wide range of bio-process solutions for the food, beverage, and animal nutrition industries. Enmex, which has a long history of partnering with global customers, has a complementary enzyme portfolio and a strong manufacturing infrastructure, which will allow Kerry to expand its fermentation and enzyme manufacturing capabilities into Latin America.	NA

Source: Company Website

13.2.8.4 Others

TABLE 316 KERRY GROUP PLC: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
March 2022	Kerry Group Plc	Expansion	New e-Commerce platform	Kerry launched a new e-commerce platform, KerrySelect.com, in North America, featuring a targeted portfolio of Kerry products. KerrySelect.com's portfolio includes dairy powders and sauces, edible decorations, seasoning blends, coatings, beverage mixes, flavor ingredients, and emulsifiers. It will help the company increase its market presence in North America.

Source: Company Website

13.2.8.5 MnM view

In the past few years, the company has acquired various other businesses like Natreon (US) to strengthen its position in the market and increase its global presence. The company is committed to long-term viability. It is dedicated to reducing its total carbon footprint and reaching net zero emissions by 2050. The company faces pressure from market rivals and has a small market share in Asia.

13.2.9 DÖHLER

13.2.9.1 Business overview

Döhler is a global producer, marketer, and provider of technology-based natural ingredients, ingredient systems, and integrated solutions for the food & beverage industry. The company's product portfolio ranges from flavors, colors, health & nutrition ingredients, cereal ingredients, dairy ingredients, specialty ingredients, and fruit & vegetable ingredients to ingredient systems.

Döhler is a global food business group that includes the following companies: Africa Juice, Bevisol, Fruit Tech Natural, Juviar, Natural Origins, Nutrafood, Ruland Engineering, and Thabico. Natural Origins provides a comprehensive range of natural plant-based ingredients (extracts, bee products, herbs, and powders) to the food supplements or nutraceutical, food and beverage, teas, herbal infusions, and spirits & beers markets.

Döhler operates globally through its 23 production sites, 50 sales offices, and application centers as well as sales activities in over 130 countries. It offers products and solutions in over 160 countries. Its operations are spread across North America, Latin America, Europe, the Middle East & Africa, and the Asia Pacific.

TABLE 317 DÖHLER: BUSINESS OVERVIEW

DÖHLER-2021	
Founded	1838
Country	Germany
City	Dortmund
Ownership	Private

Source: Company Website

13.2.9.2 Products/Solutions/Services offered

TABLE 318 DÖHLER: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Botanical Extracts	Oregano, coriander, rosemary, caraway, sage, chili, pepper, thyme, basil, and ginger	Herbs & Spices	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Food & Beverages
Coffee Concentrated Extracts	Ideal for a variety of food and beverage applications	Fruits	- Flavonoids - Alkaloids	Food & Beverages
Tea Concentrated Extracts and Tea Powder Extracts	Black tea, rooibos, guayusa, green tea, mate, herbal tea, white tea, matcha, and fruit tea	Herbs & Spices	- Oleoresins - Essential Oils	Food & Beverages

Citrus Taste Solutions	The portfolio includes terpenes, extracts, distilled oils, and cold pressed oils.	Herbs & Spices	▪ Oleoresins ▪ Essential Oils	Food & Beverages
Natural Flavors & Extracts	Lotus, lemon, citrus, rosemary, apple, red fruits, vanilla, tea, coffee cherries	Fruits Herbs & Spices	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Food & Beverages
African Botanical Extracts	Aloe vera, baobab, hibiscus, rosemary, honey bush, hoodia, irvingia, kigelia, moringa, marjoram, vanilla, rosehip, rooibos, sutherlandia, basil, kola nut, and ginger	Herbs & Spices	▪ Flavonoids ▪ Alkaloids ▪ Oleoresins ▪ Essential Oils	Food & beverages Cosmetics
Extracts by Natural Origins	The product folio includes extracts like organic lavender flower, organic echinacea purpurea aerial, organic acerola, and organic white ginseng root.	Flowers Herbs & Spices Fruits	▪ Oleoresins ▪ Essential Oils	Food & beverages Cosmetics
Natural Color Solutions	Fruits, vegetables, and other naturally colored plants and raw materials are the sources of natural colors. A clean label declaration can be made by manufacturers with the help of this product.	Fruits & Vegetables		Food & beverages Cosmetics

Source: Company Website

13.2.9.3 Recent developments

13.2.9.3.1 Deals

TABLE 319 DÖHLER: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
April 2021	Partnership	Döhler	Flavourtech (Australia)	As part of its expansion strategy, Döhler partnered with Flavourtech. This collaboration aims to provide high-quality products to its customers. Flavourtech is a global leader in the production of aroma, concentrates, and extracts.	NA

Source: Company Website

13.2.9.4 MnM view

Döhler is one of the major competitors in the flavor and ingredient space, offering customized products and service solutions to all its clients. The company has very strategically acquired companies, such as in 2016, Teawolf (US), to target inaccessible audience and enhance its product portfolio of natural ingredients and plant-based extracts. In 2021, it partnered with Flavourtech; this collaboration aims to provide the best products to its customers. The company has a wide geographical reach in North America, Latin America, Europe, the Middle East & Africa, and the Asia Pacific, which has helped it to be in tandem with competitors.

13.2.10 CARBERY GROUP

13.2.10.1 Business overview

Carbery Group is a global nutrition and dairy company. It operates under three business segments: dairy, nutrition, and taste. Plant extracts are provided under the taste segment.

Synergy Flavors (US) operates as a subsidiary of Carbery Group. Synergy Flavors is one of the leading suppliers of flavors, extracts, and essences. End users for its products include the food & beverage industry. The company offers coffee, tea, botanical, herbal, fruit, and vegetable essences and extracts. Its products find applications in nutritional, dairy, bakery, confectionery, savory products, and beverages.

TABLE 320 CARBERY GROUP: BUSINESS OVERVIEW

CARBERY GROUP-2021	
Founded	1965
Country	Ireland
City	Cork
Total No. Of Employees	882
Overall Annual Revenue	USD 633.2 Million
Ownership	Private

Source: Company Website

13.2.10.2 Products/Solutions/Services offered

TABLE 321 CARBERY GROUP: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Synergy Pure Coffee Extracts and Essences	Organic, non-GMO product verified, and decaffeinated versions available	Fruits	Alkaloids	Food & Beverages
Synergy Herbs & Botanicals	It includes product portfolio like basil, dill, hibiscus, lemongrass, and peppermint.	- Herbs & Spices - Flowers	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Food & Beverages
Synergy Tea Extracts and Essences	Authentic flavor and aroma; suitable for clean label requirements. It includes black tea, green tea, white tea, and rooibos tea.	Herbs & Spices	- Flavonoids - Alkaloids - Oleoresins - Essential Oils	Food & Beverages
Vanilla Extract	Vanilla flavoring solutions for the food, beverage, and nutrition industries	Herbs & Spices	- Oleoresins - Essential Oils	Food & Beverages

Fruit & Vegetable Extracts	The range includes watermelon, cucumber, and tomato	Fruit & Vegetables	- Flavonoids - Alkaloids	Food & Beverages
Synergy Hop	True-to-nature taste, label-friendly, easy-to-use format for brewers	Flowers	- Flavonoids - Alkaloids - Essential Oils	Food & Beverages

Source: Company Website

13.2.10.3 Recent developments

13.2.10.3.1 Deals

TABLE 322 CARBERY GROUP: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
May 2021	Acquisition	Carbery Group	Innova Flavors (US)	Carbery Group acquired Innova Flavors, a supplier of savory flavors and ingredients. The company will be part of Carbery's international flavors business, Synergy. This strategic acquisition broadens synergy US' savory capabilities and complements Synergy's global offerings.	NA

Source: Company Website

13.2.10.4 Others

TABLE 323 CARBERY GROUP: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
January 2022	Carbery Group	Expansion	Opening of new R&D center in Illinois, US	The company opened a new R&D facility in Synergy headquarters. The expansion includes space for nutrition laboratory, beverage application space, beverage thermal processing pilot plant, flavor creation lab, and for flavor research and technology lab. This will boost the company's research and help launch new products.

Source: Company Website

13.2.10.5 MnM view

The acquisition of Innova Flavors expanded the company's capabilities and savory flavor range, creating new opportunities and new avenues into new markets. The company continues to take steps toward sustainability, such as the Monitor Farms and SignPost Farms projects, which are both aimed at reducing farming's carbon footprint. The company can expand its product offerings to other applications as well.

13.2.11 SYNTHITE INDUSTRIES LTD

13.2.11.1 Business overview

Synthite Industries Ltd. is India's largest manufacturer and exporter of spice oleoresin and essential oils. The company's product portfolio has value-added spice extracts (oleoresins), essential oils, lutein and marigold, green spice extracts, herbal extracts, and tea & coffee extracts. These products are used in liquid seasoning enhancers, flavors, and perfumery ingredients. The product also comes in different forms like liquid, powder, granules, and encapsulated.

The company has a wide presence with its products supplied in approximately 11 countries like Australia, China, and the US. Some of the company brands include SPRIG, Necol, Neaox, NatXtra, Kitchen Treasures, and Straight Extracts or Straights.

TABLE 324 SYNTHITE INDUSTRIES LTD: BUSINESS OVERVIEW

SYNTHITE INDUSTRIES LTD-2021	
Founded	1972
Country	India
City	Kolenchery, Kerala
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.11.2 Products/Solutions/Services offered

TABLE 325 SYNTHITE INDUSTRIES LTD: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Oleoresins	Products available under this category include asafetida, black pepper, capsicum, cardamon, cassia, celery, clove, cocoa, coriander, cumin, decalapis, dill seeds, garlic, ginger, mace, mustard, nutmeg, onion, pimento, paprika, rosemary, tamarind, turmeric, and white pepper.	Herbs & Spices	Oleoresins	- Food & Beverage - Pharmaceuticals - Dietary - Supplements - Cosmetics
Essential oils	Products available under this category include aniseed, basil, black pepper, carom, calamus, cardamon, cassia, cedar wood, celery seed, cinnamon bark, cinnamon leaf, citronella, clove bud, clove leaf, coriander, cumin, curry leaf, davana, dill seed, fennel, garlic, ginger, ginger	Herbs & Spices	Essential oils	- Food & Beverage - Pharmaceuticals - Dietary - Supplements - Cosmetics

	grass, green pepper, asafetida, horse radish, juniper berry, lemon grass, mace, marjoram, mustard, nutmeg, olibanum, palmarosa, parsley seed, patchouli, pimento, rosemary, spearmint, thyme, turmeric, valerian root, vettrivert, and white pepper.			
Green Spice Extracts	Green chili, coriander leaf, curry leaf, garlic, ginger, onion, and pepper	Herbs & Spices	- Alkaloids - Oleoresins	- Food & Beverage - Pharmaceuticals - Dietary - Supplements - Cosmetics
Herbal Extracts	Basil, lavender, laurel, mint, organum, rosemary, sage, and thyme	Herbs & Spices	- Alkaloids - Oleoresins - Essential Oils	- Food & Beverage - Pharmaceuticals - Dietary - Supplements - Cosmetics
Coffee & Tea Extracts	Rich in antioxidants, energizing effect. Some of the products include instant tea (high clarity), brewed black tea top note, tea extract concentrate, and black tea aroma.	Herbs & Spices	Flavonoids	Food & Beverage
Floral Extracts	Jasmine absolute, Jasmine concrete, Jasmine grandiflorum, Jasmine sambac, Mimosa, Tuberose	Flowers	- Alkaloids - Essential Oils	Cosmetics
Vanilla Extracts	Its depth of flavor, fullness, and fruity aroma reflects its tropical origins.	Herbs & Spices	- Oleoresins - Essential Oils	- Food & Beverage - Cosmetics
Dried Fruit Extract	Apricot, Dates, Fig, Plum, Raisins, Tamarind	Fruits & Vegetables	- Alkaloids - Essential Oils	- Food & Beverage - Pharmaceuticals - Dietary - Supplements
Phytochemicals	Caffeine, capsaicin/capsicum oil, capsaicin/capsicum, piperine, curcumin, and lutein are just a few of the botanical extracts available from Synthite for the health supplement and functional food industries.	- Fruits & Vegetables - Herbs & Spices	- Alkaloids - Oleoresins - Essential Oils	- Food & Beverage - Pharmaceuticals - Dietary - Supplements

Source: Company Website

13.2.11.3 Recent developments

There are no recent developments specific to the market during the forecast period.

13.2.11.4 MnM View

Synthite has a strong plant extract portfolio to establish itself as one of the leading players in the market. This company has a robust consumer base in all the major countries. Also, Synthite Industries is technologically advanced and uses all the latest technologies of plant extraction like encapsulated & spray drying. The company is facing competition from established players and other new entrants in the market.

13.2.12 INDESSO

13.2.12.1 Business overview

Indesso is a producer of aroma chemicals, botanical extracts, and essential oils. The company produces products like cheese powders, seasonings, savory ingredients, and also is a food & wellness solution provider. It sources its raw materials mainly from Indonesia and Southeast Asia for its rich biodiversity. The company has two manufacturing facilities, each in West Java and Central Java (Indonesia). The firm exports its products to more than 40 countries and serving over 2,000 customers.

Indesso's products are mainly used as a flavor ingredient in food & beverages, aroma & fragrance in several consumer products, pharmaceuticals. and nutraceuticals. It collaborated with several world-class companies like Firmenich Flavors & Tastes, MartinBauer, and Oterra (previously Chr. Hansen Natural Colors) to distribute products in Indonesia.

TABLE 326 INDESSO: BUSINESS OVERVIEW

INDESSO-2021	
Founded	1968
Country	Indonesia
City	Central Java
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.12.2 Products/Solutions/Services offered

TABLE 327 INDESSO: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE		APPLICATION
Essential oils	The product range offered by Indesso includes anti-aging, balancing, bergamot arte, bergamot oil, black pepper oil, breath, cajeput oil, calming lavandin, cananga oil, cedar altas oil, citronella oil, clove bud oil, clove oil, clove stem oil, clove terpenes, digestion support oil, geranium arte, geranium oil, ginger oil, immunoboost, jojoba oil, kaffir lime arte, kaffir lime leaf oil, lavender oil, lemon arte, lemon eucalyptus oil, lemon oil, lemongrass oil, mood balance, mosquito go away, nutmeg oil, olive oil, orange oil, organic cassia bark oil, organic citronella oil, organic clove bud oil, organic patchouli oil, and organic red ginger oil.	Herbs & Spices	Essential Oil	- Food & Beverages - Pharmaceuticals & Dietary Supplements - Cosmetics
Extracts	The product range offered by Indesso includes apple arte, argan oil,	Herbs & Spices	- Flavonoids - Alkaloids	Food & Beverages Pharmaceuticals & Dietary

benzoin absolute, benzoin resinoid, black pepper oleoresin, black tea aroma, black tea extract, black tea extract powder, caffeine anhydrous, capsicum oleoresin, cassia arte, cassia extract, chamomile arte, clove bud extract, cocoa extract, cocoa tp, coffee arabica extract, coffee arabica extract powder, coffee arabica tp, coffee aroma, coffee oil, coffee robusta extract, coffee robusta extract powder, cubeb oil, cubeb pepper berries, cubebol extract, cured vanilla bean, galanga extract, ginger arte, ginger oleoresin, green coffee extract, green coffee extract powder, green patchouli absolute, and green tea aroma.

- Carotenoids Supplements
- Oleoresins Cosmetics

Source: Company Website

13.2.12.3 Recent developments

13.2.12.3.1 Deals

TABLE 328 INDESSO: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
May 2021	Partnership	Indesso	Biospringer (France)	Biospringer, a producer of yeast extract and yeast-derived products, announced its partnership with Indesso. This partnership focuses on innovation and development of value-added ingredients, eventually increasing the brand's name in product and technology development.	NA
May 2021	Indesso	Investment	EQT Private Equity (Sweden)	By utilizing EQT's internal expertise and international advisory network, EQT Private Equity will assist Indesso in achieving its goals of continued research-driven product innovation, geographic expansion, improvement of digital capabilities, and sustainability.	NA
December 2019	Partnership	Indesso	MartinBauer (Germany)	Martin Bauer, a botanical extract producer with more than 20 production sites, announced its partnership with Indesso. This partnership will enrich the company's product portfolio and opens new market opportunities.	NA

Source: Company Website

13.2.12.4 Others

TABLE 329 INDESSO: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
June 2022	Indesso	Expansion	Opening of Indesso Singapore operations center	Indesso announced the opening of its Singapore business operations. This center would manage all its other business operations in Malaysia and Tanzania. Expansion of their operating centers would build a stronger bond with its clients and partners.
December 2021	Indesso	Expansion	Opening of clover leaf oil field distillation unit in Pemba Island of Zanzibar, Tanzania	The company announced the opening of a new distillation unit in Tanzania. Tanzania is known for its clove production, and hence the company decided to start its clover leaf oil production unit in Tanzania. This is part of its strategy to expand its production.
May 2021	Indesso	Expansion	Opening of Senseria Collaboration Space	The company announced its Senseria Collaboration Space in Jakarta. This center would help to interact with clients by presenting products by “taste, smell, and wellness.” This is a one-of-a-kind engagement center where collaboration, interactions, and workshops can be conducted.

Source: Company Website

13.2.12.5 MnM View

Indesso over the years has established its strong presence in the plant extract industry, domestically and globally. The company had continuously worked on its product innovation and research, which transformed it to become a solution provider for the ingredients industry.

The company has taken initiatives in sustainable production and provides transparency and traceability of raw materials sourcing through responsible sourcing programs. Indesso's main success strategy is communicating with its clients and understanding their need. The company also partnered with several other global market players to innovate products and technologies. The partnership with MartinBauer clearly portrayed its strong growth strategies in expanding its product portfolio. EQT, a global investment organization, invested in Indesso Aroma on continuing its research and geographical expansion. This will accelerate the company's process of product development, innovation, and business expansion.

13.2.13 VIDYA HERBS

13.2.13.1 Business overview

Vidya Herbs is a leading manufacturer and supplier of plant extracts, essential oils, oleoresins, and natural carotenoids. The company uses conventional and Supercritical Fluid Extraction process for the extraction of compounds, providing different grades of products. It also holds a US patent for a plant extract composition and methods for its skin care brand, SKIN-CERA™. All its products meet the quality requirements, such as ISO 9001 & 22000, HACCP, NSF-GMP Kosher, and Halal.

The company is based in India, having its manufacturing facilities in over 5 locations in Karnataka and Telangana. The company also has a global presence in Europe, the UK, the US, Japan, South Korea, the UAE, Russia, and Australia and 5 warehouse worldwide.

TABLE 330 VIDYA HERBS: BUSINESS OVERVIEW

VIDYA HERBS-2021	
Founded	1999
Country	India
City	Bangalore, Karnataka
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.13.2 Products/Solutions/Services offered

TABLE 331 VIDYA HERBS: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Essential oils	The company manufactures essential oil using super critical fluid extraction technology which delivers higher grade of essential oils. The product range offered by Vidya Herbs includes ajowan, aniseed, basil, black pepper, calamus, cardamon, cassia, cedar wood, celery seeds, cinnamon bark, cinnamon leaf, citronella, clove bud, clove leaf, coriander, cumin seeds, curry leaf, davana, dill seed, fennel, garlic, ginger grass, ginger, green pepper, asafetida, marjoram, mustard, nutmeg, olibanum, palma rosa, parsley seeds, patchouli, pimento, rosemary, spearmint, thyme, turmeric, valerian root, vettrivert, and white pepper.	Herbs & Spices	Essential Oils	- Food & Beverages - Pharmaceuticals & Dietary Supplements - Cosmetics

Herbal Extracts	The product range offered by Vidya Herbs includes acerola extract, ashwagandha extract, amla extract, apple extract, bacopa extract, bilberry extract, black pepper extract, bitter melon extract, blue berry extract, boswellia extract, centella asiatica extract, cissus extract, chili extract, cinnamon extract, chamomile extract, clove extract, cocoa bean extract, coleus extract, fenugreek extract, garcinia extract, gardenia extract, garlic extract, ginger extract, and green coffee bean extract.	- Herbs & Spices - Fruits & Vegetables	- Flavonoids - Alkaloids - Carotenoids - Oleoresins	- Food & Beverages - Pharmaceuticals & Dietary Supplements - Cosmetics
Oleoresin	The product range includes black pepper, capsicum, cardamom, cassia, celery, cinnamon, clove, coriander, cumin, curry leaves, dill seeds, garlic, ginger, and mustard.	- Herbs & Spices - Fruits & Vegetables	Oleoresins	- Food & Beverages - Pharmaceuticals & Dietary Supplements - Cosmetics
Carotenoids	The product range includes beta carotene, lycopene, lutein, and zeaxanthin.	Fruits & Vegetables	Carotenoids	- Food & Beverages - Pharmaceuticals & Dietary Supplements - Cosmetics

Source: Company Website

13.2.13.3 Recent developments

13.2.13.3.1 New product launches

TABLE 332 VIDYA HERBS: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
May 2019	Vidya Herbs	Flower Extract	Sunca	Vidya Herbs launched Sunca, a sunflower extract with 40% chlorogenic acids. The extract is clinically proven and helps in regulating lipid metabolism. It is caffeine-free and a sustainable ingredient. This expands the company's herbal extract portfolio.
May 2019	Vidya Herbs	Plant Extract	Skin Cera™	Vidya Herbs launched Skin Cera, a Japanese Konjac extract standardized with five percent ceramides and clinically proven for its beauty benefits. The product is clinically proven to treat dryness, hyperpigmentation, redness, itching, and oiliness. This expands the company's cosmetics extract portfolio.

Source: Company Website

13.2.13.4 MnM View

Vidya Herbs has a global reach for its quality of products, and for its extensive focus on research & development in herbal & botanical extracts. It filed a US patent on the composition of amorphophallus konjac and methods for their use in skin care in December 2015. The company highly focuses on organic growth strategies, such as new product launches and is consistently investing in R&D activities to formulate new products and cater to the growing demand for plant extracts.

13.2.14 KALSEC INC.

13.2.14.1 Business overview

Kalsec Inc. is a US-based company, producing spice and herb extracts, antioxidants, colors, zeaxanthin, and advanced hop extracts. These extracts are delivered in the form of liquid products, which is ideal for most food formulations. The company serves across the US, European, and Asian markets.

The company has 2 manufacturing sites in Michigan (US) and Texas (US), and 4 R&D centers in the US, the UK, China, and Malaysia. The facility in Texas specializes in the cultivation, harvest, dehydration, and extraction of paprika, rosemary, and carrot, and specialty herbs and spices.

TABLE 333 KALSEC INC.: BUSINESS OVERVIEW

KALSEC INC.-2021	
Founded	1958
Country	US
City	Kalamazoo, Michigan
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.14.2 Products/Solutions/Services offered

TABLE 334 KALSEC INC.: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Food Protection	Food protection extracts are plant extracts with specific characteristics like antimicrobial or antioxidant properties. These blends help the food or beverage taste better and last longer by increasing their shelf-life and food safety.	Herbs & Spices	▪ Oleoresins ▪ Essential Oils	Food & Beverages
Garlic and Onion Extracts, Spice and Herb Extracts, And Capsicum Extracts	These products are delivered in liquid form. Commonly used for food applications. These extracts are also available in blends which can serve different culinary tastes.	Fruits & Vegetables	▪ Flavonoids ▪ Alkaloids ▪ Essential Oils ▪ Oleoresins	Food & Beverages
Essential Oils and Oleoresins	Kalsec has essential oils and oleoresins for spice, herb, and citrus flavors ranging from anise to lemon and oregano to thyme.	Herbs & Spice	▪ Essential Oils ▪ Oleoresins	Food & Beverages

Source: Company Website

13.2.14.3 Recent developments

13.2.14.3.1 New product launches

TABLE 335 KALSEC INC.: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
March 2021	Kalsec Inc.	Essential Oil	Lupulock encapsulated hops oil	Kalsec Inc. partnered with Firmenich for the launch of a natural alternative to hop cones and pellets. This new form of the product would help the manufacturer by delivering consistent flavor throughout the processing and reducing wastage. Kalsec Inc. provides the extract, while Firmenich supplies encapsulation technology for creating the product. This product has a shelf-life of 2 years and can be stored at ambient temperature.
June 2020	Kalsec Inc.	Plant Extracts Blend	DuraShield	Kalsec Inc. launched a new blend of antimicrobials to naturally protect food - DuraShield Antimicrobial Natural Food Protection Blends. This natural antioxidant combined with natural antimicrobials improves both shelf-life and ensures food safety in a variety of applications, including meat and poultry, deli salads, dips, sauces, and dressings. This further expands the extract portfolio of the company.

Source: Company Website

13.2.14.3.2 Deals

TABLE 336 KALSEC INC.: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
May 2022	Partnership	Kalsec Inc.	Univar Solutions (Mexico)	The company announced its partnership with Univar Solutions as its distribution partner. This strategic partnership with Univar Solutions benefits Kalsec Inc. by reaching local markets.	NA
July 2021	Collaboration	Kalsec Inc.	Codexis Inc. (US)	Kalsec expanded its research collaboration with Codexis, Inc., an enzyme engineering company. It created a supply agreement for enzyme to produce Kalsec's natural hop acid. This will help the company meet the increasing demand for hop extracts.	NA

September 2019	Joint venture	Kalsec Inc.	Diana Pet Food (US)	Kalsec, inc. and Diana Pet Food announced their joint venture, Videka, to produce the best natural antioxidant solutions for the pet food industry. This is aimed at increasing its pet food product portfolio and strengthening its position in this market.	NA
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Source: Company Website

13.2.14.4 Others

TABLE 337 KALSEC INC.: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
July 2021	Kalsec Inc.	Expansion	Opening of Kalsec's new pilot plant in Michigan, US	The company opened its new 6500 sq ft pilot plant facility in Michigan, US. The Don Berdahl Center for New Technology unit has been named after its retired company CTO Don Berdahl. This center will support new product discovery, advanced application testing, and process engineering advancements.
December 2020	Kalsec Inc.	Expansion	Opening of upgraded workspace and laboratory	Kalsec officially opened its new headquarters in Shanghai. This will support the growth of its business in China and provide workspace for more employees along with deeper laboratory capabilities.
November 2020	Kalsec Inc.	Expansion	Opening of new warehouse in the UK	The company opened its new warehouse facility in Mildenhall, UK. This new facility will enhance the supply of products to customers across continental Europe.

Source: Company Website

13.2.14.5 MnM View

Kalsec Inc. has been serving food ingredients and solutions for over six decades, making it a key player in the market. The company has extensively invested in research on developing new products like encapsulated hop oils, antimicrobial blends, and expanded its collaboration with companies like Firmenich to produce innovative products. The company provides solutions for the food & beverage industry in accordance with their processing needs, making it a trusted brand in the ingredient industry.

13.2.15 MARTINBAUER

13.2.15.1 Business overview

MartinBauer is a leading manufacturer of botanical extracts. The company is specialized in botanical products for various applications in food, beverages, and pharmaceuticals industry. They produce products like tea, herbal and fruit extracts, strong infusions, botanicals, and botanical powders in various grain sizes, including its extremely fine powders for the food & beverage industry. The company manufactures its products in 20 facilities globally. Its markets products in continents like Europe, North America, South America, and Asia.

TABLE 338 MARTINBAUER: BUSINESS OVERVIEW

MARTINBAUER-2021	
Founded	1930
Country	Germany
City	Vestenbergsgreuth, Bayern
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.15.2 Products/Solutions/Services offered

TABLE 339 MARTINBAUER: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Tea	These products are delivered in tea bags, loose blends, capsules, ready-to-drink, and fresh brew	Herbs & Spices	Flavonoids	- Pharmaceuticals & Dietary Supplements - Food & Beverages
Botanical Extracts & Tinctures	These products are used in various applications such as confectionery, ice creams, baked goods, cereals, and dairy products.	- Fruits & Vegetables - Flowers	- Essential Oils - Oleoresins - Carotenoids	Food & Beverages

Source: Company Website

13.2.15.3 Recent developments

13.2.15.3.1 Deals

TABLE 340 MARTINBAUER: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
March 2021	Acquisitions	MartinBauer	Power Brands (US)	MartinBauer acquired majority of the stakes in Power Brands. Power Brands is a leading beverage producer and marketing service company based in California, US. This strategic acquisition will help the company deliver innovative products in plant-based foods and beverages.	NA
September 2020	Acquisition	MartinBauer	Core Botanica (US)	The company announced the acquisition of Core Botanica, one of the leading manufacturers of customized mint leaf products in North America. The acquisition expands the group's portfolio of raw materials and botanical ingredients, servicing the food and dietary supplement industries, along with the tea and beverage markets.	NA
March 2019	Acquisition	MartinBauer	BI Nutraceuticals (US)	The company announced the acquisition of BI Nutraceuticals, a leading full-service manufacturer and supplier of plant-based ingredients in the US. The acquisition creates an extensive portfolio of botanical ingredients and a versatile manufacturing hub servicing the US tea, beverage, food, and dietary supplements industries.	NA

Source: Company Website

13.2.15.4 Others

TABLE 341 MARTINBAUER: OTHERS

MONTH & YEAR	COMPANY NAME	DEVELOPMENT TYPE	DEVELOPMENT	DESCRIPTION
January 2021	MartinBauer	Expansion	New warehouse	The company built a new automated pallet warehouse. To ensure smooth supply processes, the company announced its upgradation.
December 2020	MartinBauer	Expansion	The company expanded its strong infusion production unit	This expansion enables the company to respond to the increased consumer demand for premium beverages.
September 2019	MartinBauer	Expansion	The company announced its opening of a new application lab	The company announced the opening of its new application lab in New Jersey, US.

Source: Company Website

13.2.15.5 MnM View

MartinBauer focuses on inorganic growth strategies, such as acquisitions, expansions, and partnerships, to cater to a wider audience. The company made some strategic investments over the past few years for its business growth, such as acquiring Power Brands (US) in 2021, Core Botanica (US) in 2020, and BI Nutraceuticals (US) in 2019. The company has a limited product portfolio but delivers products with different application forms (like liquid, blends, loose packs, powders, etc.).

13.2.16 KANGCARE BIOINDUSTRY CO. LTD.

13.2.16.1 Business overview

Kangcare Bioindustry Co. Ltd. is a bio-science technology-based company. Its products are categorized under four categories: food & human nutrition, cosmetic & personal care, pharmaceutical, and animal nutrition. The company produces and markets innovative bioactive ingredients for industry applications in food, cosmetics, and pharmaceutical industries. It has a wide product portfolio for the food & beverage industry and offers plant extracts derived from fruits, flowers, and leaves.

TABLE 342 KANGCARE BIOINDUSTRY CO. LTD.: BUSINESS OVERVIEW

KANGCARE BIOINDUSTRY CO. LTD. -2021	
Founded	2007
Country	China
City	Nanjing, Jiangsu
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.2.16.2 Products/Solutions/Services offered

TABLE 343 KANGCARE BIOINDUSTRY CO. LTD.: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Extracts	These extracts can be used as an ingredient for products like bakery, beverage, baby food, cereals, candy & confectionary, dairy & desserts, health & sports nutrition, snack foods, or other processed foods. Some extracts offered by the company are gamma oryzanol, herbal extracts, and roselle extract.	- Herbs & Spices - Fruits & Vegetables	- Flavonoids - Alkaloids - Essential Oils - Carotenoids - Oleoresins	- Pharmaceuticals - Dietary Supplements - Food & Beverages
Extract	These extracts can be used as an ingredient for products like decorative cosmetics, skin, hair care, fragrances, bath & shower, natural cosmetics, cosmeceuticals, and nutricosmetics. Some of the extracts offered by the company are gamma oryzanol, licorice roots extract, soy isoflavones, and resveratrol.	Herbs & Spices	- Essential Oils - Oleoresins	Cosmetics

Extract	These extracts can be used as active pharmaceutical ingredients (APIs), auxiliary materials and intermediates for the pharmaceutical, veterinary, and healthcare industries. Some of the extracts offered by the company include dipsaci radix, frankincense, millettia dielsiana, and passionflower.	▪ Herbs & Spices ▪ Flowers	▪ Flavonoids ▪ Essential Oils ▪ Carotenoids ▪ Oleoresins	▪ Pharmaceuticals ▪ Dietary Supplements
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Source: Company Website

13.2.16.3 Recent developments

There are no recent developments specific to the market during the forecast period.

13.2.16.4 MnM View

The company provides a wide product portfolio for the food & beverage industry and offers plant extracts derived from fruits, flowers, and herbs. The company is highly dependent on Asia Pacific for its revenue but has opportunities to enter developed markets, such as the US and Europe. Furthermore, its focus on developing and marketing innovative and proven bioactive ingredients and delivering science-based health and beauty solutions to customers has helped it gain a leading position in the industry.

13.3 OTHER PLAYERS (SMES/START-UPS)

13.3.1 NATIVE EXTRACTS PTY LTD

13.3.1.1 Business overview

Native Extracts Pty. Ltd. is a manufacturer of botanical extracts of various indigenous plants in Australia and other plant extracts for the pharmaceutical, cosmetics, and food & beverage industries. This Australian company uses cellular extraction method to retain the phytochemical properties of plants, instead of conventional methods. The company uses raw materials that are organic certified and wild harvested.

The company's product portfolio has over 300 extracts which feature unique Australian species, and "True to Nature" versions of traditional botanicals. The company produces a unique product - Native EssenXces, which is a highly concentrated and nutrient-dense plant extracts in an ingestible application.

TABLE 344 NATIVE EXTRACTS PTY LTD: BUSINESS OVERVIEW

NATIVE EXTRACTS PTY LTD -2021	
Founded	2012
Country	Australia
City	Alstonville, New South Wales
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.3.1.2 Products/Solutions/Services offered

TABLE 345 NATIVE EXTRACTS PTY LTD: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Extracts	Native extracts have a wide range of plant cellular extracts. These cellular extracts are produced using purified, deionized rain-harvested water and less plant material, which makes them true to nature extract. Some of these include agapanthus cellular extract, aloe vera inner leaf juice extract, amla berry cellular extract, and amur bark cellular extract.	Fruits & Vegetables	- Flavonoids - Alkaloids - Essential Oils - Carotenoids - Oleoresins	- Food & Beverages - Pharmaceuticals - Dietary - Supplements - Cosmetics

Extract-Indigenous	Native extracts have a wide range of indigenous Australian plant cellular extracts. These cellular extracts are produced using purified, deionized rain-harvested water and less plant material, which makes them true to nature extract. Some of the extracts include aniseed myrtle cellular extract, banksia flower cellular extract, blackwood wattle cellular extract, blue cypress leaf cellular extract, bush tomato cellular extract, centipeda cellular extract, and davidson plum cellular extract.	- Fruits & Vegetables - Herbs & Spices	- Flavonoids - Essential Oils - Oleoresins	- Food & Beverages - Pharmaceuticals - Dietary - Supplements - Cosmetics
Rainforest Blue Essential Oil	Australian Rose Mahogany is a large tree native to New South Wales and Queensland, Australia. This essential oil provides a base tone of fresh forest odor, making them a one-of-a-kind product.	Herbs & Spices	Essential Oils	Cosmetics
Essential Oil-Indigenous	Native Extracts produce their essential oil by cold pressed or CO2 extraction, which gives high-quality pure essential oils. Some of the examples are grapeseed oil, hemp seed oil, and kakadu plum seed oil.	- Fruits & Vegetables - Herbs & Spices	Essential Oils	Cosmetics

Source: Company Website

13.3.1.3 Recent developments

13.3.1.3.1 New product launches

TABLE 346 NATIVE EXTRACTS PTY LTD: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
May 2022	Native Extracts Pty Ltd	Essential oil	Rainforest Blue	The company announced the launch of its newest blue essential oil, extracted by steam distillation of Australian rosewood. This essential oil provides a base tone of fresh forest odor, making it a one-of-a-kind product.
August 2021	Native Extracts Pty Ltd	Plum extract and an ingestible format product	Queen Garnet Plum Extract and EssenXce	The company joined force with Nutrafruit, a licensed grower of Garnet Plum, to create two kinds of product from one fruit. Topical

format Concentrated Cellular Extracts and INGESTIBLE format: 4:1 Liquid EssenXces. Launching this product increases the company's market share.

Source: Company Website

13.3.1.3.2 Deals

TABLE 347 NATIVE EXTRACTS PTY LTD: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY NAME 1	COMPANY NAME 2	DESCRIPTION	DEAL SIZE
August 2021	Partnership	Native Extracts Pty Ltd	Purenature Group (New Zealand)	The company announced its partnership with Purenature Group, a bulk cosmetics ingredient distributor and importer in New Zealand. This partnership will increase the market reach of Native Extracts.	NA

Source: Company Website

13.3.1.4 MnM View

The company focuses on R&D and innovative new products like EssenXce for the consumer markets. It extensively invested in plant profiling, designing new technology, and developing new protocols and methodologies. The company has a vast product portfolio and a strong presence in the Asia Pacific market. With a strong focus on continual improvement in process efficiencies for better solutions for people and planet, the company has strengthened its position in the plant extracts market.

13.3.2 ARJUNA NATURAL

13.3.2.1 Business overview

Arjuna Natural manufactures active compounds from plants. It was the first Indian company to produce mustard essential oil in 1990. Furthermore, in 2014, it became the first company in the world to produce turmeric extract using only solar energy. Apart from India, it has a sales presence in the US. It has manufacturing facilities in India as well as an innovation center in Kerala.

TABLE 348 ARJUNA NATURAL: BUSINESS OVERVIEW

ARJUNA NATURAL-2021	
Founded	1989
Country	India
City	Aluva, Kerala
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.3.2.2 Products/Solutions/Services offered

TABLE 349 ARJUNA NATURAL: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
BCM-95 (Curcugreen)	BCM-95 (CURCUGREEN) is a 100% pure turmeric extract with improved bioavailability and efficacy.	Herbs & Spices	- Essential Oils - Oleoresins	- Food & Beverages - Pharmaceuticals - Dietary Supplements
AKBAMAX (BosPure)	It is a one-of-a-kind extract derived from the resin of the wild <i>Boswellia serrata</i> tree. Natural oils, gums, and the medicinally active and fragrant terpene group are found in the tree resin.	Herbs & Spices	- Essential Oils - Oleoresins	- Food & Beverages - Pharmaceuticals - Dietary Supplements
Shoden	Shoden is a purified extract of the Ashwagandha plant.	Herbs & Spices	- Essential Oils - Oleoresins	- Food & Beverages - Pharmaceuticals - Dietary Supplements
Herbagut	It is a poly botanical extract that has been clinically proven to keep the gut healthy by maintaining the integrity of its internal lining. Herbagut combines the potency of 14 different herbs to improve digestive health and immunity.	Herbs & Spices	- Essential Oils - Oleoresins	- Pharmaceuticals - Dietary Supplements
Oxystorm	Oxystorm is the world's first	Herbs &	Essential Oils	Pharmaceuticals

	nitrate-enriched extract of Amaranthus, which is one of the richest sources of nitrates in the diet.	Spices	▪ Oleoresins	▪ Dietary Supplements
Tri-Low	Tri-Low is a full spectrum extract of bioactive ingredients found in Indian gooseberry, also known as Amla (Emblica Officinalis). This fruit is an excellent cardiac health supplement and has long been used to restore health, immunity, and vitality.	Fruits & vegetables	▪ Alkaloids ▪ Essential Oils ▪ Oleoresins	▪ Pharmaceuticals ▪ Dietary Supplements
Rhuleave-K	Rhuleave-K is Arjuna's clinically proven pain support formula which has been proven to provide pain relief.	Herbs & Spices	▪ Essential Oils ▪ Oleoresins	▪ Pharmaceuticals ▪ Dietary Supplements
Nigrin	Nigrin is a standardized natural extract of Brassica nigra (black mustard) containing 25% glucosinolate.	Herbs & Spices	▪ Essential Oils ▪ Oleoresins	▪ Pharmaceuticals ▪ Dietary Supplements
Ellagirush	Ellagirush is a pomegranate fruit extract that has been standardized. In addition to being a powerful antioxidant, it has been shown to provide powerful cardiovascular and prostate health support.	Fruits & vegetables	▪ Alkaloids ▪ Essential Oils ▪ Oleoresins	▪ Food & Beverages ▪ Pharmaceuticals ▪ Dietary Supplements
Flavors & Fragrances	The range includes mustard essential oil, boswellia essential oil, turmeric essential oil, olibanum oil, and olibanum resinoid fragrance.	Herbs & Spices	▪ Essential Oils ▪ Oleoresins	▪ Food & Beverages ▪ Cosmetics

Source: Company Website

13.3.2.3 Recent developments

13.3.2.3.1 New product launches

TABLE 350 ARJUNA NATURAL: NEW PRODUCT LAUNCHES

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
June 2021	Arjuna Natural	Pharmaceuticals & Dietary Supplements	Rhuleave-K	Rhuleave-K was introduced as a revolutionary pain management solution. It is a one-of-a-kind product made of extracts of curcuma longa and boswellia serrata. This broadens the company's specialty extracts product portfolio.

Source: Company Website

13.3.2.4 MnM View

The company focuses on R&D and innovative products like Rhuleave-K. It also focuses on research; for example, a study done by the company in 2021 suggests that turmeric extract and essential oil of turmeric together may enhance spatial learning and memory and provide neurotoxic protection up to a certain degree. The company is expanding its market presence, and in 2017, Arjuna Natural LLC was established in Texas, US.

13.3.3 BLUE SKY BOTANICS

13.3.3.1 Business Overview

Blue Sky Botanics manufactures herbal extracts. End users for its vast product line include personal care, beverage, and food industries. The company is a member of the Union for Ethical BiTrade and aims at ethical sourcing, quality, and traceability of Blue Sky Extracts provenance. It is also associated with a range of ethical trade standards such as the Fairtrade Foundation, Soil Association, and FairWild. The company has its 200-acre organic farm in Herefordshire, England.

TABLE 351 BLUE SKY BOTANICS: BUSINESS OVERVIEW

BLUE SKY BOTANICS-2021	
Founded	2005
Country	UK
City	Herefordshire, England
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.3.3.2 Products/Solutions/Services offered

TABLE 352 BLUE SKY BOTANICS: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Extract	The company provides a wide range of natural extracts, flavors, juices & infusions using herbs, flowers, fruits, vegetables, and seaweeds. Some of the examples are acai, alexanders, allspice, aloe vera, amaranth, red amla, and gotu kola.	- Herbs & Spices - Fruits & Vegetables	- Alkaloids - Flavonoids - Essential Oils - Carotenoids - Oleoresins	- Food & Beverages - Cosmetics

Source: Company Website

13.3.3.3 Recent developments

There are no recent developments specific to the market during the forecast period.

13.3.3.4 MNM view

The company manufactures a wide range of natural extracts, flavors, juices & infusions using herbs, flowers, fruits, vegetables, and seaweeds. The company's presence is mostly across the UK, with no such global presence. To increase its market share, it must focus on expansions and acquisitions.

13.3.4 TOKIWA PHYTOCHEMICAL CO., LTD.

13.3.4.1 Business Overview

Tokiwa Phytochemical Co., Ltd. is a manufacturer of botanicals extracts. The products offered by Tokiwa are used for various industrial applications like pharmaceutical ingredients, cosmetic ingredients, health food ingredients, and pure compounds. The company has over 21 patents and several research papers in collaboration with multiple universities and research institutes. It also offers contract manufacturing to use its technology for production purpose. Tokiwa offers a wide range of support for OEMs and ODMs, where clients get to consult with experts on creating and manufacturing the product.

TABLE 353 TOKIWA PHYTOCHEMICAL CO., LTD.: BUSINESS OVERVIEW

TOKIWA PHYTOCHEMICAL CO., LTD.-2021	
Founded	1949
Country	Japan
City	Sakura-shi, Chiba
No. Of Employees	110
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.3.4.2 Products/Solutions/Services offered

TABLE 354 TOKIWA PHYTOCHEMICAL CO., LTD.: PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Supplement Ingredients (Powder Form)	Bilberry, Black Currant, Black turmeric, Echinacea, Ganoderma Mushroom, Ginger, Ginkgo Biloba, Green tea Polyphenols, Laurel, Maca, Mulberry, Rafuma, Soybean Isoflavones, Soybean Saponin, TW-KK	Herbs & Spices	▪ Oleoresins ▪ Flavonoids ▪ Alkaloids	▪ Pharmaceuticals ▪ Dietary ▪ Supplements
Food Additives (Powder Form)	Bilberry Powder, Black Currant Powder, Licorice, Matcha, Rutin, Stevia, Yuzu	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	▪ Food & Beverages
Pharmaceutical Ingredients (Powder Form)	Calcium Sennosides 20 Tpc, Calcium Sennosides 60 Tpc, Calcium Sennosides 80 Tpc	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	▪ Pharmaceuticals ▪ Dietary ▪ Supplements
Cosmetics ingredients (liquid form)	Ashwagandha Leaf Extract, Green Tea Extract	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	Cosmetics

Cosmetics Ingredients (Powder Form)	Licorice derivatives: Carbenoxolone, Glycyrrhetic acid, Stearyl Glycyrrhetinate, Glycyrrhizic acid, Monoammonium Glycyrrhizinate, Dipotassium Glycyrrhizinate	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	Cosmetics
Cosmetics Ingredients (Liquid Form)	Ashwagandha Leaf Extract, Green Tea Extract	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	Cosmetics
Research Reagents	The company produces various purified plant- derived compounds. These compounds are useful for cellular experiments and analyses.	Herbs & Spices	▪ Flavonoids ▪ Alkaloids	Cosmetics

Source: Company Website

13.3.4.3 Recent developments

There are no recent developments specific to the market during the forecast period.

13.3.4.4 MnM view

The company extensively works on R&D and has collaborated with multiple institutes like Chiba University, Advanced Industrial Science and Technology (AIST), University of Florida, Kurashiki Medical Center, Kurashiki Seijin-byou Center, Nara Medical University, and many to drive its research in plant extracts.

13.3.5 SHAANXI JIAHE PHYTOCHEM CO., LTD (JIAHERB, INC.)

13.3.5.1 Business Overview

Shaanxi Jiahe Phytochem Co., Ltd (also known as Jiaherb, Inc.) is a manufacturer of various herbal extracts that serves pharmaceutical, nutraceutical, and cosmetics industries. Jiaherb Inc. has several factories, over 100 series of products, 2 independent labs, and more than 50 authorized patents. It manufactures products through its five factories across China. Jiaherb is a CGMP, ISO series, Kosher, and Halal certified company procuring the highest quality of raw materials.

The company provides natural specialty ingredients and innovative technology to serve its client's needs. It also launched HERBALINK, a program that offers its customers the facility of product transparency and tracking. This program includes a botanical identification report, macroscopic identification report, chemical identification report, and third-party laboratory reports. Through this program, the company ensures to supply the best-quality products to its customers.

TABLE 355 SHAANXI JIAHE PHYTOCHEM CO., LTD (JIAHERB, INC.): BUSINESS OVERVIEW

SHAANXI JIAHE PHYTOCHEM CO., LTD (JIAHERB, INC.) - 2021	
Founded	2000
Country	China
City	Xi'an
Ownership	Private

Note: This is a private company, hence no financials are available in the public domain.

Source: Company Website

13.3.5.2 Products/Solutions/Services offered

TABLE 356 SHAANXI JIAHE PHYTOCHEM CO., LTD. (JIAHERB, INC.): PRODUCTS OFFERED

PRODUCT	DESCRIPTION	SOURCE	TYPE	APPLICATION
Extract	The company provides natural and organic plant extracts.	Herbs & Spices	- Flavonoids - Alkaloids	- Pharmaceuticals - Dietary Supplements - Food & Beverages - Cosmetics
	Some of these include valerian extract, griffonia seeds extract, green tea extract, and citrus aurantium extract.			

Source: Company Website

13.3.5.3 Recent developments

There are no recent developments specific to the market during the forecast period

13.3.5.4 MnM view

The company is dynamic in developing innovative products. It uses technology-based programs like creating HERBALINK, to gain trust from its clients by providing product transparency and tracking. The company's advanced material management system allows it to precisely trace all products within the supply chain.

13.3.6 ALCHEMY CHEMICALS

Founded	1993
Country	India
City	Ujjain, MP
Ownership	Private
Business Overview	Alchemy Chemicals is an herbal extracts and phytochemical manufacturer. Some of the extracts offered by the company are Acaci concinna, Adhatoda vasica, Abroma augusta root, Acacia arabica, Acacia catechu, and Acacia concinna. Essential oil includes ajwain oil, thyme oil, anise oil, basil oil, caraway oil, citronella oil, clove oil, garlic oil, and geranium oil. Alchemy Chemicals' facility is ISO certified and also has kosher & Halal certifications.
Products Offered	▪ Herbal Extracts ▪ Essential Oils ▪ Oleoresins ▪ Phytochemicals
Geographic Presence	Asia Pacific

Source: Company Website

13.3.7 PRINOVA GROUP LLC

Founded	2002
Country	US
City	Carol Stream, IL
Ownership	Private
Business Overview	Prinova Group LLC. was formerly known as Premium Ingredients International Holdings, LLC and changed its name in July 2011. The company distributes and supplies ingredients and flavors to the food & beverage industry. Some of its product offerings include acidulants, botanical extracts, flavor enhancers, nutritional blends, preservatives, and specialty food ingredients. The company has expanded its geographical presence with offices worldwide. In North America, it has offices in Canada and Mexico, and European offices in England, Benelux, France, Germany, Spain, and Turkey. Prinova has a liquid blending facility and an expanded R&D application laboratory in Illinois, US.
Products Offered	Spices, green tea extracts, and tea extracts
Geographic Presence	North America, Europe, and Asia Pacific

Source: Company Website, Press Release, and MarketsandMarkets Analysis

13.3.8 VITAL HERBS

Founded	2018
Country	India
City	Delhi
Ownership	Private
Business Overview	Vital Herbs is a leading manufacturer, exporter, wholesaler, trader, and importer of herbal extracts and essential oils. It also sells herbal extracts, pure herbal extracts, and natural herbal extracts.
Products Offered	▪ Nutraceuticals Raw Materials ▪ Herbal Extract ▪ Natural Herbal Extracts ▪ Pure Herbal Extracts ▪ Organic Herbal Extracts ▪ Medicinal Herbal Extracts ▪ Herbal Extracts Food ▪ Essential Oil ▪ Powder Extract ▪ Natural Food Colors
Geographic Presence	Asia

Source: Company Website

13.3.9 SYDLER INDIA PVT. LTD.

Founded	2004
Country	India
City	Mumbai, Maharashtra
Ownership	Private
Business Overview	Sydler has been producing and exporting herbal formulations since 2004. Pharmaceuticals, herbal/ayurvedic formulations, herbal extract/phytochemicals, nutraceuticals, agricultural products, and spices are among the company's products. It has a manufacturing plant in Aurangabad, Maharashtra.
Products Offered	▪ Herbal/Ayurvedic Formulations ▪ Herbal Extracts/Phytochemicals ▪ Nutraceuticals
Geographic Presence	Afghanistan, Costa Rica, Cyprus, the Czech Republic, Dubai, Ghana, Kuwait, Nepal, Nigeria, Philippines, Russia, Sri Lanka, Tajikistan, Uganda, Ukraine, Oman, and the UK

Source: Company Website

13.3.10 PLANTNAT

Founded	2004
Country	China
City	Chang Sha City, Hu Nan Province
Ownership	Private
Business Overview	Plantnat is a manufacturer of plant extracts. It is a natural ingredients application solution provider for the nutrition, food & beverage, and personal care industries. It is a member of the China plant extracts standards drafting committee.
Products Offered	▪ Functional Plant Extract ▪ Standard Plant Extract ▪ Pure Herbal Extract ▪ Food Color
Geographic Presence	Asia

Source: Company Website

14 ADJACENT & RELATED MARKETS

14.1 INTRODUCTION

This chapter focuses on defining the interconnection between the plant extracts market and other related markets.

The first section identifies some of the key markets related to the plant extracts market using MarketsandMarkets KnowledgeStore and provides an overview of the interrelation. Subsequent sections provide a comprehensive analysis of the most proximately connected markets to the plant extracts market, which includes the essential oil market and the botanical extracts market. Plant extracts are expected to stimulate the product demand in pharmaceutical and nutraceutical applications due to their numerous overall health advantages.

The essential oil market MarketsandMarkets study was published separately and identified plant extracts market as a related market in this chapter, as given below.

STUDY NAME	PUBLICATION YEAR	BASE YEAR	FORECAST PERIOD
Essential Oil Market	2021	2020	2021–2026
Botanical Extracts Market	2017	2016	2017–2022

14.2 LIMITATIONS

The market insights from adjacent markets are for reference only and may have the following:

- Market definitions that differ from the main market definitions, with one or more overlapping segments, have different market sizes
- Market sizes and growth rates assessed in the past that may have changed as compared to the base year considered for the study
- Market sizes and forecasts for years that are different from those of the main market
- Pre-COVID-19 market sizes and forecasts
- The quantitative information for some market segments is kept confidential by the industry players. Hence, qualitative insights gathered during the study have been used to arrive at the market size for certain subsegments

14.3 ESSENTIAL OIL MARKET

14.3.1 MARKET DEFINITION

The International Organization for Standardization (ISO) in its Vocabulary of Natural Materials (ISO/D1S9235.2) defines an essential oil as “a product made by distillation with either water or steam or by mechanical processing of citrus rinds or by dry distillation of natural materials. Following the distillation, the essential oil is physically separated from the water phase.”

Essential oils are highly aromatic and concentrated botanical extracts of numerous plant materials. They are found in minute secretory structures of leaves, stems, flowers, bark, and roots. These essential oils provide plants with their characteristic smell and safeguard them from external stress conditions. They can, thus, be considered as regenerating, oxygenating, and immune-strengthening components in a plant.

Apart from imparting fragrance, essential oils have healing properties due to their potent antioxidant, anti-inflammatory, anti-viral, and anti-microbial properties. It is due to these qualities that essential oils find usage in a wide array of applications starting from food & beverages and aromatherapy to cosmetics & toiletries, home care, health care, and feed.

14.3.2 MARKET OVERVIEW

An essential oil is a potent aromatic component isolated from an odoriferous plant of a single botanical species. It is highly concentrated and volatile and is made of various chemical constituents, which give it its characteristic fragrance that is inherent to its parent plant. These naturally occurring aromatic compounds are stored as microdroplets in the glands of plants. They can be extracted from several parts such as leaves, stems, flowers, bark, roots, or other elements of a plant. They provide plants their characteristic smell and safeguard them from external stress conditions, imparting freshness and vitality to them.

Essential oils are highly volatile in the sense that they change state quickly from liquid to gas. When a bottle of essential oil is opened, its components rapidly diffuse into the air and directly interact with the olfactory sensors in the nose. This property of essential oils helps to maintain a healthy mind and body and elevate the senses.

Several essential oils have potent antioxidant, anti-inflammatory, antibacterial, antiparasitic, antiseptic, anti-fungal, antiviral, and anti-infectious properties. These properties make them highly beneficial in healthcare applications as they help in protection from microbes and external stress. Aromatherapy utilizes these properties of essential oils, along with its strong fragrance in treating numerous disorders such as mental illnesses, respiratory infections, digestive control, olfactory disorders, and sleep disorders.

Essential oils are also widely used as flavors in food & beverage applications. They can be used as flavorings in bakery goods, candies, confectionery, meat, pickles, soft drinks, and energy drinks, among others. As odorants, essential oils find application in cosmetics, toiletries, perfumes, detergents, and home cleaning items.

The global essential oils market in general and the aromatherapy application market, in particular, have expanded in the last few years. Steam distillation is the most common method of extraction of essential oils, followed by cold press extraction. Solvent extraction and carbon dioxide extraction are relatively new and are hence not practiced much.

Growing awareness about preventive healthcare, increasing double-income households, rise in depression and anxiety cases, increasing demand for aromatherapy, growing R&D in this field, and increase in end-use applications of essential oils are some of the factors driving the demand for these oils. On the other hand, depleting natural resources, side-effects associated with certain essential oils, and prevalence of synthetic, adulterated products are some of the factors restraining the growth of this market. In 2019, Europe was the largest market for essential oils, owing to their high usage as flavors and fragrances in food & beverage and cosmetics applications. The emerging economies of the Asia Pacific region are exhibiting the fastest growth owing to an increase in dual-income households and an exponentially increasing population.

14.3.3 ESSENTIAL OIL MARKET, BY TYPE

The major types of essential oil are orange, lemon, lime, peppermint, corn mint, citronella, spearmint, geranium, clove leaf, eucalyptus, jasmine, tea tree, rosemary, and lavender, of which orange oil accounted for the largest share of the global essential oils market. With the rising demand for food products with health benefits, the food & beverage industry is growing at a rapid pace. However, the growing demand for essential oils in various end-use industries has led to an increase in the production and consumption of essential oils in the global market.

TABLE 357 ESSENTIAL OILS MARKET, BY TYPE, 2017–2020 (USD MILLION)

Type	2017	2018	2019	2020	CAGR (2017–2020)
Orange oil	1,882.9	2,020.8	2,170.9	2,334.6	7.4%
Lemon oil	431.8	476.6	526.6	582.5	10.5%
Lime oil	202.8	218.6	235.9	254.9	7.9%
Peppermint oil	549.0	591.2	637.4	688.1	7.8%
Cornmint oil	527.7	571.3	619.2	671.3	8.4%
Citronella oil	412.3	445.9	482.8	523.2	8.3%
Spearmint oil	126.2	135.8	146.2	157.6	7.7%
Geranium oil	71.8	78.5	85.9	94.1	9.4%
Clove leaf oil	362.8	399.5	440.4	485.9	10.2%
Eucalyptus oil	702.0	762.4	829.0	901.1	8.7%
Jasmine oil	68.3	73.6	79.4	85.8	7.9%
Tea tree oil	735.2	811.5	896.6	995.0	10.6%
Rosemary oil	483.5	530.8	583.2	642.3	9.9%
Lavender oil	403.3	439.6	479.6	528.7	9.4%
Rose oil	143.9	152.1	160.8	177.4	7.2%
Coriander oil	91.6	99.4	108.0	117.5	8.7%
Other types*	136.1	154.9	174.7	181.2	10.0%
Total	7,331.1	7,962.3	8,656.6	9,421.0	8.7%

*Other types include sandalwood, cedar, chamomile, anise, sage, saffron, grape seed, ginger, thyme, cinnamon, ylang ylang, and basil oils.

Source: Related Associations/Institutes, Related Research Publications, Government Publications, and MarketsandMarkets Analysis

TABLE 358 ESSENTIAL OILS MARKET, BY TYPE, 2021–2026 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
Orange oil	2,508.8	2,713.7	2,938.4	3,185.0	3,455.9	3,740.6	8.3%
Lemon oil	647.3	712.0	784.0	864.1	953.3	1,049.4	10.1%
Lime oil	275.4	298.8	324.6	352.9	384.1	413.1	8.4%
Peppermint oil	742.8	806.6	876.9	954.5	1,040.1	1,152.4	9.2%
Cornmint oil	729.1	793.8	865.0	943.7	1,030.5	1,122.6	9.0%
Citronella oil	567.0	617.1	672.3	733.2	800.4	869.2	8.9%
Spearmint oil	169.7	184.2	200.1	217.6	236.8	256.3	8.6%
Geranium oil	103.1	113.5	125.0	138.0	152.4	168.7	10.4%
Clove leaf oil	537.3	592.2	653.6	722.0	798.5	879.2	10.4%
Eucalyptus oil	983.0	1,072.1	1,170.5	1,279.1	1,399.3	1,523.3	9.2%
Jasmine oil	92.7	100.4	108.8	118.1	128.2	141.6	8.8%
Tea tree oil	1,100.9	1,212.6	1,337.1	1,476.0	1,631.0	1,839.6	10.8%
Rosemary oil	707.6	776.1	852.2	936.7	1,030.6	1,163.6	10.5%
Lavender oil	573.0	627.1	687.1	753.6	827.3	908.7	9.7%
Rose oil	180.6	191.6	203.4	216.2	230.1	255.2	7.2%
Coriander oil	127.9	139.3	152.0	166.0	181.4	198.4	9.2%
Other types*	217.0	240.9	266.3	293.4	322.0	305.9	7.1%
Total	10,263.2	11,192.1	12,217.3	13,349.8	14,602.0	15,987.7	9.3%

*Other types include sandalwood, cedar, chamomile, anise, sage, saffron, grape seed, ginger, thyme, cinnamon, ylang ylang, and basil oils.

Source: Related Associations/Institutes, Related Research Publications, Government Publications, and MarketsandMarkets Analysis

The orange oil segment dominated the essential oils market. Used topically, orange essential oil is beneficial for maintaining the health, appearance, and texture of skin by promoting clarity, radiance, and smoothness, thereby reducing the signs of acne and other uncomfortable skin conditions.

TABLE 359 ESSENTIAL OILS MARKET, BY TYPE, 2017–2020 (KT)

Type	2017	2018	2019	2020	CAGR (2017–2020)
Orange oil	105.10	110.54	116.39	122.69	5.3%
Lemon oil	6.23	6.73	7.28	7.89	8.2%
Lime oil	7.23	7.66	8.13	8.64	6.1%
Peppermint oil	8.36	8.79	9.25	9.75	5.3%
Cornmint oil	8.71	9.16	9.65	10.18	5.4%
Citronella oil	6.27	6.64	7.05	7.48	6.1%
Spearmint oil	2.16	2.26	2.36	2.48	4.8%
Geranium oil	1.43	1.52	1.61	1.71	6.1%
Clove leaf oil	8.28	8.92	9.62	10.39	7.9%
Eucalyptus oil	14.66	15.61	16.64	17.75	6.6%
Jasmine oil	1.52	1.60	1.69	1.79	5.6%
Tea tree oil	10.72	11.50	12.36	13.29	7.4%
Rosemary oil	6.58	7.05	7.56	8.12	7.2%
Lavender oil	6.18	6.59	7.05	7.54	6.9%
Rose oil	3.37	3.48	3.60	3.77	3.8%
Coriander oil	0.62	0.65	0.69	0.73	5.9%
Other types*	3.81	4.05	4.29	4.48	5.6%
Total	201.21	212.76	225.23	238.67	5.9%

*Other types include sandalwood, cedar, chamomile, anise, sage, saffron, grape seed, ginger, thyme, cinnamon, ylang ylang, and basil oils.

Source: Related Research Publication, Government Publication, Company Press Release, Company Annual Report, Company Website, Company Publication, and MarketsandMarkets Analysis

TABLE 360 ESSENTIAL OILS MARKET, BY TYPE, 2021–2026 (KT)

Type	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
Orange oil	129.47	136.77	144.64	153.13	162.28	172.18	5.9%
Lemon oil	8.56	9.29	10.10	10.99	11.97	13.01	8.7%
Lime oil	9.19	9.79	10.43	11.13	11.90	12.73	6.7%
Peppermint oil	10.29	10.87	11.50	12.18	12.93	13.73	5.9%
Cornmint oil	10.75	11.36	12.02	12.73	13.49	14.32	5.9%
Citronella oil	7.95	8.46	9.01	9.60	10.25	10.95	6.6%
Spearmint oil	2.60	2.74	2.88	3.04	3.20	3.38	5.4%
Geranium oil	1.81	1.93	2.06	2.19	2.34	2.50	6.6%
Clove leaf oil	11.22	12.14	13.15	14.26	15.47	16.81	8.4%
Eucalyptus oil	18.96	20.27	21.70	23.25	24.94	26.71	7.1%
Jasmine oil	1.90	2.01	2.13	2.26	2.41	2.56	6.2%
Tea tree oil	14.31	15.42	16.64	17.97	19.43	21.03	8.0%

Rosemary oil	8.73	9.39	10.11	10.90	11.76	12.71	7.8%
Lavender oil	8.07	8.65	9.28	9.97	10.72	11.51	7.4%
Rose oil	3.87	4.02	4.17	4.34	4.51	4.78	4.3%
Coriander oil	0.78	0.83	0.88	0.94	1.01	1.08	6.8%
Other types*	4.73	4.93	5.10	5.24	5.33	5.45	2.9%
Total	253.18	268.86	285.80	304.11	323.94	345.42	6.4%

*Other types include sandalwood, cedar, chamomile, anise, sage, saffron, grape seed, ginger, thyme, cinnamon, ylang ylang, and basil oils.

Source: Related Associations/Institutes, Related Research Publications, Government Publications, and MarketsandMarkets Analysis

14.3.4 ESSENTIAL OIL MARKET, BY REGION

The global essential oil market is divided into five main regions, namely, North America, Europe, Asia Pacific, South America, and the Rest of the World (RoW). Globally, the number of patients with health issues is increasing rapidly. Consumers are now increasingly concerned about health & well-being, which has led to an increase in demand for essential oil food products, owing to the health benefits associated. Essential oils are used in different applications such as food & beverages, aromatherapy, home care, health care, and cosmetics & toiletries. Europe dominated the global essential oils market in 2020. North America was the second-largest market in the same year due to the increase in the consumption of food & beverage products. The Asia Pacific market is projected to grow at the highest rate due to the increase in demand for these products with essential oil ingredients.

Globally, the demand for essential oil products is high as food & beverage companies use these oils for the production of food products and provide quality & health benefits along with it. Thus, companies are investing to develop and commercialize products with essential oils. The major players in the market include Cargill (US), DuPont (US), and Givaudan SA (Switzerland).

TABLE 361 ESSENTIAL OILS MARKET, BY REGION, 2017–2020 (USD MILLION)

Region	2017	2018	2019	2020	CAGR (2017–2020)
North America	2,668.1	2,913.7	3,185.1	3,485.3	9.3%
Europe	2,880.0	3,099.8	3,339.8	3,602.1	7.7%
Asia Pacific	1,127.1	1,239.3	1,364.0	1,502.9	10.1%
South America	353.9	385.4	420.2	458.6	9.0%
RoW*	302.0	324.1	347.5	372.2	7.2%
Total	7,331.1	7,962.3	8,656.6	9,421.0	8.7%

*RoW includes the Middle East and Africa.

Source: Secondary Research, Expert Interviews, Related Research Publications, Government Publications, Press Releases, Company Website, Company Publications, and MarketsandMarkets Analysis

TABLE 362 ESSENTIAL OILS MARKET, BY REGION, 2021–2026 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
North America	3,825.5	4,175.0	4,561.0	4,987.6	5,459.7	6,123.3	9.9%
Europe	3,875.4	4,220.7	4,601.4	5,021.5	5,485.4	5,707.6	8.1%
Asia Pacific	1,665.3	1,819.2	1,989.3	2,177.6	2,386.0	2,765.9	10.7%
South America	501.0	547.9	599.8	657.2	720.8	791.4	9.6%
RoW*	395.9	429.2	465.8	505.9	550.0	599.5	8.7%
Total	10,263.2	11,192.1	12,217.3	13,349.8	14,602.0	15,987.7	9.3%

*RoW includes the Middle East and Africa.

Source: Related Research Publication, Government Publication, Company Press Release, Company Annual Report, Company Website, Company Publication, and MarketsandMarkets Analysis

Asia Pacific is estimated to be the largest market for essential oils, in terms of value, in 2021. The changing lifestyles, growing working-class population, and rising demand for essential oils in food & beverage products are the major factors contributing to the growth of the essential oils in the region. The dominance is attributed to the rising population, rapid urbanization, and increasing per capita income of consumers leading them to buy various fast-moving consumer goods such as packaged foods, dairy products, baked goods, and beverages.

TABLE 363 ESSENTIAL OILS MARKET, BY REGION, 2017–2020 (KT)

Region	2017	2018	2019	2020	CAGR (2017–2020)
North America	77.0	81.8	86.9	92.4	6.3%
Europe	71.9	75.7	79.9	84.3	5.5%
Asia Pacific	32.9	35.0	37.3	39.7	6.4%
South America	11.3	12.0	12.7	13.5	6.0%
RoW*	8.1	8.3	8.6	8.8	3.1%
Total	201.2	212.8	225.2	238.7	5.9%

*RoW includes the Middle East and Africa.

Source: Secondary Research, Expert Interviews, Related Research Publications, Government Publications, Press Releases, Company Website, Company Publications, and MarketsandMarkets Analysis

TABLE 364 ESSENTIAL OILS MARKET, BY REGION, 2021–2026 (KT)

Region	2021	2022	2023	2024	2025	2026	CAGR (2021–2026)
North America	98.4	104.9	111.9	119.5	127.8	136.7	6.8%
Europe	89.1	94.2	99.8	105.8	112.2	119.2	6.0%
Asia Pacific	42.3	45.2	48.3	51.6	55.3	59.2	7.0%
South America	14.3	15.2	16.2	17.3	18.4	19.0	5.8%
RoW*	9.1	9.4	9.7	9.9	10.2	11.2	4.3%
Total	253.2	268.9	285.8	304.1	323.9	345.4	6.4%

*RoW includes the Middle East and Africa.

Source: Related Research Publication, Government Publication, Company Press Release, Company Annual Report, Company Website, Company Publication, and MarketsandMarkets Analysis

14.4 BOTANICAL EXTRACTS MARKET

14.4.1 MARKET DEFINITION

Botanical extracts are used as ingredients in the food & beverage industry for the function of modifying flavors, aroma, or nutritive quality. Botanical extracts also impart properties such as enhanced flavor, extended shelf-life, and reduced microbial spoilage. They are extracted from the tissue of a plant, usually by treating them with a solvent, such as alcohol or water.

Botanical extracts are used in food items such as bakery & confectionery products and sauces & dressings. In the dairy industry, their applications include those in alcoholic beverages, carbonated soft drinks, iced tea, and flavored dairy products, while other industry applications include cosmetics and pharmaceuticals.

14.4.2 MARKET OVERVIEW

Botanical extracts are used for flavoring and seasoning in food & beverages. Botanical extracts contain antioxidant and antibacterial properties and considered as preventive health care with added nutrients.

The growth of the botanical extracts market is driven by the increased awareness about the side-effects of synthetic flavors in food. However, volatile prices of herbs and spices, coupled with the availability of substitute products, are the major restraints to the botanical extract market. In addition, identifying and marketing the new spices and herbs create more business opportunities for the botanical extracts market.

The demand for botanical extracts increases with the rise in demand for natural flavors. Many companies such as Döhler GmbH (Germany) and Frutarom Industries Ltd. (Israel) are focusing on natural herb & spice extracts and are increasingly used in the food & beverage industry. Consumers are becoming increasingly aware of the adverse effects of consuming foods and beverages containing synthetic ingredients. Moreover, many health organizations and food safety associations are recommending the prohibition of using various artificial flavorings in food and beverages. For instance, a petition was filed in 2015 with the US Food and Drug Administration (FDA) to ban eight synthetic flavorings, which are found in ice creams, baked goods, candy, and beverages. The substances used in those flavorings have been known to cause cancer in humans.

There is a significant rise in the number of incidences of health risks, including chest pain, dizziness, headaches, nausea, allergies, fatigue, seizures, and brain damage, since artificial flavors are extracted from petroleum and other chemicals. Increased awareness about the adverse effects of synthetic flavors among consumers is expected to drive the growth of the botanical extracts market during the forecast period.

14.4.3 BOTANICAL EXTRACTS MARKET, BY SOURCE

Botanical extracts are used in various food applications as natural flavoring products due to their aromatic and antimicrobial properties. The most commonly used botanical extracts include almonds, cloves, cinnamon, ginger, lemon, pistachio, spearmint, vanilla, and wintergreen. Increased demand for food & beverages, which support enhanced health functionality, is anticipated to exhibit significant growth for the botanical extracts market in the next five years.

The global botanical extracts market, by source, is segmented into spices, herbs, flowers, and tea leaves. Among all sources of extracts, the spices segment dominated the market for botanical extracts in 2016, followed by segments such as herbs and tea leaves. This is attributed to wide usage of spice extracts in various end-use industries such as food & beverage.

TABLE 365 BOTANICAL EXTRACTS MARKET, BY SOURCE, 2015–2022 (USD MILLION)

Source	2015	2016	2017	2022	CAGR (2017–2022)
Spices	1,141.4	1,248.1	1,366.3	2,170.1	9.7%
Herbs	1,010.4	1,099.1	1,196.4	1,849.8	9.1%
Flowers	370.9	402.4	437.1	667.4	8.8%
Tea leaves	795.0	855.7	921.0	1,345.2	7.9%
Total	3,317.6	3,605.3	3,920.7	6,032.5	9.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, Company Website, Company Publications, and MarketsandMarkets Analysis

TABLE 366 BOTANICAL EXTRACTS MARKET, BY SOURCE, 2015–2022 (TONS)

Source	2015	2016	2017	2022	CAGR (2017–2022)
Spices	68,365.5	73,912.6	79,968.1	119,936.2	8.4%
Herbs	60,597.0	65,165.3	70,108.8	102,359.9	7.9%
Flowers	22,109.1	23,720.8	25,468.0	36,760.3	7.6%
Tea leaves	48,003.8	51,063.2	54,302.0	74,737.7	6.6%
Total	199,075.4	213,861.9	229,846.8	333,794.2	7.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, Company Website, Company Publications, and MarketsandMarkets Analysis

14.4.4 BOTANICAL EXTRACTS MARKET, BY REGION

The global botanical extracts market was valued at USD 3,605.3 million in 2016, and is projected to grow at a CAGR of 9.0% from 2017 to 2022.

The demand for botanical extracts is increasing across the globe, owing to their growing application in various food products. The rise in preference for natural flavors among consumers is attributed to the growth of this market. There is a significant increase in the preference of natural flavors in food & beverages, supported by the rise in health concerns which in turn drives the demand for botanical extracts.

The botanical extracts market is experiencing positive growth around the globe, especially in the Asia Pacific region. This region accounted for a share of 21.0% of the global demand for botanical extracts in 2016. This can be attributed to the increase in health disorders such as diabetes, high cholesterol, and obesity in this region. For instance, according to the America Medical Association, China accounted for the highest number of diabetics in the world in 2015. Increase in demand for sugar-free confectionery items without compromising on taste is expected to create more opportunities for the botanical extracts market in the region.

The global botanical extracts market was dominated by North America in terms of value as well as volume in 2016. The Asia Pacific region is projected to exhibit rapid growth due to growing economies such as India and China.

TABLE 367 BOTANICAL EXTRACTS MARKET, BY REGION, 2015–2022 (USD MILLION)

Region	2015	2016	2017	2022	CAGR (2017–2022)
North America	1,171.7	1,276.6	1,391.5	2,178.3	9.4%
Europe	885.1	969.8	1,069.2	1,739.8	10.2%
Asia Pacific	692.0	758.4	831.2	1,331.4	9.9%
RoW	568.8	600.4	628.9	783.0	4.5%
Total	3,317.6	3,605.3	3,920.7	6,032.5	9.0%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, Company Website, Company Publications, and MarketsandMarkets Analysis

TABLE 368 BOTANICAL EXTRACTS MARKET, BY REGION, 2015–2022 (TONS)

Region	2015	2016	2017	2022	CAGR (2017–2022)
North America	67,228.9	72,884.0	79,045.2	120,697.4	8.8%
Europe	47,460.7	51,350.9	55,902.2	85,401.7	8.8%
Asia Pacific	44,844.1	48,495.3	52,440.0	78,549.2	8.4%
RoW	39,541.7	41,131.8	42,459.4	49,145.9	3.0%
Total	199,075.4	213,861.9	229,846.8	333,794.2	7.7%

Source: Related Research Publications, Government Publications, Company Press Releases, Company Annual Reports, Company Website, Company Publications, and MarketsandMarkets Analysis

15 APPENDIX

15.1 DISCUSSION GUIDE

Q. 1. According to our research, the plant extracts market has been segmented as below. Please validate the segmentation.

TYPE	APPLICATION	FORM	SOURCE	REGION
Oleoresins	Pharmaceuticals	Dry	Fruits & Vegetables	North America
Essential Oils	Dietary Supplements	Liquid	Herbs & Spices	Asia Pacific
Flavonoids	Food & Beverages	-	Flowers	Europe
Alkaloids	Cosmetics	-	-	Row
Carotenoids	Other Applications	-	-	-
Other Types	-	-	-	-

Other applications include personal care products, home care, feed, and toiletries.

Other types include anthocyanins, tannins, and lignin.

RoW includes Brazil, South America, Africa, and the Middle East

Primary source's viewpoint: _____

Q. 2. Please share your valuable inputs on the market size (2022) and CAGR (2022–2027) of the plant extracts market in terms of value and volume.

Primary source's viewpoint: _____

Q. 3. State the major drivers, restraints, opportunities, and challenges for the players involved in the plant extracts market.

Market Dynamics

Drivers

Restraints

Opportunities

Challenges

Primary source's viewpoint: _____

Q. 4. Kindly provide an approximate market share for the following types:

TYPE	MARKET SHARE (VALUE), 2021	CAGR (2022-2027)
Oleoresins		
Essential Oils		
Flavonoids		
Alkaloids		
Carotenoids		
Other Types		

Primary source's viewpoint: _____

Q. 5. Kindly provide an approximate market share for the following applications in the plant extracts market:

APPLICATION	MARKET SHARE (VALUE), 2021	CAGR (2022-2027)
Pharmaceuticals		
Dietary Supplements		
Food & Beverages		
Cosmetics		
Other Applications		

Primary source's viewpoint: _____

Q. 6. Kindly provide an approximate market share for the following sources in the plant extracts market:

SOURCE	MARKET SHARE (VALUE), 2021	CAGR (2022-2027)
Fruits & Vegetables		
Herbs & Spices		
Flowers		

Primary source's viewpoint: _____

Q. 7. Kindly provide an approximate market share for the following forms in the plant extracts market:

FORM	MARKET SHARE (VALUE), 2021	CAGR (2022-2027)
Dry		
Liquid		

Primary source's viewpoint: _____

Q. 8. Kindly rank the major producing and consuming regions (mentioned below) of plant extracts. Kindly share your views on the growth rate of five regions in the next six years (2022-2027).

REGION	RANK/PERCENTAGE CONTRIBUTION IN THE PLANT EXTRACTS MARKET	CAGR (2022-2027)
North America		
Asia Pacific		
Europe		
RoW		

Note: 0%–3% for low growth; 3%–6% for moderate growth; and 6%–10% for high growth.

Primary source's viewpoint: _____

Q. 9. Please provide your insights on the market size, and growth prospects across different countries/regions and mention their estimated market sizes. Also, please add other key countries or regions to the list if you are of the opinion that they need to be included.

COUNTRY	MARKET SHARE, 2021 (VALUE)	CAGR (2022-2027)
US		
Canada		
Mexico		
Germany		
UK		
Italy		
Spain		
France		
China		
India		
Japan		
Australia & New Zealand		

Primary's viewpoint: _____

Q. 10. Which are the regulatory authorities/organizations and acts that govern the quality standards for plant extracts in your country/region?

Primary source's viewpoint: _____

Q. 11. Name some of the key manufacturers of plant extracts and their respective market shares (either global or regional).

COMPANY	MARKET SHARE/RANKING, 2021
Givaudan (Switzerland)	
International Flavors & Fragrances, Inc. (US)	
Sensient Technologies Corporation (US)	
Symrise (Germany)	
Dohler (Germany)	
Synthite Industries Private Ltd (India)	
Indesso (Indonesia)	

Primary source's viewpoint: _____

Q. 12. Who are the main competitors of your company? How strong is your distribution base? Is it an international network or is just restricted to the domestic market?

Primary source's viewpoint: _____

Q. 13. What are the upcoming trends that will have a significant impact on this market in the future?

Primary source's viewpoint: _____

Q. 14. What are some of the regulations specific to the plant extracts market?

Primary source's viewpoint: _____

Q. 15. What is the impact of the existence of local players on the key players in the plant extracts market?

Primary source's viewpoint: _____

Q. 16. According to you, where does your organization stand in the macro quadrant?

STARS	EMERGING LEADERS	PARTICIPANTS	PERVASIVE COMPANIES
Company Name			

15.2 KNOWLEDGESTORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL

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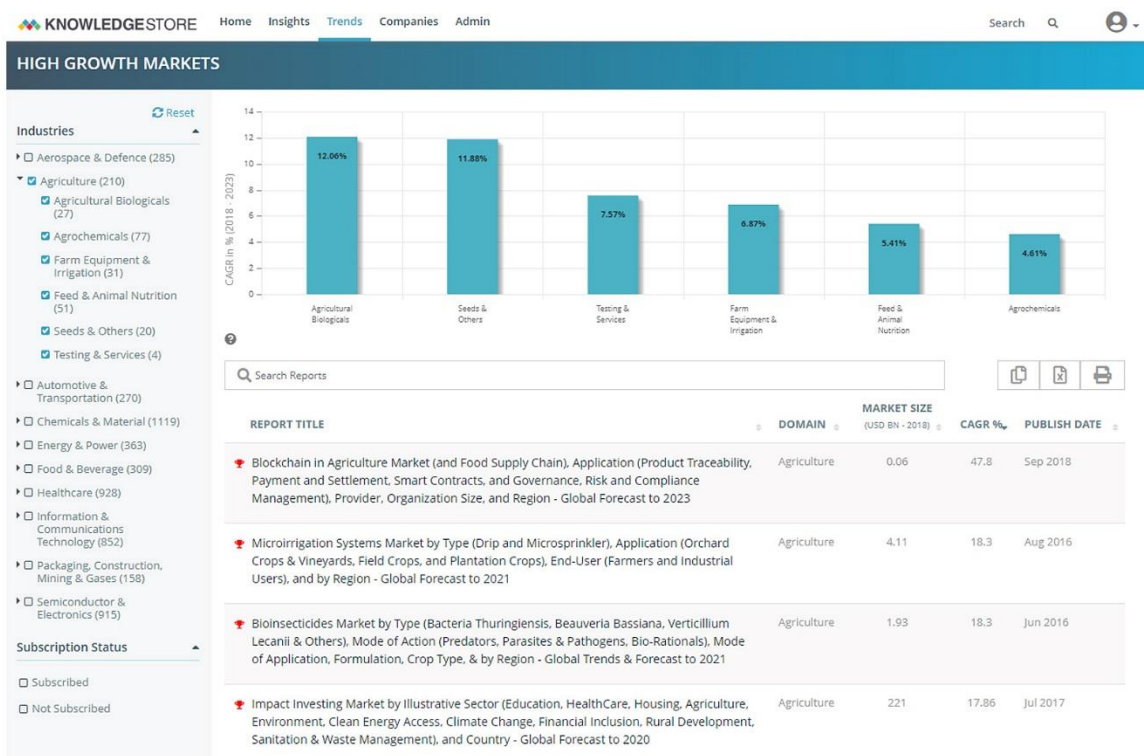
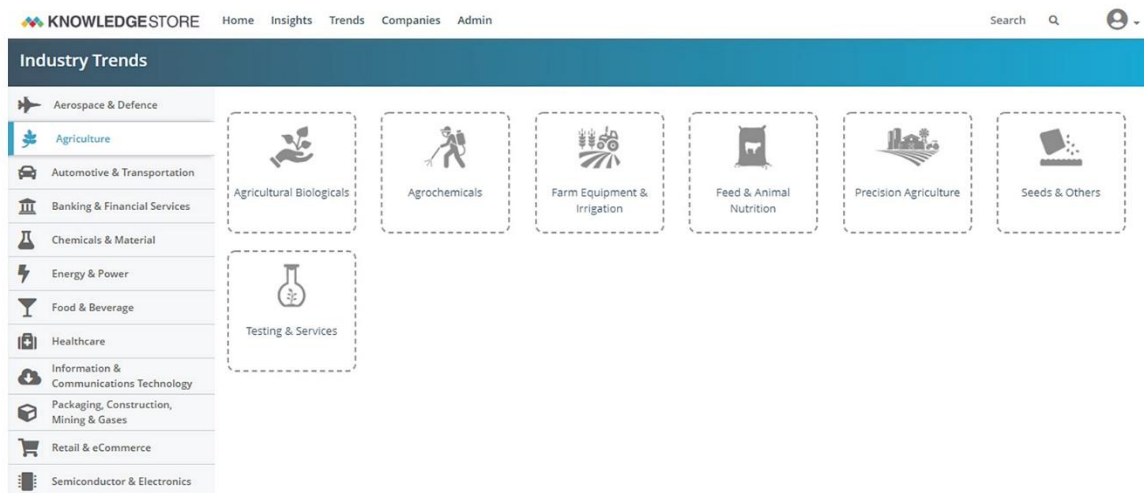
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	APAC	Europe	North America	Latin America
2021	14.15	37.85	21.10	3.28
2022	103.38	91.85	52.67	16.30
CAGR %	16.8%	19.9%	14.2%	17.8%

MARKETSANDMARKETS' KNOWLEDGESTORE: AGRICULTURE INDUSTRY SNAPSHOT



15.3 CUSTOMIZATION OPTIONS

With the given market data, MarketsandMarkets offers customizations according to the company's specific scientific needs.

The following customization options are available for the report:

Product Analysis

- Product Matrix, which gives a detailed comparison of the product portfolio of each company

Geographic Analysis

- Further breakdown of the South American plant extracts market, by key country
- Further breakdown of the Rest of European plant extracts market, by key country

Company Information

- Detailed analyses and profiling of additional market players (up to five)

15.4 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1	ESSENTIAL OILS MARKET – GLOBAL FORECAST TO 2026 By Type (Orange, Lemon, Lime, Peppermint, Citronella, and Others), Application (Food & Beverage, Cosmetics & Toiletries, Aromatherapy, Home Care, Health Care), Method of Extraction, and Region https://www.marketsandmarkets.com/Market-Reports/essential-oil-market-119674487.html	Mar 2021
2	BOTANICAL EXTRACTS MARKET - GLOBAL FORECAST TO 2022 By Source (Spices, Herbs, Flowers, Tea leaves), Application (Food: Bakery & Confectionery and Sauces & Dressings, Beverages: Alcoholic Beverages, Carbonated Soft Drinks, Iced Tea), Form, and Region https://www.marketsandmarkets.com/Market-Reports/botanical-extracts-market-3151938.html	Jun 2017

15.5 AUTHOR DETAILS

Syed H. Rizvi

Associate-Director-Food & Agriculture

Syed holds more than 13+ years of rich experience in the Food and Agriculture domain. He is actively involved in conducting strategic advisory research and spearheading turn-key projects for reputed international organizations in the agribusiness sector.

He has also been instrumental in providing solutions to clients pertaining to market sizing, supply chain analysis, competitive intelligence, market entry strategies, and so on. Syed has worked on several consulting projects sponsored by the USDA, European Export Councils, and Tier 1 Food and Agricultural Companies.

Contributors

Nishikant Yadav
Manager

Purna Chowhan
Assistant Manager

Aparna Cheela
Senior Research Analyst

Avi Sachan
Research Analyst

Poorva Gangal
Research Associate

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